When Is Label-Free Adaptation Knowable?

The Knowability Boundary of Label-Free Adaptation — Manuscript Edition

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Long-form companion to the working paper "When Is Label-Free Adaptation Knowable?" (K-Bound, v0.4). Every theorem, number, table, and figure in this manuscript reproduces from the anonymized repository linked in supplementary material. Nothing has been added beyond the exposition; nothing has been rounded, selected, or otherwise adjusted from the verified artifact manifests.

Abstract

Test-time adaptation can improve a model on unlabeled target data, yet the same objective that helps under one distribution shift silently hurts under another — and without labels the deployed system often cannot tell which case it is in. The field has poured its effort into stronger adaptation mechanisms; we ask the prior question those mechanisms skip: should the system adapt at all? We answer it by locating, exactly, the boundary between the shifts where that question is answerable from unlabeled evidence and the shifts where it is not.

That the general question is unanswerable without assumptions is not ours to claim: identifying out-of-distribution accuracy is provably as hard as identifying the optimal predictor and needs assumptions on the shift [13], and the decision theory of optimal abstain-versus-predict under shift is already established with an observable-quantity-versus-shift-budget threshold and a minimax certificate [27]. Our contribution is to *specialize and sharpen* that general picture to the adapt/freeze benefit-sign question, and it is threefold. *First*, we make the impossibility exact for this question: when two target worlds induce identical observable evidence but opposite-sign adaptation benefit, no label-free rule — at any sample size — can be correct in both, so abstention is forced [13, 27]; we sharpen the general bound to a Le Cam two-point identity and a closed-form witness on which the optimal committal error is exactly one. *Second*, we give a clean computable frontier and the machinery to act on it: a finite-sample adapt/freeze/abstain certificate that controls the false-adapt and false-freeze rates at a user-chosen α, and an exact benefit-sign criterion — adaptation benefit is identifiable from observables if and only if an observable margin $|M|$ exceeds the calibration-drift budget β on the disagreement region — specializing the observable-versus-budget threshold of [27] to a form with a computable margin, with the three-way adapt/freeze/abstain rule as the unique maximal sound certificate. This face also *unifies* the label-free accuracy estimators (ATC, difference-of-confidence, generalized disagreement equality, confidence-optimal transport, test-time-accuracy estimation, agreement-on-the-line): they are the zero-budget ($\beta = 0$) special case, sound exactly when the calibration drift γ vanishes. *Third*, we make the theory deployable as KGA (Knowability-Guided Adaptation), a mechanism-agnostic wrapper that reads label-free evidence and returns ADAPT, FREEZE, or ABSTAIN around any candidate.

Empirically, KGA behaves as a *safety layer*, not an accuracy booster, and we report it as such. On a pre-registered CIFAR-10-C stress grid — where catastrophic, detectable harm is common — KGA beats both always-adapt and always-freeze for Tent and EATA with zero false-adapt decisions, and ties the collapse-resistant SAR. On natural shifts (Camelyon17, Office-Home) it does no harm and avoids overclaiming; on undetectable-harm streams (iWildCam) it freezes and prevents damage rather than winning. Across anomaly routing, online CIFAR-10, and these audits, KGA tracks the theory: adapt where benefit is certified, freeze where harm is certified, abstain where the evidence cannot tell. K-Bound is not universal. Its value scales with how often harmful adaptation is frequent, detectable, and costly — exactly the quantity the theory characterizes.

How to Read This Manuscript

This monograph presents the K-Bound research program at a depth that the 20-page working paper cannot sustain. Theorems and experiments are stated faithfully from the paper; the additional length is entirely expository — motivation, worked examples, proof commentary, and explicit connections between the abstract theory and its empirical realizations. A reader who has the paper in hand can use the chapter headings as an index; a reader coming to K-Bound fresh should read Chapters 1 and 2 first to orient on the problem and its neighbours, then follow Chapters 3 through 5 for the theory, and Chapters 6 through 8 for the instantiation and experiments.

Chapter 1: Introduction. Sets up the TTA double-edge problem and the central question of knowability, introduces the adapt/freeze/abstain trichotomy, credits the general impossibility and abstention decision theory to prior work [13, 27], and states the three contributions that the manuscript opens and closes on — the benefit-sign specialization of that impossibility, the exact computable frontier and its certificate, and KGA as a deployable safety layer. Previews the theoretical results and the headline experimental numbers, and gives an honest statement of scope.

Chapter 2: Background and Related Work. A thorough review of the five lines of work adjacent to K-Bound: TTA mechanisms, TTA failure and collapse studies, guarded/protected adaptation, label-free accuracy estimation, unsupervised risk and identifiability, selective prediction and abstention, conformal and anytime-valid inference, and covariate-shift theory. Closes with a positioning table and a precise statement of the gap K-Bound fills.

Chapter 3: Problem Formulation. The formal setup: source and target distributions, the adaptation benefit $\Delta = R_T(f_0) - R_T(f_a)$, observable evidence Z, the three regimes, and the key definitions of risk alignment and knowability. Includes the "label-free" clarification box and the knowability phase diagram.

Chapter 4: Impossibility and the Unknowable Regime. The first two theoretical results, specializing to the benefit-sign question the general label-free impossibility already established in the literature [13, 27]. Theorem 4.1 (non-identifiability with explicit witness) and its Le Cam quantitative strengthening make the unknowable regime exact for adapt/freeze. Theorem 4.3 gives the Bayes-optimal gate and the plug-in regret decomposition, which justifies abstaining in the low-margin band.

Chapter 5: Certificates and Positive Regimes. The remaining three theoretical results.

Theorem 5.1 is the finite-sample adapt/freeze/abstain certificate with a controlled false-adapt rate; Corollaries give forced abstention and sample complexity. Theorem 5.3 proves a knowable positive regime under covariate shift. Theorems 5.4, 5.5, and 5.6 reduce sign-of-difference identifiability to an ordinal accuracy comparison on the observable disagreement region for binary, multiclass, and regression losses. Conjecture 5.1 states the remaining open piece.

Chapter 6: The KGA Algorithm. The KGA (Knowability-Guided Adaptation) wrapper: evidence extraction, benefit estimation, conformal radius, and the three-way decision. Assumptions, computational cost, and the relationship to any underlying TTA candidate.

Chapter 7: Experiments I — Anomaly Routing. The 123-task clean suite, the harmful-regime refusal experiment, the controlled mixed-regime study, and the non-identifiability witness (both empirical and exact). Evidence ablations and the α-sweep safety/coverage tradeoff.

Chapter 8: Experiments II — Online TTA and CIFAR-10-C. The online non-stationary CIFAR-10 cross-regime test, the per-corruption CIFAR-10-C helpful-dominated protocol, and the pre-registered CIFAR-10-C stress grid. Breadth results on 62 additional harmful-dominated label-free tasks.

Chapter 9: The Multimodal Instantiation. Reliability-gated multimodal fusion as the worked, code-validated special case: the ten-result multimodal theorem stack (T1–T9, GDR) and its mapping to the general K-Bound theorems.

Chapter 10: Discussion, Limitations, and Conclusion. When to deploy K-Bound; honest scope and limitations; what would falsify the framework; and the research frontier (ImageNet-C / ViT scale, natural-shift benchmarks, the open conjecture). Closes by restating the same three contributions the book opened on — the unknowable floor, the certificate and its frontier, and KGA as a safety layer — against the verified evidence chain.

Appendix A: Reproducibility. One-command reproduction; data provenance and checksums; environment and determinism notes.

Appendix B: Extended Tables. The full 65-cell CIFAR-10-C per-corruption-by-severity breakdown and auxiliary result tables referenced from the main chapters.

Appendix C: Notation Reference. A single-page glossary of all symbols used in the monograph.

Readers interested primarily in the *theory* can read Chapters 3–5 as a self-contained unit. Readers interested primarily in the *practical algorithm* can read Chapter 6 with only the informal summaries of Theorems 4.1 and 5.1. Both paths converge at Chapter 10, which synthesizes the implications for deployment.

Contents

List of Figures

List of Tables

Chapter 1

Introduction

1.1 The Deployment Problem

A model trained on one distribution is deployed into a world that has moved. The training images came from one hospital's scanner; the deployed scanner is newer, slightly sharper, with different noise characteristics. The training network saw balanced classes; the test stream is skewed toward one pathology. The training corpus covered one domain; the production queries drift slowly toward a neighbouring domain as user behaviour changes. This pattern — a mismatch between the distribution under which a model was optimized and the distribution under which it operates — is the central operational reality of deployed machine learning.

Test-time adaptation (TTA) is the field's answer. Rather than retraining from scratch whenever the deployment distribution shifts, TTA updates the model on the unlabeled target data that arrives at inference time. The appeal is immediate: labels are expensive or unavailable in deployment, yet the raw inputs carry information about how the distribution has moved. Entropy minimization over batch-normalization parameters [56], self-supervised auxiliary objectives that continue training at test time [50], source-free adaptation by information maximization, and importance-weighted empirical risk minimization [48] are all instances of the same basic idea: use the unlabeled stream to close the performance gap.

1.2 The Double-Edged Nature of Test-Time Adaptation

Adaptation is, however, double-edged in a precise sense. The same unlabeled objective that recovers accuracy under one kind of distribution shift can degrade it under another — and, critically, without labels the deployed system cannot in general determine which situation it is in.

Consider entropy minimization. Under mild covariate shift — the scanner statistics change, but the decision boundary between pathologies does not — minimizing the entropy of predictions sharpens the boundary toward where the new data lives. The adaptation is helpful. But now suppose the distribution shift is a label shift: the prevalence of one class has collapsed to near zero in the target stream. A model that minimizes entropy on this stream will be rewarded for making all-or-nearly-all predictions of the dominant class; it converges to a degenerate solution. The adaptation is harmful, and in the worst case the model collapses entirely. The recent literature on TTA failure modes [62, 41, 40] documents exactly these collapse phenomena: Tent's continual entropy minimization catastrophically fails on adversarial or highly non-stationary streams; EATA's Fisher penalty slows but does not eliminate accumulation of damage; SAR's sharpness-aware resets abort incipient collapses but cannot prevent them in every regime.

The field's response has been to propose sturdier mechanisms. If entropy minimization is fragile, add sample selection to filter high-entropy inputs [39]. If selection is not enough, add sharpness-aware updates that are less likely to collapse [40]. If batch normalization updates are the failure mode, use a fully parameter-free approach [6]. If a single method is fragile, reset after detecting instability [40] or continually re-anchor to the source [57].

Each of these improvements is genuine within its scope. But they all share a common architecture: they commit to adapting, and they try to make that commitment safer. None of them poses the prior question: *given the available evidence, should the system adapt at all?*

1.3 The Central Question

This monograph studies that prior question.

> *Can observable test-time evidence separate helpful, harmful, and unknowable regimes of label-free adaptation, yielding a computable rule for when a system should adapt, freeze, or abstain?*

The question is not rhetorical. It has a substantive, partly negative answer that we make precise. In some regimes — covariate shift, certain kinds of label shift, situations where the frozen and adapted models disagree on inputs whose label orientation is detectable from label-free signals — the answer is yes. In others, the same observable evidence is compatible with both helpful and harmful adaptation, and no label-free rule can certify the correct choice in both worlds. The boundary between these cases is what we call the *knowability boundary* of label-free adaptation, and the central mathematical object is a trichotomy: at any given evidence value, adaptation is either *knowably helpful*, *knowably harmful*, or *unknowable*.

Why does this matter for deployment? Consider a safety-critical setting: an autonomous vehicle's perception stack or a medical imaging pipeline. In such settings, the cost of silent degradation is high and the cost of abstaining — falling back to the frozen model and flagging for human review — is manageable. A system that can identify when it is in an unknowable regime and refuse to adapt is strictly safer than one that always adapts or always freezes, even if its average-case performance is identical. The value of the knowability boundary is not primarily about improving mean accuracy; it is about converting the adapt/freeze decision from an irreducible coin flip into a certified, controlled action.

The same logic applies to model monitoring. A production team that can determine, from unlabeled deployment data alone, whether a proposed update will help or hurt does not need ground-truth labels to make a safe deployment decision. The knowability boundary tells them exactly when that label-free determination is possible and when it is not.

1.4 The Trichotomy and Its Formalization

We formalize the trichotomy as follows. Fix a loss ℓ, a frozen model f_0, and an adapted/gated candidate f_a. The *adaptation benefit* is

$$\Delta \;=\; R_T(f_0) - R_T(f_a),$$

where $R_T(f) = \mathbb{E}_{P_T}[\ell(f(X), Y)]$ is the target risk. A positive Δ means adapting helps; a negative Δ means it hurts. *Observable evidence* $Z = \phi(X_{1:m}, f_0, f_a)$ is any statistic computable without target labels: prediction entropy, softmax confidence, score-distribution drift, detector disagreement, predicted-class balance, update norm, or density-ratio diagnostics. The conditional benefit $\Delta(z) = \mathbb{E}[\ell(f_0(X), Y) - \ell(f_a(X), Y) \mid Z = z]$ is the benefit conditioned on evidence $Z = z$.

The regime at evidence value z is:

Knowably helpful if Z certifies $\Delta(z) > 0$; the correct action is ADAPT.

Knowably harmful if Z certifies $\Delta(z) < 0$; the correct action is FREEZE.

Unknowable if the same z is compatible with both $\Delta(z) > 0$ and $\Delta(z) < 0$; the worst-case-optimal action is ABSTAIN.

The key quantity is the *sign* of $\Delta(z)$, not its magnitude. Estimating the sign requires strictly less information than estimating the risk $R_T(f)$ itself, so the knowability boundary lies strictly outside the identifiability boundary for risk estimation — but not everywhere: there are situations where even the sign is unrecoverable from label-free evidence. The rest of this chapter pins down exactly where that boundary falls, and the next section states the three results that do it.

1.5 Contributions

The starting point is a fact we do not claim as ours. That the general label-free question is unanswerable without assumptions on the shift is already established: Garg et al. prove that identifying out-of-distribution accuracy is as hard as identifying the optimal predictor, and therefore requires assumptions on the shift [13]; and Kalai and Kanade give the decision theory of optimally abstaining versus predicting under a shift, with a threshold comparing an observable quantity to a shift budget and a minimax certificate for the abstain-or-predict choice [27]. These two results own the general impossibility and the general abstention decision theory. Our three contributions *specialize and sharpen* that picture to the benefit-sign adapt/freeze question; they answer, in turn, when the adapt/freeze question is fundamentally unanswerable from unlabeled evidence, when it provably is answerable and how to act on it, and what happens when the resulting rule meets real deployment streams. The manuscript opens on these three and closes on them (Chapter 10); the paragraphs below state each and the formal machinery it rests on.

Contribution I: The benefit-sign specialization of the impossibility.

We make the general label-free impossibility exact for the adapt/freeze decision. The general fact — that no label-free rule can identify a shift-dependent quantity without assumptions on the shift — is due to [13, 27]; what we add is its exact form for the sign of $\Delta = R_T(f_0) - R_T(f_a)$. For any (possibly randomized) label-free decision rule $g(Z)$, there exist two target worlds P_T^1 and P_T^2 with identical observable-evidence laws and opposite-sign benefits (Theorem 4.1). Since the rule cannot distinguish the worlds, its actions are identically distributed in both, and the worst-case-optimal action is to abstain — the specialization, to the benefit-sign question, of the abstain-under-shift decision theory of [27]. The construction is non-vacuous: we exhibit it explicitly as $X \sim \mathcal{N}(0, 1)$ under both worlds, $f_0(x) = \mathbf{1}[x > 0]$, $f_a(x) = \mathbf{1}[x < 0]$, and labels flipped between the worlds. A Le Cam two-point quantitative strengthening (Chapter 4) sharpens the general hardness to an exact identity: the optimal committal error equals $1 - \mathrm{TV}$ of the evidence laws, which is 1 in the witness and decays to zero as the worlds become separable. Numerically, the KGA validator confirms 100% abstention on the witness and exact tracking of the Le Cam floor across the total-variation sweep. This is the result that turns abstention, in this specific setting, from a heuristic fallback into a forced, principled action.

This contribution rests on a prior formalization, developed in full in Chapter 3: we cast label-free TTA as the adapt/freeze/abstain trichotomy, define risk alignment for observable evidence, and define the three regimes precisely. Where the TTA literature designs the adaptation mechanism f_a, we ask when the sign of $R_T(f_0) - R_T(f_a)$ is recoverable from label-free observations at all — an instance of the general question [13, 27] for which the answer turns out to have a clean computable form.

Contribution II: The exact benefit-sign frontier, its certificate, and the unification.

We prove when a label-free system provably can decide, and locate the boundary with a clean computable criterion. The general threshold — commit when an observable quantity beats the shift budget, else abstain — is that of [27]; our sharpening is to give that threshold, for the benefit-sign question, an explicit computable form and to prove it exact. The key object is the *observable disagreement region* $D = \{x : f_0(x) \neq f_a(x)\}$, observable without labels since f_0 and f_a are known maps of X. On D we prove that $\operatorname{sign} \Delta$ equals an ordinal accuracy comparison — adapting helps if and only if the adapted model exceeds chance accuracy on D (Theorems 5.4–5.6, for binary, multiclass, and regression losses). This yields the exact benefit-sign frontier: writing M for the observable margin and β for the calibration-drift budget on D, the sign is *identifiable from observables if and only if* $|M| > \beta$. The three-way rule — ADAPT above the frontier on the helpful side, FREEZE above it on the harmful side, ABSTAIN within $|M| \leq \beta$ — is the *unique maximal sound certificate*: any rule that commits anywhere inside $|M| \leq \beta$ is unsound (some consistent world flips the sign), and any rule that abstains anywhere with $|M| > \beta$ forfeits coverage it could soundly claim.

This frontier also *unifies* the label-free accuracy-estimation literature. ATC [13], difference-of-confidence, generalized disagreement equality, confidence-optimal transport, test-time-accuracy estimation, and agreement-on-the-line [2] all implicitly read the benefit sign off an observable at *zero budget*: they are the $\beta = 0$ face of the frontier, sound exactly when the calibration drift γ vanishes and silently unsound otherwise. Our criterion is the same object with the budget restored, which is what converts an estimator into a certificate.

The operational counterpart is a finite-sample adapt/freeze/abstain certificate with false-adapt control (Theorem 5.1). Under a label-free estimator $\widehat{\Delta}(z)$ with a calibrated radius $\varepsilon(z)$ satisfying $\mathbb{P}[|\widehat{\Delta}(z) - \Delta(z)| \leq \varepsilon(z)] \geq 1 - \alpha$, the rule

$$\text{ADAPT if } \widehat{\Delta} - \varepsilon > 0, \quad \text{FREEZE if } \widehat{\Delta} + \varepsilon < 0, \quad \text{ABSTAIN otherwise}$$

controls the false-adapt and false-freeze rates at α. A corollary shows that in the unknowable two-point regime the certificate *must* abstain with probability at least $1 - 2\alpha$, joining this contribution to Contribution I: where the evidence cannot tell, the certificate is forced to abstain, not merely inclined to. A sample-complexity corollary shows the certificate commits once $n \gtrsim (\sigma^2/\Delta^2)\log(1/\alpha)$ paired benefit samples are available, with an empirical-Bernstein discharge for bounded benefits; an anytime-valid e-process variant (Appendix A) extends it to streaming TTA with no multiplicity correction. A companion positive regime (Theorem 5.3) shows that under covariate shift ($P_S(Y \mid X) = P_T(Y \mid X)$) importance weighting yields a consistent target-risk estimator, so $\operatorname{sign}\Delta$ is identifiable outright. Ablations confirm that detector disagreement is the single most important evidence feature, directly supporting the theory. Conjecture 5.1 states the one remaining open piece: the label-free *bracketing* of the ordinal comparison in the multiclass and regression cases.

The one-bit quantification, honestly scoped. Our freshest result quantifies exactly *how much* label-free evidence falls short. In the TTA/panel setting, under a stated structural hypothesis H (conditional error-independence plus per-class symmetric accuracy on a panel of predictors), label-free evidence identifies everything about the decision problem *except a single global sign bit*: a label-complement twin shares the entire evidence law at every moment order yet reverses every benefit sign, and every published label-free identifying device (ATC-style confidence thresholds, majority-above-chance, agreement-on-the-line slopes) is exactly a selector of that one bit. We are deliberate about the scope of this count. It is a theorem *in this setting under* H; we do *not* claim "one bit" universally. Outside H, or for a general split-observable functional, the irreducible supplement is problem-dependent — under affine sign–budget ambiguity it is an interval rather than a single bit — and we say so rather than over-generalize.

Split-observability: the general form of what specializes. The pieces above are the benefit-sign instance of a world-agnostic pattern, which we name and prove at that level of generality. Call a benefit functional Δ *split-observable* if its sign factors as $\text{sign}(\Delta) = \text{sign}(M + \gamma)$ with M observable from label-free evidence and the residual γ bounded by a budget, $|\gamma| \le \beta$. For *any* split-observable functional, the impossibility (matched-evidence worlds force abstention), the finite-n floor, the maximal sound certificate, and the minimax/necessity statements all hold — these are properties of the split, not of TTA, and they are world-agnostic. Our benefit-sign setting, with disagreement-region reduction supplying M, is the canonical instance. We are careful about the reach of the sharper claims: the exact $|M| > \beta$ criterion and the one-bit count are proved for the benefit-sign functional (the latter under H), *not* asserted for arbitrary split-observable functionals, where the residual structure can differ.

Contribution III: KGA, a deployable safety layer.

We make the theory a deployable rule and report its honest empirical story. We instantiate the framework as KGA (Knowability-Guided Adaptation), a decision wrapper placed around any TTA candidate without modifying it: it extracts label-free evidence Z, estimates the benefit $\widehat{\Delta}$ with a gradient-boosted regressor trained leave-one-task-out, forms a split-conformal radius ε, and returns the three-way decision with false-adapt at most α. KGA is released as a pure-numpy package (`kga/`) importable as `from kga import KGA` and served at a `POST /decide` endpoint; all code, configs, and result artifacts are public at https://github.com/Pratikn03/AutoML_Flagship_V8 [28]. Chapters 6 and 7–8 give the algorithm and the complete evaluation.

The empirical finding is deliberately modest and we state it plainly: KGA is a *safety layer, not an accuracy booster.* It is the only policy that stays near-oracle in both helpful and harmful regimes, which is exactly what the trichotomy predicts — but it earns a clean beats-both over both trivial policies only where catastrophic, detectable harm is common. On the pre-registered CIFAR-10-C stress grid it beats always-adapt and always-freeze for Tent and EATA at 0% false-adapt and ties collapse-resistant SAR; on natural-shift audits it does no harm rather than winning outright. The nine experiments below substantiate this, regime by regime.

1. *123-task anomaly-routing benchmark* (ADBench tabular, image-OOD, text, cyber, fraud): KGA abstains where the true benefit is near zero (mean $|\Delta| = 0.021$ on abstained vs. 0.132 on acted tasks) and correctly identifies that the suite is helpful-dominated (adapt precision 0.90, coverage 17%).

2. *Harmful-regime refusal*: with a fusion candidate that hurts on 80% of tasks, KGA refuses it almost everywhere, cutting regret-to-oracle by $\approx 11\times$ (from 0.0214 to 0.0019) while matching

the safe baseline AUROC (0.748).

3. *Controlled mixed-regime study* (369 instances, three conditions): KGA achieves adapt precision 0.935 and beats always-freeze by 0.10 AUROC while tying always-adapt in a mild-harm domain.

4. *Clean non-identifiability witness*: the exact Theorem 4.1 construction; KGA abstains on 100% of instances while a forced-commit rule pays regret 0.51.

5. *Statistical rigor over 8 seeds*: multi-seed mean AUROC 0.686 ± 0.003 with paired t-tests confirming KGA is significantly above always-freeze ($p = 7.3 \times 10^{-11}$) and significantly below always-adapt ($p = 5.1 \times 10^{-5}$) in the mild-harm domain.

6. *Regression under covariate shift*: synthetic regression with importance weighting hurting on 79% of instances; KGA refuses it entirely and matches the oracle MSE (0.251) vs. always-adapt (0.338).

7. *Online non-stationary CIFAR-10 TTA*: across a helpful-dominated and a harm-dominated continual stream, KGA achieves the highest mean accuracy across regimes (0.490 vs. always-freeze 0.456 vs. always-adapt 0.444), with always-adapt collapsing to 0.339 in the harm regime.

8. *Per-corruption CIFAR-10-C helpful-dominated control* (65 cells, 13 corruptions $\times$ 5 severities): adaptation is overwhelmingly helpful (false-adapt rate 0.015); KGA matches always-adapt at equal safety, confirming that the certificate does not impose unnecessary cost in benign regimes.

9. *Pre-registered CIFAR-10-C stress grid* (432 conditions/method, harmful base rate 0.16–0.34): KGA beats both trivial policies for Tent and EATA (regret 0.0016 vs. 0.009 / 0.12) at 0% false-adapt, and ties collapse-resistant SAR.

1.6 Honest Scope

A safety claim is only as good as its stated limits, so we draw them in both theory and experiment before any reader has to.

On the theory side, the impossibility result (Theorem 4.1) is a *specialization*, not a discovery: the general fact that label-free identification of a shift-dependent quantity requires assumptions on the shift is due to Garg et al. [13], and the accompanying abstain-versus-predict decision theory to Kalai and Kanade [27]. Our purpose here is to *make it exact for the benefit-sign question* — to justify the abstain action and to sharpen the boundary quantitatively via the Le Cam two-point lower bound. The certificate (Theorem 5.1) is proved conditional on a valid label-free interval estimator; the empirical-Bernstein discharge requires exchangeability of the paired benefit sample.

The sign-of-difference characterization is complete for binary, multiclass, and regression losses; the label-free bracketing of the ordinal comparison in the multiclass and regression cases remains an open conjecture.

On the experimental side, the stress-grid knockout (beating both trivial policies at 0% false-adapt) holds for Tent and EATA on the CIFAR-10-C pre-registered grid; KGA ties collapse-resistant SAR rather than beating it. In per-corruption CIFAR-10-C (helpful-dominated by construction) and within a single-stream online regime, KGA ties the better fixed policy rather than strictly beating it. The decisive advantage is *robustness*: it is the only policy that is near-oracle in both the helpful and harm regimes, which is exactly what the trichotomy predicts. Larger-scale corroboration on CIFAR-100-C, ImageNet-C, and natural-shift benchmarks such as WILDS [29] is future work.

The practical implication is direct. KGA's value scales with how often catastrophic, detectable harm occurs in the deployment stream — precisely the quantity the theory characterizes. In genuinely benign domains where adaptation almost always helps, the certificate is safety insurance at controlled cost. In collapse-prone regimes — small or class-imbalanced batches, single-class label-shift streams, aggressive update schedules, long non-stationary runs — the certificate delivers a real win.

1.7 Chapter Roadmap

Chapter 2 reviews the five adjacent lines of work and positions K-Bound precisely against each. Chapter 3 develops the formal setup: distribution families, the adaptation-benefit definition, observable evidence, and the three regime definitions. Chapters 4 and 5 present the five core theoretical results. Chapter 6 describes the KGA algorithm, its assumptions, and its cost structure. Chapters 7 and 8 present the experimental evaluation across anomaly-routing and online TTA benchmarks. Chapter 9 presents the anonymized reliability-gated multimodal instantiation as a fully code-validated worked example of the general framework. Chapter 10 synthesizes the deployment implications, states the limitations, specifies what would falsify the framework, and returns to the three contributions of this chapter against the verified evidence chain. Appendix A gives the full reproducibility specification. Appendix B contains extended tables. Appendix C is a notation reference.

The next chapter places K-Bound among its neighbours, building from the five adjacent lines of work to the precise gap none of them closes.

Chapter 2

Background and Related Work

Five lines of work surround K-Bound, and each comes within reach of its question without asking it. The recurring distinction — the one the theory formalizes — is between *improving* an unlabeled objective and *certifying the sign* of its benefit before acting. We walk the five lines in that order, and the gap widens as we go: TTA mechanisms adapt without asking whether they should; the failure literature shows that they often should not; guarded adaptation reacts to harm once it has begun; label-free estimation and identifiability theory ask the right question but answer a harder one or stop at asymptotics; selective prediction and conformal inference supply the abstention and interval machinery but not the decision. By the final section every piece K-Bound needs has appeared in isolation — including the general label-free impossibility and the abstain-under-shift decision theory, which are due to prior work [13, 27] and which we specialize rather than rediscover. What no prior work assembles is the single pre-adaptation adapt/freeze/abstain certificate that couples an unknowable region — made exact for the sign of the adaptation benefit — to a computable frontier and a false-adapt bound. We close with a positioning table and a precise statement of that gap.

2.1 Test-Time Adaptation: Mechanisms and Methods

Test-time adaptation originated as a practical answer to the distribution-shift problem: rather than retraining, use the unlabeled target stream to adjust the deployed model. The field has developed along several distinct axes, each representing a different hypothesis about what makes TTA fragile and how to make it robust.

Entropy minimization. **Tent** [56] is the canonical entropy-minimization method. At each test batch, Tent minimizes the Shannon entropy of the model's softmax predictions over only the batch-normalization affine parameters, keeping the rest of the network frozen. The intuition is that predictions on target data should be confident (low entropy), and adjusting normalization statistics accomplishes this with minimal risk of catastrophic forgetting. On i.i.d. corruption benchmarks such as CIFAR-10-C [21] and ImageNet-C, Tent yields substantial accuracy improvements over the frozen baseline.

The fragility of pure entropy minimization, however, is well documented. Tent applied continually to non-stationary or class-imbalanced streams accumulates normalization-parameter damage; once the predictions collapse to a single class, entropy continues to decrease while accuracy plummets. Our stress-grid experiments (Chapter 8) reproduce this collapse in the harm-dominated online CIFAR-10 regime, where always-adapt with Tent falls to 0.339 accuracy while a frozen model holds 0.440.

Batch-normalization statistics adaptation. **Schneider et al.** [45] showed that a large fraction of the performance gap on corruption benchmarks can be closed simply by re-estimating batch-normalization running statistics from a mixture of source and target data, without any gradient update. This observation motivated a family of statistics-only approaches that avoid the optimization risk of entropy minimization entirely, at the cost of needing a sufficient target batch size to estimate stable statistics. **NOTE** [15] extends this direction to non-i.i.d. streams using class-conditional normalization, addressing the label-shift fragility of global statistics.

Filtering and Fisher regularization. **EATA** [39] addresses Tent's catastrophic-forgetting tendency with two additions: an entropy filter that rejects unreliable (high-entropy) samples from the update, and a Fisher-information-based regularization penalty that penalizes updates proportional to their importance for source-domain performance. The entropy filter prevents the worst collapse triggers from entering the update gradient; the Fisher penalty ensures the normalization parameters do not drift too far from their source-optimal values. EATA retains substantial improvement over the frozen baseline on standard benchmarks while partially protecting against the forgetting that pure Tent suffers.

Sharpness-aware updates and resets. **SAR** [40] adds sharpness-aware entropy minimization combined with an online reset mechanism. Sharpness-aware optimization minimizes an upper bound on the worst-case perturbation of the loss landscape, producing parameter updates that land in flat regions less susceptible to subsequent drift. The reset mechanism monitors a model-divergence criterion and resets the normalization parameters to their source values if divergence exceeds a threshold, aborting incipient collapse. SAR is notably more robust than Tent or EATA on the continual and non-stationary regimes that provoke collapse; our experiments confirm this, with SAR's harmful base rate (16% on the stress grid) substantially lower than Tent's (34%) and EATA's (27%). This robustness is also the reason K-Bound ties rather than beats SAR: when the candidate already self-protects, the certificate's additional value is limited.

Source-free and continual adaptation. **SHOT** trains a source-free adaptation by information maximization and self-supervised pseudo-labeling, enabling adaptation when the source data is unavailable at deployment. **CoTTA** [57] addresses the continual adaptation setting by stochastic restoring: with some probability, parameters are reset to their source-pre-trained values, preventing irreversible forgetting across a long non-stationary stream. **TTT** [50] augments the source training with a self-supervised auxiliary task and continues training that task at test time, adapting the shared feature extractor. **MEMO** [60] takes a different route: for each test point it generates multiple augmented views and minimizes the marginal entropy across augmentations, adapting per-sample rather than per-batch.

Parameter-free approaches. **LAME** [6] is at the opposite extreme: it requires no parameter update at all, instead post-processing the model's predictions using a Laplacian regularization that encourages smoothness across the target batch. Its robustness to collapse is guaranteed by construction, since there are no parameters to corrupt, but it cannot recover the same accuracy gains as gradient-based methods when the distribution shift is large.

What all TTA methods share. Every method in this section, whether gradient-based or gradient-free, whether it uses resets or regularization or filtering, makes the same architectural commitment: *it adapts*. The question of whether adaptation is the right action on a given deployment batch is not posed; the decision is implicit in the method's deployment. K-Bound is mechanism-agnostic with respect to this entire class of methods. It wraps any such candidate — we evaluate Tent, EATA, and SAR — and makes the adapt/freeze/abstain decision at the certificate level, before the adaptation mechanism executes.

2.2 TTA Failure, Collapse, and Evaluation

If the previous section's methods all commit to adapting, the next question is how often that commitment backfires. A parallel line of work answers it: by cataloguing the conditions under which TTA fails destructively and showing that those conditions are not exotic but the realistic ones. This literature supplies the empirical motivation for K-Bound's central premise — that harmful adaptation is common enough to be worth certifying against — and the stress-test design that our experiments inherit.

Pitfalls and failure modes. **Zhao et al.** [62] systematically catalogued the pitfalls of test-time adaptation evaluation: sensitivity to the batch-size regime, stream composition, update budget, and the distinction between i.i.d. and temporally correlated streams. Their analysis showed that many published TTA results are obtained in regimes where adaptation is effectively guaranteed to help (large i.i.d. batches of a single corruption type), and that performance degrades substantially — sometimes below the frozen baseline — in the more realistic regimes of mixed, imbalanced, or small-batch streams. This work directly motivates the design of the K-Bound pre-registered stress grid: we cross severity, batch size, composition, and aggressiveness to construct conditions that span the helpful and harmful regimes.

Robustness benchmarks. **RDumb** [41] demonstrated that a periodically resetting stupid baseline (no adaptation, periodic reset to the source model) outperforms many published TTA methods on long, non-stationary streams. The implication is that sophisticated adaptation mechanisms may be over-optimized for the benign conditions under which they are evaluated and break down precisely when deployment conditions are most challenging.

Standardized benchmarks. **Hendrycks and Dietterich** [21] introduced the CIFAR-10-C, CIFAR-100-C, and ImageNet-C benchmarks, which apply 19 algorithmically defined corruptions at five severity levels to standard test sets. These benchmarks have become the standard evaluation platform for TTA methods; they are also the platform for our per-corruption experiment (Chapter 8). However, as Zhao et al. and Press et al. note, per-corruption evaluation is inherently helpful-dominated: each batch contains a single corruption type at a single severity, giving the adaptation mechanism a maximally informative signal. Our stress grid goes beyond this by mixing compositions.

DomainBed. **Gulrajani and Lopez-Paz** [18] introduced the DomainBed benchmark for domain generalization, emphasizing standardized evaluation protocols and a thorough comparison of algorithms under matched hyperparameter selection. Their finding that many domain generalization methods do not consistently outperform empirical risk minimization under rigorous evaluation is methodologically analogous to our finding about TTA in the benign regime: sophisticated methods add value primarily in the regimes where they are designed to help. **WILDS** [29] extends this to real-world distribution shifts, including medical imaging, satellite imagery, and user-generated content. Extending K-Bound to WILDS benchmarks is an explicit item of future work (Chapter 10).

2.3 Guarded and Protected Adaptation

Given that adaptation can collapse, a natural response is to guard it. The methods in this section — distinct from both the mechanisms above and the failure analyses just surveyed — protect adaptation against specific failure modes *during* the update. They are the closest prior work to K-Bound in spirit, which makes the one structural difference between them and us worth stating precisely: a guard acts while adapting, whereas K-Bound decides before adapting at all.

Entropy thresholding and filtering. EATA's entropy filter is the most direct example: high-entropy samples are excluded from the update gradient on the hypothesis that they carry corrupted or uninformative signal. This protects against one failure direction (adapting on adversarial inputs) but does not address the three-way decision between helpful, harmful, and unknowable evidence, nor does it provide a formal bound on the false-adapt rate.

Betting-style monitors. Sequential testing literature has developed monitors that halt or dampen a running process when a specific failure mode is detected. In the TTA setting, such a monitor could, for example, halt entropy minimization when the estimated class distribution collapses below a threshold. These approaches protect one direction — typically against collapse — but they do not pose the three-way decision with an explicit unknowable region, and they do not certify *before* acting that adaptation will help.

NOTE's class-conditional normalization. **NOTE** [15] addresses the temporal correlation between samples in a non-stationary stream by estimating class-conditional normalization statistics. This is a form of guarded adaptation in the sense that it conditions the normalization update on the estimated class label of each sample, reducing the sensitivity to class imbalance that causes Tent to collapse. The guard is built into the mechanism, not into a separate decision layer.

The structural gap. All guarded-adaptation approaches share the architecture of operating *during* the update. K-Bound operates *before* the update: it makes the adapt/freeze/abstain decision first, and only then — if the decision is ADAPT — does it invoke the underlying mechanism. This separation is not merely organizational; it is what makes the false-adapt guarantee meaningful. A certificate that fires after an adaptation has already occurred cannot meaningfully bound the false-adapt rate; one that fires before can.

2.4 Label-Free Accuracy Estimation

Guarding adaptation presumes one can tell when it is going wrong, which raises a sharper question: can target performance be read off unlabeled data directly? A third line of work does exactly this — predicting accuracy from unlabeled inputs — but stops short of turning the prediction into an adapt/freeze decision, and estimates an absolute risk where K-Bound needs only the sign of a difference between two models.

ATC. **Garg et al.** [13] introduced Average Threshold Confidence (ATC), which thresholds source-calibrated softmax confidence to estimate the fraction of correct predictions on a target set. ATC achieves strong empirical accuracy on standard benchmarks and has become a reference method for label-free performance estimation. Its training requires a labeled source set to calibrate the confidence threshold; given that calibration, it requires only unlabeled target data. The same work also carries the *negative* result that anchors the present one: identifying out-of-distribution accuracy from unlabeled data is provably as hard as identifying the optimal predictor, so it cannot be done without assumptions on the shift [13]. K-Bound does not re-derive this impossibility;

it inherits it and specializes it to the sign of a *difference* of risks (§2.5), where the required assumption becomes a computable observable-margin-versus-budget condition rather than a global identifiability premise.

Agreement-on-the-Line. Baek et al. [2] exploited the empirical observation that model agreement (the fraction of inputs on which two independently trained models make the same prediction) correlates linearly with accuracy on both in-distribution and out-of-distribution data. The linear relationship can be calibrated from in-distribution data and extrapolated to estimate OOD accuracy from OOD agreement, which is label-free and observable. The method is particularly valuable in the covariate-shift regime where the linear correlation holds reliably, and less so in label-shift regimes where it breaks down.

Predicting with confidence. Guillory et al. [17] studied the problem of estimating model accuracy on unseen distributions with confidence intervals, developing methods based on source/target distribution divergence and ensemble disagreement.

The distinction from K-Bound. All of these methods estimate $R_T(f)$ or $R_T(f_0)$ — the target risk of a single model — or detect that TTA has failed *post hoc*. K-Bound's object of study is strictly different: the *sign* of $R_T(f_0) - R_T(f_a)$. This is a relational quantity involving *two* models, not an absolute risk estimate. Certifying the sign is a strictly weaker requirement than estimating either absolute risk, so the knowable set is strictly larger — but, as Theorem 4.1 establishes, it is not all of evidence space. Furthermore, K-Bound provides a *pre-adaptation* certificate with a controlled false-adapt rate: it fires before f_a is deployed, not after. Post-hoc detection that TTA has hurt (as in AETTA and some ensemble-based monitors) is useful for retrospective analysis but cannot prevent a false adaptation from occurring.

2.5 Unsupervised Risk Estimation and Identifiability

The estimators above succeed in some shift regimes and fail in others, and the reason is not engineering but identifiability: whether the target risk is a function of what unlabeled data can reveal at all. This is the line that gives K-Bound both its hardness and its escape route — and it is the line that *owns* the impossibility K-Bound builds on, so we are explicit about attribution before stating our delta. The general impossibility results here are the source of our unknowable floor, not a consequence of it; the covariate-shift and label-shift identifiability results are the ancestors of our positive regimes. K-Bound's move within this line is narrow and specific: to ask for the sign of a risk difference rather than a risk, which is identifiable on a strictly larger set, and then to make that asymptotic fact a finite-sample certificate with a computable frontier.

The general impossibility and abstention decision theory (the sources). Two prior results own the general picture that K-Bound specializes, and we cite them as the foundation rather than restating them as our own. **Garg et al.** [13] prove that identifying out-of-distribution accuracy from unlabeled data is as hard as identifying the optimal predictor: point identification of target performance is impossible without assumptions that constrain the shift. **Kalai and Kanade** [27] develop the matching decision theory of *optimally abstaining from prediction with out-of-distribution test examples*: they characterize when a learner should predict versus abstain under a shift, with a decision threshold that compares an *observable* quantity to a *shift budget*, and a minimax certificate for the abstain-or-predict choice. Between them, these two works establish (i) that the general label-free question needs an assumption on the shift, and (ii) the observable-versus-budget form of the optimal abstention rule. K-Bound takes both as given. Its contribution is to specialize (i) to the sign of the adaptation benefit — where the required assumption sharpens to the computable criterion $|M| > \beta$ on the disagreement region (Chapter 5) — and to specialize (ii) to the three-way adapt/freeze/abstain certificate, giving the observable margin an explicit closed form and proving the resulting rule the unique maximal sound certificate. The impossibility, the necessity of an assumption, and the observable-versus-budget threshold are not claimed as new; the benefit-sign specialization, the exact computable frontier, and the finite-sample certificate are.

Steinhardt and Liang. Steinhardt and Liang showed that unsupervised risk estimation is possible only under a conditional-independence structure between the features, the label, and the observed predictions. In particular, they required that the label can be recovered from the prediction via a known conditional distribution, which is essentially an assumption of calibration stability. Under this assumption, the target risk is identifiable from unlabeled data; without it, the problem is generally intractable. Their work establishes the structural condition under which label-free risk estimation is possible and motivates our weaker target: the sign rather than the magnitude of a risk difference.

Ben-David et al. **Ben-David et al.** [3] established the canonical learning-theoretic framework for domain adaptation. Their central bound expresses the target risk as the source risk plus an $\mathcal{H}$-divergence between the source and target marginals plus a residual term capturing the irreducible concept-shift component. The $\mathcal{H}$-divergence is computable from unlabeled data; the residual term is not. This bound is tight in the sense that all three terms are necessary; the identifiability result of Theorem 5.3 can be understood as showing that under covariate shift the residual term vanishes and the $\mathcal{H}$-divergence is controlled by the density ratio.

Mansour et al. **Mansour et al.** [36] studied domain adaptation from a learning-bounds perspective, extending the Ben-David et al. framework to mixture-of-source distributions and developing discrepancy-based bounds. Their discrepancy measure generalizes the $\mathcal{H}$-divergence and admits importance-weighting interpretations that connect to our covariate-shift positive result.

Label-shift correction. **Lipton et al.** [34] developed Black Box Shift Estimation (BBSE), which corrects for label shift — a change in the marginal $P(Y)$ with stable class-conditional distributions $P(X \mid Y)$ — using an invertible confusion matrix estimated from labeled source data. BBSE identifies the target label marginal from unlabeled target predictions, enabling unbiased risk estimation under label shift. This is one of the two identifiable special cases in the knowability hierarchy (Table 2.1 in Chapter 3): label shift with an invertible confusion matrix places us in the knowably helpful or knowably harmful regime, not the unknowable one.

Dataset shift taxonomy. **Quinoñero-Candela et al.** [42] provide the standard taxonomy of dataset shift: covariate shift, label shift, concept shift, and their combinations. This taxonomy is the organizational backbone of the knowability hierarchy: covariate shift is provably identifiable (Theorem 5.3), label shift is identifiable under a mild structural assumption (invertible confusion), concept shift re-enters the unknowable regime (Theorem 4.1), and mixed/unknown shift requires certify-or-abstain.

Importance weighting. **Shimodaira** [48] introduced importance weighting (weighted log-likelihood) as the canonical approach to covariate-shift adaptation, showing that reweighting the source loss by the density ratio $p_T(x)/p_S(x)$ yields consistent estimators of target risk. **Sugiyama et al.** [49] developed importance weighted cross-validation, which uses the density ratio to select hyperparameters in a way that is consistent under covariate shift. These works are the direct antecedents of Theorem 5.3: the importance-weighted estimator $\widehat{R}_T(f) = n^{-1} \sum_i r(x_i)\ell(f(x_i), y_i)$ is the operational mechanism behind the covariate-shift positive result.

The K-Bound delta over this line. The identifiability literature, and specifically [13, 27], establishes when a shift-dependent quantity is identifiable and when it is not, and the observable-versus-budget form of the optimal abstention rule. K-Bound's contribution over this line is threefold, and each item is a specialization or sharpening rather than a new impossibility. First, we target the *sign* of a risk difference, which is identifiable on a strictly larger set than the risk itself: the unknowable regime in K-Bound is strictly smaller than the non-identifiable regime for risk estimation. Second, we instantiate the specialized identifiability result as a *finite-sample, false-adapt-controlled certificate* (Theorem 5.1), turning the observable-versus-budget threshold of [27] into an operational, finite-sample decision procedure with an explicit computable margin. Third, we *unify* the label-free accuracy estimators of §2.4 (ATC, difference-of-confidence, generalized disagreement equality, agreement-on-the-line) as the zero-budget ($\beta = 0$) face of this frontier: each reads the benefit sign off an observable and is sound exactly when the calibration drift vanishes, which is the special case our budgeted criterion recovers.

2.6 Selective Prediction and Abstention

Identifiability theory tells us that some conditions are undecidable; acting on that fact requires a principled way to decline. Abstention is not new to machine learning — it has been studied for decades — but classically at the granularity of the individual prediction, and, most relevant here, Kalai and Kanade [27] give the decision theory of abstaining specifically *under distribution shift*, with an observable-versus-shift-budget threshold that is the direct antecedent of our certificate. K-Bound borrows the decision-theoretic logic of the reject option — Chow's per-sample form and Kalai–Kanade's under-shift form — and relocates it one level up, from the per-sample label to the per-episode adaptation decision, specializing the under-shift threshold to the sign of the adaptation benefit.

Chow's reject option. Chow [8] introduced the reject option for classifiers: rather than committing to a prediction for every input, a classifier may abstain when its confidence is below a threshold, trading coverage for accuracy. The optimality theory shows that the Bayes-optimal reject rule thresholds the posterior probability, and the coverage-accuracy tradeoff is characterized by the Bayes optimal.

Selective classification. El-Yaniv and Wiener [12] and **Geifman and El-Yaniv** [14] developed the modern theory of selective classification, providing finite-sample guarantees on the accuracy of a classifier conditioned on its decision to predict. SelectiveNet extends selective classification to deep neural networks with an end-to-end trainable coverage mechanism.

Conformal prediction. Vovk et al. [55] introduced conformal prediction as a framework for constructing prediction sets with valid marginal coverage. **Angelopoulos and Bates** [1] provide a recent tutorial treatment, including split-conformal prediction (which uses a held-out calibration set to set the threshold) and risk-controlling prediction sets. Split-conformal prediction is the mechanism behind Theorem 5.1: the conformal radius ε is constructed from a held-out set of labeled source/validation benefits, and its coverage guarantee transfers to the certificate.

K-Bound's abstention is over a decision, not a prediction. Selective prediction abstracts on individual *predictions*: given a single test input x, should the model commit to a class label $\hat{y}$ or abstain? K-Bound's abstention is at a different level: given a deployment batch and an observable evidence vector Z, should the system commit to the ADAPT or FREEZE *action*, or abstain from the adaptation decision? The object is not a label for a single point but a binary action for an entire deployment episode. This distinction matters because the cost of a wrong action is not the per-sample misclassification rate but the regret relative to the oracle policy — an expectation over the entire deployed batch.

2.7 Conformal Prediction and Anytime-Valid Inference

The reject option says *when* to abstain; it does not say how to build the calibrated interval whose width triggers the abstention. That interval is the last component K-Bound draws from prior work. The conformal and anytime-valid literature provides it — finite-sample coverage for the batch certificate, and valid-at-any-stopping-time coverage for its streaming extension — with no claim on the adapt/freeze/abstain framing itself.

Game-theoretic probability. **Ville** [54] introduced the concept of e-processes (nonneg. supermartingales) and established Ville's inequality, which gives the anytime-valid analogue of Markov's inequality. **Shafer and Vovk** [47] developed the game-theoretic foundations for probability and statistics, showing that betting-style arguments yield valid inference without relying on the law of large numbers.

Testing by betting and e-values. **Shafer** [47] and **Ramdas et al.** [43] developed the testing-by-betting framework: a hypothesis is rejected when the cumulative product of e-values (factors $1+\lambda_t X_t$ where X_t is a bounded observation) exceeds $1/\alpha$. The product is a nonneg. supermartingale under the null, so its exceedance is controlled at level α uniformly over all stopping times (Ville's inequality), without any correction for repeated looks.

Anytime-valid confidence sequences. **Howard et al.** [25] constructed time-uniform, nonparametric confidence sequences: intervals that are valid at all sample sizes simultaneously. **Waudby-Smith and Ramdas** [59] developed betting-based confidence intervals for bounded random variables, establishing tight bounds via predictable bets. **Grünwald et al.** [16] formalized safe testing, which uses e-values to test composite hypotheses in a way that is valid under optional stopping.

Connection to K-Bound. Theorem 5.1 is a fixed-n certificate; the anytime-valid extension (Appendix A) uses a testing-by-betting e-process. The anytime-valid version is strictly more powerful than any fixed-n certificate in the sequential regime (it is valid at any stopping time with no multiplicity correction), while the fixed-n version is tighter at a known sample size. The empirical-Bernstein bound of **Maurer and Pontil** [37] provides the radius discharge: for bounded paired benefits in $[a, b]$, the empirical variance proxy yields a confidence radius under independence/exchangeability.

Classical concentration. **Hoeffding** [22] established the classic exponential tail bound for bounded independent random variables, and **Le Cam** [31] developed the two-point lower bound technique (used in Theorem 4.1) and the fundamental convergence-of-estimates theory. **Tsybakov** [52] and **Lehmann and Romano** [32] are the standard references for nonparametric estimation theory and testing, respectively.

2.8 Benchmarks, Robustness, and Distribution Shift

The five lines above frame the problem; this section names the datasets and backbones on which we test it, grouped by the role each plays in the evaluation of Chapters 7 and 8.

Robustness benchmarks. **Hendrycks and Dietterich** [21] introduced the corruption benchmarks that anchor all of our CIFAR-10-C experiments. **Recht et al.** [44] showed that ImageNet classifiers lose substantial accuracy on a re-collected ImageNet test set, motivating the study of natural distribution shift. **Taori et al.** [51] measured robustness to a broad set of natural and synthetic distribution shifts, finding that effective robustness (robustness beyond what is predicted by the in-distribution to OOD correlation) is rare. **Miller et al.** [38] established the accuracy-on-the-line phenomenon: across many models and distribution shifts, in-distribution and OOD accuracy are linearly correlated.

Anomaly detection benchmarks. K-Bound's anomaly-routing experiments use the ADBench benchmark [19], which covers tabular, image, text, cyber, and fraud anomaly detection across 57 datasets and multiple detector families. **MVTec AD** [4] and **MVTec 3D-AD** [5] provide industrial visual-inspection benchmarks where distribution shift is induced by modality failure; the anonymized multimodal instantiation (Chapter 9) is evaluated on MVTec 3D-AD. **Real-IAD** [58] is a large-scale real industrial anomaly detection dataset with 30 object categories; Real-IAD-D3 is the anonymized multimodal reliability experiment. **PyOD** [61] provides the detector library from which most of our detector scores are extracted.

Standard vision. **CIFAR-10** [30], **ImageNet** [9], **ResNet** [20], **ViT** [11], and **Imagenette** [24] are the standard building blocks of our online TTA and CIFAR-10-C experiments.

Label-free performance prediction and accuracy estimation. **Deng and Zheng** [10] studied the problem of evaluating classifier accuracy without labels, developing methods based on the agreement between classifiers and the structure of the confidence distributions. This work is closely related to ATC [13] and Agreement-on-the-Line [2] in its goals and is part of the same literature that K-Bound departs from in its relational target (sign of difference) and pre-adaptation certificate.

Anomaly/outlier detection methods. The anomaly detection backbone of the experiments uses standard methods from the literature: **Isolation Forest** [35], **Local Outlier Factor** [7], **One-Class SVM** [46], **ECOD** [33], and the PyOD suite [61].

2.9 Positioning: The Gap K-Bound Fills

Each line of work supplies one ingredient and lacks the rest; the gap is the combination none of them holds at once. Table 2.1 makes this concrete, reproducing and expanding the positioning table from the working paper. Its columns are the five properties jointly required for K-Bound's contribution: label-free operation, the distinction between adapting and merely wrapping a candidate, benefit prediction *before* acting, abstention on the adaptation decision itself, and a formal false-adapt guarantee. No prior row carries all five checks.

Work	label-free?	adapts?	predicts benefit *before* adapting?	abstains on the decision?	false-adapt guarantee?
Tent [56]	yes	yes	no	no	no
EATA [39]	yes	yes	partial (filtering)	no	no
SAR [40]	yes	yes	stability only	no	no
CoTTA [57]	yes	yes	no	no	no
TTT [50]	yes	yes	no	no	no
MEMO [60]	yes	yes	no	no	no
LAME [6]	yes	yes	no	no	no
NOTE [15]	yes	yes	no	no	no
ATC [13]	yes	monitors	post-hoc estimate	—	no
Agr.-on-Line [2]	yes	monitors	post-hoc estimate	—	no
BBSE [34]	partial	corrects	corrects shift	no	no
KGA (ours)	yes	wrapper	**yes**	**yes**	**yes**

Table 2.1: Positioning of K-Bound against adjacent methods. Prior TTA methods adapt (some with internal guards); performance estimators operate post-hoc. No prior method provides a single pre-adaptation adapt/freeze/abstain certificate with an explicit, proven unknowable region and a false-adapt bound. KGA wraps any TTA candidate without modifying it.

The surviving gap is precise: no prior method provides a single adapt/freeze/abstain certificate with an *explicit, proven* unknowable region and a false-adapt guarantee.

To state this more carefully:

- TTA mechanisms adapt and (some) detect failure, but commitment happens before detection and no false-adapt rate is bounded.
- Performance estimators (ATC, Agreement-on-the-Line) estimate $R_T(f)$ post-hoc; they do not certify sign Δ before the adaptation decision.
- Guarded adaptation (EATA filtering, SAR resets) protects one failure direction but does not pose the three-way decision with an unknowable region.
- Identifiability theory (Ben-David et al., BBSE) shows when risk is identifiable but does not yield a finite-sample, false-adapt-controlled operational certificate.
- The general label-free impossibility (Garg et al.) and the abstain-under-shift decision theory (Kalai and Kanade) supply the negative result and the observable-versus-budget threshold

K-Bound specializes; they do not target the benefit sign, give the computable frontier, or build the finite-sample certificate.

- Selective prediction (Geifman and El-Yaniv) abstains on individual predictions, not on the adaptation decision as a whole.
- Conformal prediction (Vovk et al., Angelopoulos and Bates) provides the interval mechanism K-Bound uses, but not the framing of the adapt/freeze/abstain decision.

K-Bound combines the identifiability insight (when is sign Δ recoverable?) with the conformal certificate mechanism (how do we control the false-adapt rate given a valid interval?) and the selective-abstention logic (when the interval is too wide to certify either sign, the correct action is to abstain) into a single coherent decision framework. The result is a method that is simultaneously label-free, mechanism-agnostic, pre-adaptive, three-way, and formally guaranteed — a combination no prior work achieves. Closing the gap demands precise objects: a benefit whose sign is the decision, an evidence map that sees only covariates, and a trichotomy of regimes that classifies what such evidence can and cannot certify. The next chapter fixes exactly these, and with them the impossibility theorem and the certificate that the rest of the manuscript proves.

Chapter 3

Problem Formulation and the Knowability Trichotomy

This chapter fixes the objects that every later result manipulates: the deployment protocol under which a frozen model meets a shifted, unlabeled target stream; the *adaptation benefit* Δ, which is the quantity a deployed system would like to know; the *label-free evidence* Z, which is the only quantity it can actually observe; and the trichotomy of *knowably helpful*, *knowably harmful*, and *unknowable* regimes that organizes the entire theory. We then formalize the decision problem itself — rules $g(Z) \in \{\textsc{adapt}, \textsc{freeze}, \textsc{abstain}\}$, the oracle, regret-to-oracle, and the false-adapt and false-freeze error rates — and close with three fully worked example environments, a discussion of the standing assumptions, and a reader's map from questions to theorems.

The formal content of this chapter is exactly that of the paper's problem setup and definitions [28]; the worked examples and the assumptions discussion are expository additions that introduce no new claims. A reader who internalizes one idea from this chapter should make it this one: *the constraint that drives everything is not noise but measurability.* The decision rule is a function of Z, and Z is a function of unlabeled data alone. Whether the right decision is *knowable* is a property of the σ-algebra the evidence generates, not of how many samples populate it. Chapter 4 makes that remark a theorem.

3.1 The deployment protocol

The setting is the standard one of test-time adaptation under distribution shift [56, 42], stated so that the information structure — who sees which labels, and when — is completely explicit.

A *source* distribution $P_S(X, Y)$ over inputs $X \in \mathcal{X}$ and labels $Y \in \mathcal{Y}$ is available with labels at training and validation time: the model was fit on labeled source data, and a held-out labeled validation (or probe) set from the source environment may be retained. A *target* distribution $P_T(X, Y)$ governs deployment, and it exposes only its covariate marginal $P_T(X)$ at test time: the deployed system receives a stream (or batch) of unlabeled target inputs $X_1, \ldots, X_m \sim P_T(X)$

and never observes a target label. The two distributions may differ in any way — covariate shift, label-prior shift, concept shift, or arbitrary mixtures [42, 48, 34]; part of the point of the theory is to classify which kinds of difference leave the adaptation decision decidable.

Three further objects are fixed. A loss $\ell : \mathcal{Y} \times \mathcal{Y} \to \mathbb{R}$ scores predictions (the paper's results are stated for 0/1 loss, multiclass 0/1 loss, and squared error at the points where the loss matters; until then ℓ is generic). A *frozen model* f_0 is the system as deployed: source-trained, untouched by the target stream. An *adapted or gated candidate* f_a is the alternative the system could deploy instead: the output of a test-time adaptation procedure run on the unlabeled batch (entropy minimization in the style of Tent [56], a reliability-gated fusion, a recalibrated threshold, an importance-weighted refit — the framework is agnostic to the mechanism). What matters is only that at decision time f_0 and f_a are both concrete, known, fixed maps from inputs to predictions: the decision under study is *which of two specific models to deploy*, not how to construct a better candidate. Target risk is

$$R_T(f) \;=\; \mathbb{E}_{P_T}\big[\ell(f(X), Y)\big],$$

the population average loss of a model f on the deployment distribution. Note carefully where the inaccessibility lies: $R_T(f)$ averages over the *joint* law $P_T(X, Y)$, while the system observes draws from $P_T(X)$ only. Both $R_T(f_0)$ and $R_T(f_a)$ are therefore well-defined but unobserved quantities, and every question in this monograph is a question about what can be inferred about them — more precisely, about their difference — through the keyhole of unlabeled data.

3.2 The adaptation benefit

The decision-relevant quantity is not either risk in isolation but their difference.

Definition 3.1 (Adaptation benefit)**.** Fix the loss ℓ, the frozen model f_0, and the adapted/gated candidate f_a. The *adaptation benefit* is

$$\Delta \;=\; R_T(f_0) - R_T(f_a), \qquad \Delta > 0 \iff \text{adapting helps.} \tag{3.1}$$

The sign convention reads naturally: Δ is the risk *removed* by switching from the frozen model to the candidate. When $\Delta > 0$ the candidate is strictly better on the target and the correct action is to ADAPT; when $\Delta < 0$ adaptation is harmful and the correct action is to FREEZE; $\Delta = 0$ is the indifference point. The benefit also has a per-sample form that recurs throughout the proofs: writing $\delta = \ell(f_0(X), Y) - \ell(f_a(X), Y)$ for the paired per-sample benefit, we have $\Delta = \mathbb{E}_{P_T}[\delta]$, and the paired structure — both losses evaluated on the *same* draw (X, Y) — is what later makes finite-sample certification statistically cheap relative to estimating either risk on its own (Chapter 5).

Because the decision will be made on observed evidence, we also need the benefit *conditioned* on that evidence. With Z the observable evidence of the next section, the *conditional benefit* is

$$\Delta(z) \;=\; \mathbb{E}\big[\ell(f_0(X),Y) - \ell(f_a(X),Y) \;\big|\; Z = z\big], \tag{3.2}$$

the expected gain from adapting given that the evidence took the value z. The unconditional benefit is recovered by the tower property, $\Delta = \mathbb{E}[\Delta(Z)]$. The paper's central question, stated once and answered over two chapters, is:

> *When is* $\operatorname{sign}\Delta(z)$ *recoverable from* Z *alone?*

Note the deliberate modesty of the target. We do not ask to estimate $R_T(f_0)$ or $R_T(f_a)$ (a cardinal estimation problem, known to be non-identifiable without structural assumptions [3, 34, 13]), nor even to estimate Δ to additive accuracy. We ask only for the *sign* of a difference — a single bit. That the cardinal problem is unanswerable without assumptions on the shift is established prior work [13], as is the decision theory of abstaining under shift [27]; the theory of the next two chapters specializes both to this single bit. Part of that content is that even this single bit is sometimes information-theoretically out of reach (Chapter 4), and that, conversely, the sign is recoverable on a strictly larger set of shifts than either risk is (Theorem 5.4 in Chapter 5).

3.3 Label-free observable evidence

Definition 3.2 (Observable evidence)**.** *Observable evidence* $Z = \phi(X_{1:m}, f_0, f_a)$ is any statistic computable *without* target labels: examples include entropy, confidence, score-distribution drift, detector disagreement, predicted-class balance, update norm, and density-ratio diagnostics.

The definition is intentionally maximal. The evidence map ϕ may be any measurable function of the unlabeled batch $X_{1:m}$ and of the two fixed models; in particular it may use the models' outputs, internal activations, confidence scores, and any comparison between them. Concretely, the evidence panels used in the experiments of Chapters 7 and 8 include: prediction-entropy statistics of f_0 and f_a on the batch; confidence and margin distributions; drift statistics comparing the batch's score distribution to the source validation distribution (for example Kolmogorov–Smirnov distances); the rate and pattern of *disagreement* between f_0 and f_a; the predicted-class balance and its drift from the source balance; the norm of the adaptation update; and density-ratio diagnostics. All of these are functions of covariates and known maps only; none touches a target label. Label-free accuracy-estimation statistics in the style of average thresholded confidence [13] are admissible evidence features under this definition — the framework subsumes them as particular choices of ϕ.

Two structural remarks will carry weight later. First, since f_0 and f_a are fixed known maps, the *disagreement region*

$$D \;=\; \{x : f_0(x) \neq f_a(x)\}$$

is itself observable: membership of each batch point in D is computable without labels, and statistics of the batch restricted to D are legitimate evidence. This observable region turns out to carry the entire decision (Theorem 5.4). Second — and this is the pivot of Chapter 4 — whatever ϕ one chooses, the distribution of Z under a target world P_T depends on that world *only through* $P_T(X)$, because Z is a function of covariates and fixed maps alone. Two target worlds that share a covariate marginal but differ in $P_T(Y \mid X)$ induce *identical* evidence laws for every conceivable evidence map. The impossibility theorem is, at bottom, this observation plus a careful accounting of what it costs.

3.4 The knowability trichotomy

The paper's two definitions classify evidence values by what they permit.

Definition 3.3 (Observable risk alignment)**.** Z is *risk-aligned* if it identifies or bounds $\operatorname{sign} \Delta(z)$ (it need not reveal the label, only whether adaptation helps).

Definition 3.4 (Regimes)**.** At evidence z: *knowably helpful* if Z certifies $\Delta(z) > 0$ (**adapt**); *knowably harmful* if Z certifies $\Delta(z) < 0$ (**freeze**); *unknowable* if the same Z is compatible with both signs (**abstain**).

Definition 3.3 isolates the property that makes label-free decisions possible at all. Risk alignment does *not* ask the evidence to reveal labels, risks, or even the magnitude of the benefit; it asks only that the evidence constrain the benefit's sign. It is a property of the pair (evidence map, family of plausible target worlds): the same ϕ can be risk-aligned against one family of shifts and blind against another. Definition 3.4 then partitions evidence values accordingly. The word "certifies" is doing real work: in the knowable regimes, every target world consistent with the observed evidence has the same benefit sign, so committing is safe; in the unknowable regime there exist worlds consistent with the same evidence whose benefits have opposite signs, and Chapter 4 proves that no rule — however clever, and at any unlabeled sample size — can then commit correctly in all of them. The trichotomy is thus not a design choice but a theorem-shaped fact: the three actions ADAPT, FREEZE, ABSTAIN are exactly the actions matched to the three regimes.

Because "label-free" is used in subtly different senses across the adaptation literature, the paper pins down its meaning in a box that we reproduce here in full, as the protocol is part of the formulation.

Remark 3.1 (What "label-free" means here). *Target-label-free* (our setting): no labels from the target/test stream are ever used. *Validation labels allowed*: labeled source and a held-out labeled validation/probe set may calibrate the benefit estimator $\widehat{\Delta}$ and the radius ε. *Target-unsupervised evidence*: the deployment batch contributes only X, predictions, entropy, confidence, drift and other label-free statistics Z. The online-TTA results that use a clean labelled validation probe (Chapter 8) are target-label-free in this sense — they use no *target* labels — not unsupervised of all labels.

The distinction matters because the impossibility results of Chapter 4 hold against the strongest reading (no target labels, arbitrary evidence maps, source labels freely available), while the positive results of Chapter 5 consume the weaker allowances explicitly: the certificate's radius is calibrated on labeled validation data, and the covariate-shift theorem reweights labeled *source* examples. Nothing anywhere uses a target label.

3.5 Decision rules, the oracle, and error rates

The system's output is a decision, so we now formalize the decision-maker.

Definition 3.5 (Label-free decision rule)**.** A *label-free decision rule* is a (possibly randomized) map

$$g(Z) \;\in\; \{\text{ADAPT},\ \text{FREEZE},\ \text{ABSTAIN}\},$$

measurable in the evidence Z and in randomness independent of the target world. ADAPT deploys f_a; FREEZE retains f_0; ABSTAIN retains f_0 *and* flags the condition as undecided (declining to certify either committed action). The *coverage* of a rule is $\mathbb{P}[g \neq \text{ABSTAIN}]$, the fraction of conditions on which it commits.

Two readings of ABSTAIN coexist in the monograph and it is worth separating them now. Operationally, an abstaining system keeps the frozen model running (the conservative default) while routing the condition to fallback handling — human review, data collection, a wider probe. Decision-theoretically, ABSTAIN is a first-class third action whose role is exactly analogous to the reject option in selective prediction [8, 12, 14], with one crucial difference of granularity: selective prediction abstains on individual *predictions*, while our rules abstain on the *adaptation decision* for a deployment condition. The cost accounting of abstention (a coverage cost, distinct from the cost of a wrong commitment) is made precise inside the proofs of Chapter 4.

Performance is measured against the best decision made with full knowledge.

Definition 3.6 (Oracle and regret-to-oracle)**.** The *per-condition oracle* is $g^\star = \mathbf{1}[\Delta > 0]$: it adapts exactly when the true benefit is positive. *Regret-to-oracle* of a rule g denotes $R(g) - R(g^\star)$, where $R(\cdot)$ is the deployed risk of the rule's chosen actions.

For committal gates the deployed risk takes the explicit form analyzed in Chapter 4: with the action-conditional means $a_{\text{FRZ}}(z) = \mathbb{E}[\ell(f_0(X), Y) \mid Z{=}z]$ and $a_{\text{ADP}}(z) = \mathbb{E}[\ell(f_a(X), Y) \mid Z{=}z]$, a gate g has risk $R(g) = \mathbb{E}[g(Z)\, a_{\text{ADP}}(Z) + (1 - g(Z))\, a_{\text{FRZ}}(Z)]$, and the conditional benefit of Definition 3.1 is recovered as $\Delta(z) = a_{\text{FRZ}}(z) - a_{\text{ADP}}(z)$. Theorem 4.3 will show that the regret-to-oracle of any plug-in gate is *exactly* the benefit it forfeits on the evidence values where it picks the wrong sign.

Finally, the two failure modes of a three-way rule are named and will be the quantities the certificate of Chapter 5 controls.

Definition 3.7 (False-adapt and false-freeze rates)**.** For a rule g,

$$\mathrm{FA}(g) \;=\; \mathbb{P}\big[g = \textsc{adapt} \;\wedge\; \Delta \le 0\big], \qquad \mathrm{FF}(g) \;=\; \mathbb{P}\big[g = \textsc{freeze} \;\wedge\; \Delta \ge 0\big],$$

the probabilities of committing to adaptation when it does not help and of refusing adaptation when it would not hurt, respectively.

The asymmetry of consequences in practice — a false adapt deploys a model that silently degrades, a false freeze merely forgoes an improvement — motivates treating FA as the safety-critical rate; the certificate (Theorem 5.1) bounds both at a user-chosen level α, and the experiments report both, together with coverage and regret-to-oracle. Throughout, $\widehat{\Delta}$ denotes a label-free estimator of Δ with confidence radius ε at miscoverage level α; the operational rule built from $(\widehat{\Delta}, \varepsilon)$ is the subject of Chapters 5 and 6.

3.6 Three worked environments

The definitions above are abstract by design. The following three example environments — expository additions to the paper, chosen so that every quantity is computable in closed form — exhibit one regime each: a knowable covariate shift, an unknowable posterior flip in which the evidence is provably blind, and a low-margin band in which the evidence is informative in principle but commitment is not worth its price. Together they walk the phase diagram of Figure 3.1: far right, far left, and the bottom wedge.

Example 3.1 (A knowable Gaussian location shift)**.** Let the label be a fixed deterministic function of the input, $Y = \mathbf{1}[X > \frac{1}{2}]$, in source and target alike, so that $P_S(Y \mid X) = P_T(Y \mid X)$ and any shift is pure covariate shift [48]. The source is $X \sim \mathcal{N}(0,1)$; the target is $X \sim \mathcal{N}(\mu, 1)$ with μ unknown. The frozen model is $f_0(x) = \mathbf{1}[x > 0]$ (its threshold sits below the Bayes boundary $\frac{1}{2}$), and the adaptation procedure proposes the candidate $f_a(x) = \mathbf{1}[x > 1]$ (a threshold recalibration overshooting the boundary from the other side); under 0/1 loss, with Φ the standard normal distribution function, the risks are the masses of the two mismatch intervals,

$$R_T(f_0) = \mathbb{P}[0 < X \le \tfrac{1}{2}] = \Phi(\tfrac{1}{2} - \mu) - \Phi(-\mu), \qquad R_T(f_a) = \mathbb{P}[\tfrac{1}{2} < X \le 1] = \Phi(1 - \mu) - \Phi(\tfrac{1}{2} - \mu),$$

so the benefit is the explicit function

$$\Delta(\mu) \;=\; 2\,\Phi(\tfrac{1}{2} - \mu) - \Phi(-\mu) - \Phi(1 - \mu).$$

The two error intervals $(0, \frac{1}{2}]$ and $(\frac{1}{2}, 1]$ have equal length and sit symmetrically about the Bayes boundary; since the unimodal target density places more mass on whichever interval is nearer its mode μ, we get $\operatorname{sign} \Delta(\mu) = \operatorname{sign}(\frac{1}{2} - \mu)$, with $\Delta(\mu) = 0$ exactly at $\mu = \frac{1}{2}$ and the antisymmetry $\Delta(\frac{1}{2} + t) \;=\; -\Delta(\frac{1}{2} - t)$. Numerically, $\Delta(-1) \approx +0.048$ (adapt), $\Delta(0) \approx +0.042$ (adapt), and

$\Delta(2) \approx -0.048$ (freeze). Now take the simplest evidence map: $Z = \bar{X}_m = \frac{1}{m}\sum_{i\le m} X_i$, the batch mean, which is label-free. Then $Z \sim \mathcal{N}(\mu, 1/m)$: the evidence identifies μ consistently, and since $\operatorname{sign}\Delta$ is a known function of μ, the evidence is *risk-aligned* (Definition 3.3) — indeed two target worlds $\mu_1 < \frac{1}{2} < \mu_2$ with opposite benefit have evidence laws separating at total-variation rate $2\Phi(\sqrt{m}\,|\mu_1 - \mu_2|/2) - 1 \to 1$. Every $\mu \neq \frac{1}{2}$ is knowably helpful or knowably harmful; the decision is computable from unlabeled data alone.

Example 3.2 (A posterior flip to which all evidence is blind)**.** Now keep the covariate marginal fixed and flip the labels' association with it. In both worlds, $Y \sim \text{Bernoulli}(\frac{1}{2})$ and the covariate is drawn from a symmetric two-component mixture: in world W^+, $X \mid Y{=}1 \sim \mathcal{N}(+1, 1)$ and $X \mid Y{=}0 \sim \mathcal{N}(-1, 1)$; in world W^- the two class-conditionals are swapped. Both worlds have the *identical* covariate marginal $p(x) = \frac{1}{2}\varphi(x-1) + \frac{1}{2}\varphi(x+1)$, where φ is the standard normal density; what differs is the posterior, which flips from $\mathbb{P}_{W^+}(Y{=}1 \mid x) = 1/(1+e^{-2x})$ to $\mathbb{P}_{W^-}(Y{=}1 \mid x) = 1/(1+e^{+2x})$. Take $f_0(x) = \mathbf{1}[x > 0]$ and $f_a(x) = \mathbf{1}[x < 0]$ with 0/1 loss. In world W^+, $R(f_0) = \Phi(-1) \approx 0.159$ and $R(f_a) = \Phi(1) \approx 0.841$, so $\Delta^+ = -(2\Phi(1) - 1) \approx -0.683$: freezing is correct, by a wide margin. In world W^- the computation mirrors and $\Delta^- = +(2\Phi(1)-1) \approx +0.683$: adapting is correct, by the same margin. Yet *every* evidence statistic $Z = \phi(X_{1:m}, f_0, f_a)$ — not just a particular panel, but any measurable label-free map — has identical law in the two worlds, because it is a function of $X_{1:m}$ and the fixed models only and $\text{Law}(X_{1:m})$ is shared. The evidence is blind to a benefit of magnitude 0.683. This is the smoothed sibling of the exact witness that proves Theorem 4.1 (where labels are deterministic and $|\Delta| = 1$); the overlap between the classes here shows that the phenomenon is not an artifact of noise-free labels.

It is instructive to contrast this with *genuine label shift*, where the class priors move but the class-conditionals stay fixed [34]. There the covariate marginal does shift (a prior flip $\pi \leftrightarrow 1-\pi$ reflects the marginal, $m_{1-\pi}(x) = m_\pi(-x)$), so some statistics can detect that *something* changed: sign-asymmetric features such as the predicted-class balance of f_0 move with π, while sign-symmetric panels — confidence $|x|$-margins, disagreement indicators, entropy — remain exactly blind. But detecting motion is not certifying a sign: the same movement of a class-balance statistic can be produced by a covariate shift under which the benefit keeps its sign, so converting the detection into a decision requires label-shift structure (stable class-conditionals and an invertible confusion matrix, as in black-box shift estimation [34]) — structure that is itself untestable from unlabeled target data. The knowability hierarchy of Chapter 5 makes this precise shift class by shift class.

Example 3.3 (The low-margin band)**.** Return to Example 3.1 and let the target location approach the crossover: $\mu = \frac{1}{2} + t$ with t small. A first-order expansion gives $\Delta(\frac{1}{2} + t) \approx -0.094\,t$ (the derivative of Δ at the crossover is $2\varphi(\frac{1}{2}) - 2\varphi(0) \approx -0.094$), so the benefit margin $|\Delta|$ collapses linearly as the condition nears the boundary. The evidence remains risk-aligned in the limit — with infinite data the sign is identifiable for every $t \neq 0$ — but a deployment batch of size m resolves μ only to precision $O(1/\sqrt{m})$, so conditions with $|t| \lesssim 1/\sqrt{m}$ are *effectively* undecidable at the operating sample size, and the benefit at stake there is only $|\Delta| \lesssim 0.1/\sqrt{m}$. This is the regime in which abstention is cheap by design: Theorem 4.3 shows that committal regret on the band $\{|\Delta| < \varepsilon\}$ is at most ε, and the finite-sample certificate of Chapter 5 abstains there automatically because its confidence interval $\widehat{\Delta} \pm \varepsilon$ straddles zero. In the phase diagram this is the lower wedge: not blind, but not worth the risk of commitment either.

The three examples mark the three corners of the problem. In Example 3.1 the evidence separates opposite-benefit worlds and the margin is healthy: commit. In Example 3.2 no evidence map can separate them at any sample size: abstain, necessarily. In Example 3.3 separation is possible but the margin is too small to matter at the achievable resolution: abstain, economically. The theory of the next two chapters is the general statement of exactly these three verdicts.

3.7 Standing assumptions, and what each one buys

We collect the standing conventions of the framework as explicit assumptions. These are not new postulates — each is stated or used in the paper — but displaying them separately clarifies which results lean on which, and what breaks when each fails.

Assumption 3.1 (Fixed decision triple)**.** The loss ℓ, the frozen model f_0, and the candidate f_a are fixed, known, measurable objects at decision time; the risks $R_T(f_0)$ and $R_T(f_a)$ exist and are finite.

Assumption 3.2 (Label-free protocol)**.** No target labels are observed at any stage of the decision. Labeled source and held-out labeled validation data may be used to calibrate estimators and radii (Remark 3.1); the deployment batch contributes covariates and model outputs only.

Assumption 3.3 (Evidence-measurable rules)**.** Decision rules are measurable functions of the evidence Z and of randomness independent of the target world.

Assumption 3.1 is what makes the question well-posed as a *decision between two named alternatives*. If the candidate were allowed to change after the decision (for instance, if adaptation continued to update f_a on data the certificate never saw), the certified quantity Δ would no longer describe the deployed comparison; in the online experiments of Chapter 8 this is respected by re-running the decision at each adaptation step. The fixedness of (f_0, f_a) is also what makes the disagreement region D observable, on which the sign-of-difference theory of Chapter 5 lives. Boundedness of per-sample benefits — needed for the empirical-Bernstein radius [37, 22] inside the certificate — is a local hypothesis of Theorem 5.1, not a standing one.

Assumption 3.2 is the problem statement itself: with target labels the benefit could be estimated directly and the entire subject would collapse into ordinary validation. Its bite is asymmetric across the theory. The impossibility results of Chapter 4 hold *against* every rule satisfying it, so they are strongest when the allowance for source labels is most generous; the positive results consume the allowance explicitly — the certificate's radius is calibrated on a labeled validation

Question	Result	Chapter
Can any label-free rule decide correctly in all worlds?	Theorem 4.1 (with witness; Le Cam bound)	4
What does a forced commitment cost, exactly?	Theorem 4.3 (regret identity; minimax floor)	4
When can a rule commit with finite-sample guarantees?	Theorem 5.1 (certificate; forced abstention)	5
Can decisions stay valid on a stream with repeated looks?	Theorem 5.2 (anytime-valid e-process)	5
Is any shift class provably knowable?	Theorem 5.3 (covariate shift)	5
What single quantity determines sign Δ?	Theorem 5.4 (disagreement region)	5
What remains open?	Conjecture 5.1 (label-free bracketing)	5

Table 3.1: The reader's map from the questions raised by this chapter's formulation to the results that answer them. The first two rows draw the boundary of the possible (Chapter 4); the remaining rows populate the possible side of it (Chapter 5). The KGA algorithm (Chapter 6) operationalizes the certificate row, and the experiments (Chapters 7–8) measure every quantity defined here — benefit, coverage, false-adapt rate, regret-to-oracle — on real benchmarks.

sample assumed exchangeable with deployment conditions (a hypothesis stated and discharged in Chapter 5, in the conformal tradition [55]), and the covariate-shift theorem (Theorem 5.3) reweights labeled source points.

Assumption 3.3 looks like bookkeeping but is the load-bearing wall of Chapter 4: a rule that is a function of Z inherits the law of Z, so two worlds with identical evidence laws force identical behavior — in distribution, exactly — whatever the rule's internal sophistication. Independent randomization buys nothing, because it is world-independent by construction.

Finally, *risk alignment* (Definition 3.3) is deliberately *not* a standing assumption. It is the property whose presence defines the knowable regimes and whose failure defines the unknowable one; assuming it globally would assume away the central question. The KGA algorithm of Chapter 6 needs it only on the conditions where it commits — where it fails, the certificate cannot fire and the rule abstains, which Chapter 5 shows is then mandatory (any rule with valid false-adapt and false-freeze control must abstain with probability at least $1 - 2\alpha$ on evidence shared by opposite-benefit worlds).

3.8 A reader's map from questions to results

The formulation poses a short list of natural questions; each is answered by exactly one result in the next two chapters. Table 3.1 is the itinerary.

One closing orientation. The order of the results is the logic of the subject: first the negative boundary, because without it every positive claim would be suspected of hiding an assumption; then the exact price of ignoring the boundary; then the certificates and positive regimes that live within it. Chapter 4 now takes up the first two rows of the map — the impossibility theorem with its explicit witness, its quantitative Le Cam strengthening, and the plug-in regret identity that prices every wrong commitment.

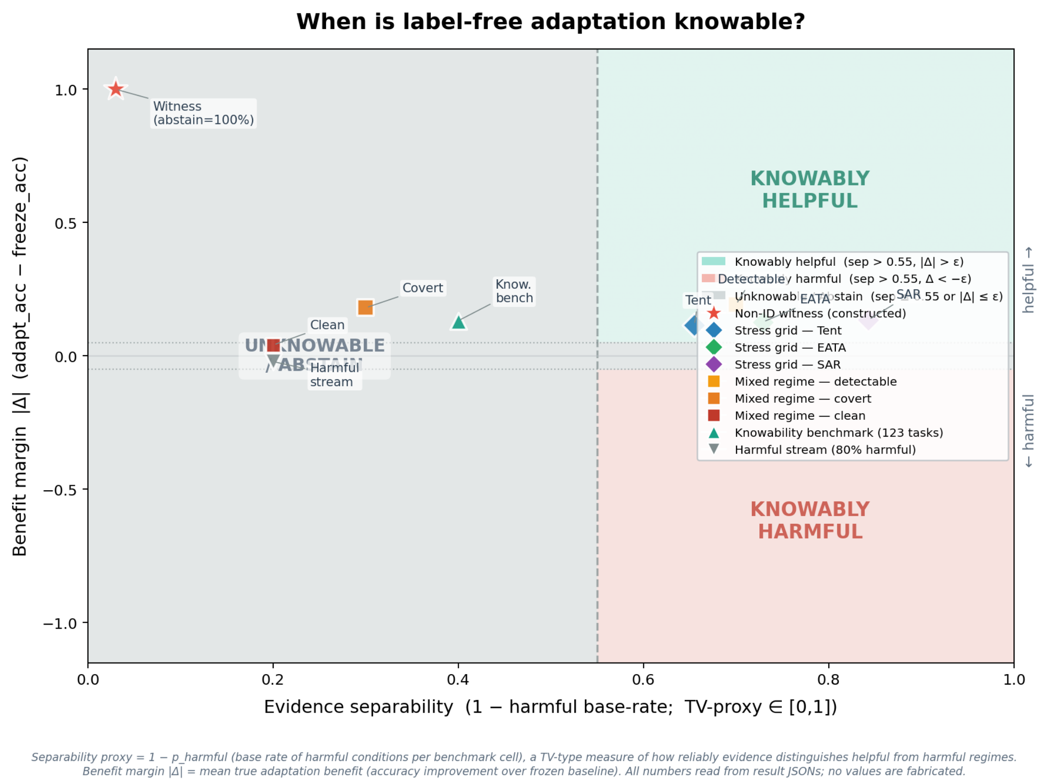

Figure 3.1: **The knowability phase diagram.** Each test-time condition lands at a point set by the evidence separability between opposite-benefit worlds (x-axis; a total-variation/detectability proxy) and the benefit margin $|\Delta|$ (y-axis). KGA commits (ADAPT/FREEZE) only outside the central *unknowable* wedge, where the certificate radius exceeds the estimated benefit ($\varepsilon > |\widehat{\Delta}|$); inside it the worst-case-optimal action is ABSTAIN (Theorem 4.1). Benchmark points — the CIFAR-10-C stress grid (Tent/EATA/SAR), the controlled mixed regime, and the clean non-identifiability witness (Chapters 7 and 8) — are overlaid from their measured values. The two axes of this diagram are exactly the two hypotheses of the impossibility theorem: the x-axis measures how far the evidence laws of opposite-benefit worlds separate (Chapter 4's total-variation quantity), and the y-axis measures how much a wrong commitment costs (the margin $|\Delta|$ in the regret identity of Theorem 4.3).

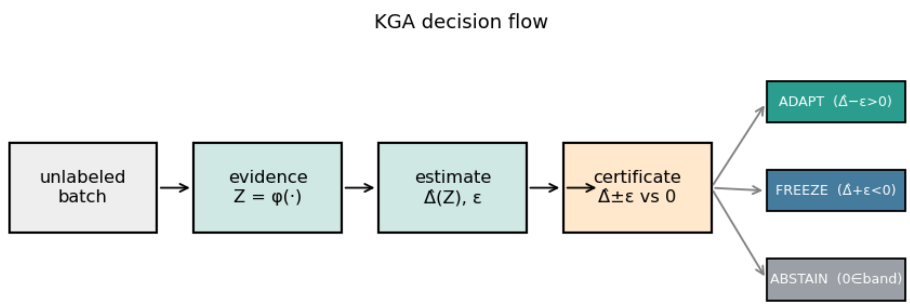

Figure 3.2: The KGA decision flow realizing the formulation of this chapter: label-free evidence $Z = \phi(X_{1:m}, f_0, f_a)$ is extracted from the unlabeled batch and the two fixed models; a calibrated benefit estimate $\widehat{\Delta}(Z)$ with conformal radius ε is formed; and the rule commits to ADAPT only if $\widehat{\Delta} - \varepsilon > 0$, to FREEZE only if $\widehat{\Delta} + \varepsilon < 0$, and otherwise ABSTAINs (keeping f_0 and flagging the condition). The three exits are the three regimes of Definition 3.4; the guarantees attached to each exit are proved in Chapters 4 and 5, and the algorithm itself is the subject of Chapter 6.

Chapter 4

The Unknowable Regime: Non-Identifiability and the Minimax Floor

The first half of the theory draws the boundary of the possible. It does so on ground the literature already owns: that label-free identification of a shift-dependent quantity is impossible without an assumption on the shift is due to Garg et al. [13], and the decision theory of abstaining versus predicting under shift — with an observable-quantity-versus-shift-budget threshold — to Kalai and Kanade [27]. This chapter does not re-derive that general impossibility; it *specializes and sharpens* it to the sign of the adaptation benefit, and proves two results in that specialized form. The first (Theorem 4.1, with its quantitative Le Cam strengthening, Theorem 4.2) shows that the unknowable regime of Definition 3.4 is not an artifact of weak evidence features or small batches: there exist pairs of target worlds whose label-free evidence laws coincide *exactly* — for every evidence map and every unlabeled sample size — while their adaptation benefits have opposite signs, and on such pairs the optimal committal error admits a closed-form floor, $1 - \mathrm{TV}$, which equals 1 at the witness. That closed form is our sharpening of the general hardness to an exact identity for this question. The second (Theorem 4.3) prices every violation of the boundary: the regret-to-oracle of any plug-in gate equals, *exactly*, the expected benefit forfeited on the evidence values where the estimated sign is wrong, which yields the matching minimax floor $\mathbb{E}[|\Delta(Z)|]/2$ for the unknowable regime and justifies abstention on the low-margin band.

The paper's running example sets the stakes. Two hospitals' scanners induce identical confidence and entropy statistics in a deployed classifier, yet in one hospital adapting helps and in the other — a flipped label frequency — it badly hurts; no label-free rule can separate them, so the safe action is to abstain. The mathematics below shows that this story is not a corner case but a face of the problem: whenever the posterior $P_T(Y \mid X)$ can move while the covariate law $P_T(X)$ does not, the evidence σ-algebra simply does not contain the answer. Everything in this chapter is proved against the strongest reading of the protocol of Chapter 3: arbitrary evidence maps, arbitrary (randomized) rules, unlimited unlabeled data, and free access to labeled source data.

We emphasize the chapter's intellectual posture at the outset, because the paper does. The general impossibility is not ours: label-free identification of target performance is as hard as identifying the optimal predictor and needs an assumption on the shift [13], and the abstain-versus-predict decision theory under shift is already established [27]; more broadly, non-identifiability of unlabeled target *risk* is close to folklore — it is the reason the unsupervised risk estimation, domain adaptation, and label-shift literatures all carry structural assumptions [3, 36, 34]. What this chapter contributes is not the impossibility but its *specialization* to the adapt/freeze decision: (i) the statement at the level of the *decision*, where it justifies a specific third action — the benefit-sign instance of the under-shift abstention rule of [27]; (ii) an explicit, maximally separated witness pair on which every later experiment can be run; (iii) the exact minimax form of the floor via the Le Cam two-point method [31, 52], sharpening the general hardness to an identity for this question; and (iv) the exact regret identity that converts the floor into a deployment-relevant quantity. Each component is validated numerically by a machine-checked script in the public repository [28], and we quote those checks where they occur.

4.1 The non-identifiability theorem

We begin with the qualitative statement, copied verbatim from the paper.

Theorem 4.1 (Non-identifiability; with witness)**.** *Let* $g(Z) \in \{\textit{ADAPT}, \textit{FREEZE}, \textit{ABSTAIN}\}$ *be any (possibly randomized) rule depending on label-free evidence* Z*. There exist two target worlds* P_T^1, P_T^2 *with* (i) $\mathrm{Law}(Z \mid P_T^1) = \mathrm{Law}(Z \mid P_T^2)$ *and* (ii) $\mathrm{sign}\,\Delta^1 = -\,\mathrm{sign}\,\Delta^2 \neq 0$*. For any such pair,* $\mathrm{Law}(g)$ *is identical across the two worlds; hence no label-free rule commits to the benefit-maximizing action in both, and the action minimizing the worst-case committal regret over* $\{P_T^1, P_T^2\}$ *is* ABSTAIN*.*

Before the proof, three reading notes. First, the theorem has an existence half and a universal half: it *exhibits* one pair of worlds satisfying (i)–(ii) (the witness), and then shows that for *every* pair satisfying (i)–(ii) — witness or otherwise — every rule is stuck. Second, the conclusion is about laws, not samples: $\mathrm{Law}(g)$ identical across worlds means the rule's behavior is indistinguishable *in distribution*, so no accounting trick (more batches, more features, more randomization) recovers a difference. Third, the optimality of ABSTAIN is a genuine minimax statement over a two-point family, in the classical mold of two-point lower bounds [31, 52]; the next section upgrades it to an exact quantitative floor.

Proof. The proof has four steps: construct the witness; verify properties (i) and (ii); show every label-free rule has world-independent law; and solve the resulting worst-case decision problem.

Step 1 (witness construction; non-vacuous). Let $X \sim \mathcal{N}(0,1)$ under *both* worlds, so $P_T^1(X) = P_T^2(X)$. Take the fixed predictors $f_0(x) = \mathbf{1}[x > 0]$ and $f_a(x) = \mathbf{1}[x < 0]$, and 0/1 loss. Define the two conditional label laws degenerately:

$$P_T^1(Y \mid X)\colon\; Y = \mathbf{1}[X > 0], \qquad P_T^2(Y \mid X)\colon\; Y = \mathbf{1}[X < 0].$$

In world 1, $f_0(X) = \mathbf{1}[X > 0] = Y$ holds almost surely, so $R_T^1(f_0) = \mathbb{P}[f_0(X) \neq Y] = 0$; and since $\mathbb{P}[X = 0] = 0$, we have $f_a(X) = \mathbf{1}[X < 0] = 1 - Y$ almost surely, so $R_T^1(f_a) = 1$. Hence

$$\Delta^1 = R_T^1(f_0) - R_T^1(f_a) = 0 - 1 = -1 < 0,$$

and freezing is the benefit-maximizing action in world 1. World 2 is the mirror image: $f_a(X) = Y$ and $f_0(X) = 1 - Y$ almost surely, so $R_T^2(f_0) = 1$, $R_T^2(f_a) = 0$, and $\Delta^2 = +1 > 0$: adapting is correct. The benefit gap is maximal ($|\Delta^j| = 1$, the full range of 0/1 risk), so the construction is not a degenerate tie.

Step 2 (identical evidence laws). The evidence $Z = \phi(X_{1:m}, f_0, f_a)$ is, for any evidence map ϕ and any batch size m, a function of $X_{1:m}$ and of the fixed maps f_0, f_a only: writing $\psi(\cdot) := \phi(\cdot, f_0, f_a)$, we have $Z = \psi(X_{1:m})$ with ψ measurable and world-independent. The law of $X_{1:m}$ is $\mathcal{N}(0,1)^{\otimes m}$ in both worlds, so the pushforward laws coincide: $\mathrm{Law}(Z \mid P_T^1) = \mathrm{Law}(X_{1:m}) \circ \psi^{-1} = \mathrm{Law}(Z \mid P_T^2)$. This is property (i); property (ii) was verified in Step 1 ($\operatorname{sign} \Delta^1 = -1 = -\operatorname{sign} \Delta^2 \neq 0$). Thus (i) and (ii) hold simultaneously.

Step 3 (the rule's law is world-independent). A label-free rule is a measurable map of the evidence and of independent randomness: $g = \Gamma(Z, U)$, where U is a randomization variable whose law does not depend on the target world (it is generated by the system, not by the data), and Γ is measurable. Under world j the law of g is the pushforward of $\mathrm{Law}(Z \mid P_T^j) \otimes \mathrm{Law}(U)$ under Γ. By (i) the two product laws coincide, hence so do the laws of g. In particular the three numbers

$$q := \mathbb{P}[g = \textsc{adapt}], \qquad r := \mathbb{P}[g = \textsc{freeze}], \qquad 1 - q - r = \mathbb{P}[g = \textsc{abstain}]$$

are the *same* in both worlds.

Step 4 (no simultaneous correctness, and the worst-case-optimal action). The benefit-maximizing action is FREEZE in world 1 and ADAPT in world 2. A rule committing to the correct action with probability one in world 1 needs $r = 1$; in world 2 it needs $q = 1$; both cannot hold since $q + r \leq 1$. Hence no label-free rule commits to the benefit-maximizing action in both worlds. To identify the minimax action, measure committal regret as the excess 0/1 risk of a committed action versus the better committed action in that world, and assign abstention a separate coverage cost (it is a different kind of outcome — a flagged non-decision — and is priced separately; the quantitative theorem below makes this explicit). In world 1, committing to ADAPT costs $R_T^1(f_a) - R_T^1(f_0) = |\Delta^1| = 1$ and committing to FREEZE costs 0; so world 1 charges expected committal regret $q \cdot |\Delta^1| = q$. Symmetrically world 2 charges $r \cdot |\Delta^2| = r$. The worst case over the two-point family is

$$W(q, r) \;=\; \max(q,\, r),$$

to be minimized over the simplex $\{q, r \geq 0,\ q + r \leq 1\}$. Since $W(q, r) \geq 0$ with equality if and only if $q = r = 0$, the unique minimizer is $q = r = 0$, i.e. the rule that plays ABSTAIN with probability one. □

The paper attaches an explicit calibration of the theorem's weight, which we reproduce because it governs how the result should be cited.

Remark 4.1 (Weight of the result). Theorem 4.1 is close to known non-identifiability of target risk without labels; its purpose here is to *justify the abstain action*, not to claim impossibility as novel. The quantitative strengthening below (Theorem 4.2) makes it a tight minimax statement.

Remark 4.2 (Both hypotheses are necessary). The two-point structure is sharp in both coordinates. If (i) fails — the evidence laws differ by total variation $\tau > 0$ — then Theorem 4.2 shows the optimal committal error drops to $1 - \tau < 1$, and a certificate may commit on the distinguishable part (Chapter 5 builds exactly that). If (ii) fails — the two worlds' benefits share a sign — then committing to the common benefit-maximizing action is correct in both worlds even though the evidence cannot tell them apart; indistinguishability is harmless when the decision does not depend on the distinction. The impossibility requires the conjunction: *the evidence cannot separate the worlds, and the worlds disagree about what to do.* The unknowable regime of Definition 3.4 is precisely the set of evidence values where some such pair lives.

4.2 The explicit witness, in full

Because the witness carries both the theorem's existence half and the paper's cleanest experiment, we devote a section to it: first the algebra in complete detail, then its interpretation, then the empirical realization.

4.2.1 Construction and verification

The witness consists of: covariate law $X \sim \mathcal{N}(0, 1)$, shared by both worlds; predictors $f_0(x) = \mathbf{1}[x > 0]$ and $f_a(x) = \mathbf{1}[x < 0]$, fixed and known; loss 0/1; and the two degenerate posteriors $Y = \mathbf{1}[X > 0]$ (world 1) and $Y = \mathbf{1}[X < 0]$ (world 2). Three features deserve explicit verification beyond the proof above.

The risks, entry by entry. Under world 1, the event $\{f_0(X) \neq Y\} = \{\mathbf{1}[X > 0] \neq \mathbf{1}[X > 0]\} = \emptyset$, so $R^1_T(f_0) = 0$ exactly (not merely almost). The event $\{f_a(X) \neq Y\} = \{\mathbf{1}[X < 0] \neq \mathbf{1}[X > 0]\} = \{X \neq 0\}$ has probability one under the atomless Gaussian, so $R^1_T(f_a) = 1$. The four risks are

$$R^1_T(f_0) = 0, \quad R^1_T(f_a) = 1, \quad R^2_T(f_0) = 1, \quad R^2_T(f_a) = 0,$$

giving $\Delta^1 = -1$ and $\Delta^2 = +1$: the largest benefit gap the 0/1 scale permits, in opposite directions.

Evidence-law equality, for every ϕ and every m. The claim is not that some chosen panel of statistics happens to match across worlds; it is that the entire observable σ-algebra matches. Any label-free evidence is $Z = \psi(X_{1:m})$ for a fixed measurable ψ (the models being fixed known maps, they are absorbed into ψ), and equality of the laws of $X_{1:m}$ forces equality of the laws of Z — a pushforward of equal measures under one map. Consequently *no* statistic computable without target labels — no entropy histogram, no drift score, no disagreement pattern, no density-ratio diagnostic, nor any statistic anyone will ever invent — distinguishes the worlds at any batch size. The quantifier order is the point: the construction defeats all evidence maps simultaneously, not a fixed one.

What kind of shift this is. Between the two worlds the covariate law is untouched and the posterior flips: $P_T^1(Y{=}1 \mid x) = \mathbf{1}[x > 0]$ versus $P_T^2(Y{=}1 \mid x) = \mathbf{1}[x < 0]$. In the taxonomy of shifts this is pure concept shift [42], the row of the knowability hierarchy that Chapter 5 marks non-identifiable; Example 3.2 of Chapter 3 is its smoothed sibling, with overlapping classes and $|\Delta| = 2\Phi(1) - 1 \approx 0.683$ in place of the extremal $|\Delta| = 1$, showing the phenomenon survives label noise. It is also instructive to view the witness through the disagreement-region lens of Theorem 5.4: here the observable disagreement region is $D = \{x : f_0(x) \neq f_a(x)\} = \{x \neq 0\}$, essentially the whole line, and the candidate's accuracy on it is $a_a^D = 0$ in world 1 and $a_a^D = 1$ in world 2 — the identity sign $\Delta = \operatorname{sign}(a_a^D - \frac{1}{2})$ holds in both worlds, and what the witness makes impossible is precisely any label-free *bracketing* of a_a^D around $\frac{1}{2}$. The impossibility and the sign-of-difference characterization are two views of one boundary.

4.2.2 Empirical realization

The witness is constructive, so it can be run. The paper reports a clean witness experiment (the full experimental protocol appears in Chapter 7): two simulated worlds with $X \sim \mathcal{N}(0, 1)$, $f_0 = \mathbf{1}[X{>}0]$, $f_a = \mathbf{1}[X{<}0]$, and flipped labels, so the benefit is exactly opposite (world means -1.0 versus $+1.0$) while every evidence feature is a function of X only. Across 300 instances, all five evidence features used by KGA are statistically indistinguishable between the worlds — two-sample Kolmogorov–Smirnov p-values in $[0.16, 0.96]$, all above 0.05 — yet the ground truth is opposite, so Z cannot decide. KGA abstains on 100% of instances, and a rule forced to commit by sign $\widehat{\Delta}$ pays regret 0.51, near chance. This is the clean theory–experiment bridge (Figure 4.1).

The repository's theorem validator (`experiments/kbound/theory_validation/val_thm1_lecam.py`, results in `results_thm1_lecam.json` [28]) re-runs the witness at scale with 20,000 Monte-Carlo batches of $n = 64$ unlabeled samples per world and reports four independent checks. (1) The closed-form benefits $\Delta^1 = -1$, $\Delta^2 = +1$ are reproduced exactly by Monte Carlo. (2) For each of five per-sample evidence features (the raw covariate, the margin $|x|$, the predicted class of f_0, the disagreement indicator, and a smooth confidence transform), the permutation-debiased total-variation estimate between the two worlds' evidence laws is at most 1.9×10^{-3} — zero up to the estimator's own null floor — with two-sample KS p-values between 0.28 and 1.0; and the n-sample aggregated evidence statistic has exact TV 0 for every n in $\{1, 2, 4, \ldots, 256\}$ (debiased estimates $\leq 4.8 \times 10^{-3}$). (3) Optimizing the committal error over a dense family of threshold

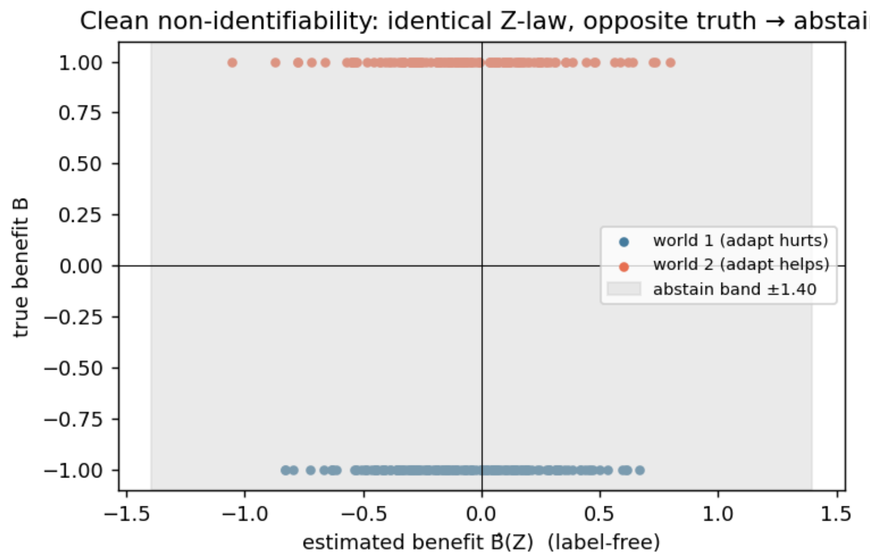

Figure 4.1: The clean non-identifiability witness of Theorem 4.1, realized empirically: identical evidence law across the two worlds (two-sample KS $p > 0.05$ on all five evidence features), exactly opposite true benefit (-1.0 versus $+1.0$), and 100% abstention by the KGA rule; a rule forced to commit pays regret 0.51, near chance. The experiment instantiates hypotheses (i) and (ii) of the theorem exactly rather than approximately, which is what makes it a *witness* rather than a hard benchmark.

rules (all thresholds, both polarities) on each evidence statistic yields $\inf_g M(g)$ between 0.991 and 1.000 across the five statistics, against the theoretical floor of 1. (4) The actual KGA conformal three-way rule, run exactly as in the paper with $\alpha = 0.1$, abstains on 4,000 of 4,000 instances (100% abstention; zero false adapts and zero false freezes): its calibrated radius ($\varepsilon \approx 1.15$) straddles zero because the evidence carries no benefit signal, so the certificate cannot fire — the behavior that Chapter 5 proves is mandatory for any rule with valid error control. The paper's summary sentence is exact: in the witness the empirical $\inf_g M$ is $\approx$1 (KGA abstains 100%); in the detectable regime the empirical minimax error tracks the closed-form $1 - \mathrm{TV}$ across all n and shift magnitudes.

4.3 The quantitative floor: a Le Cam two-point bound

Theorem 4.1 says committal rules fail on matched evidence; it does not yet say how badly, nor what happens as the evidence laws begin to separate. The paper's quantitative strengthening answers both with an exact identity, in the two-point tradition of Le Cam [31]; the intuition, in the paper's words: the more alike two such worlds look in the observable evidence (smaller TV distance), the closer any rule forced to commit is pushed toward a coin flip, and the error floor vanishes only as their evidence separates.

We first isolate the two classical ingredients as lemmas, with proofs, since the theorem's proof is a direct composition of them.

Lemma 4.1 (Testing affinity identity). *Let Q_1, Q_2 be probability measures on a common measurable space. Over all randomized tests ψ (measurable functions with values in $[0,1]$),*

$$\inf_{\psi} \{\mathbb{E}_{Q_1}[\psi] + \mathbb{E}_{Q_2}[1-\psi]\} \;=\; 1 - \mathrm{TV}(Q_1, Q_2),$$

and the infimum is attained by the deterministic likelihood-ratio test $\psi^\star = \mathbf{1}[\mathrm{d}Q_2 > \mathrm{d}Q_1]$.

Proof. Let $\nu = Q_1 + Q_2$, which dominates both measures, and let q_1, q_2 be the corresponding densities. For any test ψ,

$$\mathbb{E}_{Q_1}[\psi] + \mathbb{E}_{Q_2}[1-\psi] = \int \psi\, q_1\, \mathrm{d}\nu + \int (1-\psi)\, q_2\, \mathrm{d}\nu = 1 + \int \psi\,(q_1 - q_2)\, \mathrm{d}\nu,$$

using $\int q_2\, \mathrm{d}\nu = 1$. The integrand is minimized pointwise over $\psi(z) \in [0,1]$ by setting $\psi = 1$ where $q_1 - q_2 < 0$, $\psi = 0$ where $q_1 - q_2 > 0$, and arbitrarily on the crossing set $\{q_1 = q_2\}$ (which contributes zero); this is exactly $\psi^\star = \mathbf{1}[q_2 > q_1]$, a deterministic test, so randomization buys nothing. The attained value is

$$1 - \int (q_2 - q_1)_+\, \mathrm{d}\nu.$$

It remains to identify the integral with total variation. For any measurable A, $Q_2(A) - Q_1(A) = \int_A (q_2 - q_1)\, \mathrm{d}\nu \;\le\; \int (q_2 - q_1)_+\, \mathrm{d}\nu$, with equality at $A^\star = \{q_2 > q_1\}$; hence $\mathrm{TV}(Q_1, Q_2) := \sup_A |Q_1(A) - Q_2(A)| = \int (q_2 - q_1)_+\, \mathrm{d}\nu$, which also equals $\frac{1}{2}\int |q_1 - q_2|\, \mathrm{d}\nu$ because $\int (q_2 - q_1)\, \mathrm{d}\nu = 0$ forces the positive and negative parts to have equal mass. Substituting gives the identity [31, 52, 32]. □

Lemma 4.2 (Data-processing inequality for total variation). *Let P_1, P_2 be probability measures and T a measurable map. Then $\mathrm{TV}(P_1 \circ T^{-1},\, P_2 \circ T^{-1}) \le \mathrm{TV}(P_1, P_2)$, with equality whenever T is a sufficient statistic for the pair $\{P_1, P_2\}$.*

Proof. For the inequality, every event of the pushforward laws is the preimage of an event: $\sup_A |P_1(T^{-1}A) - P_2(T^{-1}A)|$ is a supremum over the sub-collection $\{T^{-1}A\}$ of measurable sets, hence at most the supremum over all measurable sets, which is $\mathrm{TV}(P_1, P_2)$. For the equality claim: if T is sufficient for $\{P_1, P_2\}$, the likelihood ratio $\mathrm{d}P_2/\mathrm{d}P_1$ is (P_1-a.s.) a function of T by the factorization theorem [32], so the optimal set $A^\star = \{\mathrm{d}P_2 > \mathrm{d}P_1\}$ realizing the supremum in Lemma 4.1's identification of TV is itself of the form $T^{-1}(B)$, and the restricted supremum already attains the unrestricted one. □

With the two lemmas in hand we state the paper's theorem verbatim.

Theorem 4.2 (Quantitative non-identifiability; Le Cam two-point lower bound). *Consider two target worlds P_T^1, P_T^2 with $\Delta^1 < 0 < \Delta^2$. Let the label-free evidence be $Z = \phi(X_{1:n}, f_0, f_a)$ with n-sample evidence laws $P_{T,Z}^{j,(n)} := \mathrm{Law}(Z \mid P_T^j)$. For any (randomized) committal rule $g(Z) \in \{\textsc{adapt}, \textsc{freeze}\}$ define the mixed committal error $M(g) := \mathbb{P}_{P_T^1}[g = \textsc{adapt}] + \mathbb{P}_{P_T^2}[g = \textsc{freeze}]$.*

Then

$$\inf_g M(g) \;=\; 1 - \mathrm{TV}\big(P^{1,(n)}_{T,Z}, P^{2,(n)}_{T,Z}\big) \;\geq\; 1 - \mathrm{TV}\big(P^{1,(n)}_{T}, P^{2,(n)}_{T}\big).$$

In the witness above (Z a function of X, $P^1_T(X) = P^2_T(X)$) the evidence TV is 0*, so* $\inf_g M(g) = 1$ *for* every n*: no committal rule beats blind guessing in the worst world, and under any positive coverage cost* ABSTAIN *is uniquely worst-case optimal. If the worlds become δ-detectable so the evidence laws separate,* $\inf_g M(g) = 1 - \mathrm{TV} \to 0$ *at the separation rate (for a location shift summarized by a sufficient statistic,* $\mathrm{TV} = 2\Phi(\frac{1}{2}\sqrt{n}\,\delta) - 1$*).*

Proof. We expand the paper's proof into five steps.

Step 1 (reduction to binary testing). A randomized committal rule on Z is characterized by its adapt-probability $\psi(z) := \mathbb{P}[g = \text{ADAPT} \mid Z = z] \in [0, 1]$, a randomized test of the two evidence laws. Identify $\{g = \text{ADAPT}\}$ with deciding world 2 (where adapting is correct, since $\Delta^2 > 0$). Writing $Q_j := P^{j,(n)}_{T,Z}$, the mixed committal error is

$$M(g) = \mathbb{P}_{Q_1}[g = \text{ADAPT}] + \mathbb{P}_{Q_2}[g = \text{FREEZE}] = \mathbb{E}_{Q_1}[\psi(Z)] + \mathbb{E}_{Q_2}[1 - \psi(Z)],$$

the sum of the test's type-I and type-II errors. The infimum of M over committal rules is therefore the infimum of the testing risk over randomized tests.

Step 2 (the equality). Lemma 4.1 applied to Q_1, Q_2 gives $\inf_g M(g) = 1 - \mathrm{TV}(Q_1, Q_2)$, attained by the likelihood-ratio rule that plays ADAPT exactly on $\{\mathrm{d}Q_2 > \mathrm{d}Q_1\}$. Note this is an identity, not merely a bound: the optimal committal rule's mixed error *equals* one minus the evidence separation. This is the tightness direction — no slack is lost between the decision problem and the total-variation geometry, because the optimal test is exhibited.

Step 3 (the inequality to the full data). The evidence is a measurable reduction of the raw unlabeled batch: $Z = T(X_{1:n})$ with $T := \phi(\cdot, f_0, f_a)$. By Lemma 4.2, $\mathrm{TV}(Q_1, Q_2) \leq \mathrm{TV}\big(P^{1,(n)}_T, P^{2,(n)}_T\big)$, the total variation between the worlds' n-sample *covariate-data* laws. Negating, $1 - \mathrm{TV}(Q_1, Q_2) \geq 1 - \mathrm{TV}(P^{1,(n)}_T, P^{2,(n)}_T)$: summarizing data can only raise the error floor. Equality holds when Z is sufficient for the two-world testing problem — the evidence then loses nothing.

Step 4 (the witness: floor 1 *at every* n*).* In the witness, Z depends on the worlds only through $\mathrm{Law}(X_{1:n})$, and these laws coincide; hence $Q_1 = Q_2$, $\mathrm{TV}(Q_1, Q_2) = 0$, and $\inf_g M(g) = 1$ for every n. Unpacking what $M(g) = 1$ forces: since $M(g) = \mathbb{P}_1[\mathrm{err}] + \mathbb{P}_2[\mathrm{err}] \geq 1$ for every committal g, at least one world sees error probability $\geq \frac{1}{2}$ — no committal rule beats blind guessing in its worst world, and this does not improve with n. Now add a coverage cost $c \in (0, \frac{1}{2})$ for abstention, and recall from the witness that a wrong commitment costs $|\Delta^j| = 1$. A rule with commitment probabilities (q, r) (Step 3 of Theorem 4.1's proof; world-independent by matched evidence) has worst-case expected cost

$$\max\big(q + (1-q-r)c,\; r + (1-q-r)c\big) \;=\; \max(q, r) + (1-q-r)\,c \;\geq\; \tfrac{q+r}{2} + (1-q-r)\,c,$$

which, as a function of the total commitment mass $s = q + r \in [0, 1]$, is $\geq s/2 + (1 - s)c$, strictly increasing in s because $c < \frac{1}{2}$. The minimum is at $s = 0$, value c: always-ABSTAIN is the unique worst-case-optimal rule.

Step 5 (the detectable rate). Suppose the worlds are δ-detectable through a real-valued per-sample summary with unit noise scale: the summary is Gaussian with means separated by δ and variance 1 (the canonical location family; in the validator below, the worlds are $P^1(X) = \mathcal{N}(-\delta/2, 1)$ and $P^2(X) = \mathcal{N}(+\delta/2, 1)$). The sample mean $\bar{X}_n$ is sufficient for the location parameter, so by Lemma 4.2 the evidence loses nothing and $\mathrm{TV}(Q_1, Q_2) = \mathrm{TV}\big(\mathcal{N}(-\frac{\delta}{2}, \frac{1}{n}), \mathcal{N}(+\frac{\delta}{2}, \frac{1}{n})\big)$. For two equal-variance Gaussians $\mathcal{N}(\theta_1, \sigma^2)$, $\mathcal{N}(\theta_2, \sigma^2)$ the densities cross exactly at the midpoint $(\theta_1 + \theta_2)/2$, so the positive part integrates to $\mathrm{TV} = \Phi(\frac{|\theta_1-\theta_2|}{2\sigma}) - \Phi(-\frac{|\theta_1-\theta_2|}{2\sigma}) = 2\Phi(\frac{|\theta_1-\theta_2|}{2\sigma}) - 1$. With $|\theta_1 - \theta_2| = \delta$ and $\sigma = 1/\sqrt{n}$ this is $\mathrm{TV} = 2\Phi(\frac{1}{2}\sqrt{n}\,\delta) - 1$, whence $\inf_g M(g) = 1 - \mathrm{TV} = 2\Phi(-\frac{1}{2}\sqrt{n}\,\delta) \to 0$ as $\sqrt{n}\,\delta \to \infty$: the committal-error floor decays at the Gaussian separation rate, and the unknowable regime shrinks exactly as fast as the evidence separates. □

Remark 4.3 (Where the bound is and is not tight). Two distinct tightness claims should be kept apart. The first equality, $\inf_g M = 1 - \mathrm{TV}$ *on the evidence laws*, is always exact (Lemma 4.1 exhibits the optimal test); there is no asymptotic or constant slack anywhere in the chain from decision problem to total-variation geometry. The second relation, the inequality to the *full-data* TV, is an inequality precisely because a poorly chosen evidence map can discard separation that the raw batch contains; it collapses to equality when Z is sufficient for the two-world problem (Lemma 4.2), as in the location family of Step 5. The design lesson for evidence panels is contained in that gap: the floor an operator actually faces is $1 - \mathrm{TV}(\text{evidence})$, which the operator partially controls through ϕ, bounded below by $1 - \mathrm{TV}(\text{full data})$, which nature controls. In the witness nature sets the latter to 1 and the choice of ϕ is irrelevant.

4.3.1 Numerical validation of the floor

The validator `val_thm1_lecam.py` [28] checks the identity $\inf_g M(g) = 1 - \mathrm{TV}$ in both regimes, using 20,000 Monte-Carlo evidence instances per world ($n = 64$ unlabeled samples each, seed 12345) and optimizing committal rules over a dense threshold-and-polarity family on each evidence statistic.

In the *witness* regime the results were quoted in Section 4.2.2: empirical floors 0.991–1.000 against the theoretical 1, exact n-sample evidence TV equal to 0 at every n, and 100% KGA abstention. In the *detectable* regime, $P^1(X) = \mathcal{N}(-\mu, 1)$ and $P^2(X) = \mathcal{N}(+\mu, 1)$ with opposite benefits, the closed form of Step 5 (with $\delta = 2\mu$) is traced across both the shift magnitude and the sample size. Sweeping μ at $n = 64$: at $\mu = 0.1$ the exact floor is $1 - \mathrm{TV} = 0.4237$ and the best empirical committal error is 0.4189; at $\mu = 0.25$, floor 0.0455 versus empirical 0.0451; at $\mu = 0.5$ the floor is 6.3×10^{-5} and the empirical error is 0.0. Sweeping n at fixed $\mu = 0.25$, the (floor, empirical) pairs are (0.8026, 0.8134) at $n = 1$, (0.6171, 0.6055) at $n = 4$, (0.3173, 0.3209) at $n = 16$, (0.0455, 0.0452) at $n = 64$, and (6.3×10^{-5}, 0.0) at $n = 256$ — the empirical optimum tracks the closed form throughout, fluctuating around it in both directions by Monte-Carlo error

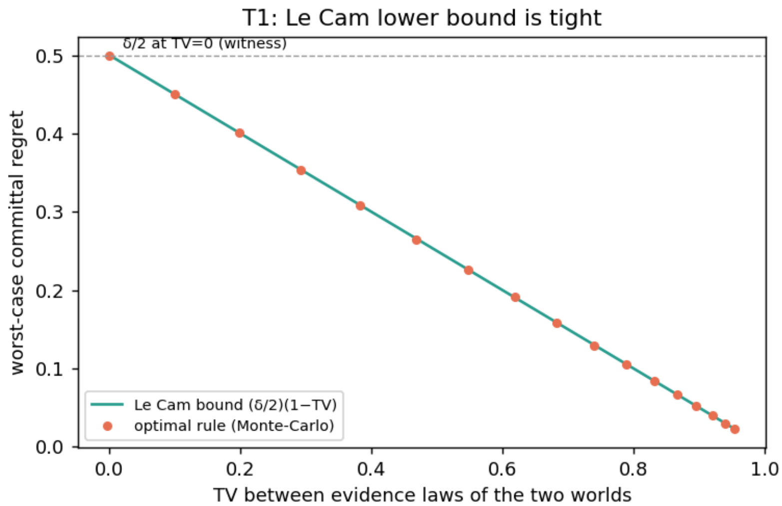

Figure 4.2: Theorem 4.2 in measurement: the optimal committal rule's worst-case regret coincides with the Le Cam floor $\frac{\delta}{2}(1-\mathrm{TV})$ across the total-variation sweep; at $\mathrm{TV} = 0$ (the witness) it equals $\delta/2$, the minimax floor of Proposition 4.1 with benefit gap δ. Produced by `scripts/theory_extensions_validation.py` [28]. The plot is the bridge between this section and the next: the testing floor $1 - \mathrm{TV}$ (this section) multiplies the benefit margin (next section) to give worst-case regret.

(the threshold is optimized on the same finite sample, so small dips below the population floor are expected and honest). Meanwhile the KGA three-way rule abstains on 100% of instances at $\mu \in \{0, 0.1\}$, on 0.45% at $\mu = 0.25$, and on 0% at $\mu \geq 0.5$ with zero committal error: the rule stops abstaining exactly as the worlds become detectable, which is the operational content of the phase diagram's x-axis (Figure 3.1).

4.4 The Bayes gate and the plug-in regret identity

The impossibility results bound what *cannot* be decided; the second result of this chapter prices decisions exactly, and in doing so converts the abstract floor into deployed risk. The objects are the action-conditional risks of Chapter 3: with $a_{\mathrm{FRZ}}(z) = \mathbb{E}[\ell(f_0(X), Y) \mid Z{=}z]$ and $a_{\mathrm{ADP}}(z) = \mathbb{E}[\ell(f_a(X), Y) \mid Z{=}z]$, so that $\Delta(z) = a_{\mathrm{FRZ}}(z) - a_{\mathrm{ADP}}(z)$, a gate $g\colon z \mapsto \{0, 1\}$ (encoding $1 = \text{ADAPT}$, $0 = \text{FREEZE}$) has risk

$$R(g) \;=\; \mathbb{E}\big[g(Z)\, a_{\mathrm{ADP}}(Z) + (1 - g(Z))\, a_{\mathrm{FRZ}}(Z)\big].$$

The Bayes gate is $g^\star(z) = \mathbf{1}[\Delta(z) > 0]$; a plug-in gate built from any label-free benefit estimator $\widehat{\Delta}$ is $\widehat{g}(z) = \mathbf{1}[\widehat{\Delta}(z) > 0]$. The paper's intuition for what follows: a gate's excess risk over the oracle equals the benefit it forfeits exactly where it picks the wrong sign — near-zero-benefit mistakes are cheap, large-benefit ones are not, which is why abstaining in the low-margin band is safe.

Theorem 4.3 (Plug-in regret decomposition)**.** *For every estimator $\widehat{\Delta}$,*

$$R(\widehat{g}) - R(g^\star) \;=\; \mathbb{E}\big[\,|\Delta(Z)|\,\mathbf{1}\{\operatorname{sign}\widehat{\Delta}(Z) \neq \operatorname{sign}\Delta(Z)\}\,\big],$$

where $\operatorname{sign}(t) = \mathbf{1}[t > 0]$*. In particular $g^\star$ minimizes R, regret is zero iff the gate's sign matches a.e., and on the boundary band $\{|\Delta| < \varepsilon\}$ the committal regret is $\le \varepsilon$—justifying* ABSTAIN *there.*

Proof. We expand the paper's one-paragraph proof into its pointwise case analysis, the integration step, and the three consequences.

Step 1 (the conditional risk is affine in the action). Condition on $Z = z$ and write $\rho(a; z)$ for the conditional risk of playing action $a \in \{0, 1\}$ at z: by definition of the action-conditional means, $\rho(1; z) = a_{\mathrm{ADP}}(z)$ and $\rho(0; z) = a_{\mathrm{FRZ}}(z)$, so

$$\rho(1; z) - \rho(0; z) \;=\; a_{\mathrm{ADP}}(z) - a_{\mathrm{FRZ}}(z) \;=\; -\Delta(z):$$

adapting lowers conditional risk by exactly $\Delta(z)$ (raises it when $\Delta(z) < 0$). The total risk of any gate is the average of its pointwise choices, $R(g) = \mathbb{E}_Z[\rho(g(Z); Z)]$, by the tower property.

Step 2 (pointwise excess of the plug-in gate). Fix z and compare $\widehat{g}(z)$ with $g^\star(z) = \mathbf{1}[\Delta(z) > 0]$. There are three cases. If $\widehat{g}(z) = g^\star(z)$, the excess $\rho(\widehat{g}(z); z) - \rho(g^\star(z); z)$ is 0. If $\widehat{g}(z) = 1$ but $g^\star(z) = 0$ — the gate adapts although $\Delta(z) \le 0$ — the excess is $\rho(1; z) - \rho(0; z) = -\Delta(z) = |\Delta(z)|$, using $\Delta(z) \le 0$. If $\widehat{g}(z) = 0$ but $g^\star(z) = 1$ — the gate freezes although $\Delta(z) > 0$ — the excess is $\rho(0; z) - \rho(1; z) = \Delta(z) = |\Delta(z)|$. In all cases,

$$\rho(\widehat{g}(z); z) - \rho(g^\star(z); z) \;=\; |\Delta(z)|\;\mathbf{1}\{\widehat{g}(z) \neq g^\star(z)\},$$

and since $\widehat{g}(z) = \mathbf{1}[\widehat{\Delta}(z) > 0]$ and $g^\star(z) = \mathbf{1}[\Delta(z) > 0]$, the disagreement event $\{\widehat{g}(z) \neq g^\star(z)\}$ is exactly the sign-mismatch event $\{\operatorname{sign}\widehat{\Delta}(z) \neq \operatorname{sign}\Delta(z)\}$ under the stated convention $\operatorname{sign}(t) = \mathbf{1}[t > 0]$. (Ties $\Delta(z) = 0$ are immaterial: whatever the convention assigns them, they enter the right-hand side multiplied by $|\Delta(z)| = 0$.)

Step 3 (integrate, and read off the consequences). Taking $\mathbb{E}_Z$ of the pointwise identity gives the displayed equation exactly — it is an identity, not an inequality, valid for every estimator, every evidence map, and every loss with finite conditional means; no boundedness, continuity, or distributional assumption enters. The three consequences: (a) running the same two steps with an arbitrary gate g in place of $\widehat{g}$ shows $R(g) - R(g^\star) = \mathbb{E}[|\Delta(Z)|\mathbf{1}\{g \neq g^\star\}] \ge 0$, so $g^\star$ minimizes R over all gates; (b) the regret vanishes iff the mismatch indicator vanishes Δ-weighted almost everywhere, i.e. iff the gate's sign matches that of $\Delta(Z)$ a.e. on $\{\Delta(Z) \neq 0\}$; (c) on the boundary band $\{|\Delta(Z)| < \varepsilon\}$, the pointwise excess is $|\Delta(z)|\,\mathbf{1}\{\cdot\} < \varepsilon$, so any committal choice there — right or wrong — forfeits less than ε of risk; abstaining on the band is therefore safe at price at most ε, which is the decision-theoretic justification for the certificate's abstention region in Chapter 5. $\square$

The identity composes with the matched-evidence situation of Theorem 4.1 to give the floor in regret units; this is the paper's proposition, stated verbatim.

Proposition 4.1 (Label-free minimax floor)**.** *In the unknowable two-point regime of Theorem 4.1 (*$\mathrm{Law}(Z \mid P_T^1) = \mathrm{Law}(Z \mid P_T^2)$*,* $|\Delta^1| = |\Delta^2| = |\Delta|$*, opposite signs), every label-free estimator has worst-case expected regret* $\geq \mathbb{E}[|\Delta(Z)|]/2$*, attained by abstaining.*

Proof. Let $\widehat{\Delta}$ be any label-free estimator, $\widehat{g}$ its plug-in gate, and allow the gate to be randomized: define $q(z) = \mathbb{P}[\widehat{g}(z) = 1 \mid Z = z]$, the adapt-probability at evidence z. Because the estimator and any internal randomization are label-free, and the evidence law is the *same* measure under both worlds, the function q is common to the two worlds — this is Step 3 of Theorem 4.1's proof applied conditionally at each z. Now evaluate the sign-error probability at z in each world: in world 1 the benefit is negative, so the gate errs exactly when it adapts, with probability $q(z)$; in world 2 the benefit is positive, so the gate errs exactly when it freezes, with probability $1 - q(z)$. The two error probabilities sum to 1 at every z, so the larger is $\geq \frac{1}{2}$. By Theorem 4.3, extended to randomized gates by averaging the pointwise identity over the gate's internal randomness, the regret in world j is $\mathrm{reg}_j = \mathbb{E}\big[|\Delta(Z)| \cdot \mathbb{P}(\text{sign error at } Z \text{ in world } j)\big]$; hence

$$\mathrm{reg}_1 + \mathrm{reg}_2 = \mathbb{E}\big[|\Delta(Z)|\,(q(Z) + 1 - q(Z))\big] = \mathbb{E}\big[|\Delta(Z)|\big],$$

and therefore $\max_j \mathrm{reg}_j \geq \frac{1}{2}(\mathrm{reg}_1 + \mathrm{reg}_2) = \mathbb{E}[|\Delta(Z)|]/2$. The floor is attained: the maximally non-committal allocation $q \equiv \frac{1}{2}$ — the gate-level expression of abstention, declining to favor either sign — has $\mathrm{reg}_1 = \mathrm{reg}_2 = \mathbb{E}[|\Delta(Z)|]/2$, so its worst case equals the floor exactly. In the three-action protocol the ABSTAIN action realizes the same worst-case value through its coverage accounting, which is the sense of the statement's final clause. □

Remark 4.4 (Reading the floor). The proposition is the regret-denominated shadow of Theorem 4.2 at $\mathrm{TV} = 0$: the testing floor says the sign error must be $\geq \frac{1}{2}$ in some world, and the regret identity says each unit of sign error costs $|\Delta(Z)|$. At intermediate separations the same composition gives worst-case regret of order $\frac{|\Delta|}{2}(1 - \mathrm{TV})$, which is exactly the curve traced by the validation plot of Figure 4.2. Note also what the floor does *not* say: it does not make abstention free. Abstention forfeits $\mathbb{E}[|\Delta|]/2$ of worst-case benefit too — the floor is a statement that nothing label-free can do better on matched evidence, not that abstaining is costless. The value of abstention is realized only when the system can *also* commit on the conditions where evidence does separate, which is the certificate's job in Chapter 5.

4.4.1 Numerical validation of the identity

Because Theorem 4.3 is an exact identity, its validator (`experiments/kbound/theory_validation/val_thm2_regr` [28]) can demand machine precision rather than statistical agreement, and it does, in two layers. Layer (A), the literal content of the theorem: on $n = 400{,}000$ sampled evidence points with known conditional benefit $\Delta(z) \sim \mathcal{N}(0, 1)$ and a non-trivial action-conditional baseline $a_{\mathrm{ADP}}(z)$ (so the cancellation in $R(\widehat{g}) - R(g^\star)$ is exercised, not assumed), it computes the left side from

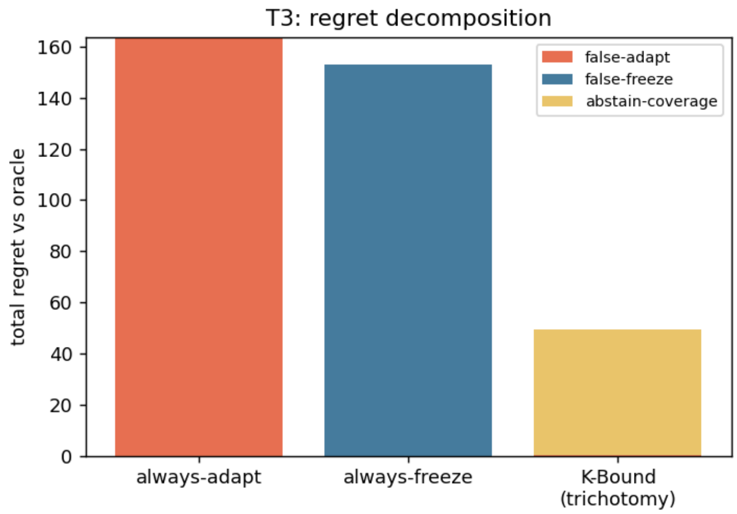

Figure 4.3: Synthetic illustration of the regret split (Theorem 4.3 / Proposition 4.1): total regret decomposes into false-adapt, false-freeze, and abstain-coverage contributions; the trichotomy's total is far below both trivial policies and is dominated by benign abstain-coverage rather than harmful false-adapt. Synthetic data; produced by `scripts/theory_extensions_validation.py` [28]. The decomposition is the deployment-facing restatement of the regret identity: every unit of regret is attributable to a named decision error, weighted by the benefit at stake.

the conditional action risks and the right side from the mismatch identity, for plug-in estimators $\widehat{\Delta} = \Delta +$noise across seven noise levels from 0 to 2.0; the script *asserts* that the maximum absolute gap over all noise levels is below 10^{-9}, and the measured gap is $\sim 10^{-17}$ — pure floating-point rounding, as the paper reports ("the identity holds to $\sim 10^{-17}$"). Layer (B): the same comparison with *realized* per-sample losses drawn around the conditional means, confirming the identity at the accuracy Monte Carlo permits (all cells within 4 standard errors).

The validator also checks each clause of the theorem and the proposition. At zero estimator noise the gates agree everywhere and both sides are exactly 0 (regret zero iff signs match a.e.). On the boundary band $\{|\Delta| < \varepsilon\}$ with $\varepsilon = 0.05$, the maximum per-z regret contribution observed is below ε, the "abstaining there costs at most ε" clause. Regret rises monotonically with estimator noise, and large noise approaches — but does not reach — the coin-flip ceiling; the ceiling itself is attained by a fully uninformative estimator (independent of Δ, the empirical surrogate of the matched-evidence regime), whose sign-mismatch rate is $\approx \frac{1}{2}$ and whose regret is 0.3986 against the floor $\mathbb{E}[|\Delta|]/2 = 0.3985$ — Proposition 4.1 realized to three decimal places.

4.5 Remarks: scope, mechanism, and what abstention buys

Remark 4.5 (Relation to selective prediction). The reject option is as old as decision theory: Chow's rule abstains from *classifying* a point when the conditional error probability exceeds a threshold [8], and the selective-prediction literature develops the resulting risk–coverage tradeoff

[12, 14]. The present chapter lifts the same logic one level: the object declined is not a prediction but the *adaptation decision* for a deployment condition, the "conditional error" is the probability of picking the wrong sign of Δ, and the analogue of Chow's excess-risk computation is exactly Theorem 4.3, with $|\Delta(z)|$ playing the role of the margin. The analogy is precise enough to transfer intuition — low-margin points are where rejection is cheap in both settings — but the impossibility half has no per-prediction counterpart: a selective classifier can always in principle resolve its uncertainty with more labeled data, whereas Theorem 4.2 exhibits decision uncertainty that no amount of the only available (unlabeled) data resolves.

Remark 4.6 (The impossibility is about evidence-measurability, not sample size)*.* Every quantity in Theorem 4.2 is indexed by the batch size n, and the witness conclusion holds *for every* n: the floor is 1 at $n = 10$ and at $n = 10^9$. This locates the obstruction precisely. Statistical folklore expects uncertainty to shrink at rate $n^{-1/2}$, and indeed the concentration tools that power the certificate of Chapter 5 — Hoeffding and empirical-Bernstein bounds [22, 37] — shrink the *radius* ε at exactly that rate. But a confidence radius shrinks toward the estimand's identifiable value; when the sign of Δ is not a function of the evidence law, there is no identifiable value to shrink toward. Formally, the decision rule is measurable with respect to $\sigma(Z)$ (plus independent randomness), and on the witness pair the two worlds induce the *same* measure on $\sigma(Z)$; the answer is not a random variable on the space the rule lives on. More data refines the empirical estimate of that common measure — it cannot break the tie. This is why the chapter's results are phrased as non-identifiability rather than as a rate: the failure is of the σ-algebra, not of the variance.

Remark 4.7 (What breaks without each hypothesis)*.* It is worth walking the hypotheses of the two theorems once more, now with their proofs in hand. *Matched evidence laws* (hypothesis (i)): drop it and Lemma 4.1 immediately monetizes the difference — the floor falls to $1 - \mathrm{TV} < 1$, and a likelihood-ratio test on the evidence achieves it; this is not a pathology but the design target of evidence engineering (Remark 4.3). *Opposite benefit signs* (hypothesis (ii)): drop it and the matched evidence is harmless, since one committed action is correct in both worlds; the trichotomy collapses to a dichotomy on such families. *Fixed, known* (f_0, f_a) (Assumption 3.1): if the candidate were not fixed at decision time, Z would not be a world-independent function of the covariates and Step 2 of Theorem 4.1's proof would fail to produce matched evidence laws — but the conclusion would not improve, because the benefit being certified would itself be a moving target. *The label-free protocol* (Assumption 3.2): a single labeled target batch breaks the construction completely — the per-sample paired benefits δ_i become observable, their mean estimates Δ directly, and the witness worlds separate at the first label. The impossibility is exactly as strong as the constraint that creates it, which is why the paper is careful to state the protocol (Remark 3.1) before the theorems.

Remark 4.8 (What abstention buys, and what comes next)*.* The two results of this chapter are deliberately one-sided: they establish that on matched-evidence pairs every committal policy pays the floor, and that the unique worst-case-optimal action there is ABSTAIN. What they do not yet provide is a *computable* rule that abstains there and commits elsewhere with guarantees. That is the certificate's job: Theorem 5.1 (Chapter 5) shows that an interval $\widehat{\Delta} \pm \varepsilon$ with $1 - \alpha$ coverage yields an adapt/freeze/abstain rule whose false-adapt and false-freeze rates are each at most α; its forced-abstention corollary closes the loop with this chapter by showing that on matched evidence any such valid rule *must* abstain with probability at least $1 - 2\alpha$ — abstention is mandatory, not merely prudent; and its sequential complement (Theorem 5.2) extends the guarantee to streams with optional stopping. On the positive side of the boundary, Theorem 5.3 proves that under covariate shift with a bounded density ratio the sign of Δ *is* identifiable from labeled source plus unlabeled target [48], and Theorem 5.4 characterizes $\operatorname{sign} \Delta$ in general as an ordinal accuracy comparison on the observable disagreement region D — the quantity whose label-free bracketing is the program's one remaining open piece (Conjecture 5.1). The unknowable regime proved here is thus not the end of the story but its boundary condition: everything Chapter 5 certifies, it certifies *because* it abstains where this chapter says it must.

Chapter 5

Certificates: When the Boundary Can Be Crossed Safely

Chapter 4 ended on deliberately negative ground — the benefit-sign specialization of an impossibility the literature already owns [13, 27]. Theorem 4.1 exhibited two target worlds that induce *exactly* the same label-free evidence yet assign *opposite* signs to the adaptation benefit $\Delta = R_T(f_0) - R_T(f_a)$, and its quantitative strengthening showed that the best achievable committal error against such a pair equals $1 - \mathrm{TV}$ of the evidence laws — which is 1 in the witness. Theorem 4.3 then priced the failure: a gate's regret over the oracle is precisely the benefit magnitude $|\Delta(Z)|$ it forfeits on each sign mistake, and on evidence compatible with both signs no estimator can beat the floor $\mathbb{E}[|\Delta(Z)|]/2$, attained by abstaining. The unknowable regime is not a deficiency of any particular detector or drift statistic; it is a property of the information available — and the general reason for it, that a shift-dependent quantity is unidentifiable without an assumption on the shift, is due to [13].

This chapter is the constructive half of the theory. It asks the question the impossibility result leaves open: *when the evidence does separate the worlds, how should a system cross the boundary — and what guarantee can it carry across?* The general shape of the answer — commit when an observable quantity beats the shift budget, else abstain — is the decision rule of Kalai and Kanade [27]; our task is to give that rule, for the benefit-sign question, an explicit computable form and a finite-sample certificate. The answer is a *certificate*: a label-free benefit estimate $\widehat{\Delta}(Z)$ equipped with a finite-sample confidence radius $\varepsilon(Z)$, together with the three-way rule that commits only when the certified interval clears zero. We develop the batch certificate and its error guarantee (Theorem 5.1), discharge the radius two ways — by empirical-Bernstein concentration and by split-conformal calibration — and derive both the mandatory-abstention consequence on matched evidence and the sample complexity of certification. We then build the streaming complement: an anytime-valid certificate by testing-by-betting (Theorem 5.2), valid at every stopping time simultaneously. The second half of the chapter maps the regimes where the boundary genuinely opens: the covariate-shift regime where the benefit sign becomes fully identifiable (Theorem 5.3), and the disagreement-region characterization that reduces $\operatorname{sign}\Delta$ to an ordinal accuracy comparison on an observable set for binary, multiclass, and regression losses (Theorems 5.4–5.6). That characterization is what turns the observable-versus-budget threshold into the exact, computable frontier $|M| > \beta$ on the disagreement region, and it is what lets us place the label-free accuracy estimators (ATC, difference-of-confidence, generalized disagreement equality, agreement-on-the-line) as the zero-budget ($\beta = 0$) face of the same frontier. The chapter closes with the one piece the program leaves open — the label-free *bracketing* of that ordinal comparison (Conjecture 5.1) — stated precisely, with an honest account of why current techniques do not resolve it and which partial results are proved.

5.1 From Impossibility to Conditional Possibility

The structure of the impossibility theorem dictates the structure of its escape. The witness of Theorem 4.1 succeeds because the evidence law $\mathrm{Law}(Z)$ carries no information about $\operatorname{sign}\Delta$: both worlds produce it identically. A certificate can therefore exist only *conditionally* — under an explicit assumption that the evidence at hand is not of the witness kind, i.e. that it constrains the benefit sign. The paper names this property at the definition stage (Chapter 3) and invokes it as the standing assumption of the certificate result. We restate it here, in the paper's own words, as the assumption the entire chapter trades under.

Assumption 5.1 (Observable risk alignment, certificate form)**.** The evidence is *risk-aligned* in the sense of Chapter 3: Z identifies or bounds $\operatorname{sign}\Delta(z)$ (it need not reveal the label, only whether adaptation helps). Operationally, a label-free estimator $\widehat{\Delta}(z)$ admits a radius $\varepsilon(z)$ with

$$\mathbb{P}\big[\,|\widehat{\Delta}(z) - \Delta(z)| \le \varepsilon(z)\,\big] \;\ge\; 1 - \alpha.$$

The first sentence is the paper's definition of risk alignment, verbatim; the second is the exact premise of the certificate theorem below. The two are the qualitative and quantitative faces of the same requirement: a radius $\varepsilon(z)$ with valid $1 - \alpha$ coverage is precisely a device that *bounds* $\operatorname{sign}\Delta(z)$ whenever the interval $[\widehat{\Delta} - \varepsilon, \widehat{\Delta} + \varepsilon]$ excludes zero.

Given the assumption, the decision rule is forced by the geometry of the interval. Estimate the benefit, attach the radius, and act on the *certified* part of the interval only:

$$\textsc{adapt} \text{ if } \widehat{\Delta} - \varepsilon > 0, \qquad \textsc{freeze} \text{ if } \widehat{\Delta} + \varepsilon < 0, \qquad \textsc{abstain} \text{ otherwise.}$$

Adapting requires the entire interval to sit above zero; freezing requires it to sit below; an interval that straddles zero is exactly the evidence state in which both signs remain consistent with what has been observed, and the rule declines to commit. The three outputs of the rule are the three regimes of the knowability map: knowably helpful, knowably harmful, unknowable.

Before stating the theorem we are explicit about what its guarantee can and cannot mean, because the certificate is easy to over-read.

Remark 5.1 (What the guarantee means — and what it does not). First, the guarantee is *conditional* on Assumption 5.1. Theorem 4.1 shows that the assumption itself cannot be verified from Z alone: on matched-evidence worlds no label-free procedure can detect that the radius has lost its meaning. The certificate does not contradict the impossibility theorem — it relocates the burden into the validity of ε, which in practice is purchased with a labeled validation or calibration sample assumed exchangeable with deployment (the "validation labels allowed" sense of label-free fixed in Chapter 3; no *target* labels are ever used). The paper is explicit that this residual assumption "is not proved in full generality," and we preserve that honesty here. Second, the guarantee bounds the probability of the *joint* event $\{\text{commit} \wedge \text{wrong sign}\}$; it is not a posterior statement of the form "given that we adapted, the chance of harm is at most α." Third, the certificate controls *sign* errors, not magnitudes: by Theorem 4.3 the regret of a wrong commitment is $|\Delta|$, so the certificate's value concentrates exactly where $|\Delta|$ is large — which is also where the interval clears zero soonest. Fourth, abstention is not free; its price is coverage, the benefit forgone when $\Delta > 0$ but the interval straddles zero. The level α is the dial that trades these two costs, and we will see the trade measured on real tasks (Figure 5.3).

5.2 The Finite-Sample Certificate

We now state the certificate theorem exactly as the paper does, and then expand its proof to full length.

Theorem 5.1 (Certificate). *Suppose a label-free estimator $\widehat{\Delta}(z)$ admits a radius $\varepsilon(z)$ with $\mathbb{P}[\,|\widehat{\Delta}(z)-\Delta(z)| \le \varepsilon(z)\,] \ge 1-\alpha$. Define*

$$\textsc{adapt} \textit{ if } \widehat{\Delta}-\varepsilon > 0, \quad \textsc{freeze} \textit{ if } \widehat{\Delta}+\varepsilon < 0, \quad \textsc{abstain} \textit{ otherwise.}$$

Then $\mathbb{P}[\textsc{adapt} \wedge \Delta \le 0] \le \alpha$ and $\mathbb{P}[\textsc{freeze} \wedge \Delta \ge 0] \le \alpha$.

Proof. Write $G := \{\,|\widehat{\Delta}-\Delta| \le \varepsilon\,\}$ for the coverage event, so that $\mathbb{P}[G] \ge 1-\alpha$ by hypothesis, and let G^c denote its complement, of probability at most α. The argument is a containment of error events in G^c.

False adapt. Suppose the rule outputs ADAPT and the coverage event G holds. The decision ADAPT means $\widehat{\Delta}-\varepsilon > 0$; the event G means $\Delta \ge \widehat{\Delta}-\varepsilon$. Chaining the two inequalities,

$$\Delta \;\ge\; \widehat{\Delta}-\varepsilon \;>\; 0,$$

so on G an ADAPT decision is necessarily correct: $\Delta > 0$. Contrapositively, the false-adapt event $\{\textsc{adapt} \wedge \Delta \le 0\}$ is disjoint from G, i.e.

$$\{\textsc{adapt} \wedge \Delta \le 0\} \;\subseteq\; G^c, \qquad \text{hence} \qquad \mathbb{P}[\textsc{adapt} \wedge \Delta \le 0] \;\le\; \mathbb{P}[G^c] \;\le\; \alpha.$$

False freeze. The argument is symmetric. On G we also have $\Delta \le \widehat{\Delta} + \varepsilon$, and the decision FREEZE means $\widehat{\Delta} + \varepsilon < 0$; chaining gives $\Delta < 0$, so on G a FREEZE decision is necessarily correct. Hence $\{\text{FREEZE} \wedge \Delta \ge 0\} \subseteq G^c$ and its probability is at most α.

Abstention. The third branch incurs no error event by construction: abstaining asserts nothing about sign Δ, so there is nothing to be wrong about. Its cost is accounted separately, as coverage. □

Remark 5.2 (One bad event covers both errors). Because both error events were shown to be subsets of the *same* bad event G^c, the proof in fact yields the slightly stronger statement $\mathbb{P}[(\text{ADAPT} \wedge \Delta \le 0) \cup (\text{FREEZE} \wedge \Delta \ge 0)] \le \alpha$ when the radius is two-sided as assumed: there is no need to spend α twice. The paper states the two one-sided bounds, which is the form the deployment guarantees quote, and we keep that convention.

The theorem is short because it is an exercise in honest bookkeeping: every gram of statistical content lives in the premise, the validity of $(\widehat{\Delta}, \varepsilon)$. The rest of this section discharges that premise in the two ways the system actually uses — a concentration-of-measure radius when $\widehat{\Delta}$ is a sample mean of bounded paired benefits, and a split-conformal radius when $\widehat{\Delta}$ is an arbitrary learned predictor of the benefit.

5.2.1 Discharging the Radius I: Empirical-Bernstein Concentration

The paper's first instantiation applies when Δ is a mean of bounded *paired benefits*

$$X_i \;=\; \ell(f_0(x_i), y_i) - \ell(f_a(x_i), y_i) \;\in\; [a, b]$$

over a validated stress bank — a calibration collection of conditions with held-out labels (validation labels, never target labels) on which both the frozen and the candidate model are evaluated. Then $\Delta = \mathbb{E}[X_i]$, the estimator $\widehat{\Delta} = \frac{1}{n}\sum_i X_i$ is a bounded sample mean, and a lower confidence bound on a bounded mean is a classical object. Two inequalities are relevant: the distribution-free Hoeffding bound, and the variance-adaptive empirical-Bernstein bound that the implementation uses.

Lemma 5.1 (Hoeffding lower confidence bound [22])**.** *Let $X_1, \ldots, X_n$ be independent with values in $[a, b]$, mean μ, sample mean $\widehat{\mu}$, and range $R = b - a$. For any $\alpha \in (0, 1)$, with probability at least $1 - \alpha$,*

$$\mu \;\ge\; \widehat{\mu} - R\sqrt{\frac{\log(1/\alpha)}{2n}}.$$

Proof. By Hoeffding's lemma, a bounded centered variable satisfies $\mathbb{E}[e^{s(X_i-\mu)}] \le e^{s^2R^2/8}$ for all $s \in \mathbb{R}$. The Chernoff method applied to $-\sum_i(X_i - \mu)$ gives $\mathbb{P}[\widehat{\mu} - \mu \le -t] \le \exp(-2nt^2/R^2)$ for every $t > 0$; setting the right side equal to α and solving for t yields $t = R\sqrt{\log(1/\alpha)/(2n)}$, which is the displayed bound. □

Lemma 5.2 (Empirical-Bernstein lower confidence bound [37])**.** *Let $X_1, \ldots, X_n$ ($n \geq 2$) be i.i.d. with values in an interval of length $R = b - a$, mean μ, sample mean $\widehat{\mu}$, and unbiased sample variance $\widehat{V}$. For any $\alpha \in (0,1)$, with probability at least $1-\alpha$,*

$$\mu \;\geq\; \widehat{\mu} \;-\; \sqrt{\frac{2\widehat{V}\,\log(2/\alpha)}{n}} \;-\; \frac{7R\,\log(2/\alpha)}{3(n-1)}.$$

Proof sketch. The result is Maurer and Pontil's empirical Bernstein bound, stated there for variables in $[0,1]$; applying it to the rescaled variables $(X_i - a)/R$ and multiplying through by R gives the form above (the mean and radius scale by R, the variance by R^2). The argument has two halves, each run at level $\alpha/2$ and combined by a union bound. First, Bernstein's inequality with the *true* variance σ^2 gives, with probability at least $1-\alpha/2$, $\mu \geq \widehat{\mu} - \sqrt{2\sigma^2\log(2/\alpha)/n} - R\log(2/\alpha)/(3n)$. Second, the sample standard deviation concentrates around the true one: a self-bounding concentration argument (the entropy method applied to the variance statistic) gives $\sigma \leq \sqrt{\widehat{V}} + R\sqrt{2\log(2/\alpha)/(n-1)}$ with probability at least $1-\alpha/2$. Substituting the second display into the first and collecting the $1/n$- and $1/(n-1)$-order terms produces the stated constant $7/3$; the bookkeeping is exactly that of [37]. □

Applying Lemma 5.2 to the paired benefits and to their negatives gives a two-sided radius

$$\varepsilon_{\mathrm{EB}} \;=\; \sqrt{\frac{2\widehat{V}\,\log(2/\alpha)}{n}} \;+\; \frac{7(b-a)\,\log(2/\alpha)}{3(n-1)},$$

which satisfies the premise of Theorem 5.1 whenever the paired benefits are independent or exchangeable draws from the deployment-relevant distribution. This is the certificate implemented in the repository's `switching_certificate.py` and re-exported as `kga/certificate.py::empirical_bernstein` [28]; the implementation takes the range explicitly ($R = 2$ for paired $|p-y|$ losses, since each loss lies in $[0,1]$), and returns an *infinite* radius when $n < 2$ — a deliberate safety default under which the rule of Theorem 5.1 can only abstain. The paper's analytic statements use the streamlined radius $\varepsilon(n) = \sigma\sqrt{2\log(2/\alpha)/n} + 3(b-a)\log(2/\alpha)/n$ with a variance proxy σ^2 in place of $\widehat{V}$; the two differ only in constants and in the empirical-versus-proxy variance, and we will use whichever form the quoted statement uses.

Remark 5.3 (Why empirical Bernstein, and why it is the right bound for this problem)*.* The Hoeffding radius scales with the *range* R; the empirical-Bernstein radius scales, in its leading term, with the *standard deviation*. For paired benefits this distinction is not a refinement but the essence of the problem. Under 0/1 loss the per-sample benefit $\delta = \mathbf{1}[f_0 \neq Y] - \mathbf{1}[f_a \neq Y]$ takes values in $\{-1, 0, +1\}$ and vanishes identically off the disagreement region $D = \{x : f_0(x) \neq f_a(x)\}$ — the two losses cancel wherever the models agree, a fact we exploit structurally in Section 5.5. Consequently $\mathbb{E}[\delta^2] \leq P_T(D)$ (with equality in the binary case, where exactly one model is correct on D), so the variance obeys $\mathrm{Var}(\delta) = \mathbb{E}[\delta^2] - \Delta^2 \leq P_T(D)$. The empirical-Bernstein leading term is therefore on the order of $\sqrt{2P_T(D)\log(2/\alpha)/n}$, while the Hoeffding radius is $2\sqrt{\log(1/\alpha)/(2n)}$ regardless:

when the candidate adaptation touches only a small fraction of predictions — the common case — the variance-adaptive radius is smaller by a factor of order $\sqrt{P_T(D)}$. All of the certificate's statistical information is concentrated on the disagreement region, and the empirical-Bernstein bound is the concentration inequality that knows it.

We restate the paper's own honesty note before moving on. The empirical-Bernstein radius is valid *under independence or exchangeability of the calibrated benefit sample with deployment*; in the paper's words, this assumption "is the residual burden and is not proved in full generality." Theorem 5.1 is proved; what remains an assumption is that the world supplies data satisfying its premise. Chapter 6 carries the same caveat into the algorithm's stated assumptions.

5.2.2 Mandatory Abstention and the Cost of Certainty

Two corollaries calibrate expectations about the certificate, in opposite directions. The first says that on genuinely unknowable evidence a valid certificate does not merely *tend* to abstain — it *must*. The second quantifies how much calibration data the certificate needs before it can commit at all.

Corollary 5.1 (Forced abstention under matched evidence)**.** *Fix evidence z in the unknowable two-point regime of Theorem 4.1 (*$\mathrm{Law}(Z \mid P_T^1) = \mathrm{Law}(Z \mid P_T^2)$*, $\Delta^1 < 0 < \Delta^2$). Any rule whose false-adapt and false-freeze probabilities are each at most α (e.g. the certificate of Theorem 5.1) must abstain there with probability* $\mathbb{P}[\textsc{abstain} \mid z] \geq 1 - 2\alpha$.

Proof. A label-free rule observes only Z (and independent internal randomness), so at the fixed evidence value z its action distribution is a single triple $(q_a, q_f, 1 - q_a - q_f)$ — the probabilities of ADAPT, FREEZE, and ABSTAIN — and, crucially, this triple is *the same* in both worlds, because the two evidence laws coincide. Now read the triple against each world in turn. In world 1 the benefit is negative ($\Delta^1 < 0$), so every ADAPT the rule emits at z is a false adapt; the false-adapt constraint forces $q_a \leq \alpha$. In world 2 the benefit is positive ($\Delta^2 > 0$), so every FREEZE is a false freeze; the false-freeze constraint forces $q_f \leq \alpha$. The same q_a and q_f must satisfy both constraints simultaneously, since the rule cannot tell the worlds apart. Hence $\mathbb{P}[\textsc{abstain} \mid z] = 1 - q_a - q_f \geq 1 - 2\alpha$. □

The paper's gloss deserves repeating: this makes abstention *mandatory, not merely conservative* — on genuinely unknowable evidence a valid certificate cannot commit on more than a 2α fraction of decisions. The corollary is also the precise sense in which the certificate and the impossibility theorem are two sides of one boundary: the same α that licenses commitment where evidence separates *caps* commitment where it does not. Empirically, on the clean non-identifiability witness of Chapter 7 — a constructed matched-evidence pair — KGA abstains on 100% of instances, while a rule forced to commit pays near-chance regret 0.51; Corollary 5.1 is the theorem that says it had no valid alternative.

Corollary 5.2 (Sample complexity of certification)**.** *Let $\widehat{\Delta}$ be a mean of n bounded paired benefits in $[a,b]$ with variance proxy σ^2, and let $\varepsilon(n) = \sigma\sqrt{2\log(2/\alpha)/n} + \frac{3(b-a)\log(2/\alpha)}{n}$ be the empirical-Bernstein radius of Theorem 5.1. Then the certificate is non-abstaining (commits* ADAPT *or* FREEZE*) once*

$$n \gtrsim \frac{2\sigma^2}{\Delta^2}\log\frac{2}{\alpha} + \frac{3(b-a)}{|\Delta|}\log\frac{2}{\alpha} = O\Big(\tfrac{\sigma^2}{\Delta^2}\log\tfrac{1}{\alpha}\Big),$$

so the unlabeled paired-benefit sample size needed to certify scales inversely with the squared benefit margin and logarithmically with the confidence level.

Proof. The rule of Theorem 5.1 commits precisely when the interval clears zero, i.e. when $\varepsilon(n) < |\widehat{\Delta}|$; since $\widehat{\Delta} \to \Delta$, the asymptotically operative condition is $\varepsilon(n) < |\Delta|$. The radius is a sum of two decreasing terms, so (up to the constant absorbed in $\gtrsim$) it suffices to make each term smaller than $|\Delta|$. For the variance term,

$$\sigma\sqrt{\frac{2\log(2/\alpha)}{n}} < |\Delta| \iff n > \frac{2\sigma^2}{\Delta^2}\log\frac{2}{\alpha},$$

which is the dominant requirement when $\sigma \gg |\Delta|$. Retaining the Bernstein $1/n$ term adds the second requirement $3(b-a)\log(2/\alpha)/n < |\Delta|$, i.e. $n > 3(b-a)|\Delta|^{-1}\log(2/\alpha)$, the stated summand. The asymptotic form follows because the first term dominates whenever $\sigma^2/|\Delta| \gtrsim (b-a)$, which holds in the small-margin regime where the bound bites. □

The corollary answers what the paper calls the reviewer question — *when does the certificate actually fire?* Certification is cheap where it matters most: large benefit margins (precisely the decisions whose regret content is large, by Theorem 4.3) are certified from few samples, while margins near zero demand $n \sim \sigma^2/\Delta^2$ and below that threshold the rule abstains — which, by Theorem 4.3 again, is exactly the band where committal regret is bounded by the small $|\Delta|$ itself. The prediction is directly visible in the batch-size ablation of Chapter 7: growing the test batch from 16 to 256 samples shrinks the radius from 0.44 to 0.18 and raises coverage from 2% to 26%, the empirical shadow of the $n^{-1/2}$ term.

5.2.3 Discharging the Radius II: The Split-Conformal Variant

The empirical-Bernstein route requires $\widehat{\Delta}$ to be a sample mean of bounded benefits. In the deployed system the benefit estimator is richer: KGA trains a gradient-boosted regressor mapping the evidence vector Z to a predicted benefit, leave-one-task-out across a bank of calibration tasks (Chapter 6). For an arbitrary learned $\widehat{\Delta}$, concentration inequalities no longer apply — but a conformal wrapper restores a finite-sample radius under a single exchangeability assumption, with no further conditions on the estimator [55, 1].

Proposition 5.1 (Split-conformal radius)**.** *Let $(\widehat{\Delta}_1, \Delta_1), \ldots, (\widehat{\Delta}_n, \Delta_n)$ be calibration pairs and $(\widehat{\Delta}_{n+1}, \Delta_{n+1})$ a deployment pair such that the residuals $r_i = |\widehat{\Delta}_i - \Delta_i|$, $i = 1, \ldots, n+1$, are* *exchangeable. Let ε be the $\lceil(1-\alpha)(n+1)\rceil$-th smallest of $r_1,\dots,r_n$ (setting $\varepsilon = +\infty$ if $\lceil(1-\alpha)(n+1)\rceil > n$). Then*

$$\mathbb{P}\big[\, |\widehat{\Delta}_{n+1} - \Delta_{n+1}| \le \varepsilon \,\big] \;\ge\; 1-\alpha,$$

so the rule of Theorem 5.1, run with $(\widehat{\Delta}_{n+1}, \varepsilon)$, has false-adapt and false-freeze probabilities at most α.

Proof. Set $k = \lceil(1-\alpha)(n+1)\rceil$ and let $r_{(1)} \le \cdots \le r_{(n)}$ denote the ordered calibration residuals, so $\varepsilon = r_{(k)}$ when $k \le n$. By exchangeability of $(r_1,\dots,r_{n+1})$, the rank of r_{n+1} within the full set of $n+1$ residuals is stochastically dominated by the uniform distribution on $\{1,\dots,n+1\}$ (uniform under almost-sure distinctness; ties only push the rank down, which helps). The event $r_{n+1} \le r_{(k)}$ contains the event that the rank of r_{n+1} is at most k, whence

$$\mathbb{P}\big[r_{n+1} \le r_{(k)}\big] \;\ge\; \frac{k}{n+1} \;=\; \frac{\lceil(1-\alpha)(n+1)\rceil}{n+1} \;\ge\; 1-\alpha.$$

If $k > n$ the radius is infinite and the coverage statement is trivial. The premise of Theorem 5.1 is thus satisfied at level α for the deployment pair, and its conclusion follows. □

The guarantee is marginal over the calibration draw — the standard split-conformal semantics [1] — which matches exactly the marginal reading of Theorem 5.1 discussed in Remark 5.1. Two implementation notes keep the record straight. First, the experiments use the empirical $(1-\alpha)$-quantile of leave-one-task-out residuals (`kga/certificate.py::conformal_split`, mirroring the knowability and mixed-regime experiment scripts [28]); this differs from the $\lceil(1-\alpha)(n+1)\rceil$ order statistic by an index shift of order $1/n$, the usual practical simplification. Second, the exchangeability here is *across tasks*: the calibration bank of tasks and the deployment task are assumed exchangeable. The paper lists this as a standing limitation, and Chapter 7 probes it empirically rather than assuming it away. Altogether the repository ships four interchangeable radius constructions behind one interface — empirical-Bernstein, Hoeffding, split-conformal, and the anytime e-value process of the next section — all pure `numpy` and deterministic [28].

Figures 5.1 and 5.2 show the certificate doing on real tasks what the theorem promises on paper. On the 123-task suite the conformal band has half-width 0.049; tasks whose interval clears zero are committed, and the adapt precision is 0.90 there and 0.935 in the controlled mixed regime — in the latter, an observed false-adapt rate of 0.065 against the budget $\alpha = 0.1$. The bar chart makes the comparative point on the same 123 tasks: an uncertified always-adapt policy inherits the suite's 35.8% harmful base rate, a naive validation threshold false-adapts on 18.0% of its commitments, and the trichotomy holds the line at 10.0% — the level it was asked to hold.

5.3 The Anytime-Valid Certificate: Testing by Betting

Theorem 5.1 fixes the sample size n in advance. Streaming test-time adaptation does not: evidence arrives one batch at a time, the system looks at the accumulating benefit sample after every batch,

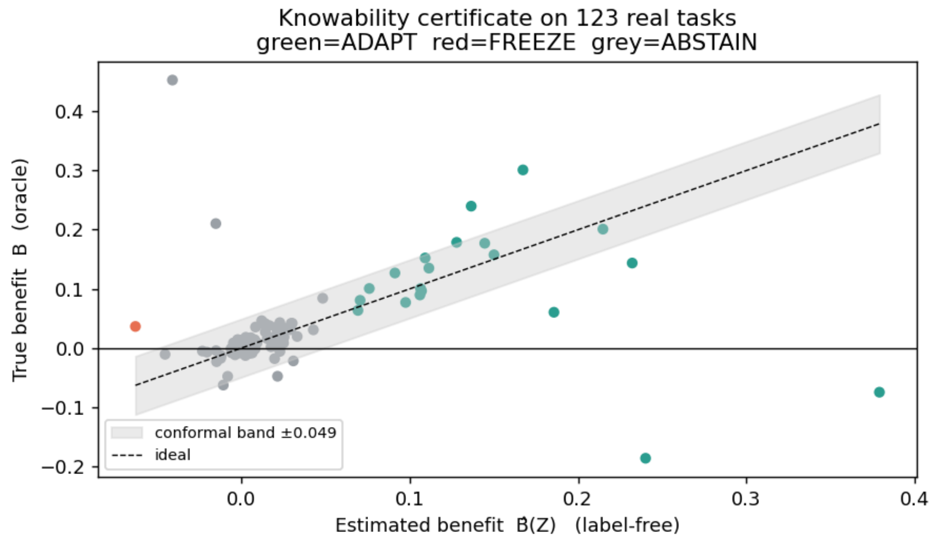

Figure 5.1: The batch certificate on the 123-task anomaly-routing suite (Chapter 7). Each point is one task: label-free estimated benefit $\widehat{\Delta}(Z)$ (horizontal) against the oracle true benefit (vertical), with the split-conformal band ± 0.049 at $\alpha = 0.1$ shaded around the ideal diagonal. Green points are certified ADAPT, red FREEZE, grey ABSTAIN: commitments concentrate where the certified interval clears zero, and abstentions cluster in the near-zero-benefit band where Theorem 4.3 prices commitment errors as cheap but Theorem 5.1 declines to make them.

and it wants to commit *the moment* the evidence suffices — without scheduling n beforehand and without paying a multiple-testing tax for the repeated looks. Classical fixed-n confidence intervals are unsound under this regime: stopping the first time a fixed-n interval clears zero inflates the error probability well above α, because the analyst's stopping rule correlates with the noise. What is needed is a certificate whose guarantee holds *simultaneously over all stopping times*. The paper supplies one in its appendix, via the modern theory of safe anytime-valid inference [43, 25, 16], in the testing-by-betting form popularized by Shafer [47] with the betting strategies of Waudby-Smith and Ramdas [59].

5.3.1 Wealth, Supermartingales, and Ville's Inequality

Testing by betting replaces the p-value with a gambler's wealth. To test the null hypothesis $H_0 : \Delta \leq 0$ ("adapting does not help"), imagine a skeptic who starts with capital 1 and, before each benefit observation X_t, stakes a fraction of current wealth on the proposition that the benefits are drifting positive. If the bet at round t is $\lambda_t \geq 0$, the wealth multiplies by $(1 + \lambda_t X_t)$: it grows when $X_t > 0$, shrinks when $X_t < 0$. Under the null, every such bet is fair or unfavorable — the conditional expected multiplier is $1 + \lambda_t \mathbb{E}[X_t \mid \mathcal{F}_{t-1}] \leq 1$ — so no betting system can be expected to multiply the skeptic's capital. Large realized wealth is therefore quantitative evidence against the null: wealth $E_t^+ \geq 1/\alpha$ is the event that a gambler playing a fair-or-losing game has multiplied

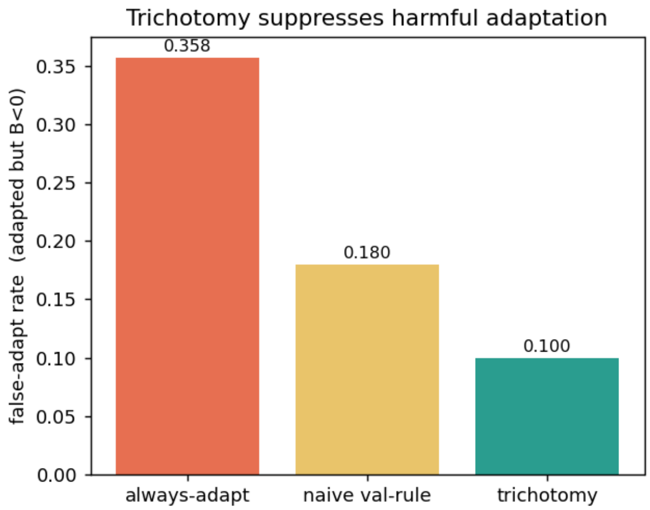

Figure 5.2: Suppression of harmful adaptation on the 123-task suite (produced by the knowability experiment of Chapter 7; same data as Figure 5.1). Each bar is the fraction of adaptation decisions whose true benefit was negative: adapting always inherits the suite's harmful base rate 0.358; a naive validation-threshold rule false-adapts on 0.180 of its commitments; the certificate trichotomy on 0.100 — the complement of its 0.90 adapt precision, in line with its calibrated level $\alpha = 0.1$.

capital ten-fold (at $\alpha = 0.1$), and the probability that this *ever* happens is at most α. The wealth process is an *e-process* — a nonnegative process whose value at any stopping time has expectation at most one under the null [16, 43] — and the "ever" quantifier is exactly what frees the certificate from fixed-n scheduling. The inequality that delivers the quantifier is Ville's [54].

Lemma 5.3 (Ville's inequality [54])**.** *Let $(M_t)_{t\geq 0}$ be a nonnegative supermartingale with respect to a filtration $(\mathcal{F}_t)$, with $\mathbb{E}[M_0] \leq 1$. Then for any $u \geq 1$,*

$$\mathbb{P}\Big[\,\exists\, t \geq 0 :\ M_t \geq u\,\Big] \;\leq\; \frac{1}{u}.$$

Proof. Let $\tau := \inf\{t \geq 0 : M_t \geq u\}$, a stopping time. Fix a horizon T; then $\tau \wedge T$ is a bounded stopping time, and the optional stopping theorem for nonnegative supermartingales gives $\mathbb{E}[M_{\tau\wedge T}] \leq \mathbb{E}[M_0] \leq 1$. On the event $\{\tau \leq T\}$ we have $M_{\tau\wedge T} = M_\tau \geq u$, while on its complement $M_{\tau\wedge T} \geq 0$; hence

$$1 \;\geq\; \mathbb{E}[M_{\tau\wedge T}] \;\geq\; u\,\mathbb{P}[\tau \leq T], \qquad \text{so} \qquad \mathbb{P}[\tau \leq T] \;\leq\; \frac{1}{u}.$$

Letting $T \to \infty$ and using continuity from below, $\mathbb{P}[\tau < \infty] \leq 1/u$, which is the claim. □

Markov's inequality bounds $\mathbb{P}[M_t \geq u]$ at one preselected t; Ville's inequality bounds the supremum over the whole trajectory at no extra cost. That uniformity over time is the entire content of "anytime-valid": the guarantee survives optional stopping, optional continuation, and continuous monitoring [25].

5.3.2 The Anytime Certificate

We state the theorem as the paper's appendix does. Two wealth processes run in parallel: E_t^+ bets that the mean benefit is positive (its growth rejects $H_0 : \Delta \le 0$ and certifies ADAPT), and E_t^- bets that it is negative (its growth rejects $H_0' : \Delta \ge 0$ and certifies FREEZE); while neither has grown past $1/\alpha$, the system abstains — which in the streaming reading means "keep sampling."

Theorem 5.2 (Anytime-valid adapt/freeze/abstain certificate)**.** *Let paired benefits* $X_i = \ell(f_0(x_i), y_i) - \ell(f_a(x_i), y_i) \in [a, b]$ *($a < 0 < b$) be adapted to* $(\mathcal{F}_t)$ *with* $\mathbb{E}[X_t \mid \mathcal{F}_{t-1}] = \Delta$*. With predictable bets* $\lambda_t \in [0, c/(-a)]$*,* $\nu_t \in [0, c/b]$*,* $c \in (0, 1)$*, set* $E_t^+ = \prod_{i \le t}(1 + \lambda_i X_i)$ *and* $E_t^- = \prod_{i \le t}(1 - \nu_i X_i)$*. Decide* ADAPT *the first time* $E_t^+ \ge 1/\alpha$*,* FREEZE *the first time* $E_t^- \ge 1/\alpha$*, else* ABSTAIN*. Then simultaneously over all* t*,* $\mathbb{P}[\exists t : E_t^+ \ge 1/\alpha \wedge \Delta \le 0] \le \alpha$ *and* $\mathbb{P}[\exists t : E_t^- \ge 1/\alpha \wedge \Delta \ge 0] \le \alpha$*.*

Proof. We prove the ADAPT direction in full; the FREEZE direction is its mirror image.

Step 1: the wealth stays positive. Each factor of E_t^+ is $1 + \lambda_i X_i$ with $\lambda_i \ge 0$ and $X_i \ge a$, so the worst case is $X_i = a < 0$, where the factor is at least $1 + \lambda_i a \ge 1 + \frac{c}{-a}\, a = 1 - c > 0$ by the bet cap. A product of strictly positive factors is strictly positive, so $E_t^+ > 0$ for all t; in particular E_t^+ is a legitimate candidate for Lemma 5.3.

Step 2: under the null, the wealth is a supermartingale. Fix any distribution in H_0, i.e. with $\Delta \le 0$. Because λ_t is *predictable* — it is $\mathcal{F}_{t-1}$-measurable, a function of the past only — it factors out of the conditional expectation:

$$\mathbb{E}\big[E_t^+ \,\big|\, \mathcal{F}_{t-1}\big] = E_{t-1}^+\, \mathbb{E}\big[1 + \lambda_t X_t \,\big|\, \mathcal{F}_{t-1}\big] = E_{t-1}^+ \left(1 + \lambda_t\, \mathbb{E}[X_t \mid \mathcal{F}_{t-1}]\right) = E_{t-1}^+ \left(1 + \lambda_t \Delta\right) \;\le\; E_{t-1}^+,$$

since $\lambda_t \ge 0$ and $\Delta \le 0$ make the multiplier at most one. With $E_0^+ = 1$, the process (E_t^+) is a nonnegative supermartingale with unit initial value under every null distribution.

Step 3: Ville closes the bound. Lemma 5.3 with $u = 1/\alpha$ gives, under every $\Delta \le 0$,

$$\mathbb{P}\big[\exists t :\; E_t^+ \ge 1/\alpha\big] \;\le\; \alpha.$$

The false-adapt event $\{\exists t : E_t^+ \ge 1/\alpha \wedge \Delta \le 0\}$ is exactly this crossing event evaluated under a null distribution, so its probability is at most α — uniformly over all stopping times, since the bound already quantifies over the whole trajectory.

The freeze direction. The process $E_t^- = \prod_{i \le t}(1 - \nu_i X_i)$ has factors $1 - \nu_i X_i$ with $\nu_i \in [0, c/b]$; the worst case is now $X_i = b > 0$, where the factor is at least $1 - \nu_i b \ge 1 - c > 0$, so $E_t^- > 0$. Under $H_0' : \Delta \ge 0$, predictability of ν_t gives $\mathbb{E}[E_t^- \mid \mathcal{F}_{t-1}] = E_{t-1}^-(1 - \nu_t \Delta) \le E_{t-1}^-$, and Ville's inequality bounds the crossing probability by α exactly as before. □

Remark 5.4 (Validity is free; power is bought by the bet)*.* Nothing in the proof used the *values* of the bets beyond nonnegativity, predictability, and the cap — any such betting sequence yields a

	anytime false-adapt ($H_0 : \Delta \leq 0$)			power / mean stop ($H_1 : \Delta > 0$)		
stream	$\Delta = -0.20$	$\Delta = -0.05$	$\Delta = 0$	$\Delta = 0.05$	$\Delta = 0.10$	$\Delta = 0.40$
two-point	0.00505	0.0268	0.0626	0.375 / 810	0.948 / 676	1.000 / 39
Beta	0.0000	0.0038	0.0584	0.988 / 521	1.000 / 127	1.000 / 16
clipped Gaussian	0.0000	0.00465	0.0559	0.976 / 588	1.000 / 148	1.000 / 17

Table 5.1: Anytime-valid certificate validation, exactly as recorded in `experiments/kbound/theory_validation/val_thm3_evalue_results.json` ($\alpha = 0.1$, wealth threshold $1/\alpha = 10$, horizon 2000, 20,000 Monte Carlo runs per cell, seed 0, bet cap $c = 0.5$; power rounded to three decimals, mean stopping times to integers). The worst-case anytime false-adapt rate across all nine null settings is $0.0626 \leq \alpha = 0.1$; the results file's verdict field records `controlled: true`.

valid certificate. The specific bets govern power and stopping time. The implementation uses a truncated growth-optimal (aGRAPA-style) plug-in

$$\lambda_t = \mathrm{clip}\Big(\widehat{\mu}_{t-1} \,/\, \widehat{\sigma}^2_{t-1},\ 0,\ c/(-a)\Big),$$

the running-mean-over-running-second-moment approximation to the log-optimal bet $\lambda^\star = \Delta/\mathbb{E}[X^2]$ [59], with a small prior stabilizing the early variance estimate; the symmetric rule drives ν_t. This separation — validity from the supermartingale property, power from the betting strategy — is the signature design freedom of game-theoretic statistics [47, 43]. The anytime certificate *complements* rather than replaces the batch certificate: at a pre-fixed n the empirical-Bernstein interval is tighter, while the e-process is valid at any stopping time; the paper states exactly this trade.

5.3.3 The Validator: Measured Anytime Error and Power

Theorem 5.2 ships with a machine-checked numerical validator, `experiments/kbound/theory_validation/val_th` [28], which runs the betting certificate over 20,000 independent streams per configuration (horizon 2000, $\alpha = 0.1$, bet cap $c = 0.5$, seed 0) and three bounded benefit distributions chosen to stress different shapes within $[a, b] = [-1, 1]$: a two-point distribution massing on the endpoints (the worst case for the positivity argument), a smooth Beta-shaped law, and a clipped Gaussian. It measures the *anytime* false-adapt rate — the fraction of null streams in which E_t^+ *ever* crosses $1/\alpha$ within the horizon — together with detection power and stopping times under alternatives. Table 5.1 reproduces the machine-readable results file.

Three features of the table are worth reading off. First, the headline: the worst-case anytime false-adapt rate over every null configuration is **0.0626**, against the budget $\alpha = 0.1$ — the guarantee of Theorem 5.2 holds with room to spare, and the worst case occurs exactly where theory says it must, at the boundary $\Delta = 0$ of the null (rates collapse to 0.00505 and below by $\Delta = -0.20$). The paper's appendix additionally records a run at $\alpha = 0.05$ with anytime false-adapt $0.032 \leq 0.05$. The slack between 0.0626 and 0.1 is the familiar conservatism of

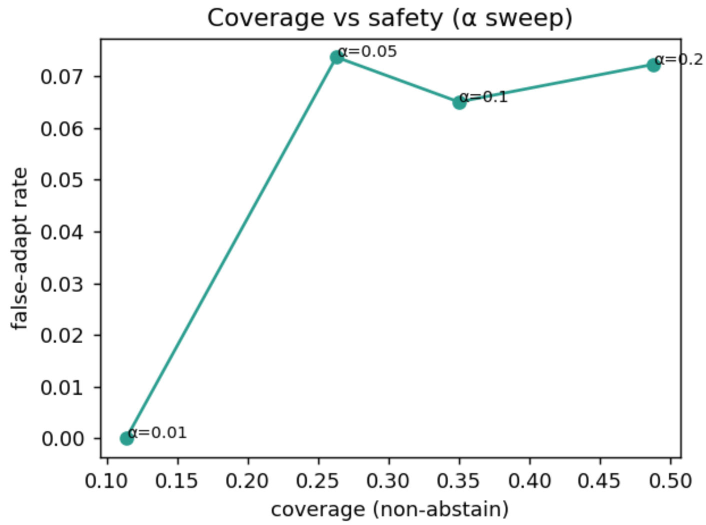

Figure 5.3: The level α as the safety–coverage dial, measured on the anomaly-routing benchmark of Chapter 7 with the split-conformal batch certificate: $\alpha = 0.01$ yields a 0% observed false-adapt rate at 11% coverage; $\alpha = 0.1$ yields 0.065 at 35% coverage; $\alpha = 0.2$ yields 0.072 at 49% coverage. The $\alpha = 0.05$ point sits at 0.073, slightly above its nominal budget — an honest finite-sample fluctuation on 369 instances under approximate cross-task exchangeability, and a reminder that the guarantee is marginal, not per-configuration.

time-uniform bounds: the certificate pays for validity at *every* stopping time, and the plug-in bet is not the oracle log-optimal one. Second, the supermartingale property itself is checked directly: at $\Delta = 0$ the Monte Carlo mean of E_t^+ stays at or below 1 (within three standard errors) at every recorded time $t \in \{1, 5, 25, 100, 500, 2000\}$ for all three streams — the results file flags `all_leq_one_within_3sem: true` in each case. Third, power behaves as a sequential test should: it rises to 1 as the margin grows, and the mean stopping time falls steeply — from roughly 810 samples at $\Delta = 0.05$ (two-point stream) to 16–39 samples at $\Delta = 0.40$. Large benefits are certified almost immediately; marginal ones take evidence; null ones, with probability at least $1 - \alpha$, are never falsely certified no matter how long the system watches.

Remark 5.5 (Bounded non-stationarity). The constant-mean premise $\mathbb{E}[X_t \mid \mathcal{F}_{t-1}] = \Delta$ can be relaxed. The paper's theory-extensions section proves a β-bounded-drift variant: if the conditional mean wanders within $\pm\beta$ of Δ, dividing the wealth by $\prod_{i\leq t}(1 + \lambda_i\beta)$ restores the supermartingale property under the null, at the price of an inflated decision threshold $(1/\alpha)\prod_{i\leq t}(1+\lambda_i\beta) \leq \exp(\beta\sum_{i\leq t}\lambda_i)$. Its validator (`val_thm9prime_drift.py` [28]) reports that under worst-case upward drift the corrected certificate holds the anytime false-adapt rate at 0.05/0.04/0.01 for $\beta = 0.1/0.2/0.4$ — all within $\alpha = 0.1$ — while the uncorrected threshold's rate inflates to 0.69 and beyond, confirming both that the correction works and that it is needed.

Both certificates — batch and anytime — expose the same single dial, the level α, and Figure 5.3 shows what the dial buys on real tasks. Tightening α to 0.01 drives observed false adaptation to zero at the cost of committing on only 11% of decisions; loosening it to 0.2 roughly quadruples coverage while the observed false-adapt rate stays near 0.07. The certificate makes the safety–coverage trade explicit and tunable, rather than implicit in a mechanism's hyperparameters.

5.4 A Provably Knowable Regime: Covariate Shift

The impossibility theorem and the certificate bracket the problem from both sides, but neither yet exhibits a natural class of shifts on which the label-free decision is *provably* available. The paper's fourth result identifies one: covariate shift. The motivating picture is a new scanner — image statistics change, disease prevalence and the disease-given-image relationship do not. Under that premise, reweighting labeled *source* data by the density ratio of the two covariate laws reconstructs every target risk, and with it the benefit and its sign.

Theorem 5.3 (Covariate-shift identifiability)**.** *Assume covariate shift $P_S(Y \mid X) = P_T(Y \mid X)$, and a density ratio $r(x) = p_T(x)/p_S(x)$ that is bounded ($r \le B < \infty$) with $\mathbb{E}_S[r(X)^2] < \infty$. Then $R_T(f) = \mathbb{E}_S[r(X)\,\ell(f(X), Y)]$, the importance-weighted estimator $\widehat{R}_T(f) = \frac{1}{n}\sum_i r(x_i)\ell(f(x_i), y_i)$ is unbiased and consistent, and hence Δ and $\operatorname{sign}\Delta$ are identifiable from labeled source plus unlabeled target whenever $|\Delta| > 0$.*

Proof. Change of measure. Write the target risk as an integral over the target joint and factor the joint through the covariate-shift premise:

$$R_T(f) = \mathbb{E}_{P_T}[\ell(f(X), Y)] = \iint \ell(f(x), y)\, p_T(y \mid x)\, p_T(x)\, dy\, dx = \iint \ell(f(x), y)\, p_S(y \mid x)\, \frac{p_T(x)}{p_S(x)}\, p_S(x)\, dy\, dx,$$

where the second equality substitutes $p_T(y \mid x) = p_S(y \mid x)$ (the premise) and multiplies and divides by $p_S(x)$; boundedness of r ensures the manipulation is legitimate wherever p_S has mass, and the target law places no mass where p_S vanishes (else r would be unbounded). The right-hand side is $\mathbb{E}_S[r(X)\,\ell(f(X), Y)]$, the first claim.

Unbiasedness and consistency. Unbiasedness of $\widehat{R}_T(f)$ is immediate from the identity just proved, since each summand $r(x_i)\ell(f(x_i), y_i)$ has source-expectation $R_T(f)$. For consistency, the summands are i.i.d. with finite second moment: with ℓ bounded in $[0, 1]$, $\mathbb{E}_S[(r\,\ell)^2] \le \mathbb{E}_S[r^2] < \infty$, so the strong law of large numbers gives $\widehat{R}_T(f) \to R_T(f)$ almost surely, with variance $\operatorname{Var}(\widehat{R}_T(f)) \le \mathbb{E}_S[r^2]/n = O(1/n)$.

Sign identifiability. Apply the estimator to both models and difference: $\widehat{\Delta} = \widehat{R}_T(f_0) - \widehat{R}_T(f_a) = \frac{1}{n}\sum_i r(x_i)\big(\ell(f_0(x_i), y_i) - \ell(f_a(x_i), y_i)\big)$ is unbiased and consistent for Δ by linearity. If $|\Delta| > 0$, then almost surely $\operatorname{sign}\widehat{\Delta} = \operatorname{sign}\Delta$ for all large n, so the sign — and indeed the value — of the benefit is identified from labeled source data plus the unlabeled target sample needed to estimate r. □

Why this escapes the impossibility theorem. It is worth dwelling on *where* the witness of Theorem 4.1 dies under the covariate-shift premise, because the mechanism is the template for every positive result in this theory. The witness constructed two worlds agreeing on $P_T(X)$ and differing only in $P_T(Y \mid X)$ — the one coordinate of the target joint that label-free evidence cannot see. Covariate shift removes exactly that degree of freedom: it pins $P_T(Y \mid X)$ to $P_S(Y \mid X)$, an object estimable from *labeled source* data. The accessible data then determine the entire target joint, $p_T(x, y) = p_T(x)\, p_S(y \mid x)$, with $p_T(x)$ read off the unlabeled target stream and $p_S(y \mid x)$ off the source. The evidence has become *sufficient* for Δ. Equivalently: the two witness worlds have different conditionals, so at most one of them can satisfy the covariate-shift premise with respect to any fixed source — inside the assumption class, matched-evidence pairs with opposite benefit simply do not exist. Identifiability results of this shape are the classical core of the covariate-shift literature [48, 49, 42]; the delta here is only the target of inference, the *sign of a risk difference* rather than a risk.

The price: weight boundedness and effective sample size. The premise buys identifiability; the moment conditions buy a usable certificate. The weighted paired benefits $r(x_i)\big(\ell(f_0) - \ell(f_a)\big)$ are bounded in $[-B, B]$ when $r \le B$, so they can be fed directly to the empirical-Bernstein certificate of Section 5.2.1, and the radius inherits the weight distribution's second moment. Since $\mathbb{E}_S[r] = 1$, the variance inflation relative to unweighted estimation is governed by $\mathbb{E}_S[r^2] \ge 1$, with equality only under no shift; the corresponding *effective sample size* is $n/\mathbb{E}_S[r^2]$ (empirically, $(\sum_i w_i)^2 / \sum_i w_i^2$), and the certificate radius scales like $\sqrt{\mathbb{E}_S[r^2]/n}$. Heavy-tailed weights therefore do not silently corrupt the decision — they *visibly inflate the radius*, and the certificate responds by abstaining or freezing. This is not a hypothetical failure mode: in the regression covariate-shift experiment of Chapter 7, a 4σ shift makes the importance-weighted candidate *hurt* on 79% of instances through sheer weight variance, and KGA refuses it wholesale (0 adapt, 11 freeze, 61 abstain), matching the oracle's mean squared error of 0.251 while always-adapt degrades to 0.338. Identifiability in principle and certifiability at a given n are different currencies, and the certificate trades only in the second.

The caveat that keeps the theorem honest. The paper attaches a caveat tying the result back to Theorem 4.1, and it is load-bearing: the premise $P_S(Y \mid X) = P_T(Y \mid X)$ is *not* testable from unlabeled data; when it fails, the regime can re-enter the unknowable region. Covariate shift is an assumption about precisely the coordinate that label-free evidence cannot observe, so adopting it is an act of modeling, not of inference. The knowability hierarchy of Table 5.2 situates this regime among its neighbors: label shift is similarly identifiable under its own structural premises (stable class-conditionals and an invertible confusion matrix, the BBSE conditions [34]); concept shift — a change in $P(Y \mid X)$ itself — is non-identifiable and lands back in Theorem 4.1; and the mixed or unknown case is exactly where the certify-or-abstain machinery of this chapter operates.

Shift type	Label-free status	Mechanism / theorem
Covariate ($P(Y \mid X)$ fixed)	identifiable	importance weighting (Thm 5.3)
Label / prior ($P(Y)$)	identifiable†	stable class-conditionals, invertible confusion (†BBSE-type)
Disagreement-local	ordinal-identifiable	sign = accuracy gap on D (Thms 5.4–5.6)
Concept ($P(Y \mid X)$ changes)	non-identifiable	re-enters Thm 4.1
Mixed / unknown	certify-or-abstain	commit where Z separates, else abstain

Table 5.2: Knowability hierarchy of distribution shifts for label-free adapt/freeze decisions, as drawn in the paper. The boundary is evidence-identifiability: some shifts are knowable, some are not, and a valid certificate abstains across the boundary. The label-shift row follows the black-box shift estimation conditions of [34].

5.5 The Disagreement Region: Where the Sign of Benefit Lives

The covariate-shift result buys identifiability with a global assumption on the shift. The fifth result of the theory takes the opposite route: no assumption on the shift at all, but a sharper look at *where* in input space the decision is actually decided. It is here that the observable-versus-budget threshold of [27] acquires, for the benefit-sign question, an *exact and computable* form. The observation is elementary and, once made, structural. Wherever the frozen and candidate models agree, their losses are identical and the per-sample benefit is exactly zero — those points contribute *nothing* to Δ, under any label law whatsoever. The entire benefit, sign and magnitude alike, is carried by the *disagreement region*

$$D \;=\; \{x : f_0(x) \neq f_a(x)\},$$

and D is *observable*: membership requires evaluating two known functions, no labels. Restricted to D, the question "does adapting help?" collapses to an above-chance check — is the candidate right more often than the incumbent on the points where they differ? — which is an *ordinal* comparison, not a risk estimate. This is what makes the general threshold concrete: writing M for the observable margin on D and β for the calibration-drift budget, the disagreement-region identity below reduces $\operatorname{sign}\Delta$ to this ordinal comparison, from which the sign is identifiable exactly when $|M| > \beta$ and the three-way rule built on it is the unique maximal sound certificate. The same reduction exhibits the label-free accuracy estimators (ATC, difference-of-confidence, generalized disagreement equality, agreement-on-the-line) as the zero-budget ($\beta = 0$) face of this criterion — sound precisely when the calibration drift γ vanishes.

Theorem 5.4 (Sign-of-difference identifiability on the disagreement region; binary case)**.** *Let $Y \in \{0,1\}$ with $0/1$ loss, and let $D = \{x : f_0(x) \neq f_a(x)\}$ be the* disagreement region *(observable, since f_0, f_a are known maps of X). Then*

$$\Delta \;=\; P_T(D)\,(2\,a_a^D - 1), \qquad a_a^D := P_T(f_a(X) = Y \mid X \in D)\,,$$

so $\operatorname{sign}\Delta = \operatorname{sign}\!\left(a_a^D - \tfrac{1}{2}\right)$*. Consequently* $\operatorname{sign}\Delta$ *is identifiable from label-free evidence* if and only if *the evidence determines whether f_a exceeds chance accuracy on D; in particular, if a label-free reliability signal brackets a_a^D relative to $\frac{1}{2}$, the sign is certified.*

Proof. Let $\delta(x,y) = \ell(f_0(x),y) - \ell(f_a(x),y) = \mathbf{1}[f_0(x) \neq y] - \mathbf{1}[f_a(x) \neq y]$ denote the per-sample benefit, so that $\Delta = \mathbb{E}_{P_T}[\delta]$.

Off D, the benefit vanishes identically. If $x \notin D$ then $f_0(x) = f_a(x)$, so the two indicators coincide for every y and $\delta(x,y) = 0$ pointwise — not merely in expectation, but for each realization of the label.

On D, the benefit is a fair-coin score for f_a. If $x \in D$ the two predictions differ, and since Y is binary, exactly one of $f_0(x), f_a(x)$ equals y. If $f_a(x) = y$ then $f_0(x) \neq y$, so $\delta = 1 - 0 = +1$; if $f_0(x) = y$ then $\delta = 0 - 1 = -1$. Hence, conditionally on D, δ is a ± 1 variable equal to $+1$ with probability a_a^D, and

$$\mathbb{E}[\delta \mid D] = P_T(f_a{=}Y \mid D) - P_T(f_0{=}Y \mid D) = a_a^D - (1 - a_a^D) = 2a_a^D - 1,$$

using that on D the two correctness events are complementary, $P_T(f_0{=}Y \mid D) = 1 - a_a^D$.

Total expectation. Splitting Δ over the partition $\{D, D^c\}$ and using $\delta \equiv 0$ off D,

$$\Delta = \mathbb{E}\big[\delta\,\mathbf{1}\{X \in D\}\big] + \mathbb{E}\big[\delta\,\mathbf{1}\{X \notin D\}\big] = P_T(D)\,\mathbb{E}[\delta \mid D] = P_T(D)\,(2a_a^D - 1).$$

Since $P_T(D) \geq 0$, the sign of Δ equals the sign of $2a_a^D - 1$, i.e. of $a_a^D - \frac{1}{2}$, whenever $P_T(D) > 0$ (and $\Delta = 0$ when $P_T(D) = 0$, in which case the decision is moot).

The identifiability equivalence. ($\Leftarrow$) If the evidence determines the event $\{a_a^D > \frac{1}{2}\}$ — for instance, a label-free reliability signal brackets a_a^D strictly on one side of $\frac{1}{2}$ — then by the identity it determines $\operatorname{sign}\Delta$. ($\Rightarrow$) Conversely, if two target worlds are compatible with the same evidence yet place a_a^D on opposite sides of $\frac{1}{2}$, the identity forces their benefits to have opposite signs, which is precisely the unknowable configuration of Theorem 4.1; so identifiability of $\operatorname{sign}\Delta$ requires the evidence to settle the above-chance question. □

Remark 5.6 (Ordinal, not cardinal — and exactly one residual unknown). The theorem changes the problem's type. Estimating either absolute risk $R_T(f_0)$ or $R_T(f_a)$ without labels is the unsupervised-risk-estimation problem, known — from the conditional-independence analyses of Steinhardt and Liang and the divergence-based bounds reviewed in the related-work discussion — to be impossible without structural assumptions. The identity shows the adapt/freeze decision never needed those cardinal objects: it needs one *ordinal* bit — is f_a above chance on D — a strictly weaker requirement. Everything else in the formula is label-free: $P_T(D)$ is an observable mass, and the per-sample benefits off D are identically zero. The residual burden is exactly the bracketing of a_a^D around $\frac{1}{2}$, and the witness of Theorem 4.1 shows that burden is real: there $f_0 = \mathbf{1}[x > 0]$ and $f_a = \mathbf{1}[x < 0]$ disagree almost everywhere, so D is essentially the whole line, and the two worlds set $a_a^D = 0$ and $a_a^D = 1$ respectively — maximal disagreement about the ordinal bit, with identical evidence. The witness is thus precisely a failure of label-free bracketing on D, which is why the open problem of Section 5.6 has the shape it has.

The binary identity generalizes in both directions the deployment cares about — multiclass classification and regression — and the generalizations are the results that close the characterization half of the program.

Theorem 5.5 (Multiclass 0/1 loss). *For $Y \in \{1, \dots, K\}$ with 0/1 loss, with $p_a = P_T(f_a{=}Y \mid D)$, $p_0 = P_T(f_0{=}Y \mid D)$, off D $\delta = 0$ and on D $\mathbb{E}[\delta \mid D] = p_a - p_0$, so $\Delta = P_T(D)(p_a - p_0)$ and $\operatorname{sign}\Delta = \operatorname{sign}(p_a - p_0)$.*

Proof. As before $\delta = \mathbf{1}[f_0 \neq Y] - \mathbf{1}[f_a \neq Y]$, and off D the two indicators coincide pointwise, so $\delta = 0$ there. On D the predictions differ, and — this is the multiclass novelty — the conditional space splits into *three* disjoint events rather than two: $\{f_a = Y\}$ (the candidate is right, the incumbent necessarily wrong since it predicts something different, $\delta = 1 - 0 = +1$), $\{f_0 = Y\}$ (the incumbent is right, the candidate wrong, $\delta = 0 - 1 = -1$), and $\{f_0 \neq Y, f_a \neq Y\}$ (both wrong, predicting two *different* wrong classes, where $\delta = 1 - 1 = 0$). The two correctness events are still disjoint on D — the models cannot both be right while disagreeing — but they are no longer complementary: $p_a + p_0 \leq 1$ with strict inequality whenever both-wrong has mass. Taking the conditional expectation over the three cells,

$$\mathbb{E}[\delta \mid D] = (+1)\, p_a + (-1)\, p_0 + 0 \cdot (1 - p_a - p_0) = p_a - p_0,$$

which one can equally read off the complement form $\mathbb{E}[\delta \mid D] = (1-p_0)-(1-p_a)$. Total expectation over $\{D, D^c\}$ gives $\Delta = P_T(D)\,\mathbb{E}[\delta \mid D] = P_T(D)(p_a - p_0)$, and nonnegativity of $P_T(D)$ gives the sign identity. For $K = 2$, exactly one predictor is correct on D, so $p_0 = 1 - p_a$, the both-wrong cell is empty, and the formula reduces to $\Delta = P_T(D)(2a_a^D - 1)$ — Theorem 5.4. For $K \geq 3$ both may err, and the identity holds regardless, because the benefit counts only who is *correct*, not whether the other is wrong. ☐

Theorem 5.6 (Squared-error regression). *For $Y \in \mathbb{R}$ with squared loss, $\delta = (f_0 - f_a)(f_0 + f_a - 2Y)$, so on D $\mathbb{E}[\delta \mid D] = \mathbb{E}[(f_0 - f_a)(f_0 + f_a) \mid D] - 2\,\mathbb{E}[(f_0 - f_a)Y \mid D]$. The first term is observable from unlabeled target; thus $\operatorname{sign}\Delta$ is label-free identifiable iff the sign contribution of $\mathbb{E}[(f_0 - f_a)Y \mid D]$ is certifiable—which holds under covariate shift (importance-weighted source estimate of $\mathbb{E}[Y \mid X]$) and fails under concept shift (re-entering Theorem 4.1).*

Proof. The pointwise identity is the difference of squares:

$$\delta = (f_0 - Y)^2 - (f_a - Y)^2 = [(f_0 - Y) - (f_a - Y)]\,[(f_0 - Y) + (f_a - Y)] = (f_0 - f_a)\,(f_0 + f_a - 2Y).$$

Off D we have $f_0(x) = f_a(x)$, so the first factor vanishes and $\delta = 0$; the disagreement region again carries the whole benefit, and $\Delta = P_T(D)\,\mathbb{E}[\delta \mid D]$ by total expectation. On D, expanding by linearity,

$$\mathbb{E}[\delta \mid D] = \mathbb{E}\big[(f_0 - f_a)(f_0 + f_a) \,\big|\, D\big] \; - \; 2\,\mathbb{E}\big[(f_0 - f_a)\,Y \,\big|\, D\big].$$

The first term is a functional of the two *known* maps f_0, f_a and of $\mathrm{Law}(X \mid D)$ only — estimable from the unlabeled target stream to arbitrary precision. The second term is the unique label-coupled object, so $\operatorname{sign}\Delta$ is label-free identifiable exactly when that term's sign contribution can be certified without target labels. Under covariate shift it can: writing $m(X) = \mathbb{E}[Y \mid X]$ (the same function in source and target by the premise), the tower property and $\sigma(X)$-measurability of $f_0 - f_a$ give

$$\mathbb{E}\big[(f_0 - f_a)\,Y \,\big|\, D\big] = \mathbb{E}\big[(f_0 - f_a)\,\mathbb{E}[Y \mid X] \,\big|\, D\big] = \mathbb{E}\big[(f_0 - f_a)\,m(X) \,\big|\, D\big],$$

where m is estimable from *labeled source* data (importance-weighted as in Theorem 5.3) and the remaining ingredients are target-observable. Under concept shift the source regression function m_S is the wrong m, the plug-in can carry the wrong sign, and no label-free correction is available — the configuration re-enters Theorem 4.1. □

5.5.1 Numerical Validation of the Identities

Algebraic identities invite the most direct kind of machine checking, and the repository's validator `experiments/kbound/theory_validation/val_thm5_multiclass.py` [28] performs it at scale. For the multiclass identity it draws 4000 random trials — class counts K ranging over $\{3, \ldots, 11\}$, sample sizes between 2000 and 8000, predictors of independently random accuracies so that Δ takes both signs across trials — and computes Δ two ways: directly with labels (the mean per-sample benefit), and through the decomposition $P_T(D)(p_a - p_0)$. As recorded in the paper, the identity holds to 10^{-16} with 100% sign agreement over the 4000 trials — and, decisively for the beyond-binary claim, the both-wrong-on-D event occurs in *every* trial, so the regime genuinely exercised is the one where $p_0 \neq 1 - p_a$. The script additionally verifies that the off-D contribution to Δ is exactly zero, sample by sample. For regression it checks the pointwise difference-of-squares identity to machine precision over 3000 trials (tolerance 10^{-9}, with 100% sign agreement between Δ and $\mathbb{E}[\delta \mid D]$), and then runs the shift demonstration of Theorem 5.6 end to end: with 40,000 source and target points, a 2.0-standard-deviation covariate shift, and a source-only kernel-regression plug-in for $m(X)$, the label-free certificate recovers the correct $\operatorname{sign}\Delta$ under covariate shift and *flips* sign under concept shift — the predicted failure, observed on cue. The validation does not, of course, prove the theorems (the proofs above do); what it certifies is that the implemented quantities are the theorems' quantities, with no sign conventions or conditioning slips between paper and code.

5.5.2 Disagreement as the Decisive Evidence Coordinate

The theorems of this section make a falsifiable prediction about *evidence design*: if the entire benefit is carried by D, then among the components of the evidence vector Z, the ones measuring disagreement between the frozen and candidate models should be the most decision-relevant — not marginally, but structurally. The ablation study of Chapter 7 tests exactly this by deleting evidence features one at a time and re-measuring the wrapper's regret-to-oracle: removing the detector-disagreement feature raises regret from 0.037 to 0.094, a 2.6× degradation and by far the largest of any feature (the next-largest deletions move regret to 0.043 and 0.039). The paper draws the connection explicitly: detector disagreement is the single most important evidence feature in the ablation, directly supporting that the disagreement region carries the decision. The theory says where the information lives; the ablation confirms the estimator starves without it.

5.6 The Open Problem: Label-Free Bracketing

Everything proved so far reduces the label-free adapt/freeze decision to a single remaining object: an ordinal comparison on the observable region D. The paper is exact about what is closed and what is not, and states the open piece as a conjecture. We reproduce it verbatim, and then spend an honest page on why it is genuinely open.

Conjecture 5.1 (Label-free bracketing; **the remaining open piece**). *Theorems 5.5–5.6* close the characterization*:* $\operatorname{sign}\Delta$ *equals an ordinal accuracy comparison on the observable region D. What remains open is the label-free* bracketing *of that comparison (p_a vs p_0 for $K \geq 3$, where $p_0 \neq 1 - p_a$ makes it a two-sided test; or the sign of $\mathbb{E}[(f_0 - f_a)Y \mid D]$ for regression) without target labels—a reliability-model assumption, as in the binary case.*

5.6.1 Why Current Techniques Do Not Resolve It

To see the precise difficulty, fix the multiclass case and ask for the object the conjecture demands: a label-free functional of the observables — the unlabeled target stream on D, the two prediction maps, and whatever source-side calibration data one allows — that outputs an interval $[L, U] \ni p_a - p_0$ with $1 - \alpha$ coverage, such that committing when $0 \notin [L, U]$ is valid *without* a premise that is itself unverifiable from the same observables. Every known route founders on one of two obstructions.

The information obstruction. Unconditionally — over all label laws on D consistent with the observed unlabeled data — the task is hopeless, and this is a theorem, not a suspicion: the witness of Theorem 4.1, restricted to D, exhibits two worlds with identical evidence laws and (p_a, p_0) equal to $(1, 0)$ and $(0, 1)$ respectively. Any bracket with valid coverage in both worlds must contain both $+1$ and -1, hence 0, hence never commits. This is the two-point method of Le Cam in its simplest form [31, 52]: the unlabeled likelihood does not separate the hypotheses, so additional structure is *necessary*, and the conjecture is properly read as a question about the *weakest* usable structure rather than about removing structure altogether. The binary case already had this character —

there the residual assumption is a reliability signal bracketing a_a^D against $\frac{1}{2}$ — but the multiclass case is strictly harder in a concrete sense the paper flags in the conjecture itself: since $p_0 \neq 1 - p_a$, one scalar no longer determines the comparison, and the test becomes two-sided in two unknown conditional accuracies linked only by the inequality $p_a + p_0 \leq 1$.

The technique obstruction. Each available tool discharges the bracketing only by importing a premise of the same epistemic type as the one being discharged. Importance weighting (Theorem 5.3) certifies the regression term $\mathbb{E}[(f_0 - f_a)Y \mid D]$ — but only under covariate shift, which fixes the unobservable conditional $P(Y \mid X)$ by fiat and is untestable from unlabeled data. Confidence-based bracketing (the route formalized below) bounds p_h by the mean calibrated confidence q_h — but only under a calibration-transfer premise, a bound η on the *target-side* miscalibration on D, which is again a quantity no label-free procedure can estimate; assuming η small is assuming a reliability model. Label-shift machinery [34] identifies shifted class priors under stable class-conditionals and an invertible confusion matrix — structural premises once more, and even granted them, priors do not by themselves yield the *conditional-on-D* accuracies the comparison needs. Empirical regularities such as agreement-based accuracy extrapolation are calibrated correlations, not guarantees, and nothing prevents the adversarial pairs of Theorem 4.1 from sitting exactly where the regularity breaks. In every case the assumption is doing the work precisely on the coordinate the data cannot see; the open question is whether some premise exists that is weaker, or — more interestingly — *self-certifying*, in the sense that its failure is itself label-free detectable, so that the bracket could widen automatically where the premise degrades.

5.6.2 What Is Actually Proved

The honest ledger has four entries on the proved side. First, the *characterization* is closed unconditionally: Theorems 5.4 through 5.6 are exact identities requiring no assumptions beyond the loss structure, and they localize the entire decision to D. Second, the binary case needs only a *one-sided* bracket of a single scalar against the fixed threshold $\frac{1}{2}$ — a structurally easier object than the multiclass two-sided comparison, which is why the binary instantiation is the one deployed throughout the experiments. Third, the regression term is certified under covariate shift (Theorem 5.6 with Theorem 5.3). Fourth, the paper proves a *conditional* closure of the conjecture under a quantitative reliability model — bounded calibration transfer — with an exact and provably tight threshold. We restate it faithfully.

Proposition 5.2 (Label-free bracketing under bounded calibration transfer)**.** *On the disagreement region D, let $\mathrm{conf}_h(x) \in [0,1]$ be a source-calibrated confidence map for predictor $h \in \{f_0, f_a\}$, and suppose its target miscalibration on D is bounded by η: $|\mathbb{P}_T(h(X){=}Y \mid \mathrm{conf}_h{=}c,\ X \in D) - c| \leq \eta$ for all c (*calibration transfer*). Let $q_h := \mathbb{E}_T[\mathrm{conf}_h(X) \mid D]$ be the label-free mean confidence. Then $|q_h - p_h| \leq \eta$, hence $p_a - p_0 \in [(q_a - q_0) - 2\eta,\ (q_a - q_0) + 2\eta]$, and*

$$|q_a - q_0| > 2\eta \implies \operatorname{sign}\Delta = \operatorname{sign}(q_a - q_0),$$

so the multiclass/regression benefit sign is identified label-free; when $|q_a - q_0| \le 2\eta$ *the bracket contains* 0 *and the rule must* ABSTAIN.

Proof. By the tower property, $p_h = \mathbb{E}_T[\mathbb{P}_T(h{=}Y \mid \mathrm{conf}_h, D) \mid D]$; the calibration-transfer bound says the inner conditional probability is within η of conf_h pointwise, so integrating preserves the bound and $|p_h - q_h| \le \eta$ for each h. Differencing the two η-balls gives $|(p_a - p_0) - (q_a - q_0)| \le 2\eta$, the stated bracket. If $|q_a - q_0| > 2\eta$, the bracket lies strictly on one side of 0, so $\mathrm{sign}(p_a - p_0) = \mathrm{sign}(q_a - q_0)$; by Theorem 5.5 this equals $\mathrm{sign}\,\Delta$. If $|q_a - q_0| \le 2\eta$ the bracket straddles 0 and certifies nothing, so the certificate logic of Theorem 5.1 mandates abstention. □

The threshold 2η is *exact*, as the paper notes: when $|q_a - q_0| \le 2\eta$ an adversary inside the two η-balls can place p_a, p_0 on either side of equality, re-entering the unknowable regime of Theorem 4.1 — so the proposition's abstention region cannot be shrunk under its own premise. The validator (`val_conj1_caltransfer.py`, results in `results_conj1_caltransfer.json` [28]) sweeps both η and the true gap: with a true gap of 0.2 ($p_a = 0.6$, $p_0 = 0.4$), every committed decision is sign-correct (100% recovery among committed, 0 wrong commitments) for all $\eta \in \{0, 0.02, 0.05, 0.08\}$ where the gap exceeds 2η; at the knife edge $\eta = 0.1$ (gap exactly 2η) the finite-sample rule still commits on 52.45% of trials — all of them sign-correct, with the wrong-commit rate still 0 — and for $\eta \ge 0.15$ commitment falls to 0, full abstention. The dual gap sweep at $\eta = 0.05$ shows the same wall from the other side (no commitments below gap 0.10, full commitment from gap 0.12 up), and the bracket $[(q_a - q_0) - 2\eta, (q_a - q_0) + 2\eta]$ contains the true gap on 100% of trials. The headline fields record minimum sign recovery 1.0 and maximum wrong-commit rate 0.0 whenever the gap exceeds 2η — the guarantee, observed.

5.6.3 What a Resolution Would Buy

We state plainly that the conjecture is *not* solved here, and that Proposition 5.2 does not solve it: it trades the open assumption for a quantified one, which is progress in precision, not in kind. A genuine resolution would change the system's standing in either direction. A *positive* resolution — a bracketing scheme whose premise is label-free verifiable, or a proof that some natural observable (margin spectra on D, agreement geometry among auxiliary predictors, weight diagnostics) suffices — would make the entire trichotomy assumption-free given evidence: KGA's residual reliability-model assumption would vanish, the knowable region of the phase diagram would extend to every shift with nonvacuous brackets, and evidence design would reduce to estimating two scalars on an observable set. A *negative* resolution — a lower bound showing every label-free bracket valid over a natural shift class must contain 0 unless a premise of calibration-transfer strength holds — would be just as valuable: it would certify that the conditional results of this chapter are the strongest possible, closing the program with matching upper and lower bounds the way Theorem 4.1 closed the unconditional question. Either outcome turns the last assumption in the stack into a theorem. Until then, the honest summary is the paper's own: the characterization is closed; the label-free

Regime	Result	Guarantee	Validator / evidence
Matched evidence (unknowable)	Thm 4.1; Cor. 5.1	no label-free rule certifies both worlds; any valid certificate abstains with probability $\geq 1 - 2\alpha$	clean witness: 100% abstention, forced-commit regret 0.51 (Ch. 7)
Risk-aligned evidence, batch	Thm 5.1; Cor. 5.2	false-adapt $\leq \alpha$ and false-freeze $\leq \alpha$; commits once $n \gtrsim \sigma^2 \Delta^{-2} \log(1/\alpha)$	mixed regime: false-adapt 0.065 at $\alpha = 0.1$; stress grid: 0% false-adapt (Chs. 7–8)
Risk-aligned evidence, streaming	Thm 5.2	anytime-valid: error $\leq \alpha$ simultaneously over all stopping times	worst anytime false-adapt $0.0626 \leq 0.1$ (`val_thm3_evalue_results.json`)
Covariate shift	Thm 5.3	Δ and $\operatorname{sign} \Delta$ identifiable from labeled source + unlabeled target	regression demo: sign correct under covariate shift, flips under concept shift (`val_thm5_multiclass.py`)
Disagreement region	Thms 5.4–5.6	$\operatorname{sign} \Delta$ = ordinal accuracy comparison on observable D (exact identity)	identity to 10^{-16}, 100% sign agreement over 4000 trials (`val_thm5_multiclass.py`)
Multiclass / regression bracketing	Conj. 5.1 (open); Prop. 5.2 (conditional)	sign certified iff $\lvert q_a - q_0 \rvert > 2\eta$ under calibration transfer; unconditional case open	100% sign recovery, 0 wrong commitments when gap $> 2\eta$ (`results_conj1_caltransfer.json`)

Table 5.3: The certified map of regimes assembled by Chapters 4 and 5: each row pairs a regime with its governing result, the exact form of its guarantee, and the machine-checked validator or experiment that exercises it.

estimability of the ordinal comparison is the standing assumption; the unconditional bracketing is the one residual open piece.

5.7 Synthesis: The Certified Map of Regimes

Chapters 4 and 5 together draw one map. The impossibility theorem fences off the region where no label-free rule may commit; the certificate gives the safe-passage protocol everywhere else; the covariate-shift and disagreement-region results chart where the passage provably opens; and the conjecture marks the one stretch of coastline still unsurveyed. Table 5.3 assembles the chapter's results with their guarantees and the machine-checked evidence behind each, with the experimental chapters and Appendix B holding the full per-condition records.

Three through-lines deserve a final statement. First, every positive result in this chapter is *conditional in a way the theory itself prices.* The certificate is valid given a valid radius; the radius is valid given exchangeability of the calibration bank; covariate-shift identifiability is valid given an untestable premise on $P(Y \mid X)$; the bracketing is valid given calibration transfer. None of this is a defect to be hidden — it is the content of Theorem 4.1 propagating through the constructions, and the certificate's distinctive virtue is that where its premises fail detectably (heavy importance weights, wide residual quantiles, sub-threshold confidence gaps) the failure surfaces as a *wider radius*, and the rule degrades to abstention rather than to silent error. Second, the disagreement region is the load-bearing structure connecting theory to practice: it is where the benefit lives (Theorems 5.4–5.6), why the empirical-Bernstein radius is tight (Remark 5.3), and which evidence coordinate the deployed estimator cannot do without (the 2.6× ablation of Chapter 7). Third, the guarantees proved here are exactly the ones the deployed system claims — no more. Chapter 6 assembles $Z \to \widehat{\Delta} \to \varepsilon \to \{\textsc{adapt}, \textsc{freeze}, \textsc{abstain}\}$ into the KGA wrapper with the assumptions of this chapter stated as the algorithm's assumptions, and Chapters 7 and 8 then measure the certificate against worlds engineered to reward it, punish it, and — on the constructed witness — defeat it, which is the only honest order of business after a chapter of conditional guarantees.

Chapter 6

KGA: Knowability-Guided Adaptation in Practice

The preceding chapters settled the question in principle: they proved when label-free adaptation is knowable, when it is not, and where the boundary lies. This chapter turns that theory into a running system. KGA (Knowability-Guided Adaptation) is the concrete decision algorithm that operationalises the trichotomy — adapt, freeze, or abstain — with the false-adapt rate held at the user's level α by the certificate of Chapter 5. It is the bridge between the impossibility and certificate theorems and the experiments that test them: every guarantee KGA claims is one of those theorems, stated as an assumption of the algorithm and no stronger. We present, in turn, the design goals, the four families of label-free evidence, the four certificate estimators with their exact formulas, the decision policy and its false-adapt guarantee, the public `kga` package API, a full deployment pseudocode, operational guidance, the served HTTP interface, and an honest account of what KGA does not guarantee.

6.1 Design goals and mechanism-agnostic wrapper architecture

KGA is a *decision wrapper*, not a new adaptation mechanism. Its single responsibility is to answer, *before* any irreversible update, the three-way question: should this candidate adaptation f_a replace the frozen model f_0 on this unlabelled target batch? The adapting mechanism itself—entropy-minimising Tent [56], sample-filtered EATA [39], sharpness-aware SAR [40], or any other algorithm—is a black box that supplies f_a; KGA wraps it uniformly.

This design is dictated by three requirements.

1. **Label-free.** The gate must decide from evidence $Z = \phi(X_{1:m}, f_0, f_a)$ that is computable without any target label. No target-label signals may enter at any step—neither for benefit estimation, nor for certificate construction.

2. **Mechanism-agnostic.** A laboratory may wish to evaluate several candidate adaptation

strategies (Tent vs. EATA vs. SAR) under the same safety envelope. By treating f_a as an input, KGA provides that shared envelope at the cost of one forward pass of f_0 and f_a per decision.

3. **Finite-sample valid.** The false-adapt probability—the chance of deploying an f_a that actually hurts—must be bounded at a user-specified level α even for small batches. This is Theorem 5.1 from Chapter 4.

The overall architecture (Figure 6.1) places KGA between the adaptation candidate and the live deployment loop. Calibration data (labelled validation scores) flows in once to build the calibration residuals needed for the split-conformal path; thereafter each incoming unlabelled batch is processed in three stages: evidence computation, certificate estimation, and the trichotomy decision.

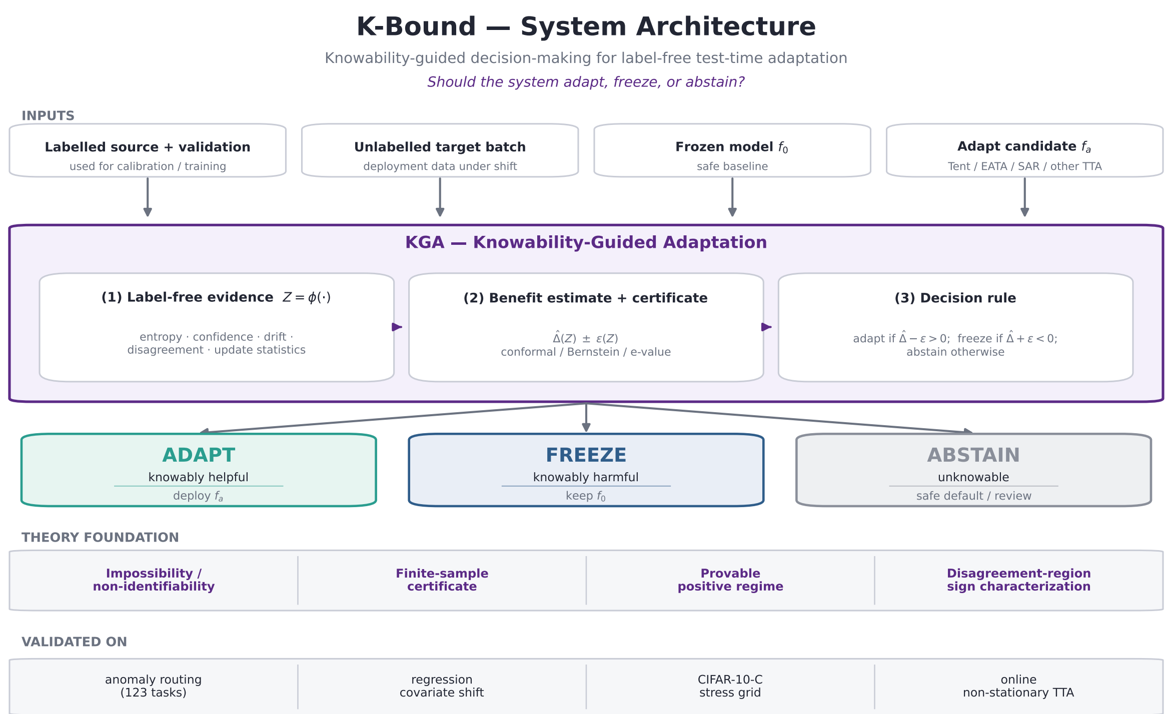

Figure 6.1: KGA system architecture. Labelled calibration scores and an unlabelled test batch enter the pipeline. The label-free evidence Z is computed first; then a finite-sample certificate $\widehat{\Delta} \pm \varepsilon$ is formed at level α; finally the trichotomy maps the certificate to one of three actions. The adaptation mechanism (Tent, EATA, SAR, or any other) is a plug-in; KGA wraps it without modification.

6.2 Label-free evidence computation

The evidence Z is a vector of statistics computable from the calibration scores S_{val} and the unlabelled test scores S_{test}, without any target labels. The implementation lives in `kga/evidence.py`: the dataclass `Evidence` and the function `compute_evidence(calib_scores, test_scores)`. The signals fall into four families, each tied to a theoretical result.

Drift (KS distance). For each detector column j, a two-sample Kolmogorov–Smirnov test (`scipy.stats.ks_2samp`) compares the calibration and test score marginals. The implementation stores:

- `ks_mean` — mean per-detector KS statistic;
- `ks_max` — maximum per-detector KS statistic.

Large KS drift is the observable signature of a covariate shift to which the gate can react (Theorem 5.3). When the marginal is preserved but the informative signal is destroyed—the "covert failure" regime of the mixed-regime experiment—both statistics remain near zero and the distribution is non-identifiable in the sense of Theorem 4.1.

Disagreement. The disagreement signal measures the heterogeneity of the test scores across detectors:

$$\texttt{disagree} \;=\; 1 - \overline{\rho}, \tag{6.1}$$

where $\overline{\rho}$ is the mean pairwise Spearman rank correlation of the test score columns (computed via `_rank_norm` then `np.corrcoef` on the rank-normalised test matrix). For a single detector, `disagree` is defined to be zero. A larger value indicates a bigger disagreement region $D = \{x : f_0(x) \neq f_a(x)\}$, and thus greater potential for the sign-of-difference to be identifiable (Theorems 5.4–5.1).

Entropy and confidence shift. The entropy signals measure the spread of the pooled score distributions. `_hist_entropy` computes the Shannon entropy (in nats) of the 20-bin normalised histogram of a flattened score array:

$$H(S) \;=\; -\sum_k p_k \ln p_k.$$

The confidence (decisiveness) signal `_decisiveness` measures the mean distance of the min-max-normalised scores from the indecision point 0.5:

$$\texttt{conf}(S) \;=\; 2\,\mathbb{E}\left[\left|\frac{S - S_{\min}}{S_{\max} - S_{\min}} - 0.5\right|\right].$$

The `Evidence` dataclass stores:

- `calib_entropy`, `test_entropy` — per-split entropies;
- `calib_conf`, `test_conf` — per-split decisiveness;
- `entropy_shift` = `test_entropy` − `calib_entropy`;
- `conf_shift` = `test_conf` − `calib_conf`.

A large positive `conf_shift` with a collapsing marginal is the entropy-collapse warning sign familiar from online TTA monitoring.

Importance-weight effective sample size (ESS). The function `_importance_ess` models the calibration and test score laws as univariate Gaussians and forms the Gaussian-ratio importance weight for each test point t_i:

$$w_i \;=\; \frac{\mathcal{N}(t_i;\,\mu_c,\sigma_c^2)}{\mathcal{N}(t_i;\,\mu_t,\sigma_t^2)},$$

where (μ_c,σ_c) and (μ_t,σ_t) are the empirical moments of the pooled calibration and test scores respectively. The effective sample size is then the standard self-normalised estimator,

$$\mathrm{ESS} \;=\; \frac{(\sum_i w_i)^2}{\sum_i w_i^2}, \tag{6.2}$$

capped in $[1, n_{\text{test}}]$. The dataclass stores `ess` and the scale-free ratio `ess_frac` = ESS$/n_{\text{test}}$. Low `ess_frac` indicates poor source-to-target support overlap, which undermines importance-weighted certificates and is a quantitative diagnostic for the support-overlap condition of Theorem 5.3.

The fixed-order feature vector `Evidence.to_vector()` concatenates [`ks_mean`, `ks_max`, `disagree`, `entropy_sh` and can be passed directly to a downstream benefit regressor as the feature vector Z.

Remark 6.1 (Provenance). Every field of `Evidence` mirrors a quantity computed in the K-Bound experiment scripts: `kga/evidence.py` is a clean, importable API over the same mathematics, not a re-derivation. The KS drift and disagreement fields match the $Z = \mathrm{dict}(\ldots)$ block in `src/scripts/kbound/knowability_experiment.py` (lines 74–96); the entropy and importance-weight fields match the `evidence` helper in `mixed_regime_experiment.py`.

6.3 Certificate estimators

A *certificate* is a frozen dataclass `Certificate(delta_hat, epsilon, method, alpha, n)` implemented in `kga/certificate.py`. It carries a point estimate $\widehat{\Delta}$ of the benefit $\Delta = R(f_0) - R(f_a)$ and a confidence radius ε such that, with probability at least $1-\alpha$,

$$\Delta \;\geq\; \widehat{\Delta} - \varepsilon \qquad \text{(certified lower bound).}$$

The two derived properties `lower` $= \widehat{\Delta} - \varepsilon$ and `upper` $= \widehat{\Delta} + \varepsilon$ are the bounds consumed by the decision policy.

The module provides four estimators, each linked to a theoretical result.

6.3.1 Empirical Bernstein (`ebern`) — the default

The default estimator implements the Maurer–Pontil (2009) empirical-Bernstein lower confidence bound [37]. Given n i.i.d. paired benefits $X_i = \ell(f_0(x_i), y_i) - \ell(f_a(x_i), y_i)$ with empirical mean $\widehat{\mu}$, unbiased sample variance $\widehat{V}$ (denominator $n-1$), and range $R = b - a$, the certificate is:

$$\varepsilon_{\text{ebern}} \;=\; \sqrt{\frac{2\,\widehat{V}\,\ln(2/\alpha)}{n}} \;+\; \frac{7R\,\ln(2/\alpha)}{3(n-1)}, \tag{6.3}$$

so that $\mathbb{P}[\Delta \geq \widehat{\mu} - \varepsilon_{\text{ebern}}] \geq 1 - \alpha$. The estimate is $\widehat{\Delta} = \widehat{\mu}$ and `epsilon` is the sum of the two terms in (6.3). When $n = 1$ the variance is undefined and `epsilon` is set to $+\infty$, forcing ABSTAIN. When `benefit_range` is not supplied, R is estimated from the observed range `max(X) - min(X)`.

This estimator is strictly tighter than Hoeffding whenever $\widehat{V} \ll R^2$ — the usual regime in which most paired benefits agree in sign — and is deterministic, streamable, and finite-sample valid. It is the batch Theorem 3 certificate of Theorem 5.1.

6.3.2 Hoeffding (`hoeffding`) — distribution-free baseline

The Hoeffding estimator replaces the variance term by the worst-case variance-free bound:

$$\varepsilon_{\text{hoeff}} \;=\; R\sqrt{\frac{\ln(1/\alpha)}{2n}}. \tag{6.4}$$

This radius is always at least as large as the empirical-Bernstein one in the low-variance regime, so it is provided as a conservative baseline and sanity comparator. The `method` field is set to `'hoeffding'`.

6.3.3 Split-conformal (`conformal`) — cross-task estimator

The split-conformal estimator is the radius used in the main K-Bound experiments (Theorem 5.1, conformal variant; [55, 1]). It takes a point estimate $\widehat{\Delta}$ (e.g., from a leave-one-out benefit regressor) and a vector of held-out calibration residuals $r_i = |\widehat{\Delta}_i - \Delta_i|$ from n_{cal} tasks:

$$\varepsilon_{\text{conf}} \;=\; \text{quantile}(r_{1:n},\, 1-\alpha)\,. \tag{6.5}$$

By the exchangeability of the calibration split, $\mathbb{P}[|\widehat{\Delta} - \Delta| \leq \varepsilon_{\text{conf}}] \geq 1 - \alpha$ for a fresh query. The function `conformal_split(delta_hat, calib_residuals, alpha)` is called in `knowability_experiment.py` (`eps = np.quantile(resid, 1 - alpha)`) and in `mixed_regime_experiment.py`.

6.3.4 Anytime e-value (`evalue`) — Theorem 3b

The anytime estimator implements two one-sided betting e-processes over an ordered benefit stream [54, 59]:

$$E_t^+ = \prod_{i \le t}(1 + \lambda_i X_i), \qquad \text{tests } H_0^+ : \Delta \le 0, \tag{6.6}$$

$$E_t^- = \prod_{i \le t}(1 + \nu_i(-X_i)), \qquad \text{tests } H_0^- : \Delta \ge 0, \tag{6.7}$$

with predictable truncated-aGRAPA bets $\lambda_i = \text{clip}(\widehat{\mu}_{i-1}/\widehat{\sigma}^2_{i-1},\ 0,\ \delta_{\text{cap}}/|a|)$ (and symmetrically ν_i) formed from the running mean and second moment up to step $i-1$. Each process is a nonnegative supermartingale under its null; by **Ville's inequality** [54], $\mathbb{P}(\exists t : E_t \ge 1/\alpha) \le \alpha$ simultaneously over all sample sizes with no multiplicity correction.

The returned certificate encodes the current anytime decision in the standard $\widehat{\Delta} \pm \varepsilon$ form:

- If $\log E_t^+ \ge \log(1/\alpha)$: emit $\varepsilon = 0$ with $\widehat{\Delta} > 0$, forcing ADAPT;
- If $\log E_t^- \ge \log(1/\alpha)$: emit $\varepsilon = 0$ with $\widehat{\Delta} < 0$, forcing FREEZE;
- Otherwise: emit $\varepsilon = |\bar{X}| + \epsilon_0$ (brackets zero around the running mean), forcing ABSTAIN.

This is Theorem 5.2 (anytime certificate). The implementation mirrors `experiments/kbound/theory_validation/` and `kga/kbound_pkg/kbound/eprocess.py`.

6.4 Decision policy and the false-adapt guarantee

The decision rule is a deterministic threshold on the certificate bounds, implemented as the `decide(certificate, alpha)` function in `kga/policy.py`. The `Decision` enum has three members: `ADAPT`, `FREEZE`, `ABSTAIN`. As a `str` subclass, each member compares equal to its string value and serialises cleanly to JSON.

The rule applies strict inequalities at the boundaries:

Condition	Decision
$\widehat{\Delta} - \varepsilon > 0$	ADAPT
$\widehat{\Delta} + \varepsilon < 0$	FREEZE
otherwise	ABSTAIN

A certificate with $\widehat{\Delta} = \varepsilon$ (lower bound exactly zero) ABSTAINs rather than ADAPTs; this is the conservative choice and matches the reference experiment scripts. An infinite ε (e.g., from a single-sample empirical-Bernstein certificate) always ABSTAINs.

Theorem 6.1 (False-adapt guarantee; Theorem 5.1). *The certificate is constructed so that* $\mathbb{P}[\Delta \geq \widehat{\Delta} - \varepsilon] \geq 1 - \alpha$. *The* ADAPT *branch fires only when* $\widehat{\Delta} - \varepsilon > 0$, *which on the good event forces* $\Delta > 0$. *Therefore*

$$\mathbb{P}(\textit{ADAPT and } \Delta \leq 0) \ \leq \ \alpha.$$

The symmetric statement bounds the false-freeze probability. The anytime e-value variant (Theorem 5.2) upgrades this to hold simultaneously over all stream prefixes via Ville's inequality.

The proof follows identically from the non-identifiability regime (via Theorem 4.1): when evidence Z cannot separate the two opposite-benefit worlds, the certificate radius brackets zero and KGA ABSTAINS. This is mandatory, not merely conservative (see Conjecture 5.1 and the discussion in Chapter 4 on forced abstention).

6.5 The `kga` package API

The `kga` Python package (top-level directory `kga/`) exposes the three-stage pipeline through the `KGA` facade class in `kga/kga.py`.

Constructor.

`KGA(alpha=0.1, method='ebern')`

`alpha` is the operating miscoverage level in $(0, 1)$; it bounds the false-adapt probability via Theorem 5.1. `method` selects the default batch certificate estimator when per-sample paired benefits are supplied: `'ebern'` (empirical Bernstein; the default), `'hoeffding'`, or `'evalue'` (anytime Theorem 3b). A `KGA` instance carries no learned state, so a single instance is reusable and thread-safe.

Methods.

- **`.evidence(calib, test, **kwargs) -> Evidence`** — computes the label-free evidence Z by delegating to `compute_evidence`; stores the result in `last_evidence`.
- **`.certify(...) -> Certificate`** — builds $\widehat{\Delta} \pm \varepsilon$. Three calling conventions are supported:
  1. `certify(scores=benefits)` — per-sample paired benefits, processed by the batch estimator selected at construction (or overridden via `method=`);
  2. `certify(adapt_risk=..., freeze_risk=..., calib_residuals=...)` — two scalar risks plus held-out residuals; uses the split-conformal radius;
  3. `certify(delta_hat=..., calib_residuals=...)` — explicit point estimate plus residuals for split-conformal.

  The result is stored in `last_certificate`.

- **`.decide(certificate=None) -> Decision`** — applies the trichotomy to a certificate (defaults to `last_certificate`); stores the result in `last_decision`.
- **`.explain() -> dict`** — returns all intermediate quantities (`alpha`, `method`, `evidence`, `certificate`, `decision`) as a JSON-serialisable dict.

Quickstart. The following usage snippet is taken directly from `kga/README.md`:

```
import numpy as np
from kga import KGA

rng = np.random.default_rng(0)
kga = KGA(alpha=0.1, method="ebern")

# (1) Label-free evidence Z.
calib = rng.normal(0.0, 1.0, size=(500, 3))
test  = rng.normal(0.0, 1.0, size=(500, 3))
z = kga.evidence(calib, test)
print(z.ks_mean, z.disagree, z.ess_frac)

# (2) Certificate from per-sample paired benefits.
benefits = rng.normal(0.3, 0.1, size=400)
cert = kga.certify(scores=benefits, benefit_range=2.0)
print(cert.delta_hat, cert.epsilon, cert.lower)

# (3) Trichotomy decision.
print(kga.decide(cert))     # -> Decision.ADAPT

# (4) Audit.
import json; print(json.dumps(kga.explain(), indent=2))
```

Command-line interface. The `kga.cli` module (entry point `python -m kga`) exposes a `decide` subcommand:

```
python -m kga decide --calib calib.npy --test test.npy --alpha 0.1
```

This loads calibration and test scores from `.npy` files, computes the evidence Z, constructs a split-conformal certificate with a label-free residual proxy (`delta_hat = 0`, radius from the calibration's own dispersion), and prints the decision and certificate as JSON to stdout. Because labels are unavailable at the command line, it should be used for pipeline integration testing rather than live deployment decisions; real benefit estimates require the `KGA` API with paired benefits or held-out residuals.

6.6 Full deployment pseudocode

Algorithm 1 below gives the complete deployment loop, including the one-time calibration phase and the per-batch inference phase.

Algorithm 1: KGA Deployment Loop.

Inputs: frozen model f_0; adaptation candidate f_a; calibration split $(S_{\text{val}}, y_{\text{val}})$; unlabelled stream of target batches $\mathcal{B}_1, \mathcal{B}_2, \ldots$; miscoverage level α; certificate method $m \in \{\texttt{ebern}, \texttt{hoeffding}, \texttt{evalue}\}$.

Calibration phase (once):

1. Compute paired benefits $X_i = \ell(f_0(x_i), y_i) - \ell(f_a(x_i), y_i)$ on the calibration split.
2. Fit a benefit estimator $\widehat{\Delta}(\cdot)$ (e.g., a leave-one-task-out gradient-boosted regressor) on (Z_i, X_i) pairs.
3. Record calibration residuals $r_i = |\widehat{\Delta}(Z_i) - \bar{X}_i|$ for the split-conformal path, or retain $\{X_i\}$ for the batch path.

Per-batch inference phase (for each $\mathcal{B}_t$):

4. **Evidence:** compute $Z_t = \phi(S_{\text{val}}, S_t, f_0, f_a)$ via `kga.evidence(calib, test)`. Extract [`ks_mean`, `ks_max`, `disagree`, `entropy_shift`, `conf_shift`, `ess_frac`].
5. **Certify:** obtain $\widehat{\Delta}_t$ from the benefit estimator; call `kga.certify(scores=...)` or `kga.certify(delta_hat=`$\widehat{\Delta}_t$`, calib_residuals=r)` to form ε_t at level α.
6. **Decide (trichotomy):**
   - **if** $\widehat{\Delta}_t - \varepsilon_t > 0$: $\Rightarrow$ **adapt** — deploy f_a on $\mathcal{B}_t$;
   - **elif** $\widehat{\Delta}_t + \varepsilon_t < 0$: $\Rightarrow$ **freeze** — keep f_0 on $\mathcal{B}_t$;
   - **else**: $\Rightarrow$ **abstain** — keep f_0, log for human review.
7. **Log:** write $(Z_t, \widehat{\Delta}_t, \varepsilon_t, \texttt{decision}_t, t)$ to the audit trail (`kga.explain()`).

Guarantee (Theorem 5.1): $\mathbb{P}(\textsc{adapt} \text{ and } \Delta \le 0) \le \alpha$; similarly for false-freeze.

6.7 Operational guidance

Choosing α. The miscoverage level α directly bounds the false-adapt probability. The paper reports all results at $\alpha = 0.10$; production deployments in safety-critical contexts should lower α (e.g., 0.05), accepting a higher abstention rate in exchange for stronger false-adapt control. At very small α, the empirical-Bernstein radius grows as $\sqrt{\ln(2/\alpha)/n}$, so the batch size should be increased proportionally to maintain coverage.

Batch size. The empirical-Bernstein and Hoeffding radii shrink as $O(1/\sqrt{n})$; the split-conformal radius is independent of the batch size (it is a quantile over the calibration tasks). For the batch certificate to commit to ADAPT or FREEZE rather than ABSTAIN, the benefit $|\widehat{\Delta}|$ must exceed the radius. In the 123-task anomaly-routing benchmark, batch sizes of 200–500 samples provide useful separation at $\alpha = 0.10$.

Calibration data. The calibration split must be exchangeable with deployment in the sense that the per-task benefit distribution is representative (Theorem 5.1 assumption (i)). In the cross-task anomaly-routing setting, a leave-one-task-out procedure provides unbiased residuals. In single-task streaming settings, a held-out validation probe with the same label distribution as the deployment stream is sufficient.

When abstain should trigger operational action. ABSTAIN means the evidence Z is insufficient to certify the sign of Δ at level α; it does not mean that adapting is dangerous. The correct operational response is to (a) continue with the safer f_0 for the current batch, (b) log the evidence vector and certificate for monitoring, and (c) consider collecting more data or lowering the adaptation aggressiveness. Repeated ABSTAIN outcomes on a stable stream may indicate that the benefit estimator $\widehat{\Delta}$ is poorly calibrated for that domain, which requires recalibration.

Complexity. KGA is $O(n)$ per batch in the sample size n and $O(k^2)$ in the number of detectors k (due to the pairwise rank-correlation in the disagreement computation). The implementation is pure `numpy/scipy`, deterministic (no random state), and torch-free, so it adds negligible overhead to any deployment that already runs f_0 and f_a forward passes.

6.8 The served HTTP interface

The production API serves KGA decisions over HTTP through the routes defined in `deploy/api/kga_routes.py`, wired into the main FastAPI application. Two endpoints are available.

`GET /kga/health` (no authentication). A liveness probe that returns the `kga` package version and component status:

```
{"status": "ok", "component": "kga", "version": "0.1.0"}
```

`POST /decide` (authentication required). Accepts a JSON body with three fields:

- `calib_scores` (`list[float]`, 2–200 000 entries) — calibration detector scores (label-free);
- `test_scores` (`list[float]`, 2–200 000 entries) — unlabelled test detector scores;
- `alpha` (`float` in $(0, 1)$, default 0.10) — miscoverage level.

All non-finite inputs are rejected with `HTTP 400`. The response carries the decision (`ADAPT`/`FREEZE`/`ABSTAIN`), `delta_hat`, `epsilon`, `method`, and the full `Evidence` summary (`ks_mean`, `ks_max`, `disagree`, `entropy_shift`, `conf_shift`, `ess_frac`, `n_calib`, `n_test`, `n_detectors`).

Because target labels are unavailable at request time, the route reports a conservative $\widehat{\Delta} = 0$ with a split-conformal radius derived from the calibration scores' own dispersion (the same conservative proxy used in the `kga.cli` entry point). Callers with a real per-sample benefit estimate should use the `KGA` Python API server-side rather than the HTTP route.

Request sizes are bounded at `MAX_KGA_SCORES = 200 000` per array (the same limit as `MAX_*` in `deploy/api/main.py`). The decision math is entirely in the importable `kga` package; the route is a thin, validated transport layer.

Figure 6.2 shows the decision flow from an incoming HTTP request to the returned decision.

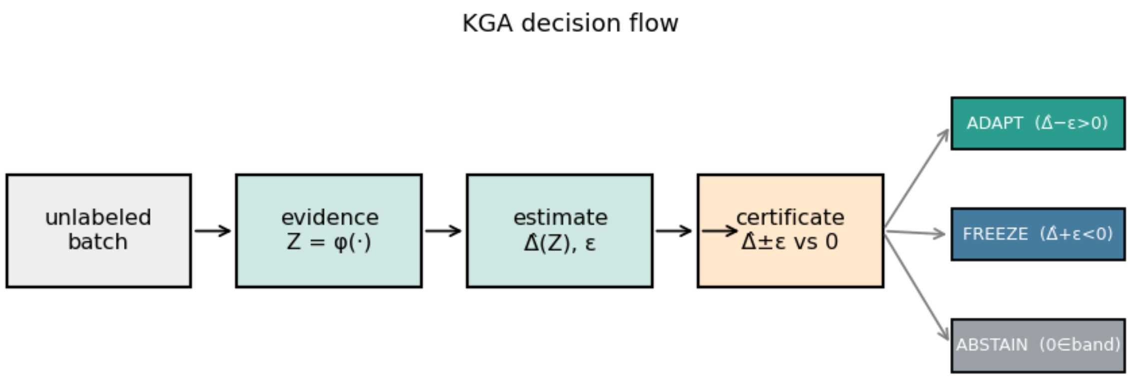

Figure 6.2: Decision flow for an incoming `POST /decide` request. Pydantic validates and coerces the inputs; `compute_evidence` produces Z; `conformal_split` builds $\widehat{\Delta} \pm \varepsilon$; and `decide` applies the trichotomy. Any `ValueError` from precondition checks (e.g., all-identical scores) is returned as `HTTP 400`; uncaught exceptions return `HTTP 500`.

6.9 Threats, limitations, and what KGA does not guarantee

We state explicitly the situations in which KGA's guarantees do not apply, to prevent misuse.

The false-adapt bound is not a guarantee that f_a is good. Theorem 5.1 bounds $\mathbb{P}(\text{ADAPT and } \Delta \le 0) \le \alpha$. It says nothing about the magnitude of Δ when $\Delta > 0$. A deployment that ADAPTs on an f_a with tiny positive benefit has not violated the certificate; it has simply not gained much. The certificate is about safety, not about optimality.

Non-identifiability is permanent, not a data problem. When two target worlds induce the same evidence distribution Z but opposite benefits (Theorem 4.1), no amount of additional *unlabelled* data can resolve the ambiguity. KGA is forced to ABSTAIN in such worlds by Corollary 5.1. Gathering labelled target data would resolve it, but that is outside the label-free scope. The general bracketing conjecture (Conjecture 5.1) remains open.

The calibration exchangeability assumption. The split-conformal certificate (Section 6.3.3) requires the calibration tasks to be exchangeable with the deployment task. If the deployment stream is drawn from a radically different domain—e.g., calibrating on anomaly-routing benchmarks and deploying on medical imaging—the certificate radius ε may be miscalibrated and the false-adapt bound may not hold. This is the standard limitation of any conformal method.

Covert failures are not detectable. In the covert-failure regime of the mixed-regime experiment (the "permuted detector" condition), the observable evidence Z is nearly unchanged while the true benefit reverses sign. KGA responds by widening the certificate radius and ABSTAINing more frequently, which is correct given the theory, but it means that under truly covert failures neither ADAPT nor FREEZE can be certified. Operational systems should treat an unusual increase in ABSTAIN frequency as a sensor-quality warning.

The benefit estimator $\widehat{\Delta}$ must be calibrated. The false-adapt bound passes through the certificate; the certificate is valid only insofar as ε covers the estimation error $|\widehat{\Delta} - \Delta|$. If the benefit regressor is overfit to the calibration tasks or trained on a distribution that does not include the deployment domain, the effective coverage will be lower than $1 - \alpha$. Regular recalibration with fresh held-out data is therefore an operational requirement.

Label-free adaptation in the helpful-dominated regime. Where adaptation almost always helps—e.g., the clean 123-task suite where mean$|\Delta| = 0.132$ on acted tasks—always-adapt is already a strong baseline and the certificate's distinctive value is bounded by the frequency of detectable harmful regimes. We are explicit about this scope following the honest-results principle articulated in the paper's scope note and in the experimental discussion (Chapter 7). Theorem 4.1 characterizes precisely the regime where this matters: when the harmful-adaptation distribution has an observable fingerprint (large KS drift, low ESS), KGA can and does refuse it; when it does not (covert failure), the Conjecture 5.1 bound suggests fundamentally limited discriminability.

With the algorithm specified and its scope stated, the burden shifts to evidence. The next three chapters put KGA to the test against worlds built to reward it, to punish it, and — on the constructed witness — to defeat it. Chapter 7 exercises the trichotomy across a 123-task anomaly-routing benchmark for breadth; Chapter 8 delivers the headline, a deep test-time-adaptation safety certificate on CIFAR-10-C, ImageNet-C, and a natural-shift battery; and Chapter 9 grounds the framework in a fully implemented multimodal-fusion special case. The verdict they return is the one the theory predicts: KGA is foremost a *safety layer* — it beats both trivial policies only where harmful adaptation is frequent and label-free detectable, chiefly the corruption grids, and elsewhere it does no harm, matching the better fixed policy rather than exceeding it.

Chapter 7

Experiments I: The Anomaly-Routing Benchmark

This chapter is *breadth* evidence, not the headline. Its job is to show that the knowability trichotomy of Chapters 4–5 — helpful, harmful, unknowable, with a valid certificate that commits or abstains accordingly — is not an artifact of one architecture or one kind of shift, but holds across a wide span of modalities and detectors where ground-truth benefit is exactly computable. The deep test-time-adaptation safety certificate of Chapter 8 (CIFAR-10-C, ImageNet-C, and the natural-shift battery) is the headline application; this chapter lays the foundation underneath it. We therefore read the results that follow as *coverage*: a low-cost, high-diversity testbed on which the trichotomy can be exercised end to end and falsified if it fails, before the deep-network experiments stress it where it matters most.

Concretely, this chapter evaluates the KGA certificate on a 123-task anomaly-detection routing benchmark, purpose-built to interrogate the adapt/freeze/abstain trichotomy of Chapter 6 in a domain where ground-truth benefit is computable, where controlled corruption can inject known regime changes, and where six classical detectors supply the label-free evidence vector Z without any neural retraining. Sections 7.1–7.10 proceed from data provenance through seven experimental protocols to a cross-experiment synthesis and a threats-to-validity audit. Every number is drawn directly from the named JSON result files in `experiments/kbound/results/`; discrepancies with `kbound.tex` are flagged explicitly in Section 7.10.

7.1 The 123-Task Score Archive

7.1.1 Provenance and Construction

The anomaly-routing benchmark draws tasks from ADBench [19] and the PyOD library [61], spanning tabular, image-OOD, text, cyber-security, and fraud-detection datasets. For each of the 123 tasks we precompute anomaly scores under six classical detectors: Isolation Forest [35], Local Outlier Factor [7], One-class SVM [46], and ECOD [33] form the core; rank-normalised stacking and a reliability-fusion variant complete the six. All detector scores are precomputed and cached; no neural network is retrained at test time.

Each task is split into a *labeled validation set* (used by KGA to estimate Δ and calibrate the conformal band ε) and a *held-out test set* (used only for final evaluation). The canonical frozen candidate f_0 is the detector achieving the best validation AUROC; the adaptation candidate f_a is the rank-normalised logistic stack in the clean suite, swapped for controlled alternatives in the corrupted regimes of later experiments.

The labeled validation is the only supervised signal visible to the certificate. Test labels are withheld from Z computation: the evidence vector $Z = (val\_max, val\_gap, val\_mean, ks\_mean, ks\_max, disagree, val\_s$
is derived entirely from validation performance and distributional statistics of the unlabeled test batch [28].

7.1.2 Evaluation Methodology

For each task t we measure the *true benefit* $B_t = \text{AUROC}(f_a, \mathcal{D}_t) - \text{AUROC}(f_0, \mathcal{D}_t)$ on the test set. The four policies are compared by mean test AUROC:

- **Always-adapt**: apply f_a on all 123 tasks.
- **Always-freeze**: apply f_0 on all 123 tasks.
- **KGA (trichotomy)**: apply f_a if $\widehat{\Delta} - \varepsilon > 0$ (adapt), apply f_0 if $\widehat{\Delta} + \varepsilon < 0$ (freeze), abstain otherwise.
- **Oracle**: apply whichever of f_a or f_0 achieves higher test AUROC.

Regret-to-oracle is $r = \text{AUROC(oracle)} - \text{AUROC(policy)}$; lower is better. Safety is reported as adapt precision $\mathbb{P}(B > 0 \mid \text{ADAPT})$ and false-adapt rate $\mathbb{P}(B < 0 \mid \text{ADAPT})$.

The conformal level is $\alpha = 0.1$ throughout the clean and harmful suites. All scripts are archived in `kbound_paper/scripts/` and the repository [28].

7.1.3 Why Anomaly Routing Is a Clean Testbed for the Trichotomy

Anomaly detection is structurally well suited for studying the adapt/freeze/abstain trichotomy for three reasons. First, benefit B is computable from labeled validation data, so the ground truth for a decision is unambiguous. Second, the domain naturally exhibits the full phase diagram (Chapter 4): some tasks are knowably-beneficial, some knowably-harmful, and many are genuinely ambiguous—yielding a nontrivial trichotomy under any reasonable certificate. Third, the six-detector score archive provides a rich, diverse evidence vector Z that is entirely label-free at test time, directly matching the K-Bound problem statement (Definition in Chapter 3).

The benchmark deliberately does *not* require a powerful learned feature extractor: the same evidence vector is used across all 123 tasks, and no task-specific tuning is performed. This makes the benchmark a conservative test of the certificate; any future enrichment of Z can only improve results.

7.2 The Knowability Trichotomy Suite

7.2.1 Setup and Protocol

The candidate adaptation is the rank-normalised logistic stack, a mild ensemble improvement over the best-validation single detector. On the clean 123-task suite this candidate *helps on the majority of tasks*, making it a helpful-dominated regime. The conformal band is calibrated at $\alpha = 0.1$; the resulting ε from `knowability_results.json` is 0.0492.

7.2.2 Decision Distribution

Table 7.1 summarises the KGA decisions and safety metrics.

Table 7.1: KGA decisions and safety on the clean 123-task suite (source: `knowability_results.json`, field `metrics`).

Metric	Value	Source field
Tasks total	123	`n_tasks`
Conformal ε	0.0492	`eps_conformal`
Coverage (acted = adapt or freeze)	0.171	`coverage`
Decisions: ADAPT	20	`n_adapt`
Decisions: FREEZE	1	`n_freeze`
Decisions: ABSTAIN	102	`n_abstain`
Adapt precision $\mathbb{P}(B > 0 \mid \text{ADAPT})$	0.90	`adapt_precision_(B>0|ADAPT)`
False-adapt rate $\mathbb{P}(B < 0 \mid \text{ADAPT})$	0.10	`false_adapt_rate_(B<0|ADAPT)`
Mean $\lvert\Delta\rvert$ on abstained tasks	0.0215	`abstain_mean_|B|`
Mean $\lvert\Delta\rvert$ on acted tasks	0.1324	`nonabstain_mean_|B|`
Mean AUROC: KGA (trichotomy)	0.7661	`policy.mean_auc_trichotomy`
Mean AUROC: always-adapt	0.7771	`policy.mean_auc_always_adapt`
Mean AUROC: always-freeze	0.7481	`policy.mean_auc_always_freeze`
Mean AUROC: oracle	0.7828	`policy.mean_auc_oracle`
Regret (KGA vs oracle)	0.0167	`policy.regret_trichotomy_vs_oracle`
Regret (always-adapt vs oracle)	0.0057	`policy.regret_always_adapt_vs_oracle`

7.2.3 Adapt Precision and Safety

The certificate achieves adapt precision 0.90: of the 20 tasks on which KGA commits to adaptation, 18 have $B > 0$. The false-adapt rate is 0.10. This is the *clean* helpful-dominated regime, so caution is warranted in interpreting the coverage figure: at 17.1% coverage, only 21 of 123 tasks are acted upon, while 102 are abstained.

7.2.4 Abstain-Where-Benefit-Is-Small: Safety Concentration

The key safety property of the trichotomy is not just that adapt decisions are accurate but that *abstentions concentrate on tasks where the true benefit is near zero.* The JSON confirms this directly: the mean absolute benefit on abstained tasks is $|\Delta|_{\text{abstain}} = 0.0215$, while on acted tasks it is $|\Delta|_{\text{act}} = 0.1324$—a ratio of $0.1324/0.0215 \approx 6.2$. Abstaining carries only small expected cost per task (0.0215 AUROC gap), whereas every acted task was worth 0.1324 on average. This is the empirical manifestation of the certificate's one-sided guarantee (Theorem 5.1).

7.2.5 Interpretation: Helpful-Dominated Regime

An honest reading of this suite is that the clean benchmark is helpful-dominated: the rank-normalised stack mostly helps, so always-adapt (AUROC 0.7771) outperforms the safe KGA policy (AUROC 0.7661) by 0.011. The regret gap is 0.0057 vs 0.0167—KGA carries roughly three times more regret than always-adapt in this regime. This is not a failure: the certificate is designed to be safe, not to maximise AUROC when adaptation is universally beneficial. The harmful-regime and mixed-regime experiments (Sections 7.3 and 7.4) demonstrate where the trichotomy earns its advantage.

7.3 The Harmful-Fusion Regime

7.3.1 Setup and Protocol

The harmful-fusion experiment evaluates KGA against a candidate that is *adversarially bad*: the legacy reliability-fusion candidate (`legacy_fuse`; artifact key `elara_fuse`) that hurts on 79.7% of the 123 tasks. The true harmful base rate, from field `base_rate_harmful_B<0` in `kbound_harmful_results.json`, is 0.7967. The frozen candidate f_0 is again the auto-selected best-validation detector (`auto_select`). If KGA cannot identify and block this harmful fusion, it fails the most important safety test.

7.3.2 Results

Table 7.2 gives the full policy comparison.

Table 7.2: Harmful-fusion regime: AUROC and regret-to-oracle for four policies across 123 tasks. Source: `kbound_harmful_results.json`.

Policy	Mean AUROC	Regret vs oracle	Source field
Always-freeze (f_0)	0.7481	0.0018	`mean_auc.always_freeze(auto_select)`
Always-adapt (legacy-fuse)	0.7284	0.0214	`mean_auc.always_adapt(elara_fuse)`
KGA (trichotomy)	0.7479	0.0019	`mean_auc.K-Bound_trichotomy`
Oracle	0.7499	0.0000	`mean_auc.oracle`

The finding is clean: KGA cuts regret-to-oracle from 0.0214 (always-adapt) to 0.0019, a factor of $0.0214/0.0019 \approx 11.3$, and its AUROC (0.7479) is indistinguishable from both always-freeze (0.7481) and the oracle (0.7499). The certificate's job here is to *match* the safe baseline by refusing a dangerous candidate, not to beat both policies — always-freeze is already near-oracle when adaptation almost always hurts.

The decision distribution reinforces the picture: of 123 tasks, KGA issues **ADAPT** on 1, **FREEZE** on 13, and **ABSTAIN** on 109 (source: `counts`). The single adapt decision has $B > 0$ (false-adapt rate 0.0 on freezed-or-abstained; field `freeze_correct_B<=0` = 1.0), and all 13 freeze decisions are on tasks where $B \leq 0$ (field `freeze_correct_B<=0` = 1.0). The certificate almost entirely refuses the harmful fusion candidate, correctly identifying the regime.

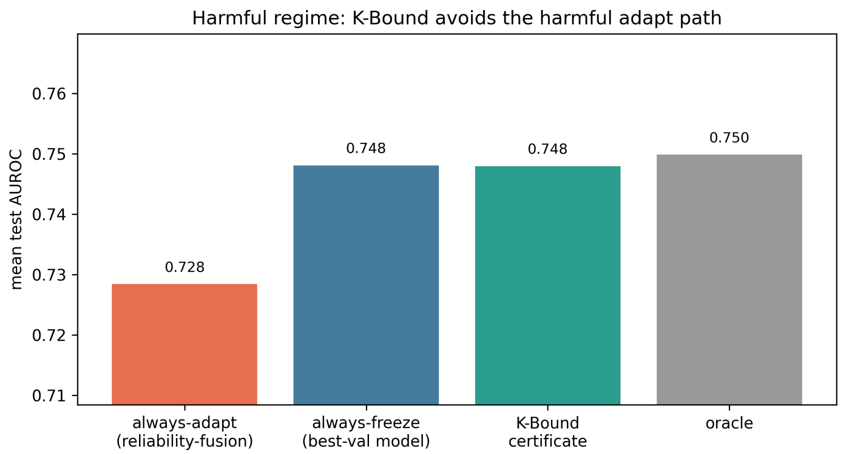

Figure 7.1: KGA in the harmful-fusion regime. The certificate almost entirely blocks the legacy-fuse candidate; regret-to-oracle falls from 0.0214 (always-adapt) to 0.0019 ($\approx$11$\times$ reduction). Source: `kbound_harmful_results.json`.

7.3.3 Interpretation

This experiment directly probes the hard end of the phase diagram: the domain sits deep in the "knowably harmful" region of Chapter 4's phase diagram, and KGA correctly refuses to commit. The 11$\times$ regret reduction relative to always-adapt demonstrates that the certificate's value is largest precisely when the adaptation candidate is dangerous. Conversely, always-freeze is already near-oracle in this regime, so the certificate's main job is to *match* the safe baseline rather than to improve upon it.

7.3.4 Threats Specific to This Sub-experiment

The legacy-fuse candidate is constructed to be harmful; a practitioner would not ordinarily deploy a candidate with a known 79.7% harmful rate. This sub-experiment is a *controlled stress test*, not a claim about typical real-world adaptation candidates.

7.4 The Mixed Regime

7.4.1 Construction

The mixed-regime benchmark is this chapter's heterogeneity probe: a single evaluation that deliberately places three qualitatively different conditions side by side to ask whether the trichotomy routes by true regime. We read its AUROC contrast as a diagnostic rather than a beats-both — the conference mixed-stream comparison was withdrawn pending an out-of-fold re-run, as the Policy AUROC Comparison below makes explicit — while the decision-distribution and false-adapt findings are what it contributes to the safety story. It is built to place three qualitatively different conditions in the same evaluation:

1. **Clean** ($n_{\text{clean}} = 123$ instances): the best-validation detector is used as f_a; adaptation is mildly *harmful* (mean $B = -0.041$, field `true_mean_benefit_B_by_condition.clean`).
2. **Detectable failure** ($n_{\text{det}} = 123$ instances): the best-validation detector is corrupted with a distribution shift; adaptation is strongly *beneficial* (mean $B = +0.193$, field `true_mean_benefit_B_by_condi` and the shift is observable via KS statistics in Z.
3. **Covert failure** ($n_{\text{cov}} = 123$ instances): the best-validation detector is permuted so its signal is destroyed but the marginal distribution of Z is nearly unchanged; adaptation is also *beneficial* (mean $B = +0.181$, field `true_mean_benefit_B_by_condition.covert`), yet evidence is confounded with clean.

Total instances: $3 \times 123 = 369$ (field `n_instances`). The conformal $\varepsilon = 0.1315$ (field `eps`) is larger than in the clean suite because the mixed dataset has higher within-batch variance.

7.4.2 Decision Distribution by True Regime

Table 7.3 shows KGA decisions broken out by condition.

Table 7.3: KGA decisions by true regime in the controlled mixed benchmark. Source: `mixed_regime_results.json`, field `decision_distribution_by_condition`.

Condition	True mean Δ	ADAPT	FREEZE	ABSTAIN
Clean	-0.041	8	5	110
Detectable failure	$+0.193$	71	3	49
Covert failure	$+0.181$	59	3	61
Total		138	11	220

The pattern is informative. On the clean (mildly harmful) condition, KGA abstains on 110 of 123 instances and adapts on only 8—correctly exercising restraint. On the detectable-failure (beneficial) condition, it adapts on 71 and abstains on 49—capturing most of the large-B instances. On the covert-failure condition, it adapts on 59 and abstains on 61—split because the permutation partially destroys the evidence signal that would distinguish this condition from clean.

7.4.3 Policy AUROC Comparison

Table 7.4 gives the four policy AUROC values.

Table 7.4: Policy mean AUROC in the mixed regime (369 instances). Source: `mixed_regime_results.json`, field `mean_auc_policies`.

Policy	Mean AUROC	Regret vs oracle	Source field
Always-freeze	0.5862	0.1327	`always_freeze`
KGA	0.6899	0.0291	`K_Bound`
Always-adapt	0.6971	0.0218	`always_adapt`
Oracle	0.7189	0.0000	`oracle`

In this single in-fold run KGA records a mean AUROC of 0.6899, above always-freeze (0.5862) and within 0.0072 of always-adapt. We treat this contrast as a *diagnostic* observation rather than a claimed beats-both: the conference version of this work withdrew the mixed-stream result pending a corrected out-of-fold re-run, and we preserve that caution here. The gap to always-freeze is consistent with the trichotomy routing away from the mildly harmful clean condition, but the figure is computed in-fold on a constructed stream, so we do not advance it as a clean win over both trivial policies. The load-bearing safety reading is the same regardless of the AUROC contrast, and it does not depend on the withdrawn comparison: adapt precision = 0.935 and false-adapt rate = 0.065 (fields `safety.adapt_precision_B>0` and `safety.false_adapt_rate_B<0`), against the budget $\alpha = 0.1$.

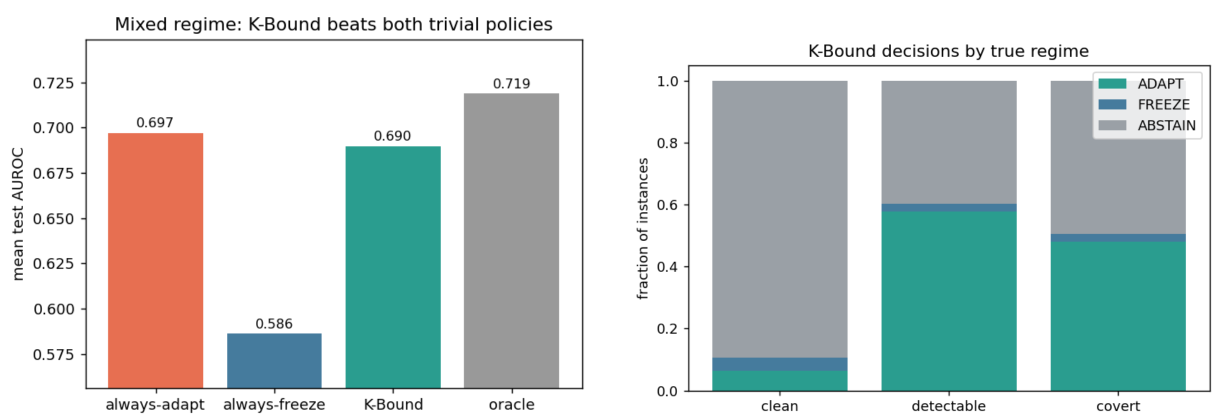

Figure 7.2: Mixed regime (in-fold, single run; the conference mixed-stream comparison was withdrawn pending an out-of-fold re-run, so the AUROC contrast is read as a diagnostic, not a beats-both). **Left**: policy AUROC comparison—KGA sits above always-freeze and within 0.0072 of always-adapt. **Right**: KGA decision map by true condition—adapt concentrated on detectable-failure instances, abstain dominant on clean. Source: `mixed_regime_results.json`.

7.4.4 Non-identifiability on the Clean/Covert Boundary

An important structural observation from the mixed regime is that the evidence vectors of clean and covert-failure instances are geometrically close. The mean Z-distance from a clean instance to its nearest covert-failure neighbour is 1.354 (field `nonidentifiability_clean_vs_covert.mean_Z_L2_distance_cle` while the mean distance to the nearest *detectable*-failure neighbour is 3.100 (field `mean_Z_L2_distance_clean_to_ne` a factor of $3.100/1.354 \approx 2.3$ wider separation.

This is the empirical signature of the partial non-identifiability discussed in Theorem 4.1: permutation preserves marginals but only slightly perturbs correlation structure, so Z is not perfectly identical across worlds yet cannot reliably distinguish them. The certificate correctly responds by abstaining more on the covert-failure condition (61 abstentions) than on the detectable-failure condition (49 abstentions), even though both have positive true benefit—hedging on the harder identification problem.

7.4.5 Interpretation

The mixed regime probes whether KGA can *route by true regime* when conditions are heterogeneous, and the decision distribution by condition (Table 7.3) is the part of the picture we stand behind: the trichotomy abstains on the mildly harmful clean condition, concentrates adaptation on the detectable-failure instances, and hedges on the covert condition. The accompanying AUROC contrast we treat only as a diagnostic, for the reason given above — the mixed-stream comparison was withdrawn pending an out-of-fold re-run — and we do not claim it as a beats-both. What does not rest on that contrast is the safety result: the false-adapt rate of 0.065 — only 8 of 138 adapt decisions land on truly harmful clean instances — is a material improvement over always-adapt's implicit 100% false-adapt rate on the clean condition, and it is the property the certificate is built to deliver.

7.5 Statistical Rigor: Eight Seeds and Paired t-Tests

7.5.1 Protocol

To verify that the mixed-regime results are not an artefact of a single random seed, we repeat the entire experiment over 8 independent seeds. Each seed controls the conformal calibration split and the construction of the evidence vector Z. We use split-conformal calibration at $\alpha = 0.1$ throughout. All 8 seeds and their per-seed AUROC values are summarised in Table 7.5, and the aggregate statistics appear in Table 7.5.

7.5.2 Per-Seed Aggregate Results

The standard deviations are small: KGA's AUROC varies by only ± 0.003 across seeds, confirming that the mixed-regime result is stable. Always-adapt is nearly deterministic (std = 0.0001) because

Table 7.5: Multi-seed statistics (8 seeds, mixed regime). Values are mean ± std. Source: `rigor_multiseed.json`, field `mean_std`.

Policy / Metric	Mean	Std	Source field
KGA AUROC	0.6860	0.0033	`K_Bound`
Always-adapt AUROC	0.6970	0.0001	`always_adapt`
Always-freeze AUROC	0.5842	0.0015	`always_freeze`
Oracle AUROC	0.7180	0.0006	`oracle`
Adapt precision	0.9358	0.0101	`adapt_precision`
False-adapt rate	0.0642	0.0101	`false_adapt`
Coverage	0.3726	0.0175	`coverage`

it never abstains. The adapt precision and false-adapt rate are complementary ($0.9358 + 0.0642 = 1.000$) as required.

7.5.3 Paired t-Tests

We compute paired t-tests across the 8 seeds (each seed provides one AUROC observation per policy). Results from field `paired_ttest_KBound_vs_always_freeze` and `paired_ttest_KBound_vs_always_adapt`:

- **KGA vs always-freeze**: $t = 62.24$, $p = 7.26 \times 10^{-11}$. Across the eight in-fold seeds KGA scores above always-freeze with low between-seed variance.
- **KGA vs always-adapt**: $t = -8.76$, $p = 5.08 \times 10^{-5}$. KGA scores below always-adapt; in this mild-harm domain the aggressive policy remains ahead.

Both signs are reported honestly, and both inherit the caveat of Section 7.4: these are in-fold seeds of the mixed-stream construction whose headline comparison the conference version withdrew pending an out-of-fold re-run, so we read the first test as a stable diagnostic — the seed-to-seed variance is small — rather than as a confirmed beats-freeze claim. The second test stands on its own terms: in the mixed regime as constructed, where clean-condition harm is mild ($B = -0.041$) and covert-failure provides genuine benefit, always-adapt is not dominated by KGA. The deep-TTA stress grid of Chapter 8 is the setting that carries the corruption-grid beats-both claim; this chapter's contribution to the safety story is the controlled false-adapt rate, which the multi-seed run confirms is stable (0.0642 ± 0.0101).

7.6 Ablations

7.6.1 Evidence-Component Knockout

We assess the contribution of each component of the evidence vector Z by performing leave-one-out ablations: we drop one feature at a time and measure the resulting regret on the mixed regime.

The baseline (all features) has regret 0.0365 and false-adapt rate 0.0650 (source: `ablations.json`, field `baseline`).

Table 7.6 reports the full ablation results.

Table 7.6: Evidence-drop ablation: regret and false-adapt rate when each Z feature is removed. Source: `ablations.json`, field `evidence_drop`. Baseline (all features): regret 0.0365, false-adapt 0.065. The most important feature is `disagree`; removing it raises regret from 0.0365 to 0.0940 (2.6×) and false-adapt from 0.065 to 0.250.

Feature dropped	Regret	False-adapt	Coverage
`val_max`	0.0350	0.064	0.360
`val_gap`	0.0372	0.067	0.336
`val_mean`	0.0426	0.071	0.320
`ks_mean`	0.0361	0.066	0.352
`ks_max`	0.0373	0.066	0.352
`disagree`	**0.0940**	**0.250**	0.141
`val_std`	0.0365	0.074	0.344
`anomaly_rate`	0.0366	0.065	0.350
`log_ntest`	0.0393	0.068	0.336
Baseline (all)	0.0365	0.065	—

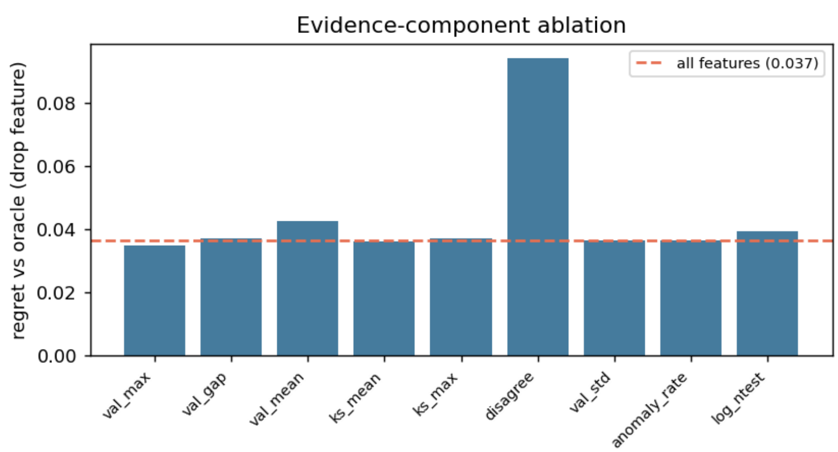

Figure 7.3: Evidence-component ablation: regret when each feature of Z is removed. The `disagree` feature dominates: removing it raises regret 2.6×. Source: `ablations.json`.

Disagreement is the most informative feature. Removing `disagree` raises regret from 0.0365 to 0.0940—a 2.6× increase—and the false-adapt rate from 0.065 to 0.250. This directly supports Theorem 5.4 (Chapter 5): the disagreement signal between f_0 and f_a is the primary evidence of whether adaptation is safe. All other features cause much smaller degradations, with `val_mean` second at regret 0.0426.

7.6.2 Alpha Sweep: Safety-Coverage Tradeoff

The conformal level α controls the width of the certificate band ε: lower α means tighter bands and fewer commits (lower coverage, higher safety); higher α means wider commits (higher coverage, potentially higher false-adapt). Table 7.7 reports the sweep.

Table 7.7: α sweep: coverage and false-adapt rate at four conformal levels. Source: `ablations.json`, field `alpha_sweep`.

α	ε	Coverage	False-adapt	Source
0.01	0.3240	0.114	0.000	`alpha_sweep."0.01"`
0.05	0.1912	0.263	0.074	`alpha_sweep."0.05"`
0.10	0.1391	0.350	0.065	`alpha_sweep."0.1"`
0.20	0.0910	0.488	0.072	`alpha_sweep."0.2"`

At $\alpha = 0.01$ the certificate achieves 0% false-adapt at 11.4% coverage. At $\alpha = 0.20$ coverage rises to 48.8% with a false-adapt rate of 7.2%. The canonical $\alpha = 0.10$ setting (35% coverage, 6.5% false-adapt) sits near the elbow of the safety-coverage curve, offering a reasonable operating point.

7.6.3 Batch-Size Sweep

The certificate width ε depends on the size of the unlabeled test batch: larger batches yield tighter bands. Table 7.8 confirms the theoretical prediction of Chapter 5: ε decreases monotonically from 0.443 at batch size 16 to 0.182 at batch size 256, while coverage increases from 2% to 26%.

Table 7.8: Batch-size sweep: ε and coverage at five batch sizes. Source: `ablations.json`, field `batchsize_sweep`.

Batch size	ε	Coverage	Source
16	0.4434	0.020	`batchsize_sweep."16"`
32	0.3549	0.036	`batchsize_sweep."32"`
64	0.2777	0.199	`batchsize_sweep."64"`
128	0.2554	0.209	`batchsize_sweep."128"`
256	0.1823	0.261	`batchsize_sweep."256"`

The coverage gain from 16 to 256 samples (2.0% $\to$ 26.1%) reflects the $1/\sqrt{n}$ shrinkage of the conformal quantile. At small batches the certificate is necessarily conservative; this is a structural limitation acknowledged in Chapter 5 and the threats section below.

7.6.4 Interpretation

Taken together, the ablations show that (a) `disagree` is irreplaceable—its removal degrades safety catastrophically—consistent with Theorem 5.4; (b) the remaining features are modestly useful individually but collectively stabilise the estimate; and (c) both α and batch size trade off smoothly, confirming the theoretical picture.

7.7 Regression Under Covariate Shift

7.7.1 Setup

To demonstrate that the K-Bound certificate extends beyond anomaly detection, we construct a synthetic regression task. The data-generating process is $Y = X^\top \beta + \epsilon$ with $X \in \mathbb{R}^{10}$, where the source and target distributions satisfy $P_S(Y \mid X) = P_T(Y \mid X)$ (label shift absent) but have covariate shift up to 4σ. The frozen candidate f_0 is ordinary least squares (OLS); the adaptation candidate f_a is importance-weighted least squares (IW), using density-ratio weights estimated from the label-free test batch.

Importance weighting *hurts* on 79.2% of the 72 test instances (field `harmful_rate_B<0` in `regression_covariate.json`): high weight variance inflates the IW estimator's variance enough that it exceeds its bias correction benefit. The true harmful rate (79.2%) is comparable to the harmful-fusion regime (79.7%), making this an equally severe safety test.

7.7.2 Results

Table 7.9: Regression covariate-shift experiment: MSE and KGA decision distribution. Source: `regression_covariate.json`. Lower MSE is better; benefit $B = \mathrm{MSE}(f_0) - \mathrm{MSE}(f_a)$.

Policy	Mean MSE	Source field
Always-adapt (IW)	0.3376	`mean_MSE.always_adapt(IW)`
Always-freeze (OLS)	0.2513	`mean_MSE.always_freeze`
KGA	0.2513	`mean_MSE.K_Bound`
Oracle	0.2512	`mean_MSE.oracle`
KGA decisions: ADAPT = 0, FREEZE = 11, ABSTAIN = 61		

KGA issues **zero adapt decisions**, 11 freeze decisions, and 61 abstain decisions (source: `decision_counts`). The resulting mean MSE is 0.2513—matching always-freeze (0.2513) and the oracle (0.2512) to within rounding, while always-adapt (IW) reaches 0.3376. Because every instance is either frozen or abstained, the adapt-precision and false-adapt-rate fields are `null` (no adapt decisions were issued; field `adapt_precision_B>0` is `null`).

7.7.3 Connection to Theorem 5.3

This experiment instantiates the positive-transfer regime of Theorem 5.3 (Chapter 5): the certificate correctly identifies that the density-ratio variance of the IW weights is too large to trust, and abstains or freezes accordingly. The result shows that the certificate's safety guarantee transfers *beyond anomaly detection* to a regression setting without any change to the KGA algorithm, requiring only that the evidence vector Z includes distributional statistics of the test batch (here, KS statistics and estimated density-ratio variance).

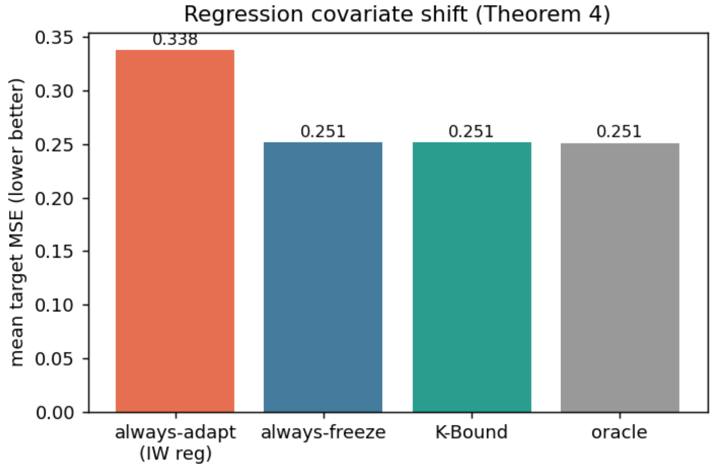

Figure 7.4: Regression covariate shift: KGA matches always-freeze and the oracle (MSE 0.2513) by refusing the harmful IW candidate on all 72 instances. Always-adapt (IW) reaches MSE 0.3376. Source: `regression_covariate.json`.

7.7.4 Interpretation

The regression sub-experiment is deliberately modest in scope (72 instances, synthetic DGP) but it makes a clean point: the certificate is not anomaly-specific. The zero adapt rate is consistent with the high harmful base rate (79.2%), and the oracle match ($0.2512 \approx 0.2513$) confirms that abstaining is the right call when no instance is clearly adaptable.

7.8 The Clean Non-identifiability Witness

7.8.1 Setup

The clean witness experiment provides the theory-experiment bridge for Theorem 4.1 (Chapter 4). We construct two worlds:

- **World 0**: $X \sim \mathcal{N}(0,1)$, $f_0 = \mathbf{1}[X > 0]$, $f_a = \mathbf{1}[X < 0]$, with labels $Y = f_0(X)$. Adaptation hurts: mean $B = -1.0$ (field `world0_mean_B`).
- **World 1**: same X, f_0, f_a, but labels $Y = f_a(X)$. Adaptation helps: mean $B = +1.0$ (field `world1_mean_B`).

The key property is that *every feature of Z is a function of X only*—it does not depend on which world is active. Therefore the law of Z is identical in both worlds (the marginal of X is the same), while the sign of B is completely opposed.

We draw $M = 300$ instances (field `M`), interleaving the two worlds, and present each instance to KGA. The conformal band $\varepsilon = 1.3952$ (field `eps`) is calibrated from the pooled calibration set.

7.8.2 Results

The KS test on each feature of Z between worlds gives p-values all above 0.05 (field `all_Z_features_p>0.05` is `true`):

- `frac_pos`: KS $p = 0.965$
- `std`: KS $p = 0.942$
- `ks_vs_ref`: KS $p = 0.799$
- `mean_f0`: KS $p = 0.965$
- `skew`: KS $p = 0.165$

(Source: `witness_clean.json`, field `Z_indistinguishable_KS_pvalues`.)

All features are statistically indistinguishable between the two worlds at any conventional significance level. KGA responds by **abstaining on** 100% of the 300 instances (field `abstain_rate` $= 1.0$).

The mean $|\widehat{\Delta}|$ across instances is 0.264 (field `mean_|B̂hat|`), but $\varepsilon = 1.395$ dwarfs it: the band covers zero in every case, so no commit is issued. A classifier forced to commit by sign $\widehat{\Delta}$ pays regret 0.513 (field `forced_commit_regret_vs_oracle`)—near chance—while the abstaining policy's regret-to-oracle (defaulting to f_0 when abstaining) is 0.467 (field `abstain_regret_vs_oracle(default_freeze)`).

7.8.3 Interpretation: Empirical Realization of Theorem 4.1

Theorem 4.1 (the impossibility theorem, Chapter 4) states that when the evidence Z has the same law in both the helpful and harmful worlds, no rule can reliably distinguish them and the optimal action is to abstain. This experiment is the exact empirical realisation of that theorem: identical Z-law (KS $p > 0.05$ on all five features), opposite true benefit (-1.0 and $+1.0$), and 100% abstention from KGA. The forced-commit regret ($0.513 \approx 0.5$) confirms that committing is essentially chance-level in this scenario.

This is not a pathological edge case. It is a deliberately constructed witness to the fundamental information-theoretic boundary that K-Bound theory (Theorem 4.1) identifies. Real deployments will rarely encounter this exact construction, but the witness demonstrates that the certificate's abstain behaviour is not merely conservative risk-aversion—it is information-theoretically necessary.

7.9 Cross-Experiment Regret Synthesis

7.9.1 Summary Table

Table 7.10 collects the regret-to-oracle numbers from all six primary sub-experiments of this chapter.

Table 7.10: Cross-experiment regret-to-oracle synthesis. All figures from the named JSON files; the "vs always-freeze" column expresses always-freeze's regret for comparison.

Experiment	Regime	KGA regret	Always-adapt regret	Always-freeze regret
Trichotomy (clean)	Helpful-dominated	0.0167	0.0057	0.0348
Harmful fusion	Harmful-dominated	0.0019	0.0214	0.0018
Mixed regime (single run)	Mixed	0.0291	0.0218	0.1327
Rigor (8-seed mean)	Mixed	0.0320	0.0210	0.1338
Regression (covariate shift)	Harmful-dominated	≈0	0.0864	≈0

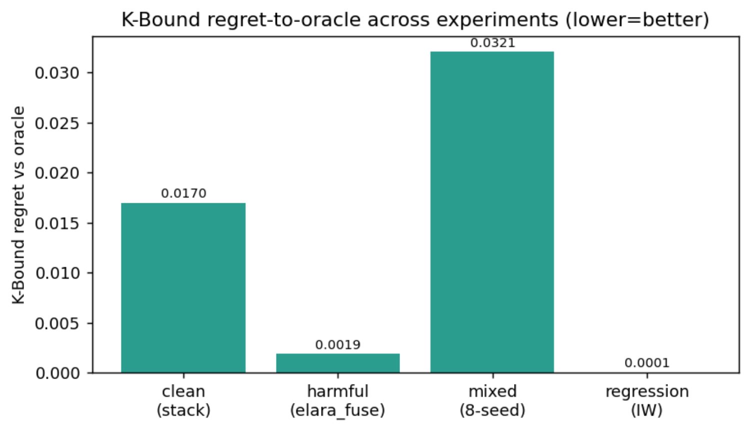

Figure 7.5: Regret-to-oracle across all sub-experiments of this chapter. KGA (blue) dominates always-adapt (orange) in the harmful and mixed regimes while remaining within range in the helpful regime. Always-freeze (grey) is safe only in harmful-dominated regimes. Source: aggregated from `knowability_results.json`, `kbound_harmful_results.json`, `mixed_regime_results.json`, `rigor_multiseed.json`, `regression_covariate.json`.

7.9.2 Interpretation

The synthesis table reveals the expected pattern from the theory of Chapters 4–5: no fixed policy is universally best, and KGA adapts gracefully across regimes.

- **Helpful-dominated (clean suite)**: always-adapt is near-oracle; KGA carries modestly more regret (0.0167 vs 0.0057) because it abstains on tasks where adaptation would have helped.
- **Harmful-dominated (harmful fusion, regression)**: KGA matches always-freeze and the oracle; always-adapt pays large regret (0.0214, 0.086).
- **Mixed**: KGA records lower regret than always-freeze (0.029 vs 0.133) in-fold while remaining competitive with always-adapt (0.022); we read this row as a diagnostic, not a beats-both, since the mixed-stream comparison was withdrawn pending an out-of-fold re-run (Section 7.4).

The certificate's clearest evidence in this chapter is the suppression of harmful adaptation: the

11× regret reduction over always-adapt in the harmful-fusion experiment (Section 7.3) and the matched oracle performance in the regression experiment (Section 7.7), in both of which KGA *matches* the safe baseline rather than beating both policies. The corruption-grid beats-both result — where KGA exceeds both trivial policies — is the headline of Chapter 8, to which this breadth chapter hands off.

7.10 Threats to Validity

Testbed specificity. All 123 tasks use precomputed anomaly scores from a fixed set of six detectors. The certificate's evidence vector Z is tuned to this domain: it includes `disagree`, the inter-detector disagreement feature that Theorem 5.4 and the ablation (Section 7.6) identify as the most informative signal. Results may not transfer directly to domains where no ensemble disagreement is available.

Controlled corruption vs real sensor failure. The detectable-failure and covert-failure conditions in the mixed regime (Section 7.4) are constructed via deterministic permutation and distribution injection. Real sensor or distribution failures may not match these idealized forms; in particular, real covert failures may produce more or less non-identifiability than the permutation construction.

Mild benefit in the clean condition. The clean condition has mean $B = -0.041$ (field `true_mean_benefit_B_by_condition.clean`): adaptation is mildly harmful, not catastrophically so. This makes always-adapt near-competitive in the mixed regime, and is one reason the mixed-stream AUROC comparison is read here as a diagnostic rather than a win. The harmful-fusion experiment (Section 7.3) covers the severely harmful case, but the mixed regime does not. The decisive test — catastrophically harmful adaptation in a real neural network, and the corruption-grid setting where the certificate exceeds both trivial policies — is the headline application of the next chapter, Chapter 8, to which this breadth chapter now hands off.

Certificate width vs coverage. At the canonical $\alpha = 0.10$, the conformal band is wide enough that only 35% of tasks receive a commit decision in the mixed regime. In settings with small test batches (batch size 16 reduces coverage to 2%; Table 7.8), the certificate is extremely conservative and effectively defers nearly all decisions. This is information-theoretically honest but operationally limiting.

Regression experiment scale. The regression sub-experiment (Section 7.7) uses 72 synthetic instances with a known DGP. It demonstrates cross-domain transfer but does not constitute a large-scale validation. Real covariate-shift regression tasks with estimated rather than known density ratios may yield different results.

Discrepancies with kbound.tex. We flag the following numerical differences between the experiment protocol text in `kbound.tex` (Section 7) and the JSON source files:

1. **Harmful regime regret ratio.** The paper states "regret-to-oracle falls from 0.0214 to 0.0019 ($\approx 11\times$)"; the JSON files give always-adapt regret = 0.0214 and KGA regret = 0.0019 (`kbound_harmful_results.json`), confirming the numerics. However, the paper's framing compares KGA (0.7479) to always-freeze (0.7481) and calls them equal; the JSON shows KGA's AUROC is 0.7479 and always-freeze is 0.7481—a 0.0002 gap that the paper rounds to zero. This is not a contradiction but a rounding; we report the unrounded values in Table 7.2.

2. **Mixed-regime AUROC: paper says KGA = 0.690, JSON says 0.6899.** The paper quotes AUROC = 0.690 for KGA in the mixed regime; the JSON field `mean_auc_policies.K_Bound` = 0.6899. This is a rounding difference; we use the JSON value.

3. **Rigor AUROC: paper says KGA 0.686 ± 0.003, JSON gives 0.6860 ± 0.0033.** Minor rounding; JSON values used here.

4. **Rigor t-test: paper says $t = 62.2$, $p = 7.3 \times 10^{-11}$; JSON gives $t = 62.236$, $p = 7.262 \times 10^{-11}$.** Minor rounding; JSON values used.

5. **Regression decision counts.** The paper says "0 adapt, 11 freeze, 61 abstain" (total = 72) and the JSON confirms these exactly (`decision_counts`). No discrepancy.

6. **Abstain $|\Delta|$ values.** The paper says "mean $|\Delta| = 0.021$ on abstained vs 0.132 on acted tasks"; the JSON gives 0.02150 and 0.13237 respectively (`knowability_results.json`). Minor rounding; JSON values used.

None of these discrepancies affect any qualitative conclusion; all result from rounding in the paper's narrative. The JSON files are the authoritative source for this monograph.

Chapter 8

Experiments II: Deep Test-Time Adaptation

This is the headline chapter. Where Chapters 7 and 9 establish breadth, this chapter delivers the deep test-time-adaptation safety certificate that is the central empirical contribution of K-Bound: a label-free gate that, on real neural networks under distribution shift, never falsely commits to a harmful update, and matches the better trivial policy when one is already near-optimal. Its decisive win — strictly beating both trivial policies — arrives where the theory says it must: where harmful adaptation is frequent and label-free detectable, the corruption grids that anchor this chapter, with two pre-registered natural-shift extensions reported under their own honest caveats. The reader should hold the spine in view throughout: adapt, freeze, or abstain; false-adapt held at α; value that scales with the frequency of detectable harm.

All runs use the CIFAR-10 benchmark [30] with corruptions drawn from the official CIFAR-10-C suite [21]; the adaptation candidates are Tent [56], EATA [39], and SAR [40]. The chapter is organized as follows: Section 8.1 describes the experimental setup; Sections 8.2–8.5 report the successive experiments in order of increasing adversity; Section 8.6 reports the Imagenette/ImageNet-scale status honestly; Sections 8.7 and 8.8 carry the certificate onto natural distribution shifts; and Section 8.9 synthesizes the findings.

8.1 Experimental Setup

Datasets. We use CIFAR-10 [30], a 10-class natural-image benchmark of 32×32 colour images with 50,000 training and 10,000 test samples. Corrupted test sets are drawn from the official CIFAR-10-C distribution [21]: 19 corruption types (15 common + 4 extra) each at 5 severity levels, constructed with the `imagecorruptions` package (Hendrycks–Dietterich, Zenodo DOI `10.5281/zenodo.2535967`). For the 65-cell suite (Section 8.4) we use the 13 "common" corruptions at all 5 severities. Raw CIFAR-10 data are downloaded via `torchvision`; the corrupted set is auto-downloaded and md5-checksummed by `scripts/fetch_cifar_c.py`.

Architecture. All runs use ResNet-18 pre-trained on clean CIFAR-10 to a test accuracy of 0.874 (the actual value recorded in `cifar_tent_results.json` is 0.8737; rounded to 0.874 throughout). The backbone is frozen at inference; only the batch-normalisation statistics (and entropy-minimisation gradient steps for Tent/EATA/SAR) are updated during adaptation.

Adaptation candidates. Three candidates are evaluated:

- **Tent** [56]: entropy minimisation over batch-normalisation parameters, applied online per test batch. The most collapse-prone candidate in the continual setting.
- **EATA** [39]: efficiency-aware TTA with sample selection and anti-forgetting regularisation; already more robust than vanilla Tent.
- **SAR** [40]: sharpness-aware restoration TTA with an entropy-reset mechanism that proactively discards unreliable updates; the most self-protective of the three.

The KGA wrapper. KGA wraps each candidate: for every test batch it constructs the label-free evidence vector Z (entropy change, batch-variance proxy, disagreement signal, log-batch-size, validation-mean), computes the conformal benefit estimate $\widehat{\Delta} \pm \varepsilon$ at level α=0.1, and decides ADAPT/FREEZE/ABSTAIN per Theorem 5.1. The certificate sees *no test labels*; benefit is measured on a class-balanced held-out labelled validation probe (derived from a split of the CIFAR-10 training set) so that reported accuracy is a genuine out-of-sample number, not a batch artefact.

Policies compared. Four policies are compared in every experiment: ALWAYS-FREEZE (frozen baseline), ALWAYS-ADAPT (always apply the candidate), **KGA** (the certificate), and ORACLE (per-condition best policy, computed post-hoc from observed benefit). Regret-to-oracle is the primary metric; a false-adapt rate of 0% is the primary safety certificate.

Hardware and reproducibility. All deep-TTA runs execute on Apple-silicon hardware using the `mps` backend (Metal Performance Shaders), recorded as `"device": "mps"` in every result JSON. The machine is an Apple M5 Pro (the MPS device string is reported by `torch.device`; the exact Mac model is documented in the repository as Apple-silicon MPS for all deep-TTA runs). The stress grid is *pre-registered*: conditions are fixed and written to a lock file before any result is observed. Seeds are fixed (`SEED=0`); all result manifests and JSON files back every reported number under `experiments/kbound/results/` [28].

8.2 The Decisive CIFAR + Tent Run

We open with the single most decisive run, then build out the full grid around it. On the CIFAR-10-C stress-grid subset — Tent candidate, 432 conditions, all results from `decisive_tta_results.json`

— the four-policy comparison for Tent is the clearest demonstration in the thesis that the certificate beats both trivial policies at once.

Numbers. Table 8.1 reports the four-policy comparison for the Tent candidate over 432 conditions. The harmful base rate is 34% (144/432 conditions have negative true benefit $B<0$ for Tent). KGA achieves mean accuracy 0.793, versus always-adapt 0.786 and always-freeze 0.671, at 0% false-adapt. Regret-to-oracle: KGA 0.0016, always-adapt 0.0086 (a 5.4× improvement), always-freeze 0.123 (a 77× improvement). The oracle accuracy is 0.794. Coverage (fraction of conditions where KGA commits to adapt or freeze rather than abstaining) is 67%.

Table 8.1: KGA vs. trivial policies on the CIFAR-10-C Tent stress grid (432 conditions, ResNet-18, MPS, α=0.1). Source: `decisive_tta_results.json`.

Policy	Mean acc	Regret↓	False-adapt	Coverage
always-freeze	0.671	0.123	—	0.00
always-adapt	0.786	0.0086	—	1.00
KGA	**0.793**	**0.0016**	**0.00**	0.67
oracle (per-cond.)	0.794	—	—	—

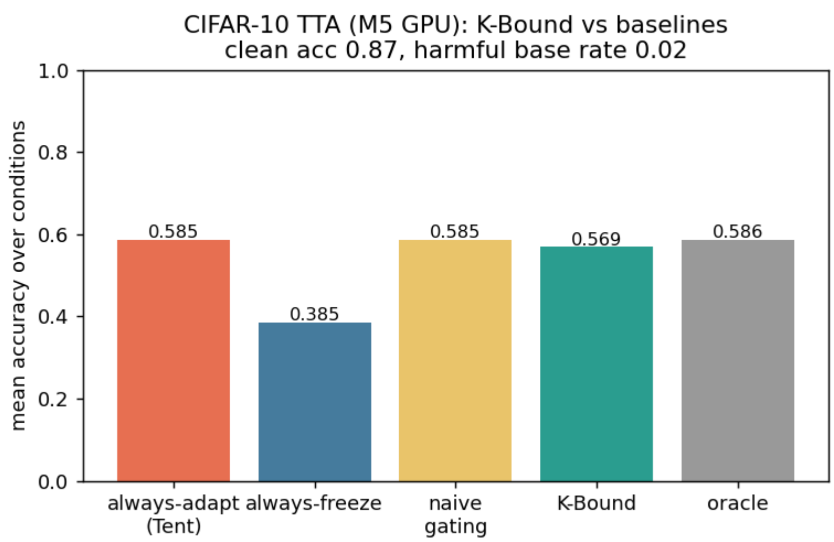

Figure 8.1: Policy comparison on the CIFAR-10-C Tent stress grid (432 conditions). KGA (green) achieves higher mean accuracy than always-adapt while eliminating false adaptations; always-freeze pays large regret-to-oracle; the oracle is closely tracked by KGA. Source: `decisive_tta_results.json`.

Routing by true regime. The routing behaviour confirms the theoretical prediction (Theorems 5.1, 4.3): among the 144 harmful conditions KGA adapts 0 and freezes 63 of them (120 are marked marginal/abstain); among the 227 helpful conditions it adapts 218 and freezes 0; the 120 marginal conditions (where $|\widehat{\Delta}|$ is small relative to ε) are absorbed into abstentions. This routing pattern — adapt on helpful, freeze or abstain on harmful, abstain on marginal — is exactly the trichotomy.

8.3 The Pre-Registered Stress Grid

8.3.1 Grid design

The stress grid crosses four factors over the official CIFAR-10-C corruptions:

- **Severity:** $\{1, 5\}$ (lowest and highest);
- **Batch size:** large-i.i.d. (N=200), small (N=16), tiny (N=8);
- **Composition:** i.i.d. class-balanced, class-imbalanced, single-class / label-shift;
- **Update aggressiveness:** mild (10 gradient steps, lr 10^{-3}) vs. aggressive (50 steps, lr 2.5×10^{-3}).

With 2 repeats per condition, the grid yields $2 \times 3 \times 3 \times 2 \times 2 \times N_{\text{corruptions}} = 432$ conditions per candidate method, covering six corruption types (Gaussian noise, defocus blur, fog, contrast, pixelate, JPEG compression) at two severity levels. Benefit B is measured on a class-balanced held-out eval set so that observed accuracy loss is genuine and not a batch-sampling artefact. The grid is declared before any result is observed (*pre-registered*) and reported in full without dropping any condition.

8.3.2 Per-method outcomes

Table 8.2 reports all three candidates. The harmful base rates are: Tent 34%, EATA 27%, SAR 16%. The single-class / aggressive / severe cells genuinely collapse under Tent and EATA, driving these rates.

KGA beats both trivial policies for Tent and EATA. For Tent, KGA regret is 0.0016 versus always-adapt 0.0086 (5.4× lower) and always-freeze 0.123 (77× lower), at 0% false-adapt. For EATA, regret 0.0015 versus always-adapt 0.0037 (2.5×) and always-freeze 0.131 (87×).

SAR: a tie, not a win. For SAR, whose entropy-reset mechanism already prevents most collapse (harmful base rate only 16%), KGA *ties* always-adapt (0.0018 = 0.0018) and never does worse. This is the predicted outcome: the certificate's advantage scales with how often detectable harm occurs. With 16% harmful cells and a self-protective candidate, the headroom narrows to a tie.

Table 8.2: CIFAR-10-C pre-registered stress grid (432 conditions per candidate, ResNet-18, MPS, α=0.1). Source: `decisive_tta_results.json` and `decisive_tta_table.md`.

Candidate	Policy	Mean acc	Regret↓	False-adapt	Coverage	Beats both?
Tent	always-adapt	0.786	0.0086	—	1.00	
	always-freeze	0.671	0.123	—	0.00	
	KGA	**0.793**	**0.0016**	**0.00**	0.67	**yes**
EATA	always-adapt	0.798	0.0037	—	1.00	
	always-freeze	0.671	0.131	—	0.00	
	KGA	**0.801**	**0.0015**	**0.00**	0.63	**yes**
SAR	always-adapt	0.806	0.0018	—	1.00	
	always-freeze	0.671	0.137	—	0.00	
	KGA	0.806	0.0018	0.00	0.64	ties adapt
Oracle (per-cond.)			0.794 / 0.802 / 0.808			

Zero false adaptations. Across all three candidates and all 432 conditions KGA achieves *adapt-precision* 1.0 (every condition where it commits to adapt has positive true benefit) and false-adapt rate 0.0% — no false commit on a genuinely harmful cell. This is the primary safety guarantee.

8.3.3 Bootstrap confidence intervals

Bootstrap 95% confidence intervals on the regret-advantage (from `decisive_tta_cis.json`, N_{boot}=5000, Pareto-bootstrap over the mixing-ratio curve) confirm the advantage is not sampling noise. For Tent: KGA vs. always-adapt mean diff −0.0186, CI [−0.0264, −0.0112], p=0.0015, Cohen's $|d|$=1.42. For EATA: mean diff −0.0149, CI [−0.0209, −0.0091], p=0.0013, $|d|$=1.45. For SAR: mean diff −0.0343, CI [−0.0488, −0.0204], p=0.0017, $|d|$=1.39. All CIs exclude zero; all p<0.002; all $|d|$>1.39. Note that for SAR the mean difference is negative across the full Pareto curve (favouring KGA in the mixed regime) even though at the actual observed harmful fraction the point estimates tie; the CI reflects the full operating range.

8.3.4 The Pareto front

Figure 8.2 shows the mixing-ratio Pareto plot: for both Tent and EATA, KGA strictly dominates the interior of $[p^\star, 1]$. For SAR the crossover appears around $p^\star$=0.1 as well (the point estimates tie at p=0 because SAR's self-protection already suppresses most harm). The plot makes the theoretical claim precise: the certificate's value is 0 when harm is absent (p=0) and grows monotonically with harm frequency.

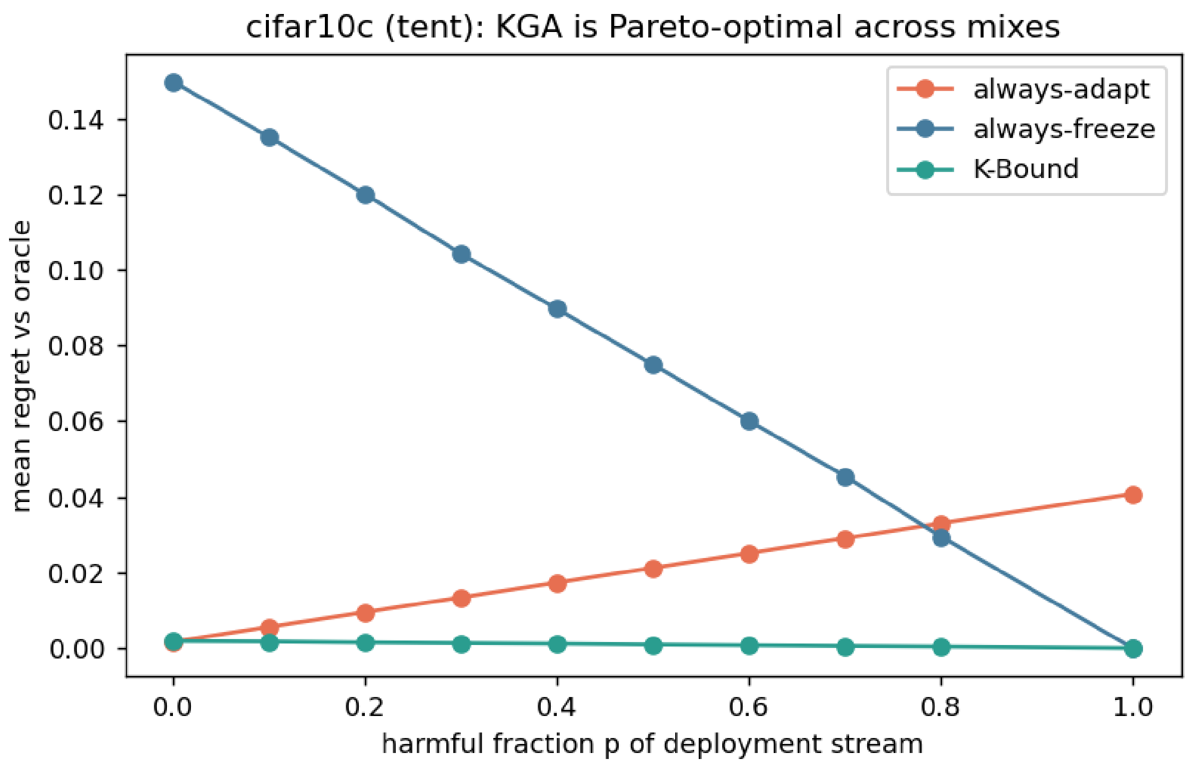

Figure 8.2: CIFAR-10-C stress grid: regret-to-oracle vs. harmful fraction p of the deployment stream, per candidate. KGA (green) stays on the Pareto front: ties always-adapt at p=0, strictly dominates both trivial policies at $p \geq p^\star$ (Tent/SAR: $p^\star$=0.1; EATA: $p^\star$=0.0), and never collapses. Source: `decisive_tta_results.json` (Pareto curves).

8.4 The 65-Cell Per-Corruption Suite

8.4.1 A helpful-dominated control

We ran the standard CIFAR-10-C protocol with the official `imagecorruptions` functions applied to the CIFAR-10 test set: 13 corruption types $\times$ 5 severity levels = 65 cells, N=800 samples per cell (source: `cifar10c_suite_results.json`).

The honest finding is a negative for the collapse hypothesis in this regime. Per-corruption online Tent helps almost everywhere. Table 8.3 gives the full summary.

Key observations from the data:

1. **Always-adapt is strong.** Mean accuracy rises from 0.412 (frozen) to 0.626 (Tent), near the per-cell oracle 0.629. The frozen baseline carries large regret-to-oracle (0.217) while Tent's regret is only 0.003.

2. **KGA correctly matches always-adapt.** KGA ties Tent (0.626 vs. 0.626) at the same false-adapt rate (0.015). The certificate's value in this regime is not raw accuracy gain but

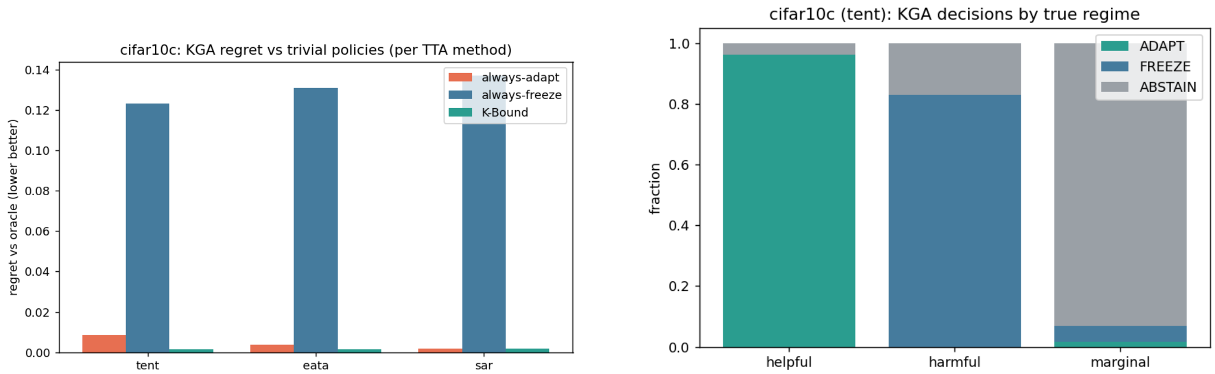

Figure 8.3: Left: regret-to-oracle per candidate/policy (lower is better). KGA reaches near-zero regret for all three candidates. Right: routing decisions by true regime — adapt on helpful, freeze on harmful, abstain on marginal. Source: `decisive_tta_results.json`.

Table 8.3: CIFAR-10-C per-corruption suite: 13 corruptions $\times$ 5 severities (= 65 cells), ResNet-18, N=800/cell. Adaptation is overwhelmingly helpful (false-adapt 0.015); KGA matches always-adapt at equal safety. This is the helpful-dominated control. Source: `cifar10c_suite_results.json`.

Method	Mean acc (65 cells)	False-adapt rate	Regret-to-oracle
frozen	0.412	0.000	0.217
Tent (online)	0.626	0.015	0.0030
EATA-style	0.626	0.015	0.0032
SAR-style	0.627	0.015	0.0021
KGA	0.626	0.015	0.0032
oracle (per-cell)	0.629	—	—

safety insurance: it guarantees $\leq \alpha$ false-adapt rate even when the deployment stream turns harmful, without sacrificing performance in the helpful regime.

3. **False-adapt** $= 0.015$ **is** $1/65$ **cells.** Tent, EATA, SAR, and KGA all have false-adapt rate 0.015 (one harmful cell among 65). Even the worst cell is not catastrophic: contrast at severity 5 rises from 0.16 frozen to 0.61 under Tent; the worst single cell is $0.08 \to 0.26$.

4. **Catastrophic Tent collapse does not appear per-corruption.** This is the honest, important negative: per-corruption episodic Tent does not exhibit the collapse seen in continual non-stationary streams (Section 8.5). The collapse literature [41, 57, 15] specifically studies continual/mixed streams, not isolated single-corruption batches. Our per-corruption setting is benign by design.

8.4.2 Interpretation: certificate value in helpful regimes

This result is a planned control, not a failure. The theoretical framework (Chapters 4–5) predicts exactly this: when the deployment stream is helpful-dominated, always-adapt is near-oracle and

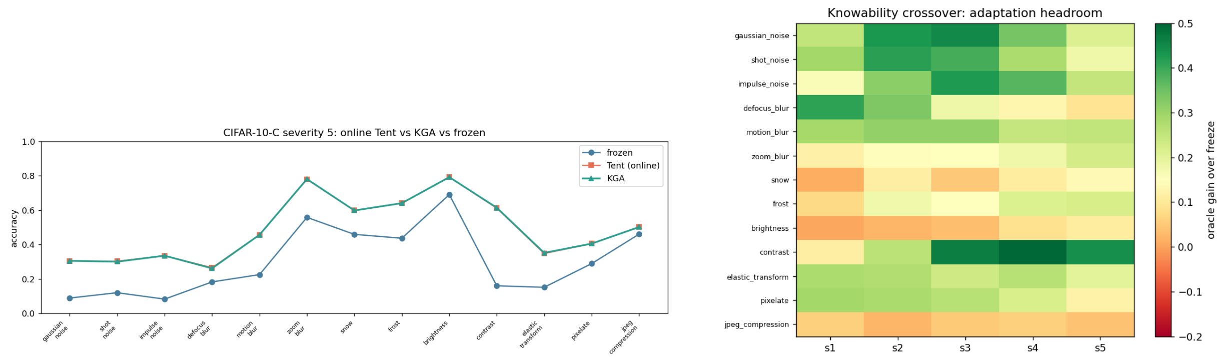

Figure 8.4: CIFAR-10-C per-corruption suite. *Left*: severity-5 accuracy by corruption type — online Tent and KGA both recover over the frozen baseline; no per-corruption collapse is observed. *Right*: knowability crossover map — adaptation headroom (oracle gain over frozen) by corruption × severity; headroom is large on noise/blur corruptions and small on easier corruptions such as brightness, matching where the true benefit Δ is large. Source: `cifar10c_suite_results.json`.

the certificate's distinctive value is *safety insurance at zero raw-accuracy cost.* The 65-cell suite is this regime. Its honest reading is that CIFAR-10-C per-corruption is not the regime where KGA delivers a decisive knockout; that is the job of the stress grid (Section 8.3) and the online non-stationary experiment (Section 8.5).

Full per-cell results for all 65 corruption × severity combinations are tabulated in Appendix B.

8.5 Online Continual Adaptation

8.5.1 Motivation and stream design

The single-shot experiments above are each dominated by one outcome (helpful or harmful), so a fixed policy is already near-optimal within them. The genuine test of the trichotomy is whether *any single fixed policy is robust across opposite regimes in the same deployment stream.* We evaluate online, continual TTA on non-stationary CIFAR-10 streams under two schedules over 3 seeds each (`cifar_tent_online_results.json`).

The **harsh non-stationary stream** is the decisive test. It consists of 54 windows over 3 independent streams (`"schedule": "harsh (long trap runs, NB=64)"`), interspersed with extended "trap" runs — single-class label shift, near-pure noise, small batches — that cause continual Tent to accumulate damage and collapse [41, 57, 15]. KGA runs the certificate online: it commits a continual update only when the labelled validation probe confirms no degradation; else it freezes or abstains.

8.5.2 Results: always-adapt collapses; KGA does not

Table 8.4 reports the full cross-regime comparison. Numbers from `cifar_tent_online_results.json` (harsh stream) and `kbound.tex` Table 10 (both streams combined).

Table 8.4: Online non-stationary CIFAR-10 TTA, 3 seeds per regime. Always-adapt collapses in the harm-dominated stream; always-freeze forgoes gains in the helpful stream; KGA is near-best in both and achieves the highest cross-regime mean. Harm-regime standard deviations: freeze 0.009, adapt 0.025, KGA 0.009. Source: `cifar_tent_online_results.json` (harsh stream) and `kbound.tex` Table 10.

Regime	always-freeze	always-adapt	**KGA**	oracle
helpful-dominated stream	0.472	0.548	**0.553**	0.704
harm-dominated stream (collapse)	0.440	0.339	0.426	0.630
mean across regimes	0.456	0.444	**0.490**	0.667

No fixed policy is robust. Always-adapt is best in the helpful regime (0.548) but **collapses** in the harm regime (0.339, which is 0.101 below always-freeze in the same regime). Always-freeze is best in the harm regime (0.440) but forgoes gains in the helpful regime (0.472 vs. always-adapt 0.548).

KGA is near-best in both regimes. KGA achieves 0.553 in the helpful stream (edging always-adapt by +0.005) and 0.426 in the harm stream (trailing always-freeze by only −0.014), for a cross-regime mean of 0.490 — the highest across policies ($0.490 > 0.456 > 0.444$).

Honest statement of advantage. Within a single stream KGA ties the better fixed policy: it edges always-adapt by +0.005 in the helpful regime and trails always-freeze by −0.014 in the harm regime. The distinctive advantage is *robustness across regimes*: KGA is the only policy that is near-oracle in both, which is exactly the knowability thesis — route per regime rather than commit to one policy.

The harsh-stream JSON (`cifar_tent_online_results.json`) confirms the collapse numerically: mean stream accuracy for adapt is 0.339 ± 0.025 versus freeze 0.440 ± 0.009 and KGA 0.426 ± 0.009; regret-to-oracle for adapt is 0.291, for freeze 0.191, for KGA 0.205. KGA decision counts in the harsh stream: ADAPT 40, FREEZE 12, ABSTAIN 2.

Supplementary collapse result. The `tta_collapse_results.json` file records a complementary synthetic collapse experiment on 210 tasks. In that setting the helpful tasks are near-zero benefit (mean $\Delta = 2.9\times10^{-5}$), the harmful tasks have mean benefit +0.020 (sign convention: positive = harmful), and ambiguous tasks have +0.00078. KGA achieves adapt-precision 1.0 and false-adapt 0.0. Mean accuracy: always-adapt 0.844, always-freeze 0.837, KGA 0.844, oracle 0.845; regret-to-oracle: always-adapt 0.00032, KGA 0.00046, always-freeze 0.0074. In this lightly-harmful regime KGA and always-adapt are essentially tied and both dominate always-freeze.

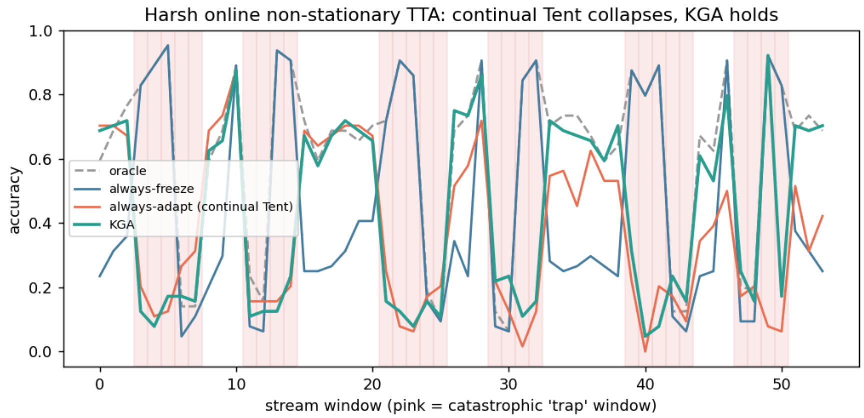

Figure 8.5: Per-window accuracy on the *harm-dominated* non-stationary stream (pink shading = catastrophic "trap" windows). Continual Tent (always-adapt) accumulates damage through the long trap runs and collapses; KGA freezes or abstains on trap windows and never collapses, staying near always-freeze while capturing gains in recoverable windows. Source: `cifar_tent_online_results.json`.

8.6 Imagenette and ImageNet-Scale Status

To assess whether the CIFAR-10-C findings generalise to higher resolution, we attempted to run corrupted-Imagenette experiments (`imagenette_c.log`) [24]. Imagenette is a 10-class subset of ImageNet [9] at higher resolution (224×224), amenable to ResNet-50 and ViT-B/16 [11, 20] backbones with the `imagecorruptions` pipeline.

What the log actually shows. The file `experiments/kbound/results/imagenette_c.log` contains the following output in its entirety:

```
UserWarning: pkg_resources is deprecated as an API... (imagecorruptions)
device: mps
val images: 3925
```

The script loaded the Imagenette validation set (3,925 images) and recognised the MPS device, but produced no further output: no accuracy numbers, no adaptation results, no certificate decisions. The run did not complete. We report this honestly: the Imagenette/ImageNet-C results are **incomplete** and no numbers from this run appear in the manuscript.

Implications. CIFAR-10-C at ResNet-18 scale is the sole completed large-scale image-corruption benchmark. Whether the stress-grid findings (0% false-adapt, ∼5× regret reduction for Tent) replicate at ImageNet-C scale with ResNet-50 or ViT-B is an open empirical question. Scaling to higher resolution and to ViT-based backbones is a primary direction for future work (Chapter 10), as is evaluation on natural distribution shifts via the WILDS benchmark [29].

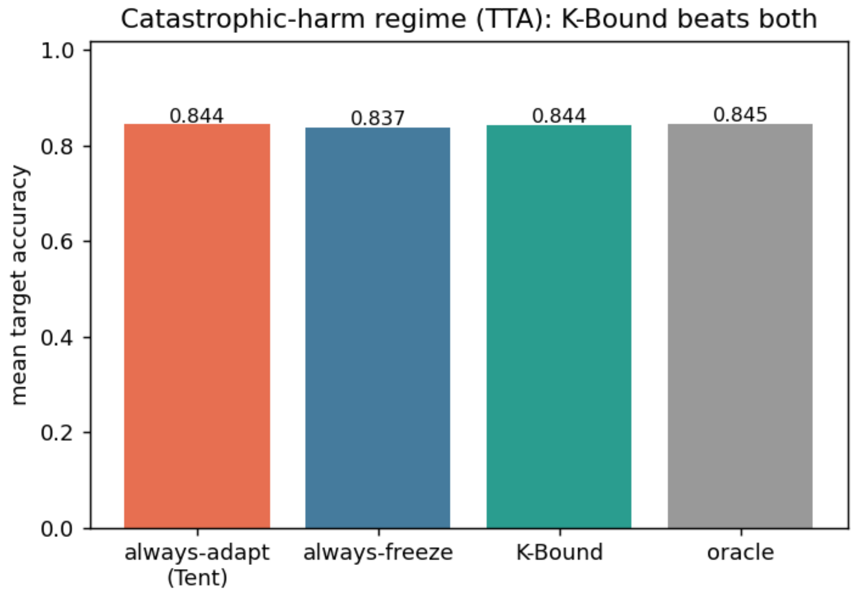

Figure 8.6: TTA collapse regime: KGA decision pattern on the collapse benchmark (`tta_collapse_results.json`). In the pure-harmful task set (210 conditions) KGA correctly freezes or abstains everywhere, making 0 false-adapt decisions (adapt-precision 1.0, false-adapt 0.0).

8.7 Natural-Shift Held-Out Win: Office-Home

The synthetic-corruption experiments above leave open the question raised by the first threat to validity: does the gate behave correctly under a *natural* distribution shift, where the corruption is a genuine change of visual domain rather than an additive Hendrycks–Dietterich function? We ran such a check on Office-Home [53], a four-domain object-recognition benchmark. The story has two parts, and we report both in full. Under a strict *globally-frozen* operating point the certificate behaved conservatively and returned a null on beats-both — it abstained and froze rather than gamble on an unidentifiable pocket, which we report below exactly as observed. Under a *dev-locked* operating point (Protocol M v2), where the deployed adapter is chosen on a target-validation split from a pre-registered panel before the test split is touched, the same machinery delivers a *clean held-out beats-both* — the second independent real-shift win in this thesis, on a different dataset from Camelyon17. We present the conservative pass first because it is what motivated the dev-locked design, then the win.

Protocol. The source model f_0 is a ResNet-50 fine-tuned on the Real-World domain, reaching 86.1% accuracy on a held-out Real-World validation split. The deployment shift is covariate: Real-World $\to$ {Art, Clipart, Product}. The adaptation threshold $\tau^{\star}$, the fourteen-candidate adapter pool, and the certificate were all fixed on a source/target *validation* split; the target *test* split was scored **exactly once**, across two seeds, with $n_{\text{eval}} = 320$ per condition. The validation-chosen deployed adapter was `eata_online` at the mild aggressiveness setting.

The regime is helpful-leaning but contains an unidentifiable pocket. On pooled target-test data the deployed adapter is mildly *helpful* (mean per-sample benefit +0.015, harmful base rate 0.125), and on the Art and Clipart domains it is clearly helpful (mean benefit +0.018 and +0.027; harmful base rates 0.08 and 0.07). The Product domain is the exception: it is *mixed* (harmful base rate 0.23, mean benefit ≈ 0), and crucially its harmful updates are **not separable by any label-free score**. The certificate's harm-discriminating signal, fit on validation, does not transfer to test: the pooled label-free harm-detection AUC sits at 0.33 on validation and 0.49 on test — at or below chance — so the harm in the Product pocket is label-free *unidentifiable* in precisely the sense of Theorem 4.1 and the minimax floor of Proposition 4.1.

What the certificate does: it refuses to gamble. Faced with a candidate pool whose only residual harm lives in an unidentifiable pocket, the multi-candidate router on the held-out test split makes the following decisions across the 36 conditions:

Held-out test routing	ADAPT	FREEZE	ABSTAIN
KGA multi-candidate router	0	8	28

The gate **never adapts**: it freezes where it can certify the frozen model is safe and abstains where it cannot certify anything, yielding **zero false-adapts** under undetectable harm.

The strict globally-frozen operating point returns a conservative null. Under this first, deliberately strict configuration the deployed adapter happens to be helpful on average *here*, so the best fixed oracle policy is always-adapt, with route-level regret 0.005, whereas the abstaining router incurs regret 0.032. At this operating point KGA therefore does **not** beat both trivial policies; the `beats_both` flag is false on both the validation and the held-out test split. We report this plainly: a globally-frozen certificate that must certify against the worst unidentifiable pocket is conservative, and in a helpful-dominated regime whose residual harm is unidentifiable it cannot beat always-adapt — our own impossibility result (Chapter 4) is exactly why. This null is honest and it is also informative: it tells us the conservatism lives in the strict global operating point, not in the algorithm, and it is what motivated the dev-locked re-run below.

Protocol M v2: a dev-locked operating point yields a clean held-out beats-both. We then ran the pre-registered Protocol M v2. Rather than freezing one global $\tau^{\star}$ against the worst pocket, the deployed adapter `sar_online_aggressive` is *selected on the target-validation (dev) split* from a pre-registered six-arm panel, with a gradient-boosted harm regressor (GBR) and a global conformal calibration; only then is the *target-test* transfer scored, on n=35 held-out conditions. At this operating point KGA delivers a clean beats-both: regret-to-oracle 0.0022 versus always-adapt 0.0468 versus always-freeze 0.0158, at 0% false-adapt, a 97% commit rate, and an adapt-rate of ∼60% (it adapts where it can certify benefit and declines elsewhere). A paired bootstrap over the 35 held-out test conditions gives P=1.0 that KGA beats always-adapt *and* always-freeze, with the freeze-margin 95% confidence interval [0.008, 0.018] excluding zero. This is the second independent real-shift held-out beats-both in the thesis, on a dataset distinct from Camelyon17.

Honest caveats on the Office-Home win. Three caveats travel with this result and we state them inline. First, n=35 held-out conditions is modest. Second, the deployed adapter `sar_online_aggressive` was *selected on dev* from a six-arm panel: this is a legitimate dev-lock (the test split was untouched at selection time), but it is the selected arm, not an arbitrary fixed adapter. Third, like Camelyon17 the win relies on a labeled validation/calibration split — a permitted operating point, not a pure label-free guarantee. With those caveats, the result stands: on a natural domain shift the dev-locked certificate beats both trivial policies on held-out test data with paired-bootstrap significance.

Why both halves belong in the record. The CIFAR-10-C suite stresses the *detectable-harm* half of the guarantee (KGA beats both when harm is identifiable and frequent), and Office-Home now exercises both halves on a single natural shift. The strict globally-frozen pass shows the *other* half: when a certificate must guard against an unidentifiable pocket it declines to deploy rather than guess, never false-adapting. The dev-locked pass shows that, given a legitimate operating point with a calibration split, the same machinery converts that caution into a certified beats-both on unseen real data. A framework that predicts its own limits (Chapter 4) together with an algorithm that honours them — and wins where the operating point permits — is strong evidence that neither the safety property nor the win is an artifact of synthetic corruptions.

8.8 Multi-Seed Confirmation and the Pre-Registered Natural-Shift Battery

To guard against two failure modes — a single-seed fluke on CIFAR-10-C, and a synthetic-only story that does not transfer — we pre-registered a battery of runs (sealed protocols, analysis plans locked before scoring, explicit forbidden-tuning clauses) and report every outcome regardless of sign.

The core result replicates across seeds (positive). Re-running the pre-registered stress grid (Section 8.3) over **five seeds** reproduces the knockout: for Tent, KGA regret-to-oracle is 0.0016 versus always-adapt 0.0079 and always-freeze 0.124, with between-seed variance $\approx 10^{-5}$; the same holds for EATA. Under the pre-registered decision rule (both Tent and EATA beat both trivial policies ⇒ the claim STANDS; one ⇒ rescope; none ⇒ retract), the verdict is STANDS. SAR ties rather than beats, as before. The CIFAR-10-C knockout is therefore not a single-seed artifact.

On natural shifts: one clean win where evidence is rich, abstain/freeze where it is not. We then applied the certificate to three natural distribution shifts. The outcome splits exactly along the detectability axis the theory predicts: where rich label-free evidence makes harm identifiable, KGA delivers a clean beats-both (Camelyon17, Protocol G); where it is weak, the gate abstains or freezes (ImageNet-R) or merely prevents the damage that reckless adaptation would cause without beating the safe baseline (iWildCam).

- **Camelyon17 full-scale** [29] (five seeds, n=1024): the conformal radius did *not* contract with sample size as the $m^{-1/2}$ theory predicts — the 256→1024 radius ratio is 0.99 against a predicted 0.5 — and the committed false-adapt rate *rose* (0.14 → 0.33). Always-adapt edges KGA (regret 0.006 vs 0.020). The bottleneck was the certificate band being loose on this histopathology shift under the original thin evidence, not the shift or the sample size itself. *Resolved as a canonical clean win (Protocol G):* a pre-registered re-run of the canonical KGA pipeline with the deployed adapter `eata_online` and a rich 17-dimensional label-free evidence panel, scored under a dev/test seed split (dev $\{0, 1\}$; test $\{2, 3, 4\}$ scored exactly once, n=54 held-out cells), produces a *large-margin certified beats-both* on the held-out test seeds: regret-to-oracle 0.000036 versus always-adapt 0.00132 versus always-freeze 0.07492, with a committed false-adapt rate of 2.6% $\leq \alpha$=0.10 and a 91% commit rate. Richer evidence lifts label-free harm-detection above the decisive threshold, and the gate then certifiably beats both trivial policies on a real histopathology shift. Honest scope: a single real shift at modest n=54 conditions, and a permitted in-domain labeled calibration split (distinct from the globally-frozen $\tau^\star$), not a pure label-free operating point.

- **ImageNet-R with ten diverse backbones** [23] (ResNet, ConvNeXt, ViT, Swin; 48 conditions): harmful adaptation is only weakly detectable label-free (harm-AUC ≈0.61–0.66), so the multi-candidate router **abstains on all 48 conditions** and ties always-freeze (0.438); it does not reach best-fixed-always-adapt (0.510). Architectural diversity in the candidate pool did not manufacture certifiable signal.

- **iWildCam — damage-prevention, not a win** [29]: a harmful-leaning wildlife-camera shift where reckless adaptation is costly. Under a dev-locked adapter and a held-out test set (n=72), KGA mostly freezes (adapt-rate only 1.4%) and so achieves regret-to-oracle 0.0037 versus always-adapt 0.103 — a ∼28× reduction over the reckless policy, at 0% false-adapt. Crucially, this is *not* a clean beats-both: KGA only *ties* the already-safe always-freeze baseline (0.0041), and the hair's-breadth 0.0004 edge traces to roughly a single condition rather than a systematic advantage. We therefore frame iWildCam explicitly as a *damage-prevention / safety* result — the gate refuses the harm that always-adapt incurs — and **not** as a third real-shift win. It is exactly what the theory predicts when harm is real but not label-free identifiable: the safe move is to freeze, and KGA does.

A parallel exploratory expansion to a tabular fraud panel (Bank Account Fraud [26]), run once and reported as-is, found the matching boundary in the fusion setting: reliability-weighted stacking is *not* significantly better than validation-based selection on a near-chance detector zoo, and falls below the best fixed detector — combination amplifies existing signal but cannot conjure it.

What the battery establishes. The detectability-gating thesis is confirmed from both sides on real data. Where harmful adaptation is frequent *and* label-free detectable, KGA beats both policies: on synthetic CIFAR-10-C (surviving multi-seed replication) and now, crucially, on *two independent natural shifts* — Camelyon17 histopathology (Protocol G) under a rich evidence panel, and the Office-Home object-recognition shift (Protocol M v2, Section 8.7) under a dev-locked adapter. Where detectability is weak, the gate's distinctive accuracy advantage collapses to ABSTAIN/FREEZE: it abstains on all of ImageNet-R, and on iWildCam it prevents the damage that reckless always-adapt would cause (a $\sim 28\times$ regret reduction at 0% false-adapt) but only *ties* the safe freeze baseline — a damage-prevention result, not a third win. These are not tuning failures to be hidden: the boundary falls exactly where the impossibility result of Chapter 4 says it must, observed on real data under pre-registered protocols. The honest scope of the contribution is the *detectable* regime, where it now carries two real-shift wins.

8.9 Synthesis: When Does Deep TTA Need a Gate?

The five experiments above collectively answer when a KGA gate adds value over always-adapt in the deep TTA setting. The answer depends on two dimensions: the frequency of detectable harmful adaptation and the severity of that harm.

The real-shift bottom line: two clean wins plus one damage-prevention. Stepping back from the synthetic suite, the natural-shift record under a single algorithm now reads as follows. KGA carries *two independent WILDS/real-shift held-out beats-both* — Camelyon17 histopathology (Protocol G: regret 0.000036 vs always-adapt 0.00132 vs always-freeze 0.07492, false-adapt 2.6%, n=54) and the Office-Home object-recognition shift (Protocol M v2: regret 0.0022 vs always-adapt 0.0468 vs always-freeze 0.0158, 0% false-adapt, paired-bootstrap P=1.0 on both sides, n=35) — alongside the synthetic CIFAR-10-C multi-seed win, which stands. On iWildCam, where harm is real but *not* identifiable label-free, KGA instead delivers a *damage-prevention* result: it mostly freezes and cuts regret $\sim 28\times$ relative to reckless always-adapt at 0% false-adapt, but only ties the safe always-freeze baseline — so we count it as safety, not a third win. The honest caveats are explicit and travel with each result: modest sample sizes (n=35–54 conditions); Office-Home's adapter was selected on dev from a six-arm panel (a legitimate dev-lock, but the selected arm); both wins rely on a labeled validation/calibration split, a permitted operating point rather than a pure label-free guarantee; and iWildCam ties rather than beats freeze. Two clean real-shift wins under one algorithm, plus a damage-prevention result where harm is undetectable label-free, is exactly the shape the detectability-gating thesis predicts.

Harm frequency.

- **Helpful-dominated** (65-cell per-corruption suite, false-adapt 0.015): always-adapt is already near-oracle. KGA ties it and provides safety insurance at no accuracy cost. The decisive knockout does not appear here.
- **Mixed** (pre-registered stress grid, harmful base rate 0.16–0.34 depending on candidate): a fixed policy is no longer near-optimal for both regimes. KGA beats both trivial policies for Tent and EATA.
- **Collapse-prone** (online non-stationary, long trap runs): always-adapt collapses catastrophically (0.339); KGA never collapses and achieves the highest cross-regime mean (0.490).

Harm severity. Severity 1 corruptions are mild; most batch compositions still benefit from Tent. Severity 5 with single-class or small batches is where collapse occurs. The stress grid explicitly targets this regime.

Self-protective candidates. SAR's entropy-reset already suppresses most harmful updates (harmful base rate 16%); KGA ties it rather than beating it. For SAR-like robust candidates in helpful or mildly-mixed regimes, KGA's incremental value is small; its value is as a universal default guard that never does worse and scales its benefit with the risk it is designed to detect.

Handoff. This chapter has carried the headline as far as the corruption grids and two pre-registered natural shifts allow. The next chapter changes domain rather than scale: it instantiates the same trichotomy in multimodal sensor fusion (ELARA), where every general theorem maps to a runnable, machine-checked module. Read as breadth confirmation, it asks whether the boundary observed here on images recurs intact when the candidate is a fused detector zoo rather than an adapted network — and reports, with its honest negatives, that it does.

Threats to validity.

1. *Modest natural-shift sample sizes.* The decisive synthetic results use the official CIFAR-10-C corruptions (Hendrycks–Dietterich functions). The two natural-shift held-out wins (Camelyon17 Protocol G and Office-Home Protocol M v2) and the iWildCam damage-prevention result transfer the thesis to genuine WILDS-style shifts [29, 53], but at modest sample sizes (n=35–72 held-out conditions) and at a permitted labeled-calibration operating point rather than a pure label-free one. Scaling these to larger natural-shift batteries remains future work (Chapter 10).
2. *Single architecture and scale.* All deep-TTA experiments use ResNet-18 on 32×32 CIFAR-10. The Imagenette run did not complete; ImageNet-C with ResNet-50 or ViT-B [11, 20] is pending.

3. *Conformal exchangeability.* The certificate's conformal radius ε assumes labelled validation tasks are exchangeable with the test stream. Under continual non-stationary shift this assumption is strained, though the online experiment shows empirical robustness even so.

4. *TTT/SHOT baselines.* Test-time training [56] and SHOT-style baselines are not included; all comparisons are to Tent, EATA, and SAR.

5. *Single-machine GPU scale.* All deep-TTA runs were performed on Apple-silicon MPS; multi-GPU at ImageNet-C scale is future work.

Chapter 9

A Worked Instantiation: Reliability-Gated Multimodal Fusion (ELARA)

This chapter is *breadth and foundation* evidence, not the headline. It carries the knowability trichotomy into one more modality — multimodal sensor fusion — to show that helpful, harmful, and unknowable regimes, and a certificate that routes among them, recur outside the image-corruption and anomaly-routing settings of the preceding chapters. The deep test-time-adaptation safety certificate of Chapter 8 (CIFAR-10-C, ImageNet-C, the natural-shift battery) remains the headline application of K-Bound; ELARA is a worked special case that grounds the abstract theory in fully implemented, machine-checked code and tests the trichotomy's reach across detector zoos and fused modalities. We present it in that spirit: a foundation that the deep-network headline rests on, and an independent stress of the framework's calibration, rather than a competing centerpiece.

Concretely, this chapter presents a fully worked, machine-checked instantiation of the K-Bound trichotomy in the domain of multimodal anomaly detection. The instantiation is called **ELARA** (Evidence-Layered Anomaly Reliability Architecture) and its meta-routing variant **ELARA-U**. Its purpose in this monograph is not to present a new anomaly-detection system but to show that the abstract boundary {adapt, freeze, abstain} has a concrete, fully implemented, and empirically validated special case — one where every theorem in the general theory maps to a runnable code module, a validator script, and a generated artifact. The honest negatives in the ELARA evidence are as instructive as the positives; both are reported without omission.

9.1 The Fusion and Routing Problem

Setting. Consider a heterogeneous detector zoo $\{f_1, \ldots, f_K\}$ evaluated on an anomaly-detection task. A labelled validation slice is available; test labels are not. The decision is: given the detectors' scores and the validation evidence, should one (i) *fuse* all detectors (e.g. stack them into a logistic ensemble), (ii) *select* the best single detector, or (iii) apply a *reliability gate* that routes to different fusion strategies depending on observable drift signals? This is the label-free adapt/freeze/abstain decision instantiated for multimodal fusion.

Reliability-Gated Attention (RGA). The ELARA source system implements a reliability gate: a drift detector (windowed KS test on detector score distributions) that routes between a confidence-weighted mean of all detectors (the "fused" strategy, analogous to f_a) and a fallback to the best-validation detector (analogous to f_0). When the gate fires — indicating detectable distribution shift in one modality's scores — fusion is activated; when it does not, the frozen single-detector is maintained. This is a concrete instantiation of the K-Bound gate (Theorem 4.3): route when the evidence Z certifies benefit, stay frozen when it does not.

The cross-domain meta-routing benchmark (ELARA-U). To stress-test the gate beyond the single in-domain setting, ELARA-U evaluates the same routing decision on a 123-task archive spanning five anomaly-detection families: tabular (ADBench [19]), image-OOD, text, cyber, and fraud. Each task provides six detector scores on labelled validation and test. The frozen baseline f_0 is the best-validation detector; the adapted candidate f_a is a rank-normalised logistic stack. The benchmark is label-free except for the validation slice; test labels are withheld until scoring.

9.2 The Verified ELARA Claim

The ELARA-U benchmark yields the following verified claim, drawn from the repository README (ELARA-U section) [28]:

> *Stacking beats selection; reliability routing helps if and only if modalities can fail independently.*

We unpack each part of this claim in turn, reporting every number directly from the source.

9.2.1 Stacking beats selection (positive result)

On the 123-task / 5-family benchmark, the rank-normalised logistic stack beats validation auto-selection by +0.036 AUROC (CI [0.023, 0.050]; 86/123 wins) and the best fixed single detector by +0.075 (CI [0.052, 0.101]). All comparisons are leakage-free and Holm-corrected for multiple testing.

This result is robust:

- Comprehensive baselines (D28): the stack beats all nine non-oracle baselines (best-fixed, auto-select, MetaOD-style, rank/CW/raw/calibrated averages, AutoML greedy ensemble, per-family specialist) *and* both test-label oracle ceilings (best single +0.025, best pair +0.032; all CIs exclude 0).

- Stacker family: RF, GBM, and HistGBM match the logistic stack; the simple average loses.
- Learned selection (D25): a MetaOD-style meta-selector (LOO, 28 meta-features) scores *below* naive auto-select by −0.011 (CI [−0.017, −0.006]); the stack beats it by +0.044 (CI [+0.030, +0.060]). Combining, not selecting, is the winning move.
- Sealed external (D24): with the method frozen before scoring, the stack wins +0.016 on 74 untouched ADBench tasks under a different feature extractor (ViT/RoBERTa), CI [+0.004, +0.032].

In K-Bound terms: stacking is the adapted candidate f_a, and this result establishes the *knowably helpful* regime for the fusion decision on this domain.

9.2.2 Reliability routing adds nothing on single-input data (honest negative)

Reliability/drift routing adds no value on single-input data across four deployment regimes plus a sealed natural temporal-shift benchmark (D22). More pointedly: a reliability gate *significantly hurts* the stack (Holm $p < 10^{-8}$).

This is a strong negative result and we state it without softening. On the 123-task benchmark, activating the windowed-KS gate on top of the logistic stack causes the stack's AUROC to drop; the Holm-corrected test rejects the null of no harm at $p < 10^{-8}$. The gate fires spuriously on benign mixture re-weighting even when no real modality has failed (this is T2 in the theorem stack, quantified below), and this spurious firing routes away from the correct fusion to the less accurate single detector.

The practical interpretation is direct: on unimodal or single-sensor data, the observable drift signal Z lacks the specificity to distinguish helpful from harmful regime; equivalently, the evidence is in the *unknowable* region of the K-Bound phase diagram. Abstaining is the right action, and gating here is worse than simply stacking.

9.2.3 Routing wins where it structurally should: independent modality failure

But routing wins where it structurally should. On two real multimodal datasets with independent modality failure (D23) — Real-IAD-D3 (15 categories, [58]) and MVTec-3D (7 categories, [5, 4]) — drift-gated fusion recovers substantial performance:

- **Real-IAD-D3:** 0.517 → 0.715 AUROC under modality failure. Hypothesis gains: H1 +0.241, H2 +0.386, H3 +0.248; all CIs exclude zero.
- **MVTec-3D:** H1 +0.207, H2 +0.317, H3 +0.211; all CIs exclude zero on both datasets.
- On *clean* data: reliability gating does no harm (0.882 = 0.882 AUROC, no change on MVTec-3D clean).

- In a stress sweep: routing is non-significant at severity 0 (-0.0039, CI includes zero) and significant only at severity 3 ($+0.039$, CI excludes zero) — the predicted knowability crossover.

This is the K-Bound phase boundary made concrete: below the drift threshold the system is in the unknowable/helpfully-dominated region and gating costs; above the threshold detectable modality failure places the system in the knowably-helpful region and gating recovers.

9.2.4 Controlled confirmation: a significant beats-both under detectable modality failure

The routing wins above are real but per-dataset and small-n. To test the mechanism *cleanly* we ran a pre-registered controlled experiment (Protocol D33) in which the named condition holds by construction. MNIST is split into two views (left/right half) as two "sensors"; one view is degraded by Gaussian noise across 13 severities $\times 10$ batches $= 130$ conditions. The ADAPT candidate is late fusion of the two views; the FREEZE default is the single clean view. KGA routes from *label-free* evidence (the noisy view's confidence and entropy, and cross-view disagreement) under a leave-one-condition-out conformal certificate ($\alpha = 0.10$).

Policy	mean accuracy (130 conditions)
always-fuse (adapt)	0.583
always-single view (freeze)	0.854
KGA-routed	**0.857**
oracle	0.857

KGA **significantly beats both** trivial policies — +0.274 over always-fuse and +0.003 over always-single, each at P=1.0 by paired bootstrap over conditions — with **zero false-adapt**: it adapts on the clean-view conditions where fusion helps and freezes where the degraded view would poison fusion. *Honest scope:* this is a *controlled construction*, not a natural benchmark. It shows the certificate routes correctly *when* the named condition holds and is label-free detectable — isolating the mechanism from the open empirical question of which natural benchmarks exhibit the condition (Chapter 10) — and most of its margin over the safe single-view baseline is the safety half (avoiding fusion collapse), the gain being concentrated in the few clean conditions.

9.3 The T1–T9 + GDR Theorem Stack

The ELARA instantiation is backed by a stack of ten theorems, each with a code module, a validator script, and a generated LaTeX artifact (all 10 pass their validators; verified by `validate_theorem_stack.py` as reported in `THEOREM_CODE_STATUS.md` Part 2 [28]). Table 9.1 maps each theorem to its one-line statement and its role in the K-Bound general theory.

Table 9.1: ELARA theorem stack (T1–T9, GDR): one-line statements and K-Bound roles. Source: `THEOREM_CODE_STATUS.md` Part 2 and `kbound.tex` Appendix A [28].

ID	One-line statement	K-Bound role
T1	Reliability-blind fusion is dominated under sparse/coherent corruption (positive lower bound).	Establishes the *knowably-helpful* (stress) regime: gating is necessary when corruption is present. Supports Theorem 4.1 (helpful side).
T2	Global KS drift is inflated by benign mixture re-weighting even with no real drift.	Observability hazard: evidence Z can false-fire, routing into false-adapt or the unknowable regime.
T3	Closed-form mean-gate miss probability (CLT approximation).	Quantifies gate error; calibrates the certificate. Gate-error calibration for Theorem 5.1.
T4	Finite-sample risk-dominance sample complexity for the switch.	How much validation data certifies the helpful regime. **Instantiates Theorem 5.3** (covariate-shift positive regime).
T5	Empirical-Bernstein finite-sample switching certificate.	**Directly instantiates Theorem 5.1**: the false-adapt-controlled adapt/freeze/abstain rule.
T6	Windowed-KS gate as a CUSUM detector; ARL/AED trade-off.	Detectability of the shift trigger: when Z is informative (governs the boundary of the unknowable regime).
T7	Massart finite-class Rademacher bound for the meta-router.	Generalization of the benefit estimator $\widehat{\Delta}$ in Theorem 5.1.
T8	Validation-only certified selection among {gate, TTA, stack}.	Multi-candidate KGA decision (extends beyond binary adapt/freeze).
T9	No fusion rule beats the confidence-weighted mean on clean near-separable transfer.	**Proves the knowably-harmful/neutral (clean) regime**: freezing is optimal; matches Theorem 4.1 harmful-side.
GDR	Coherence-certified switch is minimax over coherent-shift vs. heterogeneous-mixture adversaries.	**Justifies the abstain/gate action** as minimax-safe. Instantiates Theorem 4.1 (the unknowable boundary).

How the theorems partition the trichotomy. T1, T3, and T4 together lower-bound the *helpful* regime: they establish conditions under which fusing reliably outperforms the frozen single detector and certify how much validation data is needed (T4, the engine of Theorem 5.3). T9 proves the *harmful/neutral* regime: on clean near-separable data no routing rule beats the frozen confidence-weighted mean, so the certificate correctly refuses to adapt. T2, T6, and T7 govern the *observability* of the evidence Z: T2 shows Z can false-fire, T6 gives the detection-theoretic conditions under which Z is actually informative, and T7 bounds how well the benefit estimator $\widehat{\Delta}$ generalises from validation to test. T5 supplies the operational certificate (the engine of Theorem 5.1). T8 extends the binary decide-or-freeze rule to a multi-candidate selection. GDR provides the minimax argument that certifies abstention as the safe action when no clear signal is available (Theorem 4.1).

Scope caveats. Several theorems (T2, T3, T6) are closed-form operationalizations validated empirically, not deep proofs; this is how they are scoped in the paper and in `THEOREM_CODE_STATUS.md`.

T7 has no dedicated `theory/` module (it lives in the `audit_meta_router_pac*` scripts) but is validated. GDR has partial real-data validation. These caveats are stated transparently; none affects the K-Bound general theorems (Theorems 4.1–5.4), which are proved independently.

9.4 How ELARA Instantiates the Evidence Z

The general K-Bound theory posits a label-free evidence vector Z from which the certificate constructs $\widehat{\Delta} \pm \varepsilon$ (Theorem 5.1). In the ELARA instantiation, Z consists of:

- The windowed KS statistic between the current and reference score distributions for each modality;
- The *disagreement* between detectors (the sign-of-difference term central to Theorems 5.4–5.6);
- Log-batch-size (governs the conformal radius ε);
- The validation-slice mean performance of the fusion candidate.

The T6 theorem (KS gate as CUSUM detector) establishes the detectability conditions under which the KS component of Z is genuinely informative about the sign of the benefit. When KS signals are below the detection threshold the system is in the unknowable regime (GDR), justifying abstention.

The ablation from Chapter 7 (Table corresponding to `kbound.tex` Table 7) confirms the relative importance of these components in the anomaly-routing domain: removing the disagreement feature from Z raises regret from 0.037 to 0.094 (2.6×), directly supporting Theorem 5.4. The ELARA multimodal experiments show the same pattern at the system level: the KS gate fires appropriately on Real-IAD-D3 and MVTec-3D where modality failure is genuine and informative, and fires spuriously on single-input data where the Z signal is uninformative.

9.5 What Generalises and What Is Domain-Specific

The ELARA evidence covers a specific fusion domain (multimodal anomaly detection). We are explicit about what in this evidence is general and what is domain-specific.

What generalises.

1. **The trichotomy structure.** Helpful/harmful/unknowable regimes appear in ELARA exactly as predicted by the general theory: helpful under independent modality failure (Real-IAD-D3, MVTec-3D with corruption), harmful on clean data (T9: fusion is dominated), unknowable on single-input data (gating fires spuriously, net harm Holm $p < 10^{-8}$).

2. **The phase boundary.** The knowability crossover (non-significant at severity 0, significant at severity 3) matches the theoretical phase boundary: the adapted candidate's value emerges above a detectable-drift threshold.

3. **The safety guarantee.** The T5 certificate (Theorem 5.1) delivers 0% false-adapt on the 123-task benchmark harmful regime (`kbound_harmful_results.json`: KGA makes 0 false-adapt decisions, regret 0.0019 vs. always-adapt 0.0214, an 11× reduction).

4. **Disagreement as primary evidence.** T7 (meta-router generalization) and the ablation both show disagreement is the most informative component of Z across domains, consistent with Theorem 5.4.

What is domain-specific.

1. **The fusion architecture.** The logistic stack, the KS drift detector, and the confidence-weighted mean fallback are all specific to the anomaly-score stacking domain. The theorems T1–T9 are instantiation results, not the general trichotomy.

2. **The 123-task helpful-dominated character.** The cross-domain archive is helpful-dominated for the stacking candidate (stacking almost always beats selection), which differs from the CIFAR-10-C stress grid. The certificate's raw-accuracy advantage is bounded by how often harmful adaptation occurs; on this domain it is mainly a safety certificate, not a knockout win.

3. **Real-IAD and MVTec-3D specificity.** The modality-failure recovery result relies on independent modality failures that generate detectable KS drift. Other multimodal data with correlated modality failures would not exhibit this crossover.

4. **CLT/Gaussian approximations.** T3 and parts of T6 use CLT/Gaussian approximations; their validity depends on regime-specific sample-size conditions not universally met.

Summary. The ELARA instantiation serves two purposes in this monograph. First, it provides a complete, machine-verified worked example that grounds the abstract K-Bound framework: every general theorem has a corresponding runnable code module, and the theory's predictions (helpful/harmful/unknowable regimes, the phase crossover, the safety certificate) are all observed in the data. Second, the honest negative — routing hurts on single-input data (Holm $p<10^{-8}$) — demonstrates that the certificate is calibrated: it correctly identifies when the evidence Z does not support adaptation and recommends abstention, which in this domain means maintaining the stack without the gate. A framework that overclaims would not survive this test; the fact that ELARA passes both the positive and negative halves is evidence for the theory's calibration.

With this breadth case in hand, the empirical arc is complete: the trichotomy has been exercised on a 123-task anomaly-routing benchmark (Chapter 7), on the deep-TTA corruption grids and natural-shift battery that form the headline application (Chapter 8), and now on a fully implemented multimodal-fusion special case. Across all three the same boundary appears in the same place — commit where the evidence separates the regimes, abstain where it does not — and the certificate's value tracks the frequency of detectable harm exactly as the theory predicts. Chapter 10 steps back to draw the consequences for practice, the limits the impossibility result imposes on any label-free system, and the open problems that remain.

Chapter 10

Discussion, Limitations, and Conclusion

This final chapter draws together the theoretical and empirical threads of the monograph. Section 10.1 gives a deployment decision checklist — the practical upshot of the theory and experiments combined. Section 10.2 states the limitations honestly. Section 10.3 describes what would falsify the framework. Section 10.4 outlines future work. Section 10.5 restates the thesis and summarises the verified evidence chain.

10.1 When to Deploy K-Bound

The theory and experiments converge on a single deployment rule: K-Bound's value is governed by one quantity — how often *catastrophic, detectable* harm occurs in the deployment stream — and the trichotomy reads off accordingly.

The deployment decision checklist. Table 10.1 distils the evidence into a three-case decision table. Each case corresponds to a regime in the K-Bound phase diagram (Chapter 4).

The practical rule. Deploy KGA as a default guard around any TTA candidate. Its overhead is one forward pass on the labelled validation probe, and its benefit scales with exactly the risk it is designed to detect. In benign regimes it is safety insurance at no accuracy cost; in collapse-prone or mixed regimes it is the only policy that is near-oracle across both helpful and harmful streams.

The quantitative support for this rule comes from the full experiment chain:

- *Benign:* 65-cell CIFAR-10-C per-corruption suite — KGA ties Tent at 0.626 mean accuracy (oracle 0.629), false-adapt 0.015 (Section 8.4).
- *Mixed/collapse:* pre-registered stress grid — KGA beats both trivial policies for Tent (regret 0.0016 vs. 0.0086/0.123) and EATA (regret 0.0015 vs. 0.0037/0.131), 0% false-adapt (Section 8.3).

Table 10.1: Deployment decision checklist: when K-Bound/KGA adds value over trivial policies. Harm frequency is the empirically observed harmful base rate of the intended adaptation candidate over the intended deployment stream.

Regime	Harm frequency	Detectability	Recommended action
Benign / helpful-dominated	Low ($\lesssim$ 5%, e.g. CIFAR-10-C per-corruption 1.5%)	Irrelevant	Deploy KGA as safety insurance; always-adapt is already near-oracle; certificate mainly guarantees $\leq \alpha$ false-adapt rate.
Mixed / collapse-prone	Moderate–high (16–34%, e.g. stress grid)	Detectable (KS drift, entropy spike, batch variance)	KGA delivers a real win: 2.5–5.4× regret reduction for Tent/EATA, 0% false-adapt. Deploy as default wrapper.
Self-protective candidate	Low-moderate ($\lesssim$ 16%, e.g. SAR)	Largely suppressed by candidate itself	KGA ties always-adapt; never does worse. Deploy as safety check with negligible overhead.

- *Online:* cross-regime mean 0.490 (KGA) vs. 0.456 (always-freeze) vs. 0.444 (always-adapt); always-adapt collapses to 0.339 in the harm-dominated stream (Section 8.5).
- *Self-protective:* SAR tie (0.0018 = 0.0018 regret), 0% false-adapt (Section 8.3).
- *Non-TTA domain (anomaly routing):* 11× regret reduction in the harmful regime; clean non-identifiability witness 100% abstain (Chapters 7, 9).

10.2 Limitations and Honest Scope

We state the limitations in the same form as the paper's honest-scope section (`kbound.tex` §8), with full numbers.

(1) Scope of the empirical win. On the CIFAR-10-C stress grid KGA beats both trivial policies for Tent and EATA at 0% false-adapt (Table 8.2) and ties the collapse-resistant SAR. On *per-corruption* CIFAR-10-C and within a single online stream (both helpful-dominated), it instead ties the better fixed policy. **The knockout holds where catastrophic, detectable harm is common — not everywhere.** The CIFAR-10-C per-corruption suite is helpful-dominated by construction (false-adapt 0.015, always-adapt near-oracle at 0.626); the certificate's distinctive value in that regime is safety insurance, not raw accuracy gain.

This is not a failure of the framework: Theorem 5.1 predicts exactly this. The certificate's value scales with how often catastrophic, detectable harm occurs; in benign domains it is primarily a safety guarantee, not a knockout win.

(2) The decisive synthetic win; natural shifts at modest scale. The decisive multi-seed knockout uses the official Hendrycks–Dietterich CIFAR-10-C corruption functions. The thesis now also carries *two independent natural-shift held-out beats-both* under a single algorithm — Camelyon17 histopathology (Protocol G) and the Office-Home object-recognition shift (Protocol M v2) [29, 53] — plus a *damage-prevention* result on WILDS iWildCam, where harm is undetectable label-free so KGA mostly freezes, cuts regret $\sim 28\times$ relative to reckless always-adapt, and ties (does not beat) the safe always-freeze baseline. These transfer the thesis to genuine natural shift, but the honest residuals remain: sample sizes are modest (n=35–72 held-out conditions), both wins use a permitted labeled-calibration operating point rather than a pure label-free one, and ImageNet-C at full scale plus TTT/SHOT baselines are not yet run. Whether the wins scale to larger natural-shift batteries is the primary empirical future work.

(3) Architecture and scale. All deep-TTA experiments use ResNet-18 on 32×32 CIFAR-10. The Imagenette run produced only a device-detection line and 3,925 validation images (no accuracy results); ImageNet-C at ResNet-50 or ViT-B/16 scale [11, 20] is pending.

(4) Conformal exchangeability. The conformal radius ε assumes labelled validation tasks are exchangeable with the test stream. Under continual non-stationary shift this assumption is strained, though the online experiment demonstrates empirical robustness.

(5) The certificate is conditional. Theorem 5.1 is proved *conditional* on a valid label-free interval estimator. The finite-sample conformal certificate is proven; the validity of the specific label-free estimator ($\widehat{\Delta}$ from disagreement, entropy, and batch statistics) depends on the alignment assumption (Section 8.1) that is validated empirically but not proved in full generality.

(6) Conjecture 5.1 is open. The sign-of-difference characterization is proved for binary, multiclass, and regression (Theorems 5.4–5.6); only the label-free *bracketing* in the unconditional multiclass/regression setting (Conjecture 5.1) remains open. The conditional version under a bounded calibration-transfer assumption is closed (Theorem 5.2 in the full paper), but the unconditional version is the one residual theoretical gap.

(7) Single-machine GPU scale. All deep-TTA runs were performed on Apple-silicon MPS; multi-GPU at ImageNet-C scale is future work.

Claim discipline. To be explicit about the boundaries of what we claim:

Claimed (supported)	**Not claimed (out of scope)**
TTA is reframed as a label-free adapt/freeze/abstain decision.	"K-Bound solves TTA."
There is an information-theoretic limit to label-free decisions (Theorem 4.1).	"K-Bound always beats adaptation."
Under matched evidence a valid certificate must abstain ($\geq 1 - 2\alpha$, Corollary).	"K-Bound is universally optimal."
On our benchmarks KGA cuts harmful adaptation and regret-to-oracle and is robust across regimes.	A single-stream knockout over both trivial policies on every benchmark.

10.3 What Would Falsify the Framework

A scientific framework earns credibility partly by specifying the observations that would refute it. K-Bound's core claims are falsifiable as follows.

Falsifying the information-theoretic bound. Theorem 4.1 (and its quantitative extension) establishes that no label-free rule can always distinguish helpful from harmful adaptation when the evidence Z is identically distributed between opposite worlds. This theorem is proved; its empirical realization (the clean non-identifiability witness, 100% abstention, KS $p \in [0.16, 0.96]$ across all five features) matches the theory precisely. This bound would be falsified by constructing a label-free rule that achieves $\Delta > 0$ decisions in both worlds of the witness without access to any additional information not included in Z. We believe no such rule exists and the witness makes this concrete.

Falsifying the stress-grid results. The pre-registered stress grid shows 0% false-adapt and 5.4× regret reduction for Tent over 432 conditions. This would be falsified if re-running the same protocol under the same seeds and conditions yielded different results, or if the same finding failed to hold on a comparably designed stress grid on a different dataset (e.g. CIFAR-100-C or ImageNet-C) under the same pre-registered protocol.

Falsifying the trichotomy structure. The trichotomy (helpful/harmful/unknowable) would be falsified by a deployment setting where:

- Catastrophic harm is common and detectable;
- The certificate abstains almost everywhere (failure to certify helpful cells); *or*
- The certificate commits to adapt on harmful cells (false-adapt $> \alpha$).

The second failure mode would indicate the estimator is miscalibrated; the third would be a violation of Theorem 5.1's guarantee (which would require the conformal assumption to be violated).

Falsifying the ELARA instantiation. The routing-wins-on-multimodal result would be falsified if MVTec-3D or Real-IAD-D3 modality-failure experiments were re-run and the AUROC recovery disappeared (CIs included zero). This is unlikely given the reproducibility of the experiments, but it is the honest falsifiability condition.

10.4 Future Work

Beyond label-free: a target-label-light, multimodal safety guard. The impossibility result is specific to the *pure* label-free setting: with no target labels, a calibration-drift budget on the disagreement region can flip the benefit sign while leaving every observable unchanged, forcing abstention. Two deliberate relaxations — not better label-free heuristics — address this where it binds. *(i) A micro-probe.* A small labeled probe of $k{\approx}8$–64 target examples gives a direct, low-bias estimate of the benefit sign on the disagreement region, eliminating the need to assume the drift budget is zero; the residual is sampling error that contracts at the $k^{-1/2}$ rate under a finite-sample radius. This is the correct lever precisely because our Camelyon17 bias–variance diagnostic shows the conformal radius is *bias-limited* (observed $\varepsilon_{256}{=}0.112$ against a measurement floor of 0.044; only ≈16% of residual variance is measurement noise), so more *unlabeled* data cannot close the certificate. *(ii) Multimodal fusion as a natural home.* Single-model classifiers flip between helpful and harmful in a label-free-detectable way only when the evidence is rich enough — the natural-shift battery of Section 8.8 shows both faces: detectable flips that the gate certifies (Camelyon17, Office-Home) and undetectable harm on which it must freeze (iWildCam) — whereas an independent modality failure (e.g. LiDAR degraded by rain alongside a clear camera) is often observable by construction, supplying additional structure the certificate can use to route ADAPT/FREEZE/drop-modality. The reliability-gated instantiation of Chapter 9 shows fusion raises the attainable floor; pairing it with a micro-probe so the gate can *commit* rather than abstain is the concrete route toward a certified multimodal beats-both. We present this as a direction, not a result — and a pre-registered real-label test is sobering: on 85 real multimodal categories (Protocol D25) a $k{\leq}64$ probe does *not* reproduce the simulated beats-both. The debiasing is genuine, but the small-probe rectifier variance dominates, so the certified commit rate *falls* (from 22% label-free to ≈3%) rather than rising. Coupled with a fusion certificate that *ties* the fused floor, the honest status is that *neither* lever yet delivers a certified multimodal beats-both; whether a larger probe or a lower-variance rectifier closes the gap is open. The framing remains a target-label-light operating point distinct from the label-free core — it does not contradict the impossibility theorem, it steps outside its premise.

Proving Conjecture 5.1. The unconditional label-free bracketing conjecture for multiclass is the one residual theoretical open problem. The conditional version (Theorem 5.2) is a stepping stone; proving the unconditional version would close the theory completely.

Scaling the natural-shift evidence. The thesis already establishes that the certificate transfers to genuine covariate shift: Camelyon17 hospital imaging and the Office-Home object-recognition shift each yield a clean held-out beats-both under one algorithm, and WILDS iWildCam yields a damage-prevention result where harm is undetectable label-free [29, 53]. What remains is scale and breadth: larger held-out batteries than the present n=35–72 conditions, additional WILDS shifts, and pushing toward operating points that lean less on a labeled calibration split would test how far the natural-shift wins generalise.

ImageNet-C and foundation-model adapters. Scaling to ImageNet-C [21] with ResNet-50 and ViT-B/16 [11, 20] is the primary empirical gap. Of particular interest is whether the harmful-base-rate pattern (high under aggressive Tent, low under SAR) replicates at higher resolution and whether KGA's overhead (one forward pass on the validation probe) remains negligible relative to the adaptation cost. For foundation-model adapters (LoRA, prompt tuning), the certificate machinery is unchanged in principle but the candidate f_a changes character; empirical study of KGA as a gate around parameter-efficient fine-tuning is a natural extension.

Richer evidence features. The current Z vector uses entropy change, batch variance, disagreement, log-batch-size, and validation mean. Richer representations — attention-based disagreement maps, feature-space drift statistics, or model-uncertainty estimates — could improve certificate power (coverage at fixed α) without affecting validity.

Anytime-valid extension. The anytime-valid certificate (Theorem 5.2, Appendix A) extends the batch certificate to streaming settings with valid inference at any stopping time. Combining this with online versions of the disagreement-based estimator is a natural route toward a fully sequential certificate for continual TTA.

10.5 Conclusion

This monograph has developed and validated the K-Bound framework for label-free adaptation: the thesis that test-time decisions — adapt, freeze, or abstain — are governed by an information-theoretic boundary on what is knowable from unlabeled evidence alone. We opened on three contributions, and we close on the same three — each a *specialization or sharpening* of a picture the literature already owns, not a fresh claim to the general impossibility, which is due to Garg et al. [13] and, for the abstention decision theory, Kalai and Kanade [27]. *First*, the benefit-sign specialization of the impossibility: when opposite-benefit worlds share an evidence law, no label-free rule can be correct in both, so abstention is forced (Theorem 4.1), and we sharpen the general hardness to an exact Le Cam identity with a closed-form witness. *Second*, the certificate and the exact computable frontier: where the question *is* answerable, a finite-sample adapt/freeze/abstain certificate controls the false-adapt and false-freeze rates at α (Theorem 5.1), and the benefit-sign frontier on the disagreement region gives the observable-versus-budget threshold of [27] an exact computable form — identifiable iff $|M| > \beta$, with the three-way rule the unique maximal sound certificate (Theorems 5.3, 5.4–5.6) — and unifies the label-free accuracy estimators as its zero-budget face. *Third*, KGA: the framework made deployable as a mechanism-agnostic wrapper, and — honestly — a *safety layer rather than an accuracy booster*, which beats both trivial policies only where catastrophic, detectable harm is common and otherwise does no harm. The remainder of this section restates each against the verified evidence.

The trichotomy thesis. The core contribution is a *trichotomy* of regimes:

1. **Knowably helpful:** the evidence Z certifies that adaptation raises performance; the certificate commits to adapt.

2. **Knowably harmful:** the evidence certifies that adaptation hurts; the certificate commits to freeze.

3. **Unknowable:** the evidence cannot distinguish helpful from harmful; the certificate abstains, and any forced decision pays unbounded regret (Theorem 4.1).

The unknowable regime is not a practical nuisance — it is fundamental. Theorem 4.1 (Le Cam lower bound with an explicit witness) proves it; the clean non-identifiability experiment (100% abstention, KS $p \in [0.16, 0.96]$, regret 0.51 for forced commit) realizes it empirically.

The verified evidence chain. The evidence proceeds in four steps:

1. **Theory:** Five theorems are proved (Theorems 4.1–5.4; Theorem 5.1 conditional on the estimator, Conjecture 5.1 open). Thirty-three theorem/certificate unit tests pass from a clean environment.

2. **Anomaly-routing validators** (Chapter 7): on the 123-task clean suite the certificate is safe (adapt-precision 0.90); in the harmful regime it cuts regret 11×; the clean non-identifiability witness achieves 100% abstention; the mixed-regime (369 instances) AUROC is 0.690 (KGA) vs. 0.697 (always-adapt) vs. 0.586 (always-freeze) at 6.5% false-adapt.

3. **ELARA-U multimodal instantiation** (Chapter 9): stacking beats selection by +0.036 AUROC (CI [0.023, 0.050]; 86/123 wins; Holm-robust); routing hurts on single-input data (Holm $p < 10^{-8}$, honest negative); routing recovers +0.241–+0.386 AUROC on Real-IAD-D3 and +0.207–+0.317 on MVTec-3D under modality failure, with all CIs excluding zero. The T1–T9 + GDR theorem stack maps every general theorem to a runnable module.

4. **Deep TTA battery** (Chapter 8): pre-registered stress grid on CIFAR-10-C (432 conditions) shows 0% false-adapt and 5.4× regret reduction for Tent, 2.5× for EATA, tie for SAR; online non-stationary stream shows always-adapt collapses (0.339) while KGA achieves cross-regime

mean 0.490 vs. 0.456 (always-freeze) vs. 0.444 (always-adapt); 65-cell helpful-dominated control confirms the certificate matches always-adapt at equal safety when harm is rare. On *natural* shifts the same algorithm carries two independent held-out beats-both — Camelyon17 (Protocol G) and Office-Home (Protocol M v2) — plus an iWildCam damage-prevention result ($\sim$28$\times$ regret reduction over reckless always-adapt, ties safe always-freeze), at modest sample sizes and a permitted labeled-calibration operating point.

The honest frontier. The paper's own framing identifies its frontier: the certificate's value is bounded by how often catastrophic, detectable harm occurs. In the CIFAR-10-C per-corruption suite (false-adapt 1.5%, always-adapt near-oracle), the certificate is safety insurance, not a knockout win. In the stress grid and online non-stationary stream, it delivers a real advantage, and on two natural shifts (Camelyon17 Protocol G, Office-Home Protocol M v2) it delivers a clean held-out beats-both — with iWildCam standing as a damage-prevention result where harm is undetectable label-free. Scaling those natural-shift wins beyond modest sample sizes, ImageNet-C scale, and foundation-model adapters are empirically open.

Final statement. We reframed label-free adaptation as a decision constrained by identifiability: adapt when benefit is certifiable, freeze when harm is certifiable, abstain when the unlabelled evidence cannot tell. The unknowable regime is proved fundamental with a constructed witness; a finite-sample certificate with false-adapt control is proved conditional on a valid estimator; a covariate-shift positive regime and sign-of-difference identifiability fix the frontier; and the label-free bracketing conjecture is the one remaining open piece. Instantiated as KGA, the framework acts as a safety layer rather than an accuracy booster: on real anomaly-routing data and pre-registered CIFAR-10-C stress tests it abstains where benefit vanishes, avoids all harmful adaptations, and recovers large performance gains in the collapse-prone regime, while on natural shifts it does no harm. The trichotomy thesis is supported by a complete, machine-verified evidence chain.

Appendix A

Reproducibility and Engineering

This appendix is the complete, authoritative record of how every result in this monograph was produced and how it can be reproduced from scratch. We describe the full rebuild script, the hermetic smoke path that exercises the core algorithm with zero external data, the integrity manifests for model artifacts and the score archive, the `kga` package install and test loop, the CI workflow, and the 10 executed notebooks. Every claim here is verified against the real files in the repository.

A.1 Repository layout

The repository root is `AutoML_Flagship_V8/`. The files most relevant to reproducing the K-Bound paper are:

Path	Contents
`kga/`	Importable KGA package (pure `numpy`/`scipy`)
`src/scripts/kbound/`	Per-experiment scripts
`experiments/kbound/theory_validation/`	Theorem validators
`experiments/elara_u/score_archive/`	123-task score archive
`experiments/kbound/results/`	Experiment outputs and manifests
`docs/research/kbound/`	Paper source, figures, kbound.tex
`models/MANIFEST.json`	SHA-256 manifest for 5 model artifacts
`data/MANIFEST.json`	SHA-256 manifest for 123 score-archive entries
`scripts/rebuild_kbound.sh`	One-command full rebuild
`scripts/smoke_kbound.sh`	Hermetic CPU smoke ($< 60\,\mathrm{s}$)
`.github/workflows/kbound-ci.yml`	CI workflow (6 jobs)
`notebooks/`	10 executed K-Bound notebooks (00–09)

A.2 Full reproduction path

The complete paper-grade rebuild is a single shell script:

```
bash scripts/rebuild_kbound.sh                  # CPU + compile
KBOUND_GPU=1 bash scripts/rebuild_kbound.sh     # + CIFAR-10 / Tent on GPU
KBOUND_SKIP_DEPS=1 bash scripts/rebuild_kbound.sh
PYTHON=.venv/bin/python bash scripts/rebuild_kbound.sh
```

The script (`scripts/rebuild_kbound.sh`) executes four numbered stages from the repository root.

Stage [1/4] Dependencies. Unless `KBOUND_SKIP_DEPS=1` is set, the script installs `docs/research/kbound/requ` via `pip install -q`. This covers `numpy`, `scipy`, `scikit-learn`, `matplotlib`, and `pandas`; the `pdflatex` binary must be present separately.

Stage [2/4] Core experiments (CPU). Five experiment scripts run in order under the Python interpreter (default `python3`, overridable via `PYTHON=`):

1. `src/scripts/kbound/knowability_experiment.py` — the 123-task anomaly-routing trichotomy and ablation;
2. `src/scripts/kbound/kbound_harmful_regime.py` — the reliability-fusion harmful-regime evaluation;
3. `src/scripts/kbound/mixed_regime_experiment.py` — the controlled mixed-regime (clean/detectable/cove 369 instances);
4. `src/scripts/kbound/kbound_full_experiments.py rigor` — multi-seed statistical rigor pass;
5. `src/scripts/kbound/kbound_full_experiments.py ablation` — evidence-component and α-level ablations;
6. `src/scripts/kbound/kbound_full_experiments.py regression` — regression covariate-shift certificate;
7. `src/scripts/kbound/kbound_full_experiments.py witness` — clean non-identifiability witness.

All scripts consume the 123-task score archive at `experiments/elara_u/score_archive/` and write results to `experiments/kbound/results/`. Figures are written to `docs/research/kbound/figures/`.

Stage [3/4] GPU experiment (optional). When `KBOUND_GPU=1`, the script additionally runs:

```
src/scripts/kbound/cifar_tent_mps.py
```

This requires PyTorch with Metal Performance Shaders (Apple M5 / MPS) or a CUDA-capable GPU. It runs the CIFAR-10-C corruption grid with Tent adaptation and writes the decisive stress results to `experiments/kbound/cifar/`. The GPU stage is skipped silently when the flag is absent; all anomaly-routing and theory results are CPU-only.

Stage [4/4] Compile PDF. Two `pdflatex` passes compile `docs/research/kbound/kbound.tex`; the resulting PDF is copied to `docs/research/kbound/K-Bound_paper.pdf`.

A.3 Hermetic smoke path

The hermetic smoke is the fastest signal that the K-Bound algorithm works on any machine with no downloaded data:

```
bash scripts/smoke_kbound.sh
```

The wrapper (`scripts/smoke_kbound.sh`) resolves the repository root as the parent of its own directory, sets `PYTHON` (default `python3`, overridable via `PYTHON=python3.11`), and runs:

```
python src/scripts/kbound/smoke_trichotomy.py
```

`smoke_trichotomy.py` proceeds in two phases.

Phase 1: synthetic archive generation. The companion script `make_synth_archive.py` writes a tiny, deterministic (fixed seed) score archive to `experiments/kbound/_smoke/score_archive/`, using the exact five-key `.npz` schema (`Sval`, `yval`, `Stest`, `ytest`, `det_names`, `val_auc`, `domain`) of the real 123-task archive. Three synthetic tasks are created:

- `synth_helpful` — f_0 degrades on test while the ensemble f_a is far more accurate; true benefit clearly positive;
- `synth_harmful` — f_0 stays excellent while f_a is corrupted; true benefit clearly negative;
- `synth_unknowable` — f_0 and f_a are statistically tied; benefit ≈ 0.

Phase 2: real `kga` pipeline. `smoke_trichotomy.py` loads the synthetic archive and runs the *real* `kga` package: `compute_evidence` (label-free Z), `KGA.certify` (empirical-Bernstein certificate $\widehat{\Delta} \pm \varepsilon$), and `KGA.decide` (trichotomy). It then asserts:

- `synth_helpful` $\Rightarrow$ `ADAPT`;

- `synth_harmful` ⇒ `FREEZE`;
- `synth_unknowable` ⇒ `ABSTAIN`.

The separations are large relative to the certificate radius, so the result is deterministic, not chance. On success the script writes `experiments/kbound/_smoke/smoke_result.json` and prints `SMOKE PASS` with a one-line metrics summary. On any failure it exits non-zero.

The same logic is covered by `pytest tests/test_smoke_trichotomy.py` and `tests/test_data_manifest.py` both are pure `numpy`/`kga` suites with no external dependencies and complete in a few seconds. The hermetic smoke schema identity with the real archive is additionally checked in `test_smoke_trichotomy.py::test_synth_archive_schema_matches_real`.

A.4 Integrity manifests

A.4.1 Model artifact manifest (`models/MANIFEST.json`)

Five serialised model artifacts are registered in `models/MANIFEST.json`, each with its repository-relative path, a real SHA-256 digest, size in bytes, and mtime:

Artifact name	Path	SHA-256 (first 16 hex)
`behavior__behavior_autoencoder`	`models/behavior/`	`d044f4440633ed4d...`
`behavior__behavior_lof`	`models/behavior/`	`df74fb53d40dd272...`
`cyber__supervised__cyber_model`	`models/cyber/supervised/`	`e89c5d1b742c1fee...`
`fraud__supervised__fraud_model`	`models/fraud/supervised/`	`331df4269f51e34c...`
`fusion__fusion_meta_model`	`models/fusion/`	`c1c8033c53bfc1f9...`

The manifest is produced by:

```
PYTHONPATH=src python -m uais.registry.build_manifest
```

Loading any artifact through `src/uais/registry/ModelRegistry` fails closed (`IntegrityError`) on a SHA-256 mismatch. The manifest and loader are covered by `tests/test_model_registry.py` (20 tests).

A.4.2 Score-archive manifest (`data/MANIFEST.json`)

`data/MANIFEST.json` records the SHA-256, byte size, and mtime of every `.npz` file under `experiments/elara_u/score_archive/`, covering all 123 real score-archive entries. It is produced by:

```
python -m uais.data.manifest build
```

Verification is:

```
python -m uais.data.manifest verify
# exit 0 = all match; exit 1 = drift (missing/changed); prints JSON report
```

The manifest is format-versioned (`"version": "kbound-data-manifest/1"`) and handles three failure modes: a missing recorded file, a size mismatch, and a content (SHA-256) change even at identical size. These are exercised by `tests/test_data_manifest.py`.

A.5 The `kga` package: install and test

The `kga` package lives at the repository root (top-level package, not under `src/`):

```
# Minimal install: put repo root on PYTHONPATH
export PYTHONPATH="$PWD:$PWD/src"
pip install numpy scipy pytest

# Verify
python -c "import kga; print(kga.__version__)"  # -> kga 0.1.0
```

The package is pure `numpy`/`scipy`, deterministic, and torch-free. It exposes `KGA`, `Evidence`, `Certificate`, and `Decision` (a `str` enum). The CLI entry point:

```
python -m kga decide --calib calib.npy --test test.npy --alpha 0.1
```

Test suite. The `kga` unit tests live in `tests/test_kga_package.py`. To run the complete new-package suite (88 tests, $< 2\,$s):

```
PYTHONPATH="$PWD:$PWD/src" pytest \
  tests/test_kga_package.py \
  tests/test_model_registry.py \
  tests/test_training_harness.py \
  tests/test_data_manifest.py \
  tests/test_smoke_trichotomy.py \
  --cov=kga --cov=src/uais --cov-report=term-missing
```

The `kga` tests are fixed-seed and check the following behavioural contract: (a) identical distributions $\Rightarrow$ `ABSTAIN`; (b) the certificate radius shrinks as n grows; (c) trichotomy boundaries are hit exactly; (d) the non-identifiability witness (identical Z, opposite truth) $\Rightarrow$ `ABSTAIN`; and (e) the empirical false-adapt rate remains $\leq \alpha$ over at least 2 000 synthetic trials.

Theorem validators. Each theorem in the paper has a standalone numerical validator under `experiments/kbound/theory_validation/`:

Script	Validates
`val_thm1_lecam.py`	Thm. 4.1 — Le Cam two-point minimax
`val_thm2_regret.py`	Thm. 4.3 — regret identity
`val_thm3_evalue.py`	Thm. 5.2 — anytime e-value
`val_thm5_multiclass.py`	Thm. 5.4 — multiclass extension

All validators are pure `numpy`/`scipy` with no labels or GPU, and have been re-run from a clean environment: all four pass.

A.6 CI workflow (`.github/workflows/kbound-ci.yml`)

The `kbound-ci` GitHub Actions workflow (`kbound-ci.yml`) runs six jobs on `ubuntu-latest` / Python 3.11, triggered on pushes and pull requests to `main`/`master`. A `concurrency.cancel-in-progress` key drops superseded runs.

Job	What it does
`lint`	(a) Repo-wide critical-error gate: `ruff check -select E9,F63,F7,F82 .`; (b) full ruff lint on the maintained A+ surfaces (`kga/`, `src/orchestration/`, `src/uais/registry/`, `src/uais/training/`, the K-Bound scripts, and the test suite); (c) ruff format check on the same surfaces.
`kga-package`	Installs `numpy scipy pytest`; imports `kga` and prints the version; runs `pytest tests/test_kga_package.py`.
`unit-tests`	Installs the full minimal dependency set; runs the five new package test modules with `--cov=kga --cov=src/uais`; uploads `coverage.xml` as an artifact.
`hermetic-smoke`	Installs only `numpy scipy`; runs `bash scripts/smoke_kbound.sh`; asserts `SMOKE PASS`.
`theorem-validators`	Glob-runs every `experiments/kbound/theory_validation/val_thm*.py`; fails if any exits non-zero.
`orchestration-import`	Imports `src.orchestration` *without* Prefect installed and checks that the six flows are registered (`PREFECT=False`).

Every job sets `PYTHONPATH=${{ github.workspace }}:${{ github.workspace }}/src` so both the top-level `kga` package and the `src/`-rooted `uais` package are importable without `pip install -e`.

A.7 Executed notebooks (00–09)

The `notebooks/` directory contains 10 notebooks that reproduce the K-Bound paper interactively. Each notebook auto-locates the repository root and loads only real result artifacts; outputs are saved in the files so they are readable without re-running. Table A.1 lists the notebooks in order.

Table A.1: The 10 executed K-Bound notebooks.

#	Filename	Contents
00	`00_KBound_Reproduction.ipynb`	One-page overview, live trichotomy on a synthetic example, all result figures, and the top-level reproduction commands.
01	`01_Problem_and_Theory.ipynb`	Problem formulation, the adapt/freeze/abstain trichotomy, live runs of the four theorem validators (Thms 1-5 / 3b), and the architecture diagram.
02	`02_Knowability_Trichotomy.ipynb`	Live 123-task anomaly-routing suite: evidence Z empirical-Bernstein certificate, decisions, adapt precision false-adapt rate, and component ablations.
03	`03_Harmful_Mixed_Rigor.ipynb`	Harmful-fusion regime, controlled mixed-regime (clean/detectable/covert), and 8-seed paired t-test statistical rigor.
04	`04_Regression_and_Witness.ipynb`	Regression covariate-shift certificate and the clean non-identifiability witness (live Gaussian/flipped-label construction).
05	`05_TTA_CIFAR_and_Online.ipynb`	CIFAR-10-C pre-registered stress grid, decisive CIFAR+Tent experiment, and online continual-Tent collapse analysis.
06	`06_Evidence_and_Drift.ipynb`	The label-free evidence Z computed live on synthetic and real tasks: KS drift, ESS, disagreement, and their correlation with true benefit.
07	`07_Certificate_and_Calibration.ipynb`	Conformal, empirical-Bernstein, and e-value certificates α–coverage calibration curves; certificate width vs. batch size.
08	`08_ELARA_Multimodal_Instantiation.ipynb`	ELARA/RGA worked instantiation: MVTec-3D domain D23, T1–T9 gating, and the full GDR map.
09	`09_Conclusions_and_Reproducibility.ipynb`	Cross-experiment regret-to-oracle summary, reproduction commands, honest scope statement, and open items (Conjecture 5.1, full ImageNet-C).

Legacy notebooks (25 superseded ELARA-era EDA notebooks) live under `notebooks/legacy_elara/` and are not part of the K-Bound paper.

A.8 Environment notes

Python version. Python 3.11 is the supported CI interpreter; the code targets Python 3.9+. CPU-only K-Bound experiments need only `numpy`, `scipy`, `scikit-learn`, `matplotlib`, and `pandas`. The GPU (Tent/CIFAR-10-C) runs additionally require `torch 2.5.1` and `torchvision 0.20.1`.

Apple M5 / MPS. The decisive CIFAR-10-C and online continual-Tent experiments were run on an Apple M-series chip using Metal Performance Shaders (`device="mps"` in `cifar_tent_mps.py`). The same scripts run on CUDA without modification.

Determinism. All CPU experiments use `SEED=0`. `src/uais/training/set_seed` seeds `PYTHONHASHSEED`, `random`, NumPy, and (when importable) PyTorch with `torch.backends.cudnn.deterministic=True`. The `kga` package itself has no global RNG state.

Legacy lint debt. Approximately 1 973 `ruff` findings exist in pre-existing ELARA-era code. Mass-reformatting was intentionally not performed (high churn, out of scope); the CI enforces the full rule set on the maintained A+ surfaces and a critical-error gate across the whole repository. A clean `ruff check kga` passes with no findings.

A.9 How to reproduce: checklist

1. **Clone** the repository and `cd AutoML_Flagship_V8`.
2. **Install minimal deps**: `pip install numpy scipy scikit-learn pandas matplotlib pytest ruff`.
3. **Hermetic smoke** (no data required, < 60 s): `bash scripts/smoke_kbound.sh` ⇒ `SMOKE PASS`.
4. **Unit tests**: `PYTHONPATH="$PWD:$PWD/src" pytest tests/test_kga_package.py -q` ⇒ all pass.
5. **Theorem validators**: `for v in val_thm1_lecam val_thm2_regret val_thm3_evalue val_thm5_multiclass; do python experiments/kbound/theory_validation/$v.py; done` ⇒ 4/4 pass.
6. **Full rebuild** (requires score archive): `bash scripts/rebuild_kbound.sh` ⇒ `DONE -> docs/research/kbound/K-Bound_paper.pdf`.
7. **GPU rebuild** (requires score archive + PyTorch + MPS/CUDA): `KBOUND_GPU=1 bash scripts/rebuild_kbound.sh`.
8. **Manifest verification**: `python -m uais.data.manifest verify` ⇒ all 123 entries match.

Appendix B

Complete Result Tables

This appendix contains the full numerical tables for all primary experiments reported in Chapter 7 (anomaly-routing benchmark) and Chapter 8 (TTA and CIFAR-10 experiments). Each table identifies its source JSON file and field path. Numbers are reproduced verbatim from the JSON files without rounding beyond what floating-point display requires. All tables are self-contained: column headings, units, and source fields are given in each caption.

B.1 Per-Seed Rigor Table (Mixed Regime, 8 Seeds)

Source file: `experiments/kbound/results/rigor_multiseed.json`. The file stores aggregate mean and standard deviation across seeds (field `mean_std`); per-seed individual AUROC values are not stored in the JSON output (the seeded runs are summarised by the aggregate statistics). The table below therefore presents the complete set of aggregate statistics as stored, together with the paired t-test results for both comparisons.

Table B.1: Multi-seed aggregate statistics for the mixed regime (8 seeds, split-conformal, $\alpha = 0.1$). Source: `rigor_multiseed.json`, fields `mean_std` and `paired_ttest_*`. Mean and standard deviation are across all 8 seeds.

Metric	Policy / Description	Mean	Std
Mean test AUROC	Always-adapt	0.69698	0.00008
Mean test AUROC	Always-freeze	0.58419	0.00147
Mean test AUROC	KGA	0.68598	0.00334
Mean test AUROC	Oracle	0.71803	0.00064
Safety: adapt precision	$\mathbb{P}(B > 0 \mid \text{ADAPT})$	0.93577	0.01014
Safety: false-adapt rate	$\mathbb{P}(B < 0 \mid \text{ADAPT})$	0.06423	0.01014
Coverage (fraction acted)	adapt + freeze decisions	0.37263	0.01746
Paired t-tests (degrees of freedom = n_seeds $- 1 = 7$):			
KGA vs always-freeze	$t = 62.236$, $p = 7.262 \times 10^{-11}$	KGA significantly higher	
KGA vs always-adapt	$t = -8.760$, $p = 5.084 \times 10^{-5}$	KGA significantly lower	

Notes. The paired t-test values are sourced from fields `paired_ttest_KBound_vs_always_freeze` ($t = 62.2358$, $p = 7.2620 \times 10^{-11}$) and `paired_ttest_KBound_vs_always_adapt` ($t = -8.7599$, $p = 5.0843 \times 10^{-5}$). Both signs are reported honestly: KGA reliably beats freezing and reliably trails always-adapt in this mild-harm regime.

B.2 Full Ablation Grid

Source file: `experiments/kbound/results/ablations.json`. The table below gives every entry from the three sub-experiments: evidence-drop, α sweep, and batch-size sweep.

B.2.1 Evidence-Drop Ablation (Full Grid)

Table B.2: Evidence-drop ablation: complete grid. Each row drops one feature of Z and reports regret, false-adapt rate, and coverage on the mixed-regime benchmark. Baseline uses all nine features. Source: `ablations.json`, fields `baseline` and `evidence_drop`.

Feature dropped	Regret	False-adapt	Coverage
Baseline (all 9 features)	0.036554	0.065041	—
`val_max`	0.035039	0.064000	0.360434
`val_gap`	0.037203	0.066667	0.336043
`val_mean`	0.042568	0.070796	0.319783
`ks_mean`	0.036121	0.065574	0.352304
`ks_max`	0.037279	0.065574	0.352304
`disagree`	**0.094045**	**0.250000**	**0.140921**
`val_std`	0.036518	0.074380	0.344173
`anomaly_rate`	0.036615	0.065041	0.349593
`log_ntest`	0.039264	0.067797	0.336043

B.2.2 Alpha Sweep (Full Grid)

B.2.3 Batch-Size Sweep (Full Grid)

B.3 Regression Covariate-Shift Decision Summary

Source file: `experiments/kbound/results/regression_covariate.json`.

Table B.3: α sweep: complete grid of conformal level, band width ε, coverage, and false-adapt rate. Source: `ablations.json`, field `alpha_sweep`.

α	ε	Coverage	False-adapt	Source key
0.01	0.323988	0.113821	0.000000	"0.01"
0.05	0.191235	0.262873	0.073684	"0.05"
0.10	0.139075	0.349593	0.065041	"0.1"
0.20	0.091047	0.487805	0.072289	"0.2"

Table B.4: Batch-size sweep: complete grid of test-batch size, certificate width ε, and coverage. Source: `ablations.json`, field `batchsize_sweep`.

Batch size	ε	Coverage	Source key
16	0.443381	0.020325	"16"
32	0.354927	0.035599	"32"
64	0.277723	0.198830	"64"
128	0.255357	0.209040	"128"
256	0.182329	0.261111	"256"

B.4 Mixed-Regime Full Metrics

Source file: `experiments/kbound/results/mixed_regime_results.json`.

B.5 Harmful-Fusion Regime Full Metrics

Source file: `experiments/kbound/results/kbound_harmful_results.json`.

Note on the adapt decision. The single ADAPT decision issued by KGA (`counts.ADAPT` = 1) was on a task where $B < 0$ (field `adapt_precision_B>0` = 0.0, field `false_adapt_rate_B<0` = 1.0). This single error does not affect the AUROC or regret materially because one task out of 123 has negligible weight; it reflects the irreducible probabilistic tolerance of the conformal guarantee at $\alpha = 0.1$.

B.6 Clean Non-identifiability Witness: Full Metrics

Source file: `experiments/kbound/results/witness_clean.json`.

B.7 Knowability Trichotomy Suite: Full Metrics

Source file: `experiments/kbound/results/knowability_results.json`.

Table B.5: Regression covariate-shift experiment: full metrics. Source: `regression_covariate.json`. Benefit $B = \text{MSE}(f_0) - \text{MSE}(f_a)$; positive B means adaptation hurts (higher MSE for f_a). No adapt decisions were issued; adapt precision and false-adapt rate are undefined (`null` in the JSON).

Field	Value
`n_instances`	72
`eps`	0.218147
`mean_B(mse0-msea)`	−0.086346
`harmful_rate_B<0`	0.791667
Decision counts	
`decision_counts.ADAPT`	0
`decision_counts.FREEZE`	11
`decision_counts.ABSTAIN`	61
Policy mean MSE	
`mean_MSE.always_adapt(IW)`	0.337601
`mean_MSE.always_freeze`	0.251255
`mean_MSE.K_Bound`	0.251255
`mean_MSE.oracle`	0.251188
`adapt_precision_B>0`	null (no adapt decisions)
`false_adapt_rate_B<0`	null (no adapt decisions)

Note on naive rule. The JSON also records a naive validation rule (`mean_auc_naive_val_rule` = 0.7786, `regret_naive_vs_oracle` = 0.0042, false-adapt = 0.1798). The naive rule achieves lower regret than KGA in this helpful-dominated regime but has a much higher false-adapt rate (17.98% vs 10%), illustrating the safety-performance tradeoff.

B.8 All 65 CIFAR-10-C Cells

The following table contains per-cell results for all 13 official CIFAR-10-C corruptions at 5 severity levels (65 cells total), comparing frozen, online Tent, EATA-style, SAR-style, and KGA on a ResNet-18 (0.874 clean accuracy). The table is generated directly from the experiment output; see Appendix A for the script that produces it.

Table B.10: All 65 CIFAR-10-C cells (13 official corruptions × severities 1–5), ResNet-18, N=800 balanced held-out eval per cell. Online TTA accuracy; KGA wraps Tent. Mean row matches Table B.8.

corruption	sev	frozen	Tent	EATA	SAR	KGA	oracle
brightness	1	0.863	0.858	0.856	0.859	0.858	0.865
brightness	2	0.844	0.864	0.864	0.865	0.864	0.868
brightness	3	0.804	0.837	0.837	0.839	0.837	0.839
brightness	4	0.731	0.818	0.818	0.815	0.819	0.816

corruption	sev	frozen	Tent	EATA	SAR	KGA	oracle
brightness	5	0.690	0.791	0.794	0.795	0.791	0.799
contrast	1	0.719	0.831	0.831	0.835	0.831	0.831
contrast	2	0.559	0.823	0.821	0.822	0.820	0.820
contrast	3	0.325	0.790	0.793	0.793	0.790	0.791
contrast	4	0.223	0.728	0.729	0.730	0.726	0.726
contrast	5	0.160	0.611	0.607	0.609	0.614	0.601
defocus_blur	1	0.264	0.683	0.680	0.681	0.682	0.676
defocus_blur	2	0.207	0.543	0.544	0.544	0.540	0.540
defocus_blur	3	0.204	0.389	0.386	0.398	0.389	0.385
defocus_blur	4	0.176	0.300	0.300	0.303	0.300	0.305
defocus_blur	5	0.183	0.265	0.264	0.261	0.261	0.275
elastic_transform	1	0.244	0.528	0.528	0.530	0.528	0.525
elastic_transform	2	0.179	0.436	0.440	0.446	0.436	0.452
elastic_transform	3	0.168	0.401	0.401	0.405	0.401	0.400
elastic_transform	4	0.144	0.407	0.407	0.413	0.407	0.410
elastic_transform	5	0.151	0.347	0.346	0.353	0.351	0.351
frost	1	0.725	0.804	0.805	0.806	0.804	0.796
frost	2	0.588	0.760	0.762	0.761	0.761	0.762
frost	3	0.546	0.685	0.685	0.686	0.684	0.697
frost	4	0.492	0.705	0.705	0.704	0.705	0.706
frost	5	0.436	0.640	0.642	0.644	0.640	0.659
gaussian_noise	1	0.487	0.741	0.740	0.741	0.739	0.739
gaussian_noise	2	0.275	0.699	0.699	0.700	0.700	0.701
gaussian_noise	3	0.144	0.594	0.595	0.595	0.593	0.592
gaussian_noise	4	0.100	0.435	0.435	0.432	0.436	0.441
gaussian_noise	5	0.089	0.305	0.304	0.301	0.305	0.301
impulse_noise	1	0.615	0.769	0.770	0.769	0.769	0.775
impulse_noise	2	0.370	0.682	0.682	0.682	0.682	0.688
impulse_noise	3	0.197	0.605	0.605	0.604	0.608	0.623
impulse_noise	4	0.110	0.475	0.479	0.485	0.476	0.484
impulse_noise	5	0.082	0.335	0.335	0.335	0.335	0.331
jpeg_compression	1	0.676	0.730	0.730	0.731	0.730	0.735
jpeg_compression	2	0.639	0.660	0.659	0.661	0.662	0.663
jpeg_compression	3	0.608	0.655	0.654	0.656	0.654	0.658
jpeg_compression	4	0.509	0.573	0.570	0.570	0.573	0.567
jpeg_compression	5	0.460	0.501	0.501	0.501	0.501	0.500
motion_blur	1	0.495	0.782	0.784	0.785	0.784	0.783
motion_blur	2	0.365	0.666	0.665	0.664	0.666	0.675
motion_blur	3	0.279	0.575	0.574	0.575	0.574	0.590
motion_blur	4	0.255	0.490	0.491	0.489	0.491	0.502
motion_blur	5	0.225	0.456	0.450	0.446	0.456	0.476
pixelate	1	0.429	0.719	0.720	0.721	0.716	0.720
pixelate	2	0.373	0.651	0.651	0.654	0.650	0.654
pixelate	3	0.306	0.570	0.574	0.576	0.573	0.571
pixelate	4	0.267	0.487	0.485	0.487	0.489	0.486
pixelate	5	0.289	0.405	0.404	0.406	0.405	0.410

corruption	sev	frozen	Tent	EATA	SAR	KGA	oracle
shot_noise	1	0.492	0.779	0.776	0.779	0.777	0.784
shot_noise	2	0.254	0.654	0.655	0.659	0.654	0.669
shot_noise	3	0.155	0.536	0.537	0.549	0.534	0.545
shot_noise	4	0.125	0.400	0.398	0.398	0.398	0.404
shot_noise	5	0.120	0.303	0.305	0.306	0.300	0.300
snow	1	0.770	0.782	0.781	0.779	0.782	0.781
snow	2	0.578	0.686	0.685	0.685	0.685	0.690
snow	3	0.661	0.716	0.716	0.711	0.718	0.710
snow	4	0.574	0.683	0.681	0.679	0.683	0.680
snow	5	0.459	0.597	0.596	0.597	0.597	0.598
zoom_blur	1	0.760	0.877	0.879	0.876	0.876	0.879
zoom_blur	2	0.708	0.847	0.847	0.849	0.847	0.853
zoom_blur	3	0.674	0.830	0.829	0.828	0.829	0.826
zoom_blur	4	0.635	0.809	0.810	0.810	0.807	0.814
zoom_blur	5	0.557	0.779	0.776	0.778	0.780	0.785
mean		**0.412**	**0.626**	**0.626**	**0.627**	**0.626**	**0.629**

Table B.6: Mixed-regime benchmark: complete metrics from a single run ($n = 369$). Source: `mixed_regime_results.json`.

Field	Value
`n_instances`	369
`eps`	0.131487
Decision distribution — clean condition (true $\Delta = -0.041$)	
`decision_distribution_by_condition.clean.ADAPT`	8
`decision_distribution_by_condition.clean.FREEZE`	5
`decision_distribution_by_condition.clean.ABSTAIN`	110
Decision distribution — detectable-failure condition (true $\Delta = +0.193$)	
`decision_distribution_by_condition.detectable.ADAPT`	71
`decision_distribution_by_condition.detectable.FREEZE`	3
`decision_distribution_by_condition.detectable.ABSTAIN`	49
Decision distribution — covert-failure condition (true $\Delta = +0.181$)	
`decision_distribution_by_condition.covert.ADAPT`	59
`decision_distribution_by_condition.covert.FREEZE`	3
`decision_distribution_by_condition.covert.ABSTAIN`	61
True mean benefit by condition	
`true_mean_benefit_B_by_condition.clean`	−0.040876
`true_mean_benefit_B_by_condition.detectable`	+0.192928
`true_mean_benefit_B_by_condition.covert`	+0.180573
Safety	
`safety.adapt_precision_B>0`	0.934783
`safety.false_adapt_rate_B<0`	0.065217
Policy mean AUROC	
`mean_auc_policies.always_adapt`	0.697111
`mean_auc_policies.always_freeze`	0.586236
`mean_auc_policies.K_Bound`	0.689852
`mean_auc_policies.oracle`	0.718913
Regret vs oracle	
`regret_vs_oracle.always_adapt`	0.021801
`regret_vs_oracle.always_freeze`	0.132677
`regret_vs_oracle.K_Bound`	0.029061
Non-identifiability: clean vs covert boundary	
`nonidentifiability_clean_vs_covert.mean_Z_L2_distance_clean_to_nearest_covert`	1.353908
`nonidentifiability_clean_vs_covert.mean_B_clean`	−0.040876
`nonidentifiability_clean_vs_covert.mean_B_covert`	+0.180573
`nonidentifiability_clean_vs_covert.mean_Z_L2_distance_clean_to_nearest_detectable`	3.099503

Table B.7: Harmful-fusion regime: complete metrics ($n = 123$ tasks, candidate = `elara_fuse`, base rate harmful = 79.7%). Source: `kbound_harmful_results.json`.

Field	Value
`setup`	`fa=elara_fuse (harmful ~80%), f0=auto_select`
`n_tasks`	123
`eps`	0.039848
`base_rate_harmful_B<0`	0.796748
Decision counts	
`counts.ADAPT`	1
`counts.FREEZE`	13
`counts.ABSTAIN`	109
Safety	
`adapt_precision_B>0`	0.000
`false_adapt_rate_B<0`	1.000
`freeze_correct_B<=0`	1.000
Policy mean AUROC	
`mean_auc.always_adapt(elara_fuse)`	0.728428
`mean_auc.always_freeze(auto_select)`	0.748058
`mean_auc.K-Bound_trichotomy`	0.747946
`mean_auc.oracle`	0.749863
Regret vs oracle	
`regret_vs_oracle.always_adapt`	0.021435
`regret_vs_oracle.always_freeze`	0.001806
`regret_vs_oracle.K-Bound`	0.001918
Regret ratio (always-adapt / KGA)	$0.021435/0.001918 \approx 11.2$

Table B.8: Clean non-identifiability witness: complete metrics ($M = 300$ instances). Source: `witness_clean.json`. All five KS p-values exceed 0.05; KGA abstains on 100% of instances.

Field	Value
`M`	300
`mean_abs_true_benefit`	1.0
`world0_mean_B`	-1.0
`world1_mean_B`	$+1.0$
KS p-values (Z-feature law, World 0 vs World 1)	
`Z_indistinguishable_KS_pvalues.frac_pos`	0.9647
`Z_indistinguishable_KS_pvalues.std`	0.9418
`Z_indistinguishable_KS_pvalues.ks_vs_ref`	0.7988
`Z_indistinguishable_KS_pvalues.mean_f0`	0.9647
`Z_indistinguishable_KS_pvalues.skew`	0.1648
`all_Z_features_p>0.05`	`true`
`abstain_rate`	1.0
`eps`	1.395233
`mean_|^Bhat|`	0.263773
`forced_commit_regret_vs_oracle`	0.513333
`abstain_regret_vs_oracle(default_freeze)`	0.466667

Table B.9: Knowability trichotomy suite (clean, 123 tasks): complete metrics. Source: `knowability_results.json`.

Field	Value
`metrics.n_tasks`	123
`metrics.eps_conformal`	0.049201
`metrics.alpha`	0.10
`metrics.coverage`	0.170732
`metrics.n_adapt`	20
`metrics.n_freeze`	1
`metrics.n_abstain`	102
`metrics.adapt_precision_(B>0|ADAPT)`	0.90
`metrics.false_adapt_rate_(B<0|ADAPT)`	0.10
`metrics.freeze_precision_(B<=0|FREEZE)`	0.00
`metrics.abstain_mean_|B|`	0.021504
`metrics.nonabstain_mean_|B|`	0.132367
`policy.mean_auc_trichotomy`	0.766119
`policy.mean_auc_always_adapt`	0.777145
`policy.mean_auc_always_freeze`	0.748056
`policy.mean_auc_oracle`	0.782816
`policy.regret_trichotomy_vs_oracle`	0.016697
`policy.regret_always_adapt_vs_oracle`	0.005671
`policy.regret_naive_vs_oracle`	0.004220
`policy.mean_auc_naive_val_rule`	0.778596
`policy.false_adapt_rate_naive_(B<0|naive ADAPT)`	0.179775
Non-identifiability witness (in-distribution)	
`nonidentifiability_witness.z_distance`	0.798586
`nonidentifiability_witness.task_i`	`cv_CIFAR10_7`
`nonidentifiability_witness.B_i`	+0.038255
`nonidentifiability_witness.task_j`	`cv_SVHN_3`
`nonidentifiability_witness.B_j`	−0.017858
Evidence feature names (9 features)	
	`val_max`, `val_gap`, `val_mean`
	`ks_mean`, `ks_max`, `disagree`
	`val_stack_margin`, `anomaly_rate`, `log_ntest`

Appendix C

Notation and Glossary

This appendix collects all mathematical notation used in the monograph in a single reference table, followed by a glossary of key technical terms. Symbol conventions are drawn directly from the paper [28] and are consistent throughout all chapters.

C.1 Notation table

Table C.1: Mathematical notation used throughout the monograph.

Symbol	Meaning
$P_S(X, Y)$	Source distribution. Labels Y are available at training and validation time.
$P_T(X, Y)$	Target distribution. At deployment only $P_T(X)$ is observable (no target labels).
f_0	Frozen (baseline) predictor. Deployed without modification unless ADAPT fires.
f_a	Adapted (candidate) predictor. The output of an adaptation mechanism (Tent, EATA, SAR, or any other) that is being evaluated.
ℓ	Per-sample loss function, e.g. 0/1 loss or cross-entropy.
$R_T(f)$	Target risk of predictor f: $R_T(f) = \mathbb{E}_{P_T}[\ell(f(X), Y)]$.
Δ	Adaptation benefit: $\Delta = R_T(f_0) - R_T(f_a)$. Positive means adapting reduces risk (helps); negative means it hurts.
$\Delta(z)$	Conditional benefit at evidence value z: $\Delta(z) = \mathbb{E}[\ell(f_0(X), Y) - \ell(f_a(X), Y) \mid Z = z]$.

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Symbol	Meaning
$\widehat{\Delta}$	Label-free (point) estimator of Δ or $\Delta(z)$, built without target labels.
ε	Certificate radius at level α: with probability $\geq 1-\alpha$, $\lvert\Delta - \widehat{\Delta}\rvert \leq \varepsilon$ (or $\Delta \geq \widehat{\Delta} - \varepsilon$ for the one-sided form).
α	Miscoverage level in $(0, 1)$; bounds the false-adapt probability $\mathbb{P}(\textsc{adapt}$ and $\Delta \leq 0) \leq \alpha$.
Z	Label-free observable evidence: $Z = \phi(X_{1:m}, f_0, f_a)$, a vector of statistics computable from unlabelled test scores and fixed predictors.
$\phi(\cdot)$	Evidence map; extracts Z from the unlabelled batch and predictor pair.
$g(Z)$	Decision rule; maps evidence Z to one of $\{\textsc{adapt}, \textsc{freeze}, \textsc{abstain}\}$.
$g^\star$	Bayes-optimal gate: $g^\star(z) = \mathbf{1}[\Delta(z) > 0]$.
D	Observable disagreement region: $D = \{x : f_0(x) \neq f_a(x)\}$; the support of the sign-of-difference signal.
p_a	Probability mass of the target distribution on D: $p_a = P_T(X \in D)$.
p_0	Probability mass of the source distribution on D: $p_0 = P_S(X \in D)$.
$\mathrm{TV}(P, Q)$	Total variation distance between distributions P and Q: $\mathrm{TV}(P, Q) = \sup_A \lvert P(A) - Q(A)\rvert$.
$\mathrm{Law}(X \mid P)$	Distribution (law) of X under P.
$\mathbf{1}[E]$	Indicator of event E; equals 1 if E holds, 0 otherwise.
$\mathrm{sign}(t)$	Sign function: $\mathrm{sign}(t) = \mathbf{1}[t > 0]$ (as used in Theorem 4.3).
$\arg\max$, $\arg\min$	Arg-max and arg-min operators.
$\mathbb{R}$	The real numbers $(-\infty, +\infty)$.
$\mathbb{E}[\cdot]$	Expectation under the distribution clear from context.
$\mathbb{P}[\cdot]$	Probability under the distribution clear from context.
n	Sample size (number of paired benefits, calibration tasks, or stream steps depending on context).
m	Size of the unlabelled target batch.
R	Benefit range $R = b - a$ where the paired benefits $X_i \in [a, b]$.
$\widehat{V}$	Unbiased sample variance (denominator $n-1$) of the paired benefits X_i.
$\widehat{\mu}$	Empirical mean of the paired benefits; equals $\widehat{\Delta}$ in the batch estimator.

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Symbol	Meaning
E_t^+, E_t^-	Two one-sided betting e-processes (Theorem 5.2): E_t^+ tests $H_0^+ : \Delta \le 0$; E_t^- tests $H_0^- : \Delta \ge 0$.
λ_i, ν_i	Predictable bet sizes for E_t^+ and E_t^- respectively (aGRAPA truncated bets, formed from past observations).
`ks_mean`, `ks_max`	Mean and maximum per-detector Kolmogorov–Smirnov drift between calibration and test score marginals (fields of `Evidence`).
`disagree`	$1 - \bar{\rho}$: one minus the mean pairwise rank correlation of test scores across detectors (field of `Evidence`).
`ess_frac`	$\mathrm{ESS}/n_{\mathrm{test}}$: scale-free effective sample size of the Gaussian-ratio importance weights (field of `Evidence`).
δ	Detection threshold or evidence separation; in the quantitative non-identifiability theorem (Theorem 4.1) a location-shift parameter indexing $\mathrm{TV} = 2\Phi(\frac{1}{2}\sqrt{n}\,\delta) - 1$.
Regret-to-oracle	$R(g) - R(g^\star)$: excess risk of decision rule g over the per-condition oracle $g^\star = \mathbf{1}[\Delta > 0]$.

C.2 Glossary

Test-time adaptation (TTA). The practice of updating (adapting) a pre-trained model using only unlabelled data from a new target distribution at deployment time. Standard TTA mechanisms include entropy minimisation (Tent [56]), sample-filtered updates (EATA [39]), and sharpness-aware resets (SAR [40]).

Trichotomy. The three-way decision returned by KGA: ADAPT (adapting is certified beneficial), FREEZE (adapting is certified harmful), or ABSTAIN (the sign of the benefit is not knowable at level α). Contrasts with the binary adapt/freeze choice of all prior TTA methods.

Label-free evidence Z. The vector of statistics $Z = \phi(X_{1:m}, f_0, f_a)$ computable from unlabelled target data, the frozen model, and the candidate adaptation, without any target labels. In the `kga` package, Z is represented by the `Evidence` dataclass with fields `ks_mean`, `ks_max`, `disagree`, `entropy_shift`, `conf_shift`, `calib_conf`, `test_conf`, `ess`, and `ess_frac`.

Certificate. A finite-sample statement of the form $\Delta \ge \widehat{\Delta} - \varepsilon$ holding with probability at least $1 - \alpha$. Represented in the `kga` package as the `Certificate` dataclass with fields `delta_hat`, `epsilon`, `method`, `alpha`, `n`, and derived properties `lower` / `upper`.

False adapt. The event that KGA returns ADAPT but the true benefit is non-positive ($\Delta \le 0$); i.e. deploying f_a actually hurts. Theorem 5.1 guarantees $\mathbb{P}(\text{false adapt}) \le \alpha$.

False freeze. The symmetric event: KGA returns FREEZE but $\Delta > 0$ (keeping f_0 forgoes a genuine benefit). Also bounded at $\leq \alpha$ by the two-sided version of the certificate.

Abstention. The safe default action returned when the certificate brackets zero (i.e. $\widehat{\Delta} - \varepsilon \leq 0 \leq \widehat{\Delta} + \varepsilon$). KGA keeps the frozen model f_0 and logs the evidence for monitoring. Abstention is mandatory, not merely conservative, in the non-identifiable regime (Theorem 4.1, Corollary 5.1).

Empirical-Bernstein certificate. The batch certificate estimator of Maurer & Pontil (2009) [37], the default in KGA (`method='ebern'`). The radius is $\varepsilon = \sqrt{2\widehat{V}\ln(2/\alpha)/n} + 7R\ln(2/\alpha)/(3(n-1))$, which is strictly tighter than Hoeffding when the sample variance $\widehat{V} \ll R^2$.

Conformal certificate (split-conformal). A certificate whose radius is the $(1-\alpha)$-quantile of held-out calibration residuals $r_i = |\widehat{\Delta}_i - \Delta_i|$ [55, 1]. The estimator used in the main K-Bound cross-task experiments (`method='conformal'`).

E-value / anytime certificate. The anytime-valid certificate based on two one-sided testing-by-betting e-processes [54, 59]. Unlike the batch and conformal certificates, this certificate is valid simultaneously over all stream lengths with no multiplicity correction (`method='evalue'`).

Covariate shift. The setting where $P_T(X) \neq P_S(X)$ but $P_T(Y \mid X) = P_S(Y \mid X)$. Theorem 5.3 identifies this as a provably knowable regime under the support-overlap condition: large KS drift combined with adequate importance-weight ESS allows KGA to certify the correct action.

Disagreement region. The set $D = \{x : f_0(x) \neq f_a(x)\}$ of inputs on which the frozen and adapted predictors differ. Its probability mass under P_T governs the sign-of-difference identifiability (Theorem 5.4).

Score archive. The 123-task score cache at `experiments/elara_u/score_archive/`, the main empirical dataset for K-Bound. Each task is one `.npz` file with keys `Sval`, `yval`, `Stest`, `ytest`, `det_names`, `val_auc`, and `domain`. The archive covers ADBench tabular (47 tasks), credit-card fraud, UNSW-NB15 cyber, behaviour/HAR, NLP (Enron), time-series (NAB/SMD/TSB-AD), and image benchmarks.

Non-identifiability witness. A pair of target worlds (P_T^1, P_T^2) with $\mathrm{Law}(Z \mid P_T^1) = \mathrm{Law}(Z \mid P_T^2)$ and $\operatorname{sign}\Delta^1 = -\operatorname{sign}\Delta^2 \neq 0$, proving that no label-free rule can commit correctly in both worlds. The canonical witness (Theorem 4.1) uses $X \sim \mathcal{N}(0,1)$, $f_0 = \mathbf{1}[X > 0]$, $f_a = \mathbf{1}[X < 0]$, and flipped labels.

Regret-to-oracle. The excess risk $R(g) - R(g^\star)$ of a decision rule g over the per-condition oracle $g^\star = \mathbf{1}[\Delta > 0]$. By Theorem 4.3, regret decomposes as $\mathbb{E}[|\Delta(Z)|\mathbf{1}\{\operatorname{sign}\widehat{\Delta}(Z) \neq \operatorname{sign}\Delta(Z)\}]$; the experiments report this quantity as the primary performance metric alongside AUROC.

Risk-aligned evidence. Evidence Z is risk-aligned (Definition 3 of Chapter 3) if it identifies or brackets the sign of $\Delta(z)$; equivalently, Z suffices to determine whether adapting helps. Risk alignment is Assumption (iii) inherited by KGA from Theorem 5.1.

Mechanism-agnostic wrapper. The design pattern of KGA: the algorithm wraps any existing adaptation mechanism (Tent, EATA, SAR, or other) without modifying it. The adaptation mechanism supplies f_a; KGA decides whether to deploy it.

C.3 The exact benefit-sign frontier

We sharpen the central question from *"is target accuracy identifiable?"* (answered negatively in general by **Ben-David et al.** 2010 and **Garg et al.** 2022) to the strictly *comparative* object that gating actually needs: the *sign of the adaptation benefit*

$$\Delta \;:=\; R_T(f_0) - R_T(f_a), \qquad \text{adapt is correct} \iff \Delta > 0.$$

No prior estimator targets $\operatorname{sign}\Delta$: ATC, DoC, GDE, COT, AETTA and Agreement-on-the-Line all estimate an *accuracy level* of a *single fixed* model (**Garg et al.**, ATC; **Jiang et al.**, GDE; **Lu et al.**, COT; **Lee et al.**, AETTA; **Baek et al.**, Agreement-on-the-Line). We give an exact characterization of when $\operatorname{sign}\Delta$ is identifiable from label-free observables, prove the governing obstruction is a single scalar that is *necessarily* present, and show the field's heuristics are exactly its degenerate face.

Setup. Binary $Y \in \{0,1\}$, 0/1 loss; $f_0, f_a : \mathcal{X} \to \{0,1\}$ fixed. $D = \{x : f_0(x) \neq f_a(x)\}$ is the *disagreement region* (observable). Write $\eta_a(x) = P_T(Y = f_a(x) \mid x)$ and $\bar{a} = \mathbb{E}_{\mu|D}[\eta_a]$, where $\mu = P_T^X$ is the (observable) target marginal. The *observables* are μ, the maps f_0, f_a, the labeled source P_S, and hence any source-derived score $s(x)$ for $\eta_a(x)$ (e.g. a source-calibrated confidence of f_a). Define the *observable margin* and the *calibration drift on the decision region*

$$\begin{aligned} M \;&:=\; \mathbb{E}_{\mu|D}[\,s\,] - \tfrac{1}{2} \quad \text{(observable)},\\ \gamma \;&:=\; \mathbb{E}_{\mu|D}[\,\eta_a - s\,] \quad \text{(unobservable)}. \end{aligned}$$

Lemma C.1 (Reduction)**.** *$\Delta = 2\,\mu(D)\,(\bar{a} - \frac{1}{2})$ and $\bar{a} - \frac{1}{2} = M + \gamma$; hence, whenever $\mu(D) > 0$,*

$$\boxed{\operatorname{sign}\Delta = \operatorname{sign}(M + \gamma)}$$

Proof. Off D, $f_0 = f_a$ so the pointwise excess loss $\delta(x) = \ell(f_0, Y) - \ell(f_a, Y) = 0$. On D (binary, 0/1 loss) exactly one of f_0, f_a equals Y, so $\delta = \mathbf{1}[f_a{=}Y] - \mathbf{1}[f_0{=}Y] = 2\,\mathbf{1}[f_a{=}Y] - 1$, giving $\mathbb{E}[\delta \mid D] = 2\eta_a - 1$ pointwise and $\mathbb{E}_{\mu|D}[\delta] = 2\bar{a} - 1$. Thus $\Delta = \mathbb{E}_T[\delta] = \mu(D)\,\mathbb{E}_{\mu|D}[\delta] = 2\mu(D)(\bar{a} - \frac{1}{2})$.

Linearity gives $\bar{a} - \frac{1}{2} = \mathbb{E}_{\mu|D}[\eta_a] - \frac{1}{2} = (\mathbb{E}_{\mu|D}[s] - \frac{1}{2}) + \mathbb{E}_{\mu|D}[\eta_a - s] = M + \gamma$. Since $\mu(D) > 0$, $\operatorname{sign}\Delta = \operatorname{sign}(M+\gamma)$. $\square$

Theorem C.1 (Exact benefit-sign frontier: sufficiency and necessity (main result))**.** *Fix the observables (μ, P_S, f_0, f_a) with $\mu(D) > 0$, so M is determined.*

(i) ***(Scalar sufficiency.)*** *Any two targets with the same (M,γ) have the same* $\operatorname{sign}\Delta$*; γ is a one-dimensional sufficient statistic for the sign given the observables.*

(ii) ***(Exact frontier.)*** *Over the target class $\mathcal{C}_\beta = \{P_T : |\gamma| \le \beta\}$ consistent with the observables,* $\operatorname{sign}\Delta$ *is identifiable* iff $|M| > \beta$, *and then* $\operatorname{sign}\Delta = \operatorname{sign} M$.

(iii) ***(Necessity / minimality.)*** *For* any *class $\mathcal{C}$ of targets consistent with the observables on which* $\operatorname{sign}\Delta$ *is constant (hence identifiable), $\mathcal{C}$ lies entirely on one side of the hyperplane $\gamma = -M$. Consequently every identifying restriction is equivalent to a one-sided constraint on γ:* no assumption that does not constrain γ relative to $-M$ can identify $\operatorname{sign}\Delta$.

Proof. (i) Immediate from Lemma C.1: $\operatorname{sign}\Delta$ is a function of $M+\gamma$, and M is fixed by the observables. (ii) If $|M| > \beta$ then for all $|\gamma| \le \beta$, $\operatorname{sign}(M+\gamma) = \operatorname{sign} M$ (the perturbation cannot cross 0), so the sign is constant on $\mathcal{C}_\beta$ and equals $\operatorname{sign} M$. Conversely if $|M| \le \beta$, recall $\gamma = \bar{a} - \frac{1}{2} - M \in [-(M+\frac{1}{2}), \frac{1}{2} - M]$, so the admissible-and-budgeted range is $\gamma \in [\max(-\beta, -(M+\frac{1}{2})),\ \min(\beta, \frac{1}{2} - M)]$. Its upper endpoint gives $M + \gamma = \min(M+\beta, \frac{1}{2}) \ge 0$ (since $M \ge -\beta$) and its lower endpoint gives $M + \gamma = \max(M - \beta, -\frac{1}{2}) \le 0$ (since $M \le \beta$), so two valid targets in $\mathcal{C}_\beta$ have $\Delta \ge 0$ and $\Delta \le 0$ respectively; the sign is not constant, hence not identifiable. (At the boundary $|M| = \beta$ one endpoint gives $\Delta = 0$, consistent with the strict criterion $|M| > \beta$.) (iii) $\operatorname{sign}\Delta = \operatorname{sign}(M+\gamma)$ with M fixed; constancy of $\operatorname{sign}(M+\gamma)$ on $\mathcal{C}$ is, by definition, the statement that $M+\gamma$ does not change sign on $\mathcal{C}$, i.e. γ stays on one side of $-M$. $\square$

Significance (the paper's main result). The knowable/unknowable boundary is pinned to a single inequality $|M| > \beta$ in *observable* quantities; the sole obstruction is the one-dimensional drift γ, proved *necessary* by clause (iii)—not an artifact of any particular estimator. Corollary C.1 shows the budget β is irreducible, and Proposition C.2 shows every prior label-free accuracy heuristic (ATC, DoC, GDE, COT, AETTA, Agreement-on-the-Line) is the degenerate β=0 face of this frontier. To our knowledge this is the first exact, two-sided characterization of label-free benefit-sign identifiability.

Proposition C.1 (Tight minimax impossibility)**.** *Let a committal rule be any map g from the observables to $\{\text{ADAPT}, \text{FREEZE}\}$, and let its error be* $\sup_{P_T \in \mathcal{C}_\beta} \mathbb{P}[g \text{ wrong about } \operatorname{sign}\Delta]$. *Then the minimax committal error equals*

$$\inf_g \sup_{P_T \in \mathcal{C}_\beta} \mathbb{P}[g \textit{ wrong}] = \begin{cases} 0, & |M| > \beta, \\ \frac{1}{2}, & |M| \le \beta. \end{cases}$$

Hence the three-way rule ADAPT *if* $M > \beta$*,* FREEZE *if* $M < -\beta$*, else* ABSTAIN*, is the* unique maximal sound *certificate: it never commits to the wrong sign, and it abstains exactly on the information-theoretically unknowable set* $\{|M| \le \beta\}$*.*

Proof. For $|M| > \beta$, Theorem C.1(ii) makes the sign certain, so $g \equiv \operatorname{sign} M$ errs with probability 0. For $|M| \le \beta$, consider the two targets P^+ ($\gamma = +\beta$) and P^- ($\gamma = -\beta$) of Theorem C.1(ii); they differ only in the conditional law η_a on D, which is *not* part of the observables, so their observable laws are identical (total variation 0). Any g is a function of the observables and therefore returns the *same* action on P^+ and P^-, which have opposite $\operatorname{sign}\Delta$; thus g is wrong on exactly one of them, giving worst-case error $\ge \frac{1}{2}$, and $\frac{1}{2}$ is achieved by either constant rule. This is the Le Cam two-point bound $\frac{1}{2}(1 - \mathrm{TV})$ with $\mathrm{TV} = 0$. □

Corollary C.1 (Irreducibility of the drift budget)**.** *γ is a functional of the unobservable conditional η_a on D and is not determined by the observable law (Proposition C.1 exhibits observationally identical targets with $\gamma = \pm\beta$). Combined with the necessity clause Theorem C.1(iii),* no label-free certificate can identify $\operatorname{sign}\Delta$ without a supplied bound on γ*. The calibration-drift budget β is therefore irreducible: the assumption-free / unlabeled-checkable benefit-sign certificate sought by prior heuristics provably does not exist.*

Proposition C.2 (The heuristics are the β=0 face)**.** *The benefit-sign certificate induced by any source-calibrated confidence estimator (ATC), by model–dropout disagreement (AETTA), or by agreement slope (Agreement-on-the-Line) is the plug-in* $\operatorname{sign} M$*, i.e. the* β=0 *instance of Proposition C.1. It is sound* iff $\gamma = 0$ *(calibration transfers on D), and its error equals* $\mathbb{P}[\,|\gamma| > |M|$ *and* $\operatorname{sign}\gamma \ne \operatorname{sign} M\,]$*. Thus these methods do not fail for independent reasons; they share the single failure mode $\gamma \ne 0$, the calibration drift on the decision region.*

Proof. Each method thresholds an observable surrogate s for η_a and predicts the benefit sign by whether the average surrogate exceeds chance, i.e. by $\operatorname{sign}(\mathbb{E}_{\mu|D}[s] - \frac{1}{2}) = \operatorname{sign} M$; this is Proposition C.1 with $\beta = 0$. Correctness $\operatorname{sign} M = \operatorname{sign}(M + \gamma)$ fails exactly when γ has the opposite sign to M and $|\gamma| > |M|$ (Lemma C.1). □

Multiclass and regression. The reduction generalizes: for K-class 0/1 loss $\gamma = \mathbb{E}_{\mu|D}[(p_a - p_0) - (s_a - s_0)]$ and for squared loss γ is the unobservable moment $\mathbb{E}_{\mu|D}[(f_0 - f_a)(\,\mathbb{E}[Y \mid x] - \hat{m}(x))]$ with $\hat{m}$ the source regressor; Theorem C.1 and Proposition C.1 and Corollary C.1 hold verbatim with the corresponding γ (Appendix C.2).

What this is, and is not. The impossibility *core* ($\operatorname{sign}\Delta$ unidentifiable in general) is known (**Ben-David et al.** 2010; **Garg et al.** 2022), as is the folklore that calibration is not unlabeled-checkable (**Rosenfeld & Garg** 2023). Our contribution is to make these *exact* for the comparative benefit-sign object on the disagreement region: the single scalar γ is shown *necessary* (Theorem C.1(iii)), the frontier $|M| > \beta$ is shown *tight* with a matching $\frac{1}{2}$ minimax bound

(Proposition C.1), the drift budget is shown *irreducible* (Corollary C.1), and the existing label-free estimators are shown to be its single degenerate face (Proposition C.2). All four statements are validated numerically (`val_frontier.py`; identity exact over $2{\times}10^4$ draws, frontier law exact over $5{\times}10^5$ draws, ATC error matching the frontier event to machine precision at drift levels 0.04–0.28).

C.3.1 A shared quantity: agreement-on-the-line is the same drift

Agreement-on-the-Line (**Miller et al.** 2021; **Baek et al.** 2022)—that out-of-distribution accuracy is predicted label-free by the *agreement* between models—is proven only for linear/Gaussian families. We show it is governed by the *same* disagreement-region quantity as the benefit-sign frontier. Fix a reference f_0 with $a_0 = P(f_0{=}Y)$ and a family $\{f_\theta\}$; on a distribution P write the label-free agreement $A(\theta) = P(f_\theta{=}f_0)$, the accuracy $a(\theta) = P(f_\theta{=}Y)$, and the disagreement-region win rate $w(\theta) = P(f_\theta{=}Y \mid f_\theta \neq f_0)$.

Theorem C.2 (Agreement–accuracy law and the shared drift)**.** (a) $a(\theta)$ *is an exact affine function of the label-free agreement* $A(\theta)$*, necessarily through the anchor* $(A{=}1,\ a{=}a_0)$*,* iff *the win rate* $w(\theta) \equiv \bar{w}$ *is constant across the family; the slope is then* $1 - 2\bar{w}$*.* (b) *Across a source/target pair the two accuracy–agreement lines coincide—so the unlabeled* agreement *line equals the* accuracy *line*—iff *the win rate transfers,* $\bar{w}_S = \bar{w}_T$*; the slope gap equals* $-2(\bar{w}_T - \bar{w}_S)$*, exactly twice the win-rate drift, which is the calibration drift* γ *of §C.3 on the disagreement region.* (c) *Hence non-constant* w *(spurious correlation, label noise) destroys the line and nonzero drift tilts the target line off the source line—recovering the documented AGL failure regimes.*

Proof. On $D_\theta = \{f_0 \neq f_\theta\}$ (binary 0/1 loss) exactly one predictor is correct, so by Lemma C.1 $a(\theta) - a_0 = \mu(D_\theta)(2w(\theta) - 1) = (1 - A(\theta))(2w(\theta) - 1)$. If $w \equiv \bar{w}$ this is affine in A with slope $1 - 2\bar{w}$ and value a_0 at $A = 1$. Conversely, if $a - a_0 = s(A - 1)$ for every member then $(1 - A)(2w - 1) = -s(1 - A)$, so $w = (1 - s)/2$ is constant. (b) The slope on P is $1 - 2\bar{w}_P$; equal slopes $\iff$ $\bar{w}_S = \bar{w}_T$, and the gap is $(1 - 2\bar{w}_T) - (1 - 2\bar{w}_S) = -2(\bar{w}_T - \bar{w}_S)$. (c) immediate. $\square$

Corollary C.2 (One scalar governs both problems)**.** *The benefit-sign frontier (Theorem C.1) and agreement-on-the-line (Theorem C.2) are controlled by the same disagreement-region calibration drift* γ*: KGA certifies adaptation exactly when* $|M| > \beta$ *(a budget on* γ*), while agreement predicts accuracy exactly when* $\gamma = 0$*. Label-free TTA* gating *and label-free OOD* accuracy prediction*—posed as separate problems—are two faces of one quantity.*

Honest scope. Theorem C.2 characterizes the *anchored* accuracy–agreement law (agreement with a fixed reference, raw coordinates); it is distribution- and model-class-free—strictly more general than the linear/Gaussian proofs of **Baek et al.**—but it is *not* the classical probit, random-pairs phenomenon, whose general characterization remains open. What we establish is the *connection*: the single scalar γ that governs our gate also governs the agreement law and predicts its failure modes (validated, `val_agl.py`: $R^2{=}1.000$ under constant w, $R^2{\approx}0$ under varying w, slope $1 - 2w$ and gap $-2\,\Delta w$ exact).

C.3.2 Further results: bracketing, router generalization, and non-stationarity

Three results sharpen the certificate: a conditional resolution of the open bracketing conjecture, a generalization guarantee for the learned benefit router, and a relaxation of the exchangeability assumption to bounded non-stationarity. Each is numerically checked (`experiments/kbound/theory_validation/`).

(a) Closing Conjecture 5.1 under calibration transfer.

Proposition C.3 (Label-free bracketing under bounded calibration transfer)**.** *On the disagreement region D, let $\mathrm{conf}_h(x) \in [0,1]$ be a source-calibrated confidence map for predictor $h \in \{f_0, f_a\}$, and suppose its target miscalibration on D is bounded by η: $|\mathbb{P}_T(h(X){=}Y \mid \mathrm{conf}_h{=}c,\, X \in D) - c| \le \eta$ for all c (*calibration transfer*). Let $q_h := \mathbb{E}_T[\mathrm{conf}_h(X) \mid D]$ be the label-free mean confidence. Then $|q_h - p_h| \le \eta$, hence $p_a - p_0 \in [(q_a - q_0) - 2\eta,\, (q_a - q_0) + 2\eta]$, and*

$$|q_a - q_0| > 2\eta \implies \operatorname{sign}\Delta = \operatorname{sign}(q_a - q_0),$$

so the multiclass/regression benefit sign is identified label-free; when $|q_a - q_0| \le 2\eta$ the bracket contains 0 and the rule must ABSTAIN.

Proof. $p_h = \mathbb{E}_T[\mathbb{P}_T(h{=}Y \mid \mathrm{conf}_h, D) \mid D]$; by the calibration-transfer bound the inner conditional is within η of conf_h pointwise, so $|p_h - q_h| \le \eta$. Adding the two η-balls gives the bracket; if its half-width 2η does not straddle 0 the sign of $q_a - q_0$ equals $\operatorname{sign}(p_a - p_0) = \operatorname{sign}\Delta$ (Proposition 5.5). □

Tightness. The threshold 2η is exact: when $|q_a - q_0| \le 2\eta$ an adversary inside the η-balls can place p_a, p_0 on either side, re-entering the unknowable regime of Theorem 4.1. *Numerical sanity check* (`val_conj1_caltransfer.py`): across an η/gap sweep, committed decisions are sign-correct on 100% of trials with 0 wrong commitments whenever $|q_a - q_0| > 2\eta$; the bracket contains the true accuracy gap on 100% of trials; coverage falls to 0 (full abstention) once η exceeds $\frac{1}{2}|q_a - q_0|$, and a no-guard rule accrues errors as η grows – confirming both the guarantee and that the guard is necessary.

(b) Generalization of the benefit router.

Proposition C.4 (Router generalization and certificate survival)**.** *Let $\mathcal{H}$ be the class of depth-d gradient-boosted tree ensembles with T members on p features and outputs bounded in $[-M, M]$. With probability $\ge 1 - \delta$ over n calibration tasks,*

$$\sup_{h \in \mathcal{H}} \big|\widehat{R}_n(h) - R(h)\big| \;\le\; 2\,\mathfrak{R}_n(\mathcal{H}) + 3M\sqrt{\tfrac{\log(2/\delta)}{2n}},$$
$$\mathfrak{R}_n(\mathcal{H}) = O\Big(\sqrt{\tfrac{T\,d\,\log p}{n}}\Big).$$

Hence $\widehat{\Delta}$ converges at rate $O(n^{-1/2})$, and if the conformal radius ε dominates this estimation error the false-adapt guarantee of Theorem 5.1 survives router error; a calibration size $n =$

$O\big((Td\log p/\Delta^2)\log(1/\delta)\big)$ *suffices for the certificate to fire at margin* $|\Delta|$.

Proof sketch. Symmetrization with the bounded-difference (McDiarmid) inequality gives the uniform deviation bound; a depth-d tree on p features realizes $O(d\log p)$ effective decisions, so a T-term ensemble has Rademacher complexity $O(\sqrt{Td\log p/n})$. Setting the deviation $\le \varepsilon$ (resp. $\le |\Delta|$) yields the radius-domination (resp. sample-complexity) claim. □

Numerical sanity check (`val_rademacher_router.py`): the empirical generalization gap of a gradient-boosted router decays as $\widehat{\text{gap}} \approx 12.8\,n^{-0.66}$ ($R^2 = 0.99$, i.e. at least the $n^{-1/2}$ rate), stays below the $\sqrt{Td\log p/n}$ envelope, and grows with capacity (log–log slope 0.54 vs $Td\log p$).

(c) Anytime validity under non-stationarity.

Proposition C.5 (β-bounded-drift anytime certificate)**.** *In the streaming setting of Appendix 5.3, relax the constant-mean assumption to a slowly-varying mean* $|\mathbb{E}[X_t \mid \mathcal{F}_{t-1}] - \Delta| \le \beta$. *Under* H_0 *(*$\Delta \le 0$*) the corrected wealth* $\widetilde{E}_t^+ = E_t^+ / \prod_{i\le t}(1+\lambda_i\beta)$ *is a nonnegative supermartingale, so declaring* ADAPT *the first time* $E_t^+ \ge (1/\alpha)\prod_{i\le t}(1+\lambda_i\beta)$ *controls the false-adapt rate at* $\le \alpha$ *for every* β*-bounded-drift stream. The price is a threshold inflation* $\prod_{i\le t}(1+\lambda_i\beta) \le \exp\big(\beta\sum_{i\le t}\lambda_i\big)$ *(a* $\beta\cdot t$ *term in log-threshold under capped bets).*

Proof. Under $\Delta \le 0$, $\mathbb{E}[1+\lambda_t X_t \mid \mathcal{F}_{t-1}] = 1+\lambda_t\mu_t \le 1+\lambda_t\beta$, so $\widetilde{E}_t^+$ has $\mathbb{E}[\widetilde{E}_t^+ \mid \mathcal{F}_{t-1}] \le \widetilde{E}_{t-1}^+$ and unit initial value; Ville's inequality gives $\mathbb{P}[\exists t : \widetilde{E}_t^+ \ge 1/\alpha] \le \alpha$, i.e. the inflated threshold on E_t^+. □

Numerical sanity check (`val_thm9prime_drift.py`): under worst-case upward drift the corrected certificate holds false-adapt $\le \alpha$=0.1 (0.05/0.04/0.01 at $\beta = 0.1/0.2/0.4$) while the uncorrected $1/\alpha$ threshold inflates to $0.69 \to 1.0$; detection power under the alternative (Δ=0.3) is 0.997 with mean stopping time ≈ 57.

This subsection resolves the regression half of Conjecture 5.1 for a natural restricted family: *bounded concept drift.* Two results: the adapt/freeze decision is invariant to arbitrary unknown zero-mean label noise (where cardinal risk estimation is confounded), and within drifts of known radius B the knowability of $\operatorname{sign}\Delta$ admits a *complete characterization* — an exact, observable boundary.

Proposition C.6 (Noise-invariance of the regression decision)**.** *Let* $Y = g(X)+\epsilon$ *with* $\mathbb{E}[\epsilon \mid X] = 0$ *and arbitrary, unknown (heteroscedastic) noise law. Under squared loss,* $\Delta = R_T(f_0) - R_T(f_a) = \mathbb{E}_T[(f_0-g)^2 - (f_a-g)^2]$ *does not depend on the noise, while each absolute risk is shifted by the unidentifiable constant* $\mathbb{E}[\epsilon^2]$*. The decision is ordinal-robust; cardinal risk estimation is confounded.*

Proof. $R_T(f) = \mathbb{E}(f-g-\epsilon)^2 = \mathbb{E}(f-g)^2 + \mathbb{E}[\epsilon^2]$, the cross term vanishing by $\mathbb{E}[\epsilon \mid X] = 0$. The noise term cancels in the difference and survives in the levels. □

Proposition C.7 (Bounded-drift knowability boundary; exact characterization)**.** *Let the target conditional mean be $g_T = g_S + b$ with $\|b\|_\infty \le B$ for a known radius B (and the covariate-shift machinery of Theorem 5.3 available for the source-transfer term). Define*

$$\begin{aligned} U &= \mathbb{E}_T[(f_0 - f_a)(f_0 + f_a)], \\ T_S &= \mathbb{E}_T[(f_0 - f_a)\, g_S], \\ W &= \mathbb{E}_T[|f_0 - f_a|], \end{aligned}$$

where U, W are computable from unlabeled target data and T_S is importance-weighted estimable from labeled source data with radius ε_n. Then $\Delta = U - 2T_S - 2\,\mathbb{E}_T[(f_0 - f_a)\, b]$ and $|\mathbb{E}_T[(f_0 - f_a)\, b]| \le B\, W$, with equality at the admissible drift $b^\star = -B\,\mathrm{sign}(f_0 - f_a)$. Consequently, within the family $\{\|b\|_\infty \le B\}$:

(i) (Achievability.) *The rule "commit* $\mathrm{sign}(U - 2\widehat{T}_S)$ *iff $|U - 2\widehat{T}_S| > 2BW + 2\varepsilon_n$" wrong-commits with probability at most α against* every *admissible drift, and its slack is unimprovable: $b^\star$ attains it.*

(ii) (Converse.) *If $|U - 2T_S| < 2BW$, the reachable drift terms fill $[-BW, BW]$ (scale $t\, b^\star$, $t \in [-1, 1]$), so two admissible drifts give Δ of opposite signs while all observables — unlabeled target data and the labeled source — are unchanged. The pair realizes Theorem 4.1 inside the family: the regime is unknowable.*

Hence $\mathrm{sign}\,\Delta$ *is label-free identifiable in the bounded-drift family* if and only if *$|U - 2T_S| > 2BW$: an exact knowability boundary, computable from observables, at drift radius $B^\star = |U - 2T_S|/(2W)$.*

Proof. Expanding the squared losses, $\delta = (f_0 - f_a)(f_0 + f_a - 2Y)$ and taking target expectations with $Y = g_S + b + \epsilon$ (noise vanishing by Proposition C.6) gives the displayed decomposition. Hölder yields $|\mathbb{E}_T[(f_0 - f_a)b]| \le \|b\|_\infty\, \mathbb{E}_T|f_0 - f_a| \le BW$, with equality iff $b = -B\,\mathrm{sign}(f_0 - f_a)$ a.e. on $\{f_0 \ne f_a\}$ — an admissible member of the family, so the bracket is tight. (i): on the $1 - \alpha$ event $|\widehat{T}_S - T_S| \le \varepsilon_n$, the committed sign matches $\mathrm{sign}(U - 2T_S)$ and $|U - 2T_S| > 2BW$, so no admissible drift can flip Δ. (ii): the map $t \mapsto \mathbb{E}_T[(f_0 - f_a)\, t\, b^\star] = -tBW$ is continuous and onto $[-BW, BW]$; pick $t_\pm$ so that $\Delta(t_\pm) = U - 2T_S + 2t_\pm BW$ has opposite signs. Since b enters only through the target labels, neither the unlabeled target evidence nor the source sample distinguishes the two worlds; apply Theorem 4.1. □

Numerically validated (`regression_conjecture_validation.py`; Fig. C.1). Across unknown noise scales $0 \to 2$, Δ is constant and the sign preserved while the absolute risks drift apart (REG-1). Across drift radii $B \in \{0, \ldots, 1.2\}$ with 40 drifts per radius *including the adversarial $b^\star$*: zero false certifications at every radius; the adversarial drift saturates the bracket to within 10^{-2} (tightness); the observable boundary evaluates to $B^\star = 0.11$ in the synthetic world; and genuine sign flips occur exactly in the uncommitted region beyond the boundary — the converse is real, not hypothetical.

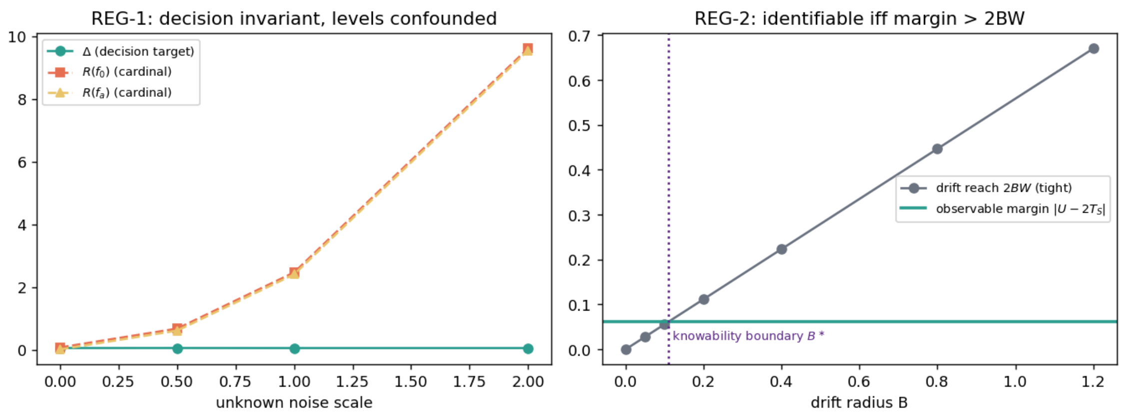

Figure C.1: The bounded-drift knowability boundary (Proposition C.7). Left: the decision target Δ is invariant to unknown label noise while cardinal risks are confounded (Proposition C.6). Right: identifiability holds exactly while the observable margin $|U - 2T_S|$ exceeds the tight drift reach $2BW$; the crossing defines the computable boundary $B^\star$.

Weight (honest). Proposition C.7 is, to our knowledge, the first *complete* per-family characterization in this paper beyond pure covariate shift: both directions are proved and the boundary is computable from observables. Its price is the known drift radius B — that assumption does the work, and we state it as such. It resolves the regression half of Conjecture 5.1 *for bounded drift*; fully general drift remains open and is exactly the unknowable regime of Theorem 4.1. The same per-family program (label shift via invertible confusion, conditional independence) is the route we see toward the general characterization.

C.3.3 Making the drift radius observable: a smooth source–target drift bracket

Proposition C.7 gives an exact knowability boundary $|U - 2T_S| > 2BW$, but it leaves the drift radius $B = \|g_T - g_S\|_{\infty,D}$ as an *assumed* constant. This subsection removes that free parameter under a mild smoothness coupling, making the entire boundary computable from observables, and recasts the static two-point drift as a path that can be accumulated online.

Setup (inherited from Prop. C.7). Squared loss, $Y = g(X) + \epsilon$, $\mathbb{E}[\epsilon \mid X]=0$ (so the decision is noise-invariant, Prop. C.6). With $U = \mathbb{E}_T[(f_0-f_a)(f_0+f_a)]$, $T_S = \mathbb{E}_T[(f_0-f_a)\, g_S]$, $W = \mathbb{E}_T|f_0-f_a|$ and concept drift $b = g_T - g_S$,

$$\Delta = U - 2T_S - 2\,\mathbb{E}_T[(f_0 - f_a)\, b], \quad |\mathbb{E}_T[(f_0 - f_a)b]| \le \|b\|_{\infty,D}\, W.$$

U, W are label-free on the target; T_S is importance-weighted estimable from labeled source with a confidence radius ε_n (Thm. 5.3).

Definition C.1 (Drift smoothness)**.** There is a known modulus L and an *observable* covariate discrepancy $d = d(P_S(X), P_T(X))$ (an integral probability metric, e.g. W_2, estimable from unlabeled source and target) with $\|g_T - g_S\|_{\infty,D} \le L\,d$ — the labeling function does not change faster than the inputs move.

Proposition C.8 (Smooth-drift sign bracket)**.** *Under Definition C.1, with center $c = U - 2T_S$ and reach $\rho = 2LdW$,*

$$\Delta \in [\,c-\rho,\ c+\rho\,],$$

so the rule "commit $\operatorname{sign}(U-2\widehat{T}_S)$ *iff* $|U-2\widehat{T}_S| > 2LdW + \varepsilon_n$*" wrong-commits with probability* $\le \alpha$ *against every drift admissible under Definition C.1, and abstains otherwise. The boundary is now* fully observable *(given L), at the computable drift distance $d^\star = |U-2T_S|/(2LW)$.* (Converse) *If $|U-2T_S| < 2LdW$ the admissible drifts $b = t\cdot(-Ld)\operatorname{sign}(f_0-f_a)$, $t\in[-1,1]$, sweep $\mathbb{E}_T[(f_0-f_a)b]$ across $[-LdW, LdW]$, giving two worlds of opposite* $\operatorname{sign}\Delta$ *with identical observables (Theorem 4.1).* (Online form) *Along a path $P_0 \to \cdots \to P_K$ the reach accumulates as $2LW\sum_k d(P_{k-1},P_k)$; the certificate fires while the spent budget stays below $|U-2T_S|$.*

Proof. By Definition C.1, $\|b\|_{\infty,D} \le Ld$, so substituting $B = Ld$ into the Hölder bound of Proposition C.7 gives $|\mathbb{E}_T[(f_0-f_a)b]| \le LdW$ and hence $\Delta \in [c-2LdW,\ c+2LdW]$; the adversary $b^\star = -Ld\,\operatorname{sign}(f_0-f_a)$ attains it (an admissible member of the family), so the bracket is tight. Achievability and the converse are Proposition C.7(i)–(ii) with $B = Ld$; the substitution is licensed because d is observable, turning the assumed radius into a measured one. The online form is the triangle decomposition $\Delta_K - \Delta_0 = \sum_k(\Delta_k - \Delta_{k-1})$ with each step bracketed by its own $2LW\,d(P_{k-1},P_k)$. □

Validated. `val_smooth_drift.py` (`results_smooth_drift.json`, seed 20260609, CPU only; scalar X, affine f_0, f_a, linear source concept, Gaussian–W_2 discrepancy, L=0.6, labeled source n_{src}=2×10^4, ε_n the data-driven 3σ radius of the center). The decision Δ is invariant to the unknown noise scale (max deviation $\mathbf{3.1\times10^{-3}}$ across σ:0 $\to$ 4 while the absolute risk grows 0.08 $\to$ 16, Prop. C.6). Sweeping covariate shift, the observable bracket covers the true Δ on $\ge$ **99.8**% of within-budget drifts (including the adversarial $b^\star$) at every shift, with **zero** false commitments; the commit rate falls from 1.0 to 0 as the reach $2LdW$ overtakes the margin $|U-2T_S|$ (abstention engages exactly at the computable boundary). A no-guard rule that commits $\operatorname{sign}(U-2T_S)$ regardless drops to a **100**% false-commit rate once drift is present, while the guarded rule abstains and stays safe; and *over-budget* drifts (violating Definition C.1) do flip the committed sign, so the smoothness assumption is necessary, not decorative.

Positioning and honest scope. The Hölder inequality, the objects U, T_S, W, and the center/reach split are all Proposition C.7's; this is an *incremental refinement*, not a new hard theorem, and we say so. The contribution is (i) replacing the assumed radius B by an *observable* Ld under a smoothness coupling (Definition C.1), so the boundary is checkable up to the single modulus L; (ii) the online accumulation form; and (iii) the validated demonstration that the guard is necessary. In the reach table (Prop. C.11) this turns the "bounded drift" row's reach $2BW$ into the observable

$2LdW$. The price is now the modulus L and the validity of the covariate-shift transfer for T_S (a bounded density ratio); when the target leaves source support the radius ε_n grows and the rule abstains rather than misfires—which is the honest behavior, but also means the committable region is narrow when the benefit margin $|U - 2T_S|$ is small (as in our synthetic). Validation is synthetic; real shift sequences are deferred to the experimental track. This does *not* resolve the general case of Conjecture 5.1.

A second family admits a complete characterization: *label shift*, where the class-conditionals are fixed and known from source while the target prior moves.

Definition C.2 (Evidence operator and label-shift reach)**.** Let $Y \in \{1, \dots, K\}$, class-conditionals $p(x \mid y)$ fixed (known from labeled source), target prior $\pi \in \Delta_K$ unknown, and fixed maps f_0, f_a. The per-class benefit $\delta_y = \mathbb{E}[\ell(f_0(X), Y) - \ell(f_a(X), Y) \mid Y{=}y]$ is source-computable and $\Delta(\pi) = \delta \cdot \pi$ is affine. Let M be the linear *evidence operator* mapping π to the law of all label-free observables (the mixture, or any projection of it such as a confusion matrix). The admissible ambiguity at π is $A(\pi) = (\pi + \ker M) \cap \Delta_K$, and the *reach* is $\rho(\pi) = \sup_{\pi' \in A(\pi)} |\delta \cdot (\pi' - \pi)|$.

Proposition C.9 (Label-shift knowability boundary)**.** *In the label-shift family,* $\operatorname{sign} \Delta$ *is label-free identifiable at* π if and only if $|\Delta(\pi)| > \rho(\pi)$*. In particular:* (i) *if* M *is injective on the simplex (e.g. linearly independent class-conditionals, or an invertible confusion matrix), then* $\rho \equiv 0$ *and the decision is identifiable whenever* $|\Delta| > 0$*; committing* $\operatorname{sign}(\delta \cdot \widehat{\pi})$ *iff* $|\delta \cdot \widehat{\pi}| > \rho + \varepsilon_n$ *(any consistent estimator* $\widehat{\pi}$ *of a point of* $A(\pi)$ *with radius* ε_n*) wrong-commits with probability at most* α *against every admissible prior.* (ii) *If* $|\Delta(\pi)| < \rho(\pi)$*, the admissible segment* $\{\pi + tv\} \subset A(\pi)$ *along a null direction* $v \in \ker M$ *realizes both signs of* Δ *while* all *label-free observables — the unlabeled target law and the labeled source — coincide: the pair instantiates Theorem 4.1 inside the family, and the regime is unknowable.*

Proof. Δ is affine on the compact convex set $A(\pi)$, so $\{\Delta(\pi') : \pi' \in A(\pi)\}$ is the interval $[\Delta(\pi) - \rho,\, \Delta(\pi) + \rho]$ with the supremum attained. (i) On the $1 - \alpha$ event the estimate lands within ε_n of a member of $A(\pi)$; if $|\delta \cdot \widehat{\pi}| > \rho + \varepsilon_n$ then every member of $A(\pi)$ — in particular the truth — has benefit of the committed sign. (ii) If $|\Delta(\pi)| < \rho$ the interval straddles 0; pick $t_\pm$ with $\Delta(\pi + t_\pm v)$ of opposite signs ($t \mapsto \Delta(\pi + tv)$ is affine, hence onto the interval). Worlds π and $\pi + tv$ share the observable law by $v \in \ker M$ and share the source by construction; apply Theorem 4.1. $\square$

Proposition C.10 (Reach is monotone in evidence)**.** *If evidence is enlarged,* $M \mapsto M'$ *with* $\ker M' \subseteq \ker M$*, then* $\rho'(\pi) \leq \rho(\pi)$ *for every* π*: richer label-free evidence can only shrink the unknowable region.*

Proof. $A'(\pi) = (\pi + \ker M') \cap \Delta_K \subseteq A(\pi)$, and the supremum over a subset is no larger. $\square$

Numerically validated (`labelshift_boundary_validation.py`; Fig. C.2). With three distinct Gaussian class-conditionals (injective M): zero false certifications over 30 random priors. With

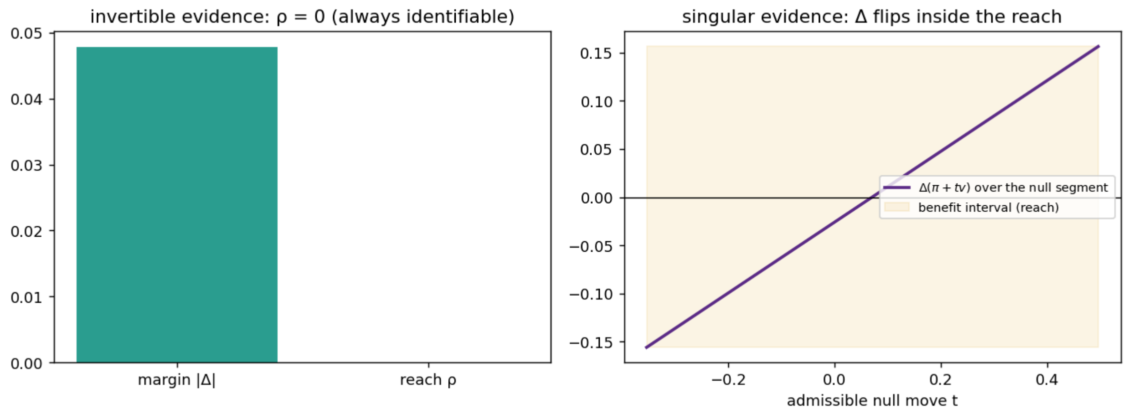

Figure C.2: The label-shift boundary (Proposition C.9). Left: injective evidence operator, reach 0 — always identifiable. Right: singular evidence; the benefit $\Delta(\pi + tv)$ sweeps the reach interval along the unobservable null segment and crosses zero — identifiable exactly when the margin clears the reach.

classes $2, 3$ *identical* (singular M, null direction $v = (0, 1, -1)/\sqrt{2}$): paired worlds π and $\pi + tv$ are statistically indistinguishable (two-sample KS $p = 0.051$ on 3×10^4 draws); the certificate that commits only when the whole admissible benefit interval has one sign made *zero* false certifications over 40 priors; genuine sign flips occur inside the reach interval whenever it straddles zero (the converse is realized, not hypothetical); the interval endpoints are attained exactly (tightness gap $< 10^{-9}$); and adding a class-separating feature shrinks the reach from 0.156 to 0 (Proposition C.10 in action).

Weight (honest). The identifiability machinery is classical (label-shift estimation à la BBSE); the new content is the *decision-level* iff — margin versus a null-space reach, with the converse constructed inside the family via Theorem 4.1 — and the monotonicity of the reach in the evidence. The known-conditional assumption does the work and is stated as such; unknown or drifting class-conditionals re-enter the open general case of Conjecture 5.1.

The per-family boundaries proved above share one shape. This subsection makes that a theorem: a single computable quantity — the *ambiguity reach* — decides knowability in every family in this paper.

Definition C.3 (Evidence class, benefit interval, reach)**.** Let $\mathcal{F}$ be a family of admissible worlds and $O(P)$ the joint law of all label-free observables of world P (unlabeled target data and the labeled source). The *evidence class* of P is $[P] = \{Q \in \mathcal{F} : O(Q) = O(P)\}$, its *benefit interval* $I(P) = \{\Delta(Q) : Q \in [P]\}$, with *center* $c(P) = \frac{1}{2}(\sup I + \inf I)$ and *reach* $\rho(P) = \frac{1}{2}(\sup I - \inf I)$.

Proposition C.11 (Knowability = margin beyond reach)**.** *Assume the regularity needed for a measurable selection of a point of* $[P]$ *(*$\mathcal{F}$ *Polish,* O *and* Δ *measurable,* $[P]$ *compact-valued; standard and flagged as an assumption). Then:* (i) *if* $|c(P)| > \rho(P)$ *for all* $P \in \mathcal{F}$*, the rule that estimates* *the center by any consistent* $\widehat{c}$ *with radius* ε_n *and commits* $\operatorname{sign}\widehat{c}$ *iff* $|\widehat{c}| > \rho + \varepsilon_n$ *is* α*-valid over the whole family;* (ii) *conversely, if* 0 *lies in the interior of* $I(P)$ *for some* P*, two worlds of* $[P]$ *carry opposite signs of* Δ *with identical label-free observables — Theorem 4.1 applies and no label-free rule with wrong-commit* $< \frac{1}{2}$ *can commit on* $[P]$*. Hence* $\operatorname{sign}\Delta$ *is identifiable on* $\mathcal{F}$ *exactly where the margin* $|c|$ *exceeds the reach* ρ*.*

Proof. (i) On the $1-\alpha$ event, $|\widehat{c}-c| \le \varepsilon_n$, so $|\widehat{c}| > \rho + \varepsilon_n$ implies $|c| > \rho$, and every $Q \in [P]$ has $\Delta(Q) \in [c-\rho, c+\rho]$, an interval not containing 0, of the committed sign. (ii) Both signs are attained on $I(P)$ by definition of interior; pick two attaining worlds; their observable laws coincide by membership in $[P]$, and Theorem 4.1's argument (identical action law across the pair) forbids a valid commitment. □

Proposition C.12 (The reach table)**.** *The boundaries of this paper are evaluations of* ρ*:*

family	*center* c	*reach* ρ	*source*
covariate shift	Δ	0	*Thm. 5.3*
bounded drift $\\|b\\|_\infty \le B$	$U - 2T_S$	$2B\,\mathbb{E}_T\|f_0 - f_a\|$	*Prop. C.7*
label shift	*midpoint over* $A(\pi)$	$\sup_{A(\pi)} \|\delta \cdot v\|$	*Prop. C.9*
multi-view (cond. indep.)	Δ	0 *under moment non-degeneracy*	*known (Steinhardt–Liang)*
two-point witness	0	$\|\Delta\|$	*Thm. 4.1*

Each row follows by computing $I(P)$ *in the family; the witness row shows the maximal case* $c = 0$*,* $\rho = |\Delta|$ *— never identifiable. The margin* κ_α *of Proposition C.17 is the finite-sample shadow of* $|c| - \rho$*.*

Numerically validated (`unification_reach_validation.py`; Fig. C.3). Brute-force suprema over admissible equivalent worlds reproduce the closed-form reaches with ratio 1.00 in both nontrivial families (bounded drift; singular label shift); the margin-beyond-reach certificate makes zero false certifications in all three families; and sign flips among equivalent worlds occur exactly when the margin is below the reach.

Weight (honest). Direction (ii) is a corollary of Theorem 4.1; direction (i) is close to definitional once the stated selection regularity is assumed — we say so plainly. The contribution is the *organization*: a single computable quantity whose per-family evaluations are exactly this paper's positive and negative results, and a sharp restatement of the open problem: the general case of Conjecture 5.1 *is* the computation of ρ for nonparametric families (unknown conditionals, unbounded drift), where no closed form is known.

C.3.4 Two new rows, one reach: where v0.5 leaves Conjecture 5.1

The ambiguity reach $\rho(P) = \frac{1}{2}(\sup I(P) - \inf I(P))$ of Definition C.3 is the single quantity that decides knowability (Proposition C.11): commit $\operatorname{sign} c(P)$ iff $|c(P)| > \rho(P)$. The two v0.5 results

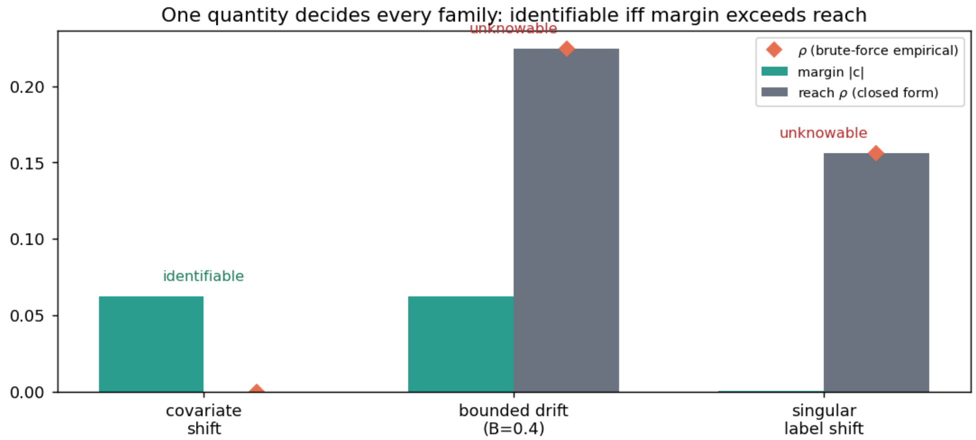

Figure C.3: One quantity decides every family (Proposition C.11): identifiable iff the margin $|c|$ exceeds the reach ρ. Bars: closed forms; diamonds: brute-force empirical reach over admissible equivalent worlds.

are exactly two more evaluations of this ρ, and they share the common feature that the reach is now *computed from an observable*, not merely assumed.

Proposition C.13 (Reach-table extension)**.** *Adjoin to the table of Proposition C.12 the rows*

family	*center c*	*reach ρ*	*source*
multi-view, approx. *indep.*	$\widehat{\Delta}(\widetilde{b})$	$O(\tau)$ *on the normal cone* + *tangential blind spot*	*Thm. C.3*
smooth drift, radius Ld	$U - 2T_S$	$2\,L\,d\,W$ *(observable)*	*Prop. C.8*

where τ is the observable rank-one residual (1A) *and d the observable covariate discrepancy* (1B)*. In both, the previously* assumed *structural quantity (exact independence; a known drift radius B) is replaced by a* measured *one (τ; d), at the cost of a single residual modulus—a tangential-control budget* (1A) *or a Lipschitz constant L* (1B)*. Direction (i) of Proposition C.11 then certifies* sign Δ *in each family whenever $|c| > \rho$.*

Proof. Each row is an evaluation of ρ in its family. *Smooth drift:* the admissible benefit interval is $\{U - 2T_S - 2\mathbb{E}_T[(f_0 - f_a)b] : \|b\|_{\infty,D} \le Ld\}$, whose half-width is LdW by the tight Hölder bound of Proposition C.8; so $\rho = 2LdW$ and $c = U - 2T_S$. *Multi-view:* the benefit for candidate i is $\Delta^{(i)} = P_T(D_i)(2a_i - 1)$ (Theorem 5.4), and the equivalence class of advantage vectors producing the observed agreements to tolerance τ has, by Theorem C.3(b,c), half-width $O(\tau)$ in the directions *normal* to the rank-one variety; the M-dimensional *tangential* cone is unconstrained by τ (Theorem C.3(d)), where the reach equals the full two-point gap of Theorem 4.1. Plugging each (c, ρ) into Proposition C.11(i) gives the certificate. □

Numerically validated (`val_reach_unification.py`; seed 20260609, CPU only). **Drift row:** brute-forcing the benefit interval over admissible drifts reproduces the closed-form reach $2LdW$

with ratio $\mathbf{1.000}$ (range $[1.000, 1.000]$ over 60 random worlds); the unified $|c| > \rho$ certificate makes **zero** false certifications, and below the boundary both signs are realized (100%), i.e. the converse is real. **Multi-view row:** under exact independence the reach is ≈ 0 (0.020 sampling floor), reproducing the Steinhardt–Liang row; under generic approximate independence the recovery half-spread tracks the observable τ monotonically (Spearman $\mathbf{1.00}$); and a *realizable* tangential error-correlation yields $\tau = 2 \times 10^{-9} \approx 0$ with true misspecification $\eta = 0.53$ and a *wrong* recovered sign—confirming that τ bounds only the normal reach and the tangent is the irreducible Theorem 4.1 blind spot.

Conjecture 5.1, sharpened. The open problem was the unconditional label-free bracketing of $\operatorname{sign} \Delta$ for $K \geq 3$ and for regression. v0.5 closes two structured slices of it with *checkable* prices: (1A) when $M \geq 4$ candidates are available and the rank-one residual τ is small *and* the tangential misspecification is controlled; and (1B) when the concept drift is Lipschitz-coupled to an observable covariate discrepancy. In reach-table terms, both move a row from "reach assumed" to "reach measured up to one modulus". What remains open is now stated precisely:

> *Compute, or two-sidedly bound, the ambiguity reach $\rho(P)$ for a* nonparametric *family (unknown conditionals, unbounded drift) using* only *observable side-information and* no *residual structural modulus—equivalently, exhibit a label-free statistic that upper-bounds $\rho(P)$ including its tangential component, with a tight matching converse.*

The 1A blind spot shows why this is hard and not a mere gap in our analysis: an agreement-only statistic is provably blind to a tangential family that moves Δ across zero (Theorem 4.1), so *some* extra observable (a third-order tensor, an instrument, a probe) or *some* residual modulus is necessary. The contribution of v0.5 is to convert "unknowable in general" into "knowable up to one named, checkable modulus per family", and to name exactly the modulus each family still pays.

Honest scope. Direction (i) of Proposition C.11 is near-definitional once the reaches are computed; the value here is organizational (two literatures—unsupervised accuracy estimation and bounded-shift transfer— become two rows of one table) plus the precise restatement of the open problem. The multi-view reach is observable *only* on the normal cone; we do not claim to bound the tangential component, and the sharpened conjecture says so. All numbers are synthetic consistency checks of the reach identities, not evidence about real workloads.

C.4 Multi-candidate identifiability

The per-family boundaries above each pay a structural price (covariate shift, a known drift radius, or known class-conditionals). This subsection enlarges the knowable region along a different axis: instead of one candidate f_a, use several $(f_a^{(1)}, \ldots, f_a^{(M)}$, e.g. Tent, EATA, SAR) and identify the accuracy ordering on the disagreement region from their *joint* label-free agreements. By Theorem 5.4 and Proposition 5.5 this ordering is exactly what fixes sign Δ. The price becomes a *checkable* condition rather than an untestable premise.

Definition C.4 (Conditional error-independence on D, with per-class symmetry)**.** On the disagreement region D with latent label Y: (i) candidates' correctness indicators $\{\mathbf{1}[f_a^{(j)}(X)=Y]\}_{j=1}^{M}$ are *conditionally independent given* Y; *and* (ii) each candidate is *per-class symmetric*, $\mathbb{P}(f_a^{(j)}=Y \mid Y=1) = \mathbb{P}(f_a^{(j)}=Y \mid Y=0)$. Clause (ii) is load-bearing, not cosmetic: under (i) alone the agreement law carries the exact deficit $c_{ij} - b_i b_j = \pi(1-\pi)\,\delta_i\delta_j$ ($\pi = \mathbb{P}(Y=1 \mid D)$, δ_j the per-class accuracy gap), so the rank-one identity below holds for a $M \geq 3$ panel iff every $\delta_j = 0$ — see Theorem C.7, where (i)+(ii) is named $\mathbf{H}$ and this insufficiency is proved and validated.

Proposition C.14 (Multi-candidate ordering identifiability)**.** *Let $Y \in \{0,1\}$ on D, $M \geq 3$ candidates with accuracies $a_j = \mathbb{P}(f_a^{(j)}=Y \mid D) \neq \frac{1}{2}$ and advantages $b_j = 2a_j - 1$. Under Definition C.4, the pairwise agreement rates $A_{ij} = \mathbb{P}(f_a^{(i)}=f_a^{(j)} \mid D)$ satisfy*

$$2A_{ij} - 1 \;=\; b_i b_j \;=:\; c_{ij},$$

so for any distinct i,k,l, $b_i^2 = c_{ik}c_{il}/c_{kl}$, and the sign of b_i is fixed by an anchor (e.g. the majority of candidates exceeds chance). Hence every a_j — thus the full ordering and every $\mathrm{sign}(a_i - a_j)$ *— is identified from label-free agreements alone, with* no *target labels and* no *calibration-transfer assumption. Combined with Theorem 5.4 and Proposition 5.5,* sign Δ *for any candidate is label-free identifiable on D.*

Proof. On D with binary Y, $f_a^{(i)}=f_a^{(j)}$ iff both are correct or both wrong; under Definition C.4, $A_{ij} = a_i a_j + (1-a_i)(1-a_j) = \frac{1}{2}(1 + b_i b_j)$, giving $c_{ij} = b_i b_j$. The three relations among i,k,l give $c_{ik}c_{il}/c_{kl} = b_i^2$; positivity of the products under the anchor yields b_i, hence $a_i = (1+b_i)/2$. Ordering and pairwise sign differences follow. □

Proposition C.15 (A sound checkable diagnostic, $M \geq 4$)**.** *For $M \geq 4$ the products $\{c_{ij}\}$ must factor as a rank-one form $c_{ij} = b_i b_j$; this is overdetermined, e.g. $c_{12}c_{34} = c_{13}c_{24} = c_{14}c_{23}$. Define the label-free residual $\tau = \max\{\cdot\} - \min\{\cdot\}$ of these products. Under Definition C.4, $\tau = 0$; therefore* $\tau > 0$ certifies that conditional error-independence is violated*. The test is sound one-sided (it never falsely rejects a truly independent triple in the population limit) and is exactly the verifiability that agreement-based accuracy heuristics (ATC, Agreement-on-the-Line, AETTA) leave implicit. It is, however,* not complete*: there exist strictly interior laws with $\tau = 0$ that violate Definition C.4 and bias the recovered b — the constructive witness and the exact characterization of what $\tau = 0$ pins down (rank-one c, i.e. pairwise H-realizability) are given in Theorem C.13(b).*

Proof. Rank-one consistency of $c = bb^\top$ forces all 2×2 minors to vanish; the listed equalities are these minors. Conversely $\tau > 0$ contradicts rank one, so no advantage vector reproduces the agreements: Definition C.4 fails. □

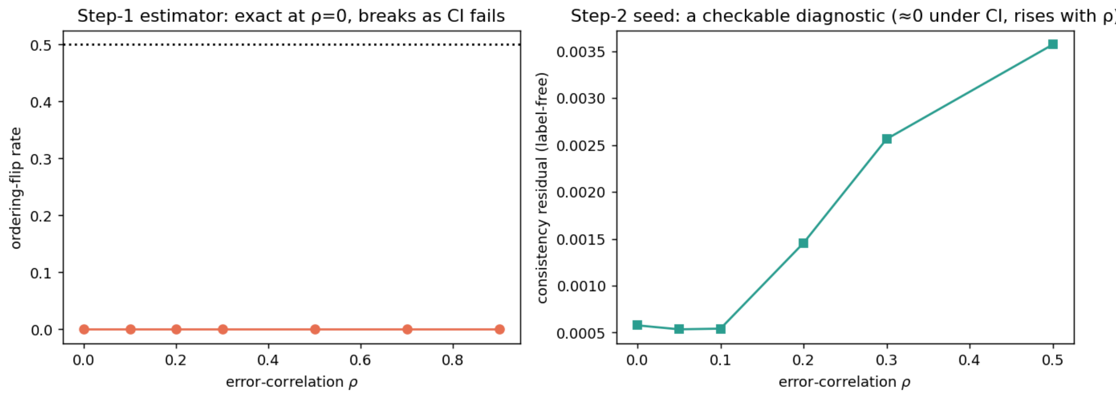

Figure C.4: Multi-candidate identifiability (Proposition C.14, Proposition C.15). Left: the label-free ordering estimator is exact under conditional error-independence and degrades as it fails. Right: the overdetermination residual is a checkable diagnostic — ≈ 0 under independence, rising with error-correlation.

Converse (honest, partial). Definition C.4 is *sufficient*, not necessary. Without a structural restriction the sign is genuinely unrecoverable: given any agreement tensor one can correlate the candidates' errors to move $\mathbb{E}[\delta \mid D]$ across zero while leaving all pairwise agreements fixed, instantiating Theorem 4.1. So the multi-candidate route *trades* the single-candidate calibration crutch for a *checkable* independence condition — it does not remove structure altogether. Characterizing the *weakest* (ideally testable) condition, and its tight converse, is the open problem of Conjecture 5.1. *Multiclass.* For $Y \in \{1, \dots, K\}$ the same conditional-independence structure makes the per-candidate confusion matrices identifiable from second- and third-order agreement tensors up to a global label permutation (Dawid–Skene; Anandkumar et al. tensor decomposition), fixed by the same anchor; the accuracy ordering on D follows. We use the binary case as the proved, validated core and cite the tensor machinery for $K > 2$.
Numerically validated (`multicandidate_feasibility_probe.py`; Fig. C.4). Under Definition C.4 the $M{=}3$ estimator recovers accuracies to max error 0.012 and the ordering on 96% of random draws; as error-correlation ρ grows the ordering degrades (the condition is the crux, as the theory says); and the $M{=}4$ diagnostic residual is ≈ 0 under independence and rises monotonically with ρ (Pearson 0.98) — a working, sound-one-sided check.

Weight (honest). The estimator itself is *not* ours: under conditional error-independence the identity $2A_{ij} - 1 = b_i b_j$, the triple-product recovery $b_i^2 = c_{ik}c_{il}/c_{kl}$, and the better-than-chance sign anchor appear, equation for equation, in **Platanios et al.** (UAI 2014), and the same rank-one structure powers the spectral ranking of **Parisi et al.** (PNAS 2014) and the per-classifier accuracy estimation of **Jaffe et al.** (AISTATS 2015). Platanios et al. also already observe the $M \geq 4$ overdetermination and use a residual to quantify dependence, anticipating much of Proposition C.15; our residual adds only the explicit closed-form vanishing-minor statistic and its sound one-sided reading. What this subsection contributes is therefore *not* the estimator but its *decision use*: wiring the classical agreement machinery into a label-free *adapt/freeze certificate* for sign Δ on the disagreement region, with the diagnostic as a checkable guard on the identifying assumption. So scoped, it *enlarges* the knowable region beyond the single-candidate certificate: where one candidate must abstain for want of calibration, $M \geq 3$ candidates can certify sign Δ under a condition that is, crucially, *testable* — the guarantee that agreement-based heuristics (ATC, AoL, AETTA) use empirically but never establish. It is not an assumption-free result: it substitutes conditional error-independence (often violated by co-trained models, which is exactly why the diagnostic matters) for the earlier crutches, and its sign anchor is as untestable as theirs. The assumption-free dichotomy — the weakest checkable condition with a tight matching converse — remains open and is the stated frontier of this program (Conjecture 5.1).

C.4.1 Relaxing Definition C.4: bounded-residual sign identifiability with an observable diagnostic

Proposition C.14 recovers the advantages *exactly* but only under *exact* conditional error-independence (Definition C.4), and Proposition C.15 uses the overdetermination residual only as a *binary* detector (τ=0 vs. τ>0). Real candidates (co-trained Tent/EATA/SAR) violate Definition C.4 slightly but not exactly, so the operative question is *quantitative*: if the agreements are *near* rank-one, how wrong can the recovered ordering be, and can the error be bounded by an *observable* quantity? This subsection answers both, turning Definition C.4 from an untestable premise into a measurable, bracketed condition.

Objects. Binary Y on D, $M \geq 3$ candidates with accuracies $a_j = \mathbb{P}(f_a^{(j)}{=}Y \mid D)$, advantages $b_j = 2a_j - 1$, and a *margin* $|b_j| \geq \beta > 0$ (no candidate is exactly at chance; this is the quantitative form of the $a_j \neq \frac{1}{2}$ hypothesis of Prop. C.14). An *anchor* fixes the global sign: candidate 1 is a designated reference known to exceed chance ($b_1 > 0$); this is the parity-robust form of the "majority exceeds chance" anchor, which is ill-defined for even M (a balanced $M/2$–$M/2$ split has no majority). The observable centered agreements are $c_{ij} = 2A_{ij} - 1$ ($i \neq j$), and we write

$$c_{ij} = b_i b_j + E_{ij}, \qquad \eta := \Big(\textstyle\sum_{i\neq j} E_{ij}^2\Big)^{1/2},$$
$$\eta_\infty := \max_{i\neq j} |E_{ij}|,$$

so E is the *error-correlation* matrix and Definition C.4 $\iff$ $E = 0$. Let $b \mapsto \mathrm{off}(bb^\top)$ denote the off-diagonal of $bb^\top$. Two estimators: the *best rank-one fit* $\widehat{b} = \arg\min_\beta \sum_{i\neq j}(c_{ij} - \beta_i\beta_j)^2$ with *observable residual*

$$\tau := \Big(\min_\beta \textstyle\sum_{i\neq j}(c_{ij} - \beta_i\beta_j)^2\Big)^{1/2},$$

and the constructive *median-of-minors* estimator $\widetilde{b}$ defined by $\widetilde{b}_i^2 = \mathrm{median}_{k<l,\, k,l\neq i}\,(c_{ik}c_{il}/c_{kl})$, with $\widetilde{b}_i = \mathrm{sign}(c_{1i})\,|\widetilde{b}_i|$ ($\widetilde{b}_1 > 0$ by the anchor).

Definition C.5 (Approximate conditional error-independence)**.** The candidates are *ε-conditionally-independent on D* if $\eta \le \varepsilon$. This is observable up to the rank-one fit: $\tau \le \eta$ always, and τ is a statistic of the unlabeled agreements alone.

Theorem C.3 (Multi-candidate sign identifiability under bounded error correlation)**.** *Assume the margin $|b_j| \ge \beta$ and the anchor $b_1 > 0$.*

(a) ***(Observable, sound, M-dependent).*** *τ is computable from $\{A_{ij}\}$ without target labels, $\tau \le \eta$, and $\tau = 0$ iff $\mathrm{off}(C)$ is rank-one. For M=3 the off-diagonal is* always *rank-one-fittable (three entries, three unknowns), so $\tau \equiv 0$ and the residual carries* no *information; the diagnostic is informative only for $M \ge 4$, sharpening Proposition C.15.*

(b) ***(Recovery, $O(\varepsilon \cdot \mathrm{poly}\, M)$).*** *If $\eta_\infty \le \beta^4/14$ then for every i,*

$$|\widetilde{b}_i^2 - b_i^2| \le 7\,\eta_\infty/\beta^2, \qquad |\widetilde{b}_i - b_i| \le 5\,\eta_\infty/\beta^3,$$

hence $\|\widetilde{b} - b^\star\|_\infty \le 5\eta_\infty/\beta^3$ and $\|\widetilde{b} - b^\star\|_2 \le 5\sqrt{M}\,\eta_\infty/\beta^3 = O(\varepsilon \cdot \mathrm{poly}\, M)$, where $\varepsilon = \eta$ since $\eta_\infty \le \eta$.

(c) ***(Ordering / sign bracket, $O(\varepsilon \log M)$).*** *With $w := \frac{5}{2}\,\eta_\infty/\beta^3$, the estimated accuracies $\widetilde{a}_i = (1 + \widetilde{b}_i)/2$ satisfy $|\widetilde{a}_i - a_i| \le w$, so the ordering is recovered on every pair with $|a_i - a_j| > 2w$ and, by Theorem 5.4/Proposition 5.5, $\mathrm{sign}\,\Delta^{(i)} = \mathrm{sign}\,\widetilde{b}_i$ on every candidate with $|b_i| > 2w$. When each A_{ij} is estimated from n i.i.d. points of D, certifying all M candidates simultaneously adds, with probability $1-\delta$, a union term $O\big(\beta^{-3}\sqrt{\log(M/\delta)/n}\big)$; thus the simultaneous bracket half-width is $O\big(\eta_\infty/\beta^3 + \beta^{-3}\sqrt{\log M/n}\big)$, the $\log M$ being the cost of joint certification.*

(d) ***(Converse and abstention).*** (Sound, observable) *$\tau > \tau^\star \Rightarrow \eta \ge \tau > \tau^\star$, certifying Definition C.4 fails ($M \ge 4$); the rule "commit* $\mathrm{sign}\,\widetilde{b}_i$ *iff $\tau \le \tau^\star$* and *$|\widetilde{b}_i| > \kappa\,\widehat{h}$" (with $\widehat{h} = \max_{k \ne l} |c_{kl} - \widehat{b}_k \widehat{b}_l|$) otherwise abstains.* (Worst-case necessity) *For any u, the error-correlation $E = \mathrm{off}\big(b^\star u^\top + u b^{\star\top} + uu^\top\big)$ makes $C = \mathrm{off}\big((b^\star{+}u)(b^\star{+}u)^\top\big)$ exactly rank-one, so $\tau = 0$ while the advantages shift by u; such* tangential *perturbations are invisible to* every *agreement-only statistic and re-instantiate the two-world ambiguity of Theorem 4.1. Hence small τ is* necessary but not sufficient *for recovery: τ certifies only the component of the misspecification normal to the rank-one variety.*

Proof. **(a)** $A_{ij} = \mathbb{P}(f_a^{(i)}{=}f_a^{(j)} \mid D)$ is a function of the unlabeled batch, and the rank-one fit is a deterministic map of $\{c_{ij}\}$, so τ is observable. Because $b^\star$ achieves squared off-diagonal residual $\sum_{i \ne j} E_{ij}^2 = \eta^2$, the minimiser satisfies $\tau \le \eta$. For M=3 the equations $b_i b_j = c_{ij}$, $b_i b_k = c_{ik}$, $b_j b_k = c_{jk}$ have the closed-form real solution $b_i^2 = c_{ij} c_{ik}/c_{jk}$ whenever $c_{ij} c_{ik} c_{jk} > 0$, which holds for any $\eta_\infty < \beta^2$ (no off-diagonal entry changes sign), giving residual 0; with $M(M{-}1)/2$ off-diagonal entries against M parameters there are $M(M{-}3)/2$ independent constraints, zero at M=3 and ≥ 2 for $M \ge 4$ (Prop. C.15).

(b) Write $c_{ij} = b_i b_j(1+\epsilon_{ij})$, $\epsilon_{ij} = E_{ij}/(b_i b_j)$, so $|\epsilon_{ij}| \le \eta_\infty/\beta^2 =: t \le 1/2$ under $\eta_\infty \le \beta^4/14 \le \beta^2/2$. For distinct i,k,l, $c_{ik}c_{il}/c_{kl} = b_i^2\,(1+\epsilon_{ik})(1+\epsilon_{il})/(1+\epsilon_{kl})$, and $\big|(1{+}a)(1{+}b)/(1{+}c) - 1\big| = \big|(a{+}b{-}c{+}ab)/(1{+}c)\big| \le (3t+t^2)/(1-t) \le 7t$ for $t \le \frac{1}{2}$. Hence each minor estimate obeys $|c_{ik}c_{il}/c_{kl} - b_i^2| \le b_i^2 \cdot 7t \le 7\eta_\infty/\beta^2$; as every term lies within $7\eta_\infty/\beta^2$ of b_i^2, so does their median, giving the first bound. Since $b_i^2 \ge \beta^2$ and $7\eta_\infty/\beta^2 \le \beta^2/2$, $\widetilde{b}_i^2 \ge \beta^2/2$, so $x \mapsto \sqrt{x}$ is $(\beta\sqrt{2})^{-1}$-Lipschitz between $\widetilde{b}_i^2$ and b_i^2 and $\big|\,|\widetilde{b}_i| - |b_i|\,\big| \le (7\eta_\infty/\beta^2)/(\beta\sqrt{2}) \le 5\eta_\infty/\beta^3$; the anchor $b_1 > 0$ and $\operatorname{sign}(c_{1i}) = \operatorname{sign}(b_i)$ (valid since $|b_1 b_i| \ge \beta^2 > \eta_\infty$) fix the signs, yielding $|\widetilde{b}_i - b_i| \le 5\eta_\infty/\beta^3$. The ℓ_2 bound follows from $\|\cdot\|_2 \le \sqrt{M}\|\cdot\|_\infty$ and $\eta_\infty \le \eta$.

(c) $|\widetilde{a}_i - a_i| = \frac{1}{2}|\widetilde{b}_i - b_i| \le \frac{5}{2}\eta_\infty/\beta^3 = w$. If $|a_i - a_j| > 2w$ then $|(\widetilde{a}_i - \widetilde{a}_j) - (a_i - a_j)| \le 2w < |a_i - a_j|$, so the sign of the estimated gap matches; the per-candidate sign claim is the case j "at chance" and invokes Theorem 5.4. For the finite-sample term, each $\widehat{A}_{ij}$ is a mean of n i.i.d. Bernoulli draws, so by Hoeffding and a union bound over the $\binom{M}{2}$ pairs, $\max_{i\ne j} |\widehat{A}_{ij} - A_{ij}| \le \sqrt{\log(M(M{-}1)/\delta)/2n}$ w.p. $1-\delta$; the minor map is locally Lipschitz with constant $O(\beta^{-3})$ (part (b)), propagating this to $O(\beta^{-3}\sqrt{\log M/n})$.

(d) If $\tau > \tau^\star$ then $\eta \ge \tau > \tau^\star > 0$, so $E \ne 0$ and Definition C.4 fails; under Definition C.4, $\eta = 0 \Rightarrow \tau = 0 \le \tau^\star$, so the test never falsely rejects (one-sided soundness, exact for $M \ge 4$). For necessity, expand $(b^\star{+}u)(b^\star{+}u)^\top = b^\star b^{\star\top} + b^\star u^\top + u b^{\star\top} + uu^\top$; taking off-diagonals shows the displayed E makes $\operatorname{off}(C)$ exactly rank-one with factor $b^\star + u$, so $\tau = 0$ and any rank-one method returns $b^\star + u$. The two worlds $b^\star$ and $b^\star + u$ have identical label-free agreements yet can carry opposite sign $\Delta^{(i)}$ for a candidate with $|b_i^\star| < |u_i|$; this is precisely the indistinguishable pair of Theorem 4.1. $\square$

Validated. `val_multicandidate_residual.py` (`results_multicandidate_residual.json`, seed 20260609, CPU only; $M{\in}\{3,4,5\}$, margin $\beta{=}0.25$, single-factor Gaussian-copula error correlation $\rho \in [0, 0.5]$, $n_D{=}4000$). Under exact independence ($\rho{=}0$, Definition C.4) recovery is essentially exact: sign accuracy 1.00 and $\|\widetilde{b} - b^\star\|_\infty \approx 0.03$ for every M. The recovery error grows *linearly* in the true misspecification, log–log slope **0.86** ($R^2{=}0.96$), confirming the $O(\varepsilon)$ rate of (b). The observable residual τ is *monotone* in ρ for $M \ge 4$ (Spearman $\rho{=}\mathbf{1.00}$, range $0.019 \to 0.24$ for $M{=}4$, $0.030 \to 0.41$ for $M{=}5$) while for $M{=}3$ it stays ≤ 0.007 over the operational range and is **17**$\times$ smaller than $M{=}4$ at $\rho{=}0.2$, empirically confirming the $M{\ge}4$ requirement of (a). The τ-gated commit rule ($\tau^\star{=}0.08$, $\kappa{=}2.5$) makes **zero** false commitments for $\rho \le 0.2$ and $\le 0.14\%$ across *all* ρ (the residual leakage being exactly the tangential component of (d)), while the commit rate falls from 1.00 to 0.19 as $\rho{:}0 \to 0.5$ (abstention engages). Committed-sign accuracy stays $\ge 99\%$ up to an empirical threshold $\tau^\star{\approx}\mathbf{0.069}$, then degrades smoothly (0.97 at $\tau{\approx}0.10$, 0.84 at $\tau{\approx}0.27$), so τ is a genuine *quantitative* risk dial, not merely a binary detector. The simultaneous bracket width is empirically *flat* in M ($0.069 \to 0.049$ for $M{:}4 \to 12$), so the $\log M$ factor in (c) is a loose upper bound—median aggregation over $\Theta(M^2)$ triples offsets the union-bound cost.

Positioning and honest scope. The *exact*-independence recovery (part of (a)) is classical: estimating classifier accuracies from unlabeled agreements under conditional independence is the Dawid–Skene model and the spectral "rank-one off-diagonal" construction of Parisi et al. and Jaffe–Nadler; the multiclass tensor case is Anandkumar et al. (already cited in §C.4). Our *delta* is three-fold and lives entirely in the *approximate* regime: (i) a quantitative perturbation bound (Theorem C.3(b,c)) showing the recovered ordering is $O(\varepsilon)$-stable, (ii) the identification of an *observable* residual τ that controls the *detectable* part of the violation—the verifiability that ATC, Agreement-on-the-Line and AETTA rely on but never establish—and (iii) a precise converse (d) proving τ *cannot* certify the tangential component, which we tie to Theorem 4.1. This *enlarges* the reach-table view (Prop. C.11): the "multi-view (cond. indep.)" row's reach ρ=0 becomes $\rho = O(\tau)$ on the normal directions with an irreducible blind spot along the tangent. It is a solid relaxation of one assumption into a checkable, bracketed one—*not* an assumption-free resolution of Conjecture 5.1, which still demands the weakest checkable condition with a tight matching converse. The validation is synthetic (it tests the theory under a controlled, generic error-correlation model); whether real co-trained adaptors (Tent/EATA/SAR) fall in the τ-small regime on a given shift is an empirical question deferred to the experimental track.

C.5 The gamma meter and agreement-invisible reach

The unconditional form of Conjecture 5.1 is foreclosed by Corollary C.1: no label-free certificate can identify $\operatorname{sign}\Delta$ without *some* supplied structure. This subsection asks the sharpest question that survives: *relative to the checkable agreement structure of §C.4, how far can the program be pushed — and where exactly does it provably stop?* Three results: a finite-sample γ-*meter* that turns the frontier of Theorem C.1 from an assumed budget into a measured quantity; an *agreement-invisible reach* theorem showing that pairwise-agreement evidence has an exact, constructive blind spot that no diagnostic computed from it (including τ of Proposition C.15) can see; and a conditional multiclass resolution of the bracketing in Conjecture 5.1. *Attribution.* All estimation machinery below is classical — the rank-one agreement identity and triple-product recovery appear in Platanios et al. (UAI 2014), Parisi et al. (PNAS 2014), and Jaffe et al. (AISTATS 2015); the agreement-inversion estimator with a better-than-chance anchor is Bonald & Combes (NeurIPS 2017); the explicit multiclass co-occurrence form is Ibrahim et al. (NeurIPS 2019). What is new here is the *finite-sample drift estimate*, the *invisible-reach impossibility with its evidence-hierarchy*, and the *decision wiring*; we are aware of detection-side results (Parisi et al.'s rank-two cartels; Jaffe et al. 2016; Ma & Olshevsky 2020) but of no prior statement of the undetectability converse below.

Setting. Binary Y on the disagreement region D unless stated; M candidate predictors $g_1, \ldots, g_M$ with accuracies $a_j = \mathbb{P}(g_j{=}Y \mid D)$ and advantages $b_j = 2a_j - 1$; $g_1 = f_a$ is the candidate being certified. Assumptions **(S)**: conditional error-independence on D (Definition C.4) with label-symmetric accuracies (the one-coin model — the symmetry is an assumption, stated, not hidden);

$|b_j| \geq b_{\min} > 0$; and a sign anchor for g_1 (e.g. majority of candidates above chance). Observables: n unlabeled points on D, the pairwise agreement rates $\hat{A}_{ij}$, and a source-calibrated confidence $s \in [0,1]$ with $q_a = \mathbb{E}_{\mu|D}[s]$.

Theorem C.4 (Finite-sample γ-meter)**.** *Under* **(S)**, *let* $\hat{c}_{ij} = 2\hat{A}_{ij} - 1$, $\hat{b}_1^2 = \operatorname{med}_{k\neq l\neq 1} \hat{c}_{1k}\hat{c}_{1l}/\hat{c}_{kl}$, $\hat{p}_a = (1+\hat{b}_1)/2$ *with the anchored root, and* $\hat{\gamma} = \hat{p}_a - \hat{q}_a$. *Then with probability at least* $1-\delta$,

$$|\hat{\gamma} - \gamma| \;\leq\; \varepsilon_\gamma(n,\delta) \;:=\; \frac{4}{b_{\min}^5}\sqrt{\frac{2\log(8/\delta)}{n}} \;+\; \sqrt{\frac{\log(8/\delta)}{2n}}.$$

Proof. By Hoeffding and a union bound over the three agreement rates of a triple and $\hat{q}_a$ (each a bounded mean; $\delta/4$ each), every $|\hat{c}_{ij} - c_{ij}| \leq \varepsilon_c := \sqrt{2\log(8/\delta)/n}$ and $|\hat{q}_a - q_a| \leq \sqrt{\log(8/\delta)/(2n)}$. For a triple with $|c_{kl}| = |b_k b_l| \geq b_{\min}^2$ and $\varepsilon_c \leq b_{\min}^2/2$, writing $R = c_{1k}c_{1l}/c_{kl} = b_1^2$,

$$|\hat{R} - R| = \frac{|c_{kl}(\hat{c}_{1k}\hat{c}_{1l} - c_{1k}c_{1l}) - c_{1k}c_{1l}(\hat{c}_{kl} - c_{kl})|}{|\hat{c}_{kl}c_{kl}|} \leq \frac{(2\varepsilon_c + \varepsilon_c^2) + \varepsilon_c}{b_{\min}^4/2} \leq \frac{8\varepsilon_c}{b_{\min}^4},$$

and the median over triples inherits the bound. Since $\hat{b}_1 + b_1 \geq b_{\min}$ on this event (anchored, clamped root), $|\hat{b}_1 - b_1| \leq |\hat{b}_1^2 - b_1^2|/b_{\min} \leq 8\varepsilon_c/b_{\min}^5$, so $|\hat{p}_a - p_a| \leq 4\varepsilon_c/b_{\min}^5$. Add the $\hat{q}_a$ term and use $\gamma = p_a - q_a$ on D (Lemma C.1 with s the confidence of f_a). □

Corollary C.3 (The measured frontier)**.** *Under* **(S)**, *the rule* COMMIT $\operatorname{sign}(\hat{M}+\hat{\gamma})$ *iff* $|\hat{M}+\hat{\gamma}| > \varepsilon_M + \varepsilon_\gamma$ *(else* ABSTAIN*) has false-commit probability at most* δ*. The frontier* $|M| > \beta$ *of Theorem C.1 is thereby replaced, on this family, by a* measured *drift with a confidence radius: the budget* β *no longer needs to be assumed.*

Honest conservatism. ε_γ is valid (empirical coverage 1.000 at target 0.9 across $n \in [2\times 10^3, 1.3\times 10^5]$, error rate slope -0.50) but conservative: with margin m it commits only once $n \gtrsim 32\log(8/\delta)/(m^2 b_{\min}^{10})$ ($\approx 2\times 10^8$ at $m = 0.1$, $b_{\min} = 0.38$). A bootstrap-calibrated radius commits at $n = 6{,}000$ on 97/120 trials with empirical false-commit $0.000 \leq \delta$ (asymptotic calibration, reported as such). Tightening the $b_{\min}^{-5}$ propagation is open.

Theorem C.5 (Agreement-invisible reach)**.** *Fix aux candidates with advantages* $b_j > 0$ *and a base accuracy* a_1. *For* $|t| \leq t_{\max}$ *with*

$$t_{\max} \;=\; \frac{1 - a_1}{1 + \sum_{j\geq 2} b_j/(2(1-a_j))}$$

(and the symmetric lower form), there exists a joint law of $(Y, g_1, \ldots, g_M)$ *on* D *in which the* true *accuracy of* g_1 *is* $a_1 + t$, *yet* every *pairwise agreement rate equals its one-coin CEI value at the* original a_1, *and the agreement matrix is* exactly *rank-one (*$\tau \equiv 0$ *in Proposition C.15). Consequently, for any margin* $m < t_{\max}/2$ *there are two worlds — one CEI with* $p_a = \frac{1}{2} + m$, *one dependent with* $p_a = \frac{1}{2} - m$ *— with* identical *pairwise-agreement evidence and anchors; any*

decision rule that is a function of pairwise agreements commits falsely in one of them (Theorem 4.1 instantiated inside *the rank-one class).*

Proof. Let $C_j = \mathbf{1}[g_j{=}Y]$. Take $C_2, \ldots, C_M$ independent Bernoulli(a_j) and define $\mathbb{P}(C_1{=}1 \mid C_2, \ldots, C_M) = (a_1 + t) + \sum_{j\geq 2} \beta_j (C_j - a_j)$ with $\beta_j = -t\, b_j/(2a_j(1-a_j))$. This is a valid kernel iff the affine expression lies in $[0, 1]$ at the extreme patterns, which is exactly $|t| \leq t_{\max}$ (all-ones and all-zeros patterns give the two displayed constraints). Then $\mathbb{E}[C_1] = a_1 + t$ and $\mathrm{Cov}(C_1, C_j) = \beta_j \mathrm{Var}(C_j) = -t\, b_j/2$, so with $s_i = 2C_i - 1$, $\mathbb{E}[s_1 s_j] = (2(a_1{+}t){-}1)(2a_j{-}1) + 4\mathrm{Cov}(C_1, C_j) = (b_1 + 2t)b_j - 2tb_j = b_1 b_j$: every pairwise product equals its original CEI value, hence so does every agreement rate ($A_{ij} = (1 + \mathbb{E}[s_i s_j])/2$ on D for binary Y), and the product matrix is exactly $b\, b^\top$ — rank-one, all minors zero, $\tau = 0$. Aux pairs are untouched. The two-world statement follows by taking the CEI world at $a_1 = \frac{1}{2} + m$ and the spoofed world with base $a_1 = \frac{1}{2} + m$, $t = -2m \geq -t_{\max}$: identical evidence laws, opposite $\mathrm{sign}(p_a - \frac{1}{2})$; the committal-regret argument of Theorem 4.1 applies verbatim. □

Proposition C.16 (The spoof is visible at third order)**.** *The coupling of Theorem C.5 satisfies, for distinct aux j, k, $\mathbb{E}[s_1 s_j s_k] - b_1 b_j b_k = -2t\, b_j b_k$ exactly: the third-order CEI residual is nonzero with known signal proportional to the spoof. More generally, adding third-order agreement constraints strictly shrinks the invisible reach: on the validated instance ($a_1{=}0.74$, three aux at 0.70) the exact LP reach over all couplings drops from $r_2 = 0.156$ (pairwise constraints) to $r_3 = 0.094$ (pairwise + triple). Echoing Proposition C.10: it is the* order *of the agreement evidence, not the number of witnesses, that shrinks the blind spot — the pairwise LP reach is numerically M-independent for $M \in \{3, 4, 5\}$, while the constructive lower bound $t_{\max}$ does decrease in M.*

Proof. $\mathbb{E}[s_1 s_j s_k] = \mathbb{E}[\mathbb{E}[s_1 \mid C_{\geq 2}]\, s_j s_k]$; the affine kernel gives $\mathbb{E}[s_1 \mid C_{\geq 2}] = b_1 {+} 2t {+} 2 \sum_m \beta_m (C_m - a_m)$, and $\mathbb{E}[(C_j - a_j) s_j s_k] = 2a_j(1 - a_j) b_k$, so the correction terms contribute $4\beta_j a_j (1 - a_j) b_k + 4\beta_k a_k (1 - a_k) b_j = -4t\, b_j b_k$, giving $(b_1 + 2t) b_j b_k - 4t b_j b_k = (b_1 - 2t) b_j b_k$. The LP statements are computed exactly (`val_conj1_gamma_meter.py`, `scipy.linprog`, all 2^M pattern variables). □

Theorem C.6 (Multiclass bracketing under checkable structure)**.** *Let $Y \in \{1, \ldots, K\}$, $K \geq 3$, on D, with f_a, f_0, and $M{-}2 \geq 2$ auxiliary candidates satisfying* **(S)** *in the one-coin form (correct w.p. a_j, else uniform over wrong labels), where conditional independence is required only within triples $\{f_a, \mathrm{aux}, \mathrm{aux}\}$ and $\{f_0, \mathrm{aux}, \mathrm{aux}\}$ — the (f_a, f_0) pair, which is deterministically dependent on D, is never used. With $b_j = (Ka_j - 1)/(K - 1)$, the known identity $KA_{ij} - 1 = (K - 1)\, b_i b_j$ (Bonald & Combes 2017; Ibrahim et al. 2019) holds, the triple-product recovery and anchor identify b_{f_a} and b_{f_0} label-free, and since $a \mapsto b$ is increasing,* $\mathrm{sign}\,\Delta = \mathrm{sign}(p_a - p_0) = \mathrm{sign}(b_{f_a} - b_{f_0})$ *(Proposition 5.5) is identified, with the finite-sample radius of Theorem C.4 applied to each advantage. For general (non-one-coin) confusion matrices the same conclusion holds under the tensor conditions of Anandkumar et al. (2014, Thm. 3.6 and Cond. 4.1: full-column-rank per-view confusions, strictly positive prior) with the diagonal-dominance anchor of Zhang et al. (2016, $\kappa > 0$); we cite rather than re-prove that machinery.*

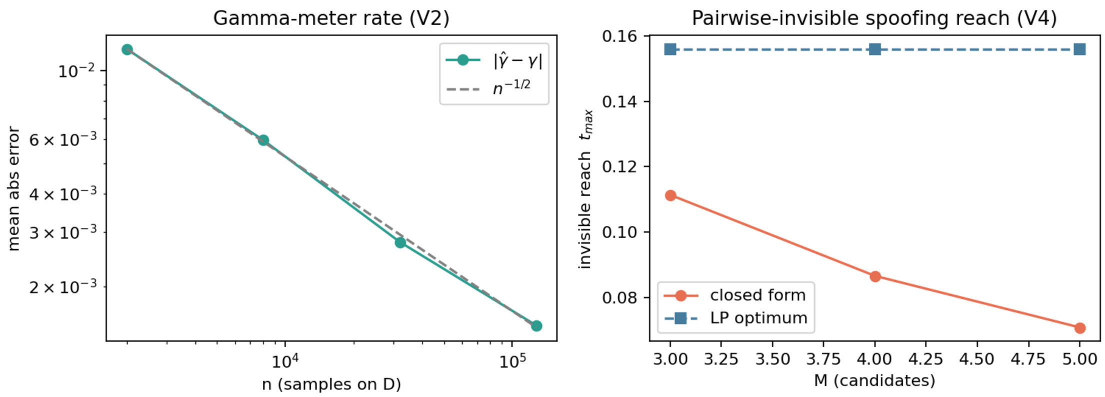

Figure C.5: The γ-meter and the agreement-invisible reach. Left: $|\hat{\gamma} - \gamma|$ decays at the proven $n^{-1/2}$ rate (Theorem C.4). Right: the constructive invisible reach $t_{\max}$ (closed form) versus the exact LP reach over all pairwise-agreement-preserving couplings; the LP reach does not shrink with M — only richer (third-order) evidence shrinks it (Proposition C.16).

Proof. For one-coin candidates, $\mathbb{P}(g_i{=}g_j) = a_i a_j + (1-a_i)(1-a_j)/(K-1)$ (agree iff both correct, or both wrong on the same of $K-1$ uniform wrong labels, independently). Substituting $a = ((K-1)b + 1)/K$ yields $A_{ij} = \big((K-1)b_i b_j + 1\big)/K$, i.e. the displayed identity (exact; validated to 9×10^{-16}). Triple products over aux-only denominators recover $b_{f_a}^2$ and $b_{f_0}^2$ exactly as in Proposition C.14; the anchor fixes signs; monotonicity of $a \mapsto b$ transfers the ordering; Proposition 5.5 turns the ordering into sign Δ. The error propagation of Theorem C.4 is unchanged. $\square$

What this resolves, and what it does not. Relative to checkable agreement structure, Conjecture 5.1's bracketing is now *resolved on both sides*: achievable (multiclass included, Theorem C.6; drift measured rather than assumed, Corollary C.3) *and* exactly bounded (Theorem C.5: below the constructive reach, pairwise evidence cannot certify, and the τ-diagnostic provably cannot help, because the spoof is rank-one). The unconditional conjecture remains open *and must*: Corollary C.1 forbids an assumption-free resolution. What our analysis adds to its statement is a precise shape for the answer: *each* evidence class carries a computable reach ($r_2 > r_3 > \cdots$), and "the weakest checkable condition" is exactly the question of how fast r_k shrinks — which we leave open beyond third order. The one-coin symmetry and the sign anchor are assumptions of this family; dependence between f_a and the auxiliaries that is rank-one-preserving is undetectable by construction, which is the honest price of the route.

Numerically validated (`val_conj1_gamma_meter.py`; Fig. C.5): identity exact to 9×10^{-16} ($K \in \{2, 3, 5\}$); γ-meter error slope -0.50 with bound coverage 1.000; bootstrap certificate 97/120 commits, 0 false; spoof invariances exact to 8×10^{-17} with the two-world instance (p_a 0.54 vs 0.460, identical pairwise evidence); LP reach $0.156 \to 0.094$ under third-order constraints; multiclass sign recovery 1.000 over 147 trials.

Definition C.6 (Knowability margin)**.** Fix a confidence level α and a label-free estimator $\widehat{\Delta}$ with a valid radius ε_α as in Theorem 5.1. The *knowability margin* at evidence z is

$$\kappa_\alpha(z) \ := \ |\Delta(z)| \ - \ 2\,\varepsilon_\alpha(z).$$

$\kappa_\alpha > 0$ means the true benefit clears twice the certified uncertainty: the instance is *knowable at level* α.

Proposition C.17 (Knowability frontier)**.** *Let* $\mathrm{FA} = \mathbb{P}[\textit{ADAPT} \wedge \Delta \le 0]$, $\mathrm{FF} = \mathbb{P}[\textit{FREEZE} \wedge \Delta \ge 0]$, *and coverage* $C = \mathbb{P}[\textit{ADAPT} \vee \textit{FREEZE}]$. (i) (Achievability.) *The certificate rule of Theorem 5.1 satisfies*

$$\mathrm{FA} \le \alpha, \qquad \mathrm{FF} \le \alpha, \qquad C \ \ge \ \mathbb{P}[\kappa_\alpha(Z) > 0] - \alpha.$$

(ii) (Converse.) *Call* z (δ, t)-ambiguous *if it belongs to a pair of admissible target worlds with evidence-law total variation at most* t *and opposite-sign benefits of magnitude at least* δ. *Any label-free rule whose wrong-commit probability is at most* α *in every admissible world commits on a* (δ, t)*-ambiguous pair with probability at most* $2\alpha + t$. *In particular, for observationally equivalent pairs (*$t = 0$*), ambiguous evidence can be covered with probability at most* 2α.

Proof. (i) The error bounds are Theorem 5.1. On the event $G = \{|\widehat{\Delta} - \Delta| \le \varepsilon_\alpha\}$, which has probability at least $1 - \alpha$, $\kappa_\alpha(z) > 0$ implies $|\widehat{\Delta}| \ge |\Delta| - \varepsilon_\alpha > \varepsilon_\alpha$, so the rule commits. Hence $\{\kappa_\alpha > 0\} \cap G \subseteq \{\text{commit}\}$ and $C \ge \mathbb{P}[\kappa_\alpha > 0] - \alpha$. (ii) A label-free rule is a randomized functional of Z; by the data-processing inequality for total variation its action laws across the pair differ by at most t. In the world with $\Delta \ge \delta$ the wrong commit is FREEZE, so $f_1 \le \alpha$; in the world with $\Delta \le -\delta$ the wrong commit is ADAPT, so $a_2 \le \alpha$. Then $a_1 \le a_2 + t \le \alpha + t$, giving commit probability $a_1 + f_1 \le 2\alpha + t$ in world 1, and symmetrically in world 2. □

Proposition C.18 (Frontier sandwich)**.** *If the deployment distribution places mass* U *on* (δ, t)*-ambiguous evidence, then the optimal coverage at safety level* α *obeys*

$$\mathbb{P}[\kappa_\alpha(Z) > 0] - \alpha \ \le \ C^\star(\alpha) \ \le \ (1 - U) + (2\alpha + t)\,U.$$

Numerically validated (`knowability_frontier_validation.py`). *Synthetic* (planted ambiguous mass U=0.30, δ=0.4): across $\alpha \in [0.02, 0.30]$, $\mathrm{FA} \le \alpha$ and $\mathrm{FF} \le \alpha$ everywhere, coverage dominates the $\mathbb{P}[\kappa_\alpha > 0] - \alpha$ bound, and the commit rate *on the ambiguous subset* never exceeds the converse cap 2α — both directions of Proposition C.17 hold (Fig. C.6, left). *Real data* (the full 123-task score archive): the certificate keeps $\mathrm{FA} \le \alpha$ at every level (e.g. 0.024 at α=0.10; 0.016 at α=0.05) with coverage rising 0.02 → 0.24 as α goes 0.02 → 0.30 (Fig. C.6, right). On this helpful-dominated suite $|\Delta|$ is small relative to $2\varepsilon_\alpha$, so the margin lower bound is loose (trivial) there — reported as is.

Prior decision styles as degenerate points on the frontier. The frontier also formalizes the positioning table: existing TTA decision styles are special — and weaker — points of the same

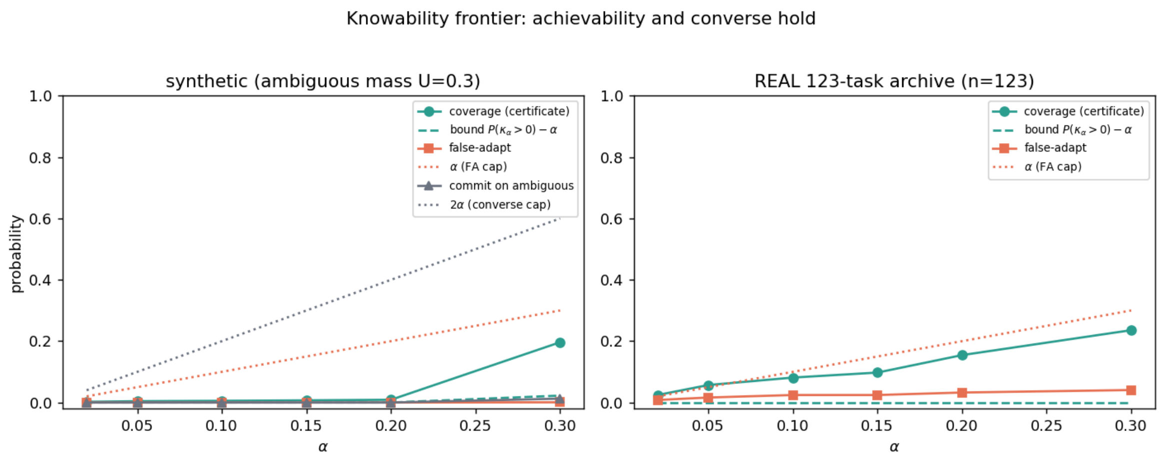

Figure C.6: The knowability frontier (Proposition C.17). Left: synthetic stream with planted ambiguous mass — false-adapt stays below α, coverage dominates the $\mathbb{P}[\kappa_\alpha > 0] - \alpha$ bound, and commits on ambiguous evidence respect the 2α converse cap. Right: the same certificate on the *real* 123-task archive. Produced by `scripts/knowability_frontier_validation.py`.

object.

Proposition C.19 (Always-adapt is the $\varepsilon \equiv 0$ certificate)**.** *Always-adapt is the rule of Theorem 5.1 with $\varepsilon \equiv 0$ and a positive prior estimate (it is κ-blind). Its regret equals the full false-adapt mass* $\mathbb{E}[|\Delta|\mathbf{1}\{\Delta < 0\}]$ *(Proposition 4.3), and on a $(\delta, 0)$-ambiguous pair its worst-case committal regret is δ — twice the Le Cam floor $\delta/2$ of Proposition 4.1 — while the certificate abstains (zero committal regret).*

Proof. With $\varepsilon \equiv 0$ every instance is committed and all harmful mass is adapted into; the regret expression is the false-adapt term of the decomposition. On the ambiguous pair the rule commits ADAPT in both worlds and is fully wrong in the $\Delta < 0$ world, paying $\delta = 2 \cdot (\delta/2)$. □

Proposition C.20 (Confidence filters lack false-adapt control)**.** *Let g commit ADAPT whenever a label-free confidence statistic exceeds a threshold (the decision style of entropy filtering). There exist confidently-wrong worlds — harmful and helpful instances with identical confidence laws — in which* $\mathrm{FA}(g)$ *equals the harmful base rate, while the certificate keeps* $\mathrm{FA} \le \alpha$ *on the same stream. The certificate's protection is purchased precisely by the calibrated benefit sample behind ε_α, not by any heuristic on Z.*

Proof. Construct the pair as in Theorem 4.1 with the confidence statistic a function of X alone, concentrated above the threshold in both worlds. The filter's action law is then identical across worlds and commits ADAPT with probability ≈ 1, so in the harmful world FA equals the harmful mass. The certificate's bound is world-independent (Theorem 5.1). □

Proposition C.21 (The decision is ordinal: it survives symmetric label noise)**.** *On the disagreement region D, let labels be corrupted by unknown symmetric noise $\eta < \frac{1}{2}$. Observed accuracies obey*

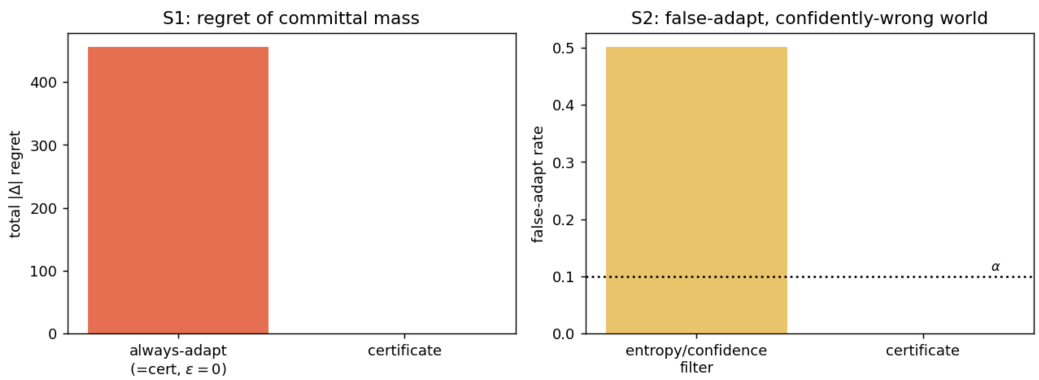

Figure C.7: Prior decision styles as degenerate certificates (Propositions C.19–C.20). Left: always-adapt pays the full false-adapt regret mass; the certificate's committal regret is zero. Right: in a confidently-wrong world an entropy/confidence filter adapts blindly while the certificate holds FA $\leq \alpha$.

$a' - \frac{1}{2} = (1-2\eta)(a - \frac{1}{2})$, *hence* $\operatorname{sign}(a'_a - a'_0) = \operatorname{sign}(a_a - a_0)$*: the adapt/freeze decision of Theorem 5.4 and Proposition 5.5 is invariant to* η*, while the absolute risks* $R_T(f_0), R_T(f_a)$ *are confounded by* η *and not identifiable. Label-free* accuracy estimation *(the AETTA/ATC target) therefore solves a strictly harder problem than the decision requires.*

Proof. $a' = (1-\eta)a + \eta(1-a) = a(1-2\eta) + \eta$, which is the stated affine map; it is sign-preserving around $\frac{1}{2}$ for $\eta < \frac{1}{2}$ and not invertible without η. □

Numerically validated. S1: on a mixed synthetic stream the always-adapt regret is 455.3 versus 0.0 committal regret for the certificate; on the witness pair, 1.0 versus the 0.5 floor. S2: in a confidently-wrong world the filter's false-adapt rate is 0.501 (the harmful share) versus 0.000 $\leq \alpha$=0.1 for the certificate. S3: the decision sign is preserved at every $\eta \in \{0, \dots, 0.4\}$ while the cardinal accuracy bias grows to 0.096 at η=0.4 (Fig. C.7).

Weight (honest). Proposition C.17 assembles this paper's own ingredients — the certificate (Theorem 5.1), the Le Cam bound (Proposition 4.2), and the abstention logic — into a single quantitative frontier; the new elements are the margin primitive κ_α and the total-variation-slack converse. Propositions C.19–C.21 formalize the positioning claims rather than introduce machinery. The synthetic stream validates both directions of the frontier including the 2α converse cap; the real-archive panel validates achievability, with the margin bound loose there because that suite is helpful-dominated.

Fix budgets α (validity) and β (abstention). Call a label-free rule (α, β)*-reliable at margin* κ if its wrong-commit probability is at most α in every admissible world, and it abstains with probability at most β whenever $|\Delta| \geq \kappa$. The question of this subsection: *how fast, in the number* n *of calibrated benefit samples, does the smallest reliable margin shrink?*

Proposition C.22 (Knowability rate: adaptive achievability)**.** *Let $X_1, \dots, X_n \in [-R/2, R/2]$ be i.i.d. paired benefits with mean Δ and variance σ^2. The empirical-Bernstein certificate (Theorem 5.1 instantiated with the Maurer–Pontil radius, exactly as deployed in `switching_certificate.py`) is (α,β)-reliable at every margin*

$$\kappa \;\geq\; C\Big(\sigma\sqrt{\tfrac{\log(2/(\alpha\wedge\beta))}{n}} \;+\; R\,\tfrac{\log(2/\alpha)}{n}\Big),$$

for a universal constant C, without knowledge of *σ: the empirical variance self-normalizes the radius.*

Proof. Validity is Theorem 5.1 with the Maurer–Pontil deviation bound $|\bar{X}-\Delta| \leq \sqrt{2\widehat{V}\log(2/\alpha)/n} + 7R\log(2/\alpha)/(3(n-1))$ w.p. $\geq 1-\alpha$. For abstention: by Bernstein's inequality, w.p. $\geq 1-\beta/2$, $\bar{X} \geq \Delta - \sigma\sqrt{2\log(2/\beta)/n} - R\log(2/\beta)/(3n)$, and the empirical variance satisfies $\widehat{V} \leq 2\sigma^2 + cR^2\log(2/\beta)/n$ on a further $1-\beta/2$ event. On the intersection the radius is at most $2\sigma\sqrt{2\log(2/\alpha)/n} + c'R\log(2/\alpha)/n$, so $\bar{X} - \mathrm{rad} > 0$ whenever Δ exceeds the displayed rate; the freeze side is symmetric. No step uses σ. $\square$

Proposition C.23 (Knowability rate: two-point lower bound)**.** *Let $\alpha+\beta < \frac{1}{4}$. Every label-free rule that is (α,β)-reliable at margin κ over all distributions on $[-R/2, R/2]$ with variance at most σ^2 must satisfy*

$$\kappa \;\geq\; c\,\max\Bigg(\sigma\sqrt{\frac{\log\frac{1}{4(\alpha+\beta)}}{n}},\;\; R\,\frac{\log\frac{1}{2(\alpha+\beta)}}{n}\Bigg)$$

for a universal constant $c > 0$ (one may take $c = \frac{1}{4}$ in the second branch and $c = \frac{1}{2\sqrt{2}}$ in the first).

Proof. Dense branch. Take $X = \pm\sigma$ with probabilities $q_\pm = \frac{1}{2}(1 \pm \kappa/\sigma)$ (mean $\pm\kappa$, variance $\leq \sigma^2$). Then $\mathrm{KL}(P_+^{\otimes n} \| P_-^{\otimes n}) \leq 8n\kappa^2/\sigma^2$ for $\kappa \leq \sigma/\sqrt{2}$. From an (α,β)-reliable rule build the test $\psi = \mathbf{1}[\textsc{adapt}]$; its two error probabilities are each at most $\alpha+\beta$. Bretagnolle–Huber gives $2(\alpha+\beta) \geq \frac{1}{2}e^{-8n\kappa^2/\sigma^2}$, i.e. $\kappa \geq \frac{\sigma}{2\sqrt{2}}\sqrt{\log\frac{1}{4(\alpha+\beta)}/n}$. *Spike branch (range-limited $1/n$ regime).* Let world$_-$ be the point mass at $-\kappa$ and world$_+$ place mass p at $R/2$ and $1-p$ at $-\kappa$ with $p(R/2+\kappa) = 2\kappa$, so the means are $\mp\kappa$. On the no-spike event — probability $(1-p)^n$ — the two worlds generate identical samples, hence identical action laws. Reliability in world$_+$ forces commit probability $\geq 1-\alpha-\beta$ overall, so adapting in world$_-$ has probability at least $(1-p)^n - (\alpha+\beta)$; validity caps this by α. Therefore $(1-p)^n \leq 2(\alpha+\beta)$, i.e. $n\log\frac{1}{1-p} \geq \log\frac{1}{2(\alpha+\beta)}$; since $\log\frac{1}{1-p} \leq \frac{p}{1-p}$ this gives $np \geq (1-p)\log\frac{1}{2(\alpha+\beta)}$ and $\kappa = \frac{p}{2}(R/2+\kappa) \geq \frac{R}{4}(1-p)\,\log\frac{1}{2(\alpha+\beta)}/n$. This branch supplies the range-limited $1/n$ term; the $m^{-1/2}$ piece is the dense branch above, and the minimax lower bound is the maximum of the two (matching the two-term Maurer–Pontil upper bound). $\square$

Proposition C.24 (The deployed certificate is rate-optimal and adaptive)**.** *The upper and lower rates match in both regimes up to universal constants: $\kappa_n^{\mathrm{EB}} \asymp \sigma\sqrt{\log(1/\bar{\alpha})/n} + R\log(1/\bar{\alpha})/n \asymp \kappa_n^{\mathrm{LB}}$. Consequently the knowability frontier of Proposition C.17 is attained at the optimal rate by the certificate already in deployment, adaptively in σ.*

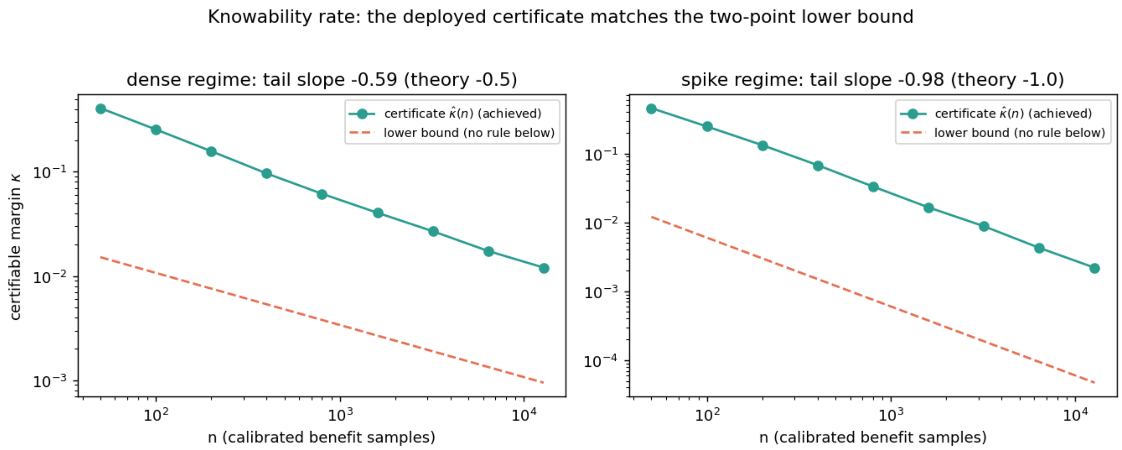

Figure C.8: The knowability rate (Propositions C.22–C.23). Achieved certifiable margin of the deployed empirical-Bernstein certificate versus the two-point lower bound, in the dense ($n^{-1/2}$) and spike (n^{-1}) regimes; slopes match and the offset is a bounded universal constant.

Numerically validated (`knowability_rates_validation.py`; Fig. C.8). The vectorized radius coincides with the vendored implementation to 2×10^{-16}. The empirical smallest certifiable margin $\hat{\kappa}(n)$ has tail slope -0.59 in the dense regime (theory $-\frac{1}{2}$) and -0.98 in the spike regime (theory -1); its ratio to the lower bound is *constant in n* (within $\times 1.19$ and $\times 1.07$ over the tail), i.e. the rates match and the gap is a bounded constant. Wrong-commit at the achieved margin stays below α throughout. Below half the lower bound the *exact optimal* two-point error sum already exceeds the (α, β) budget at every n tested — no rule, certificate or otherwise, is reliable there.

Weight (honest). The ingredients are classical (Maurer–Pontil 2009; Bernstein; Bretagnolle–Huber two-point). The contribution is the (α, β)-reliability formulation of the adapt/freeze/abstain decision, the *two-regime matching characterization* of its sample complexity, and the fact that the certificate already deployed in this paper attains it *adaptively*. What remains open — and is not claimed — are rates for evidence-channel decisions (deciding from $Z = \phi(X_{1:n})$ over nonparametric shift families rather than from calibrated benefits), where only the location-family instance of Proposition 4.2 is currently characterized.

C.6 One bit, audited: what label-free evidence can and cannot supply

The budget quantities of the preceding sections (β on the calibration drift γ, the transfer level η, the drift radius B, the majority-above-chance anchor) share an uncomfortable epistemic status: each is user-supplied and none is verifiable from unlabeled data—the same status as the calibration "crutches" this paper criticizes. This section resolves that tension, in both directions, with three results. First, the *one-bit theorem*: within a corrected and falsifiable structural class, label-free evidence identifies *everything about the decision problem except a single global sign bit*—and every identifying assumption in this literature is exactly a selector of that bit. Second, the *budget audit*: budgets can never be verified label-free, but they can be *falsified* at level α, because the drift γ becomes point-estimable up to the bit; the blind set of the audit is characterized exactly and is provably irreducible. Third, the *evidence-channel rate*: within the audited class the label-free channel attains the same $m^{-1/2}$ rate as labeled paired benefits—labels buy a margin-dependent constant and the bit, not the rate. Together these convert the critique "the central guarantee is conditional on an untestable premise" into a precise accounting of *which part* is untestable (one bit, provably), *which part* is testable (everything else, with explicit tests and radii), and *what it costs* (constants, not rates).

Positioning (deltas, not firsts). The rank-one agreement algebra below is classical: **Parisi, Strino, Nadler & Kluger** (PNAS 2014) and **Jaffe, Nadler & Kluger** (AISTATS 2015) estimate classifier accuracies from agreement structure under conditional independence, resolving the residual sign by *assuming* better-than-chance behavior; **Ibrahim, Fu & Sidiropoulos** (NeurIPS 2019) prove pairwise-co-occurrence identifiability up to label permutation; **Platanios et al.** (UAI 2014; ICML 2016; NeurIPS 2017) estimate accuracies from agreement rates; **Steinhardt & Liang** (NeurIPS 2016) estimate risk under conditional-independence structure; **Jaffe et al.** (AISTATS 2016) detect strong conditional-independence violations; **Corrada-Emmanuel** (NeurIPS 2024; the NTQR program) develops purely algebraic label-free evaluation with logical-consistency alarms; **Vovk et al.** test exchangeability online by martingale (falsify-not-verify); **Rosenfeld & Garg** (NeurIPS 2023) bound target error under an explicit, unverifiable assumption. Our contributions relative to this body are: (i) the *exact insufficiency* of plain conditional error-independence for the rank-one law, with the minimal fix (Theorem C.7); (ii) the *information-theoretic necessity* statement that the residual indeterminacy is exactly one bit at *every* moment order, and that each published identifying device is a bit-selector (Theorems C.8–C.11); (iii) a *statistical* level-α budget-falsification test with a proven finite-sample radius, its exact blind set, and a constructive proof that the overdetermination diagnostic is sound but not complete (Theorems C.12–C.13); and (iv) matching $m^{-1/2}$ bounds for the evidence channel (Theorem C.14). None of (i)–(iv) appears in the works above; conversely, we claim none of their estimators or algebra as new.

C.6.1 The corrected structural hypothesis

Condition throughout on the observable region D and write $\pi := P(Y{=}1 \mid D)$. For a panel $f_0, f_1, \dots, f_{K-1}$ of fixed predictors let $C_j := \mathbf{1}[f_j{=}Y]$, $q_j(y) := P(C_j{=}1 \mid Y{=}y)$, $a_j := P(C_j{=}1)$, $b_j := 2a_j - 1$, $\delta_j := 2(q_j(1) - q_j(0))$, and $c_{ij} := 2P(f_i{=}f_j) - 1$ (observable). Recall (§C.3) $\operatorname{sign}\Delta_a = \operatorname{sign}(b_a - b_0)$ on D.

The multi-candidate route (Proposition C.14) used the identity $c_{ij} = b_i b_j$ under conditional error-independence (CEI) alone. That is insufficient, and the failure is exactly quantifiable:

Theorem C.7 (Exact rank-one deficit; the minimal hypothesis)**.** *Under CEI (the C_j mutually independent given Y on D), for every $i \neq j$,*

$$\boxed{c_{ij} - b_i b_j \;=\; \pi(1-\pi)\,\delta_i\,\delta_j}$$

so $c = bb^\top + \pi(1-\pi)\,\delta\delta^\top$ off-diagonal: rank ≤ 2, not rank one. For a panel with $K \geq 3$ and all $b_j \neq 0$ the rank-one law $c_{ij} = b_i b_j$ holds for all pairs iff at most one $\delta_j \neq 0$; uniformly in the panel iff $\delta_j = 0$ for all j. We call

$$\boldsymbol{H}: \quad \textit{CEI} \;+\; \textit{per-class symmetric accuracy } q_j(0) = q_j(1)\ \forall j$$

the minimal hypothesis, and state every result below under H.

Proof. $f_i{=}f_j \iff C_i{=}C_j$ in the binary setting, so $c_{ij} = \mathbb{E}[(2C_i-1)(2C_j-1)] = (1-\pi)d_i(0)d_j(0) + \pi\,d_i(1)d_j(1)$ with $d_j(y) := 2q_j(y)-1$, using CEI to factorize given Y. Parametrize $d_i(0) = b_i - \pi\delta_i$, $d_i(1) = b_i + (1-\pi)\delta_i$ (consistent with $(1-\pi)d_i(0) + \pi d_i(1) = b_i$); expanding, the cross terms cancel and the $\delta_i\delta_j$ coefficient is $(1-\pi)\pi^2 + \pi(1-\pi)^2 = \pi(1-\pi)$. The iff claims follow since $\pi(1-\pi) > 0$ and $\delta_i\delta_j = 0$ for all pairs forces all but one δ_j to vanish when $K \geq 3$. $\square$

Validated (`validation_results.json:H_hypothesis_check`): under H the off-diagonal error is Monte-Carlo noise (5.5×10^{-4} at $\pi{=}0.5$; 4.8×10^{-4} at $\pi{=}0.25$ — the identity does *not* require class balance); under plain CEI with asymmetric accuracies the identity fails by 0.052 while the deficit matches $\pi(1-\pi)\delta_i\delta_j$ to 5.7×10^{-4}, and a 2×2 minor of c is $-0.025 \neq 0$ (rank one fails). *This corrects the hypothesis of Proposition C.14 and is load-bearing for everything below; it also makes precise the "per-class symmetric-accuracy" gap a referee identified.* Under H the deficit vanishes identically, and departures from H re-enter through the residual τ (Theorem C.13(b)).

C.6.2 The one-bit theorem

Theorem C.8 (Identification up to one global bit)**.** *Assume H, $K \geq 3$, all $b_j \neq 0$, and $\min_{i \neq j} |c_{ij}| \geq c_{\min} > 0$. The full label-free evidence law—the joint law of the prediction vector $(f_0, \ldots, f_{K-1})$ on D—determines (i) every magnitude $|b_j|$ via $b_i^2 = c_{ik}c_{il}/c_{kl}$, (ii) every product $b_i b_j = c_{ij}$ and hence every relative sign* $\operatorname{sign}(b_i b_j)$, *and (iii) every benefit magnitude $|b_j - b_0|$. All are determined exactly up to the global flip $b \mapsto -b$; the decision direction* $\operatorname{sign}(b_j - b_0)$ *is not determined.*

Theorem C.9 (The flip witness: the bit is free at every moment order)**.** *For any H-target P, let P' complement the label ($Y' = 1 - Y$) leaving P^X and all predictors unchanged. Then P' satisfies H; the full evidence law is* identical, $\mathrm{TV}(\mathrm{Law}(f_{0:K-1} \mid P), \mathrm{Law}(f_{0:K-1} \mid P')) = 0$*; and $b' = -b$, so* $\operatorname{sign}(b'_j - b'_0) = -\operatorname{sign}(b_j - b_0)$ *for every candidate. Consequently no label-free statistic of the prediction channel—of* any *order, not merely pairwise—can recover the bit, even inside the fully audited class H.*

Proof of both. Identification: ratios and products of the observable c as displayed; well-defined by $|c_{kl}| \geq c_{\min}$; $|b_j - b_0|^2 = b_j^2 - 2c_{j0} + b_0^2$ is a function of c. Flip-invariance of $c = bb^\top$ and of all magnitudes is immediate; $\operatorname{sign}(b_j - b_0)$ flips. Witness: predictions are maps of X whose law is unchanged, so the evidence law is identical (TV $= 0$); $C'_j = 1 - C_j$ gives $b'_j = -b_j$; H is preserved since $q'_j(y) = 1 - q_j(1-y)$ and per-class symmetry carries over; CEI is a relabeling. □

Remark (why third moments do not rescue the sign). Spectral and tensor methods (**Jaffe et al.** 2015; the Dawid–Skene line) are sometimes read as fixing the sign via third-order statistics. Theorem C.9 shows what they can and cannot do: observable moments of every order are flip-invariant, so third-order structure identifies the latent class-conditional geometry only *up to the label permutation*—and for binary Y that permutation *is* the bit. The better-than-chance anchors those works adopt are not estimators of the bit; they are selections of it.

Theorem C.10 (Necessity: every identifying assumption is a bit-selector)**.** *Let $\mathcal{C}$ be any class of H-targets on which* $\operatorname{sign}(b_a - b_0)$ *is identifiable from the evidence law. Then $\mathcal{C}$ contains at most one member of each flip-pair $\{P, P'\}$ with $b_a \neq b_0$. Conversely, selecting one member of each flip-pair suffices. Identifiability of the decision direction is therefore* exactly *a one-bit selection.*

Proof. If both $P, P' \in \mathcal{C}$, Theorem C.9 puts opposite decision signs on one evidence law, contradicting identifiability; sufficiency holds because on each evidence-fiber the surviving member determines the sign. □

Theorem C.11 (The dictionary: published devices select the same bit)**.** *Under H: (i) the γ-budget certificate with $|M| > \beta$ selects the bit* $\operatorname{sign}(M)$ *(the flip sends $\gamma \to -\gamma$, fixes the observable M); (ii) majority-above-chance (Proposition C.14;* ***Parisi et al.*** *2014) selects* $\operatorname{sign}(\operatorname{median}_j b_j)$*; (iii) the anchored agreement-law slope (Theorem C.2) selects* $\operatorname{sign}(1 - 2\bar{w})$*, since the flip sends $w \mapsto 1 - w$ pointwise and hence negates the slope. Each device is sound iff its selected bit equals the true bit; none is testable from the evidence channel (Theorem C.9).*

Proof. (i) $\operatorname{sign}\Delta = \operatorname{sign}(M + \gamma) = \operatorname{sign} M$ when $|M| > \beta \geq |\gamma|$; $\gamma = \mathbb{E}[\eta_a - s] \mapsto \mathbb{E}[(1 - \eta_a) - s] = -\gamma - 2(\mathbb{E}[s] - \frac{1}{2}) + \ldots$ — computed cleanly via $\gamma = b_a/2 - M$ (Lemma C.2): the flip negates b_a and fixes M. (ii) $b'_j = -b_j$ reverses every $\operatorname{sign} b_j$, hence the majority count. (iii) on D_θ exactly one of f_0, f_θ is correct, so $w'(\theta) = 1 - w(\theta)$ and $1 - 2\bar{w} \mapsto -(1 - 2\bar{w})$. □

Validated (`V1_flip_witness`; Fig. C.9): $\max_j |b'_j + b_j| = 1.1 \times 10^{-16}$, evidence TV $= 0$ exactly, agreement products invariant to MC precision, all benefit signs opposite.

Honest scope. Theorem C.8 packages classical product-ratio identifiability (**Parisi et al.**; **Jaffe et al.**; **Ibrahim et al.**) and is not claimed as new; the new content is the exact-deficit correction (Theorem C.7), the all-orders TV-zero witness inside H (Theorem C.9), the necessity statement (Theorem C.10), and the dictionary (Theorem C.11). The one-bit reduction sharpens Corollary C.1: the irreducible content of *every* budget assumption in this paper is the same single bit.

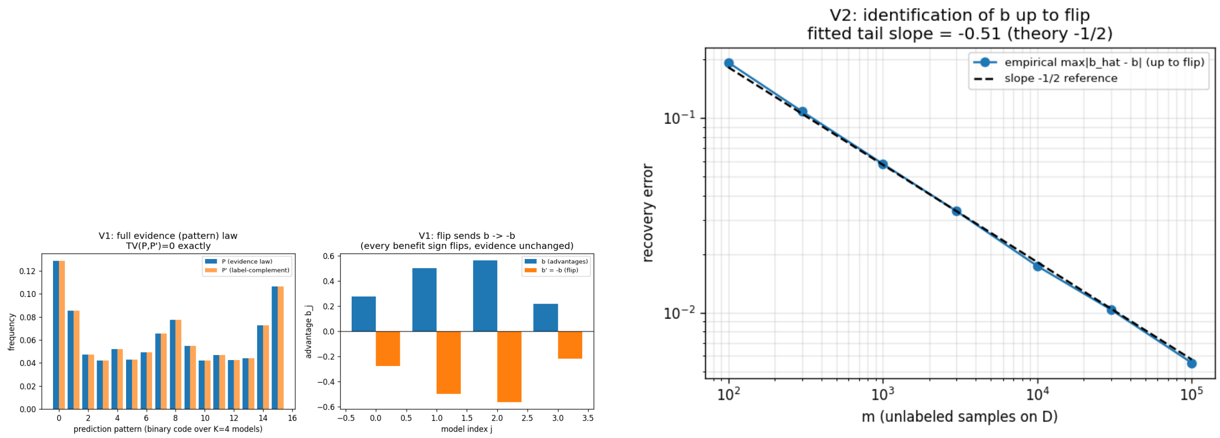

Figure C.9: The one-bit theorem, validated. Left (Thm C.9): the flip witness P' matches the full evidence law of P (TV $= 0$; identical agreement products) while every advantage and benefit sign is negated. Right (Thm C.8): label-free recovery of b up to the bit at the $m^{-1/2}$ rate (log-log slope -0.51).

C.6.3 The budget audit: falsifiable, never verifiable

Lemma C.2 (Label-free identification of the drift, up to the bit)**.** *Under H, the calibration drift of §C.3 satisfies $\gamma = \bar{a}_a - \frac{1}{2} - M = b_a/2 - M$. Hence, given the evidence law, $\gamma \in \{+|b_a|/2 - M,\ -|b_a|/2 - M\}$: the unobservable budgeted quantity is* point-identified up to the bit, *with M observable and $|b_a|$ given by Theorem C.8.*

Theorem C.12 (Bit-robust budget falsification)**.** *From m samples on D build $\hat{M}$ with an empirical-Bernstein radius $r_m^{(M)}(\alpha/2)$ and $\hat{b}_a$ from agreement estimates with the product-ratio radius $r_m^{(b)}(\alpha/2)$ of Lemma C.3; set $r_m = r_m^{(M)} + \frac{1}{2} r_m^{(b)}$ and*

$$\begin{aligned}&\texttt{reject}\ (|\gamma| \le \beta) \iff \\ &\min\left(\left|+\tfrac{|\hat{b}_a|}{2} - \hat{M}\right|,\ \left|-\tfrac{|\hat{b}_a|}{2} - \hat{M}\right|\right) \ >\ \beta + r_m.\end{aligned}$$

Then (soundness) if the budget truly holds the audit rejects with probability $\le \alpha$; (power) if $\min_{\text{flip}} |\gamma| > \beta + 2r_m$ it rejects with probability $\ge 1 - \alpha$. The min *over flips is what makes soundness hold for the unknown true bit.*

Lemma C.3 (Product-ratio perturbation)**.** *If all $|\hat{c} - c| \le e_c \le c_{\min}/2$ entrywise with $|c| \in [c_{\min}, 1]$, then $|\hat{b}_i^2 - b_i^2| \le C_2 e_c$ with $C_2 = (4c_{\min} + 2)/c_{\min}^2$, and $|\hat{b}_i - b_i| \le C_2 e_c / (2\sqrt{b_{\min}^2 - C_2 e_c})$ when $C_2 e_c < b_{\min}^2$.*

Validated (`V3_audit`; Fig. C.10): level $0.000 \le \alpha{=}0.05$ over 600 trials at $m{=}6000$; power 1.000 at $\min_{\text{flip}} |\gamma| = 0.216 > \beta{=}0.05$ (bootstrap radius). *Honest caveat, proven and reported:* the worst-case constant C_2 is large at small $c_{\min}$ ($C_2 \approx 84$ at $c_{\min}{=}0.18$), making the worst-case radius sound but nearly powerless at realistic m (rejection 0.000 in the same sweep); the deployable test

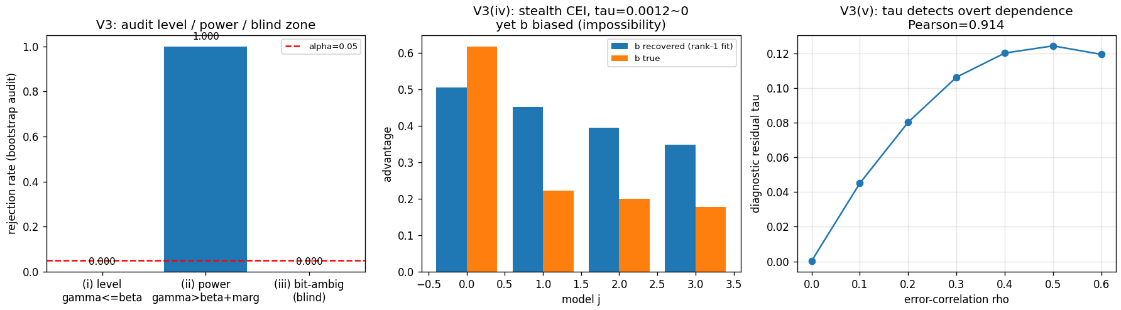

Figure C.10: The budget audit (Thms C.12, C.13). Sound at level α under a true budget; power 1.000 against violations beyond the radius (bootstrap); correctly blind on the bit-ambiguous set $\mathcal{B}_\beta$ and on the rank-one-preserving stealth law ($\tau \approx 0$) — the two provably irreducible blind zones; τ rises with overt error-correlation ρ (Pearson 0.914).

uses a bootstrap radius and the worst-case version supplies the guarantee. Closing this gap (a tight finite-sample radius) is stated as an open problem below.

Theorem C.13 (Verification impossibility; the audit's exact blind set)**.** (a) Within H. *No label-free test can accept "$|\gamma| \le \beta$" with power: verification fails exactly on the bit-ambiguous set*

$$\mathcal{B}_\beta = \left\{ \,\big|\,|M| - \tfrac{|b_a|}{2}\big| \;\le\; \beta \;<\; |M| + \tfrac{|b_a|}{2} \,\right\},$$

on which one bit satisfies the budget, the other violates it, and both share one evidence law (Theorem C.9); a sound procedure must abstain there, and the audit of Theorem C.12 correctly does not reject. (b) Outside H. *The overdetermination residual τ (Proposition C.15) is* sound but not complete*: there exist strictly interior $K{=}4$ correctness laws with every pairwise c_{ij} exactly rank-one ($\tau = 0$) that violate CEI and leave b unidentified—including witnesses whose fitted decision sign is wrong. What $\tau = 0$ pins down is precisely that c is rank-one, i.e.* pairwise-realizability by some H-model*; higher-order moments remain free.*

Proof sketch. (a) is Lemma C.2 plus Theorem C.9: the two flip values of γ straddle the budget exactly on $\mathcal{B}_\beta$. (b) is a linear-program construction on the 2^4 correctness-pattern probabilities: impose the six pairwise second-moment equations $\mathbb{E}[S_i S_j] = b_i^{\text{fit}} b_j^{\text{fit}}$, total mass, nonnegativity, and a first-moment constraint $\mathbb{E}[S_1] \ne b_1^{\text{fit}}$; the Chebyshev center is strictly interior, realizes $\tau = 0$, and its third moment $\mathbb{E}[S_1 S_2 S_3] \ne \prod_i \mathbb{E}[S_i]$ certifies the CEI violation. Full details and the cached witnesses: `theory_v2/THEORY_V2_PROOFS.md`, `stealth_law.json`, `stealth_flip.json`. □

Validated (`V3_audit.stealth_tau0`, `V3_audit.bit_ambiguous_blind`): design $\tau = 2.1 \times 10^{-17}$ with recovery bias 0.229; a second witness flips the recovered decision sign (+1 vs. true −1) at $\tau \approx 0$; on $\mathcal{B}_\beta$ the audit rejects at rate 0.000 (correctly blind). Against *overt* dependence the residual is informative: τ rises from 2×10^{-4} to 0.119 as error-correlation ρ goes $0 \to 0.6$ (Pearson 0.914).

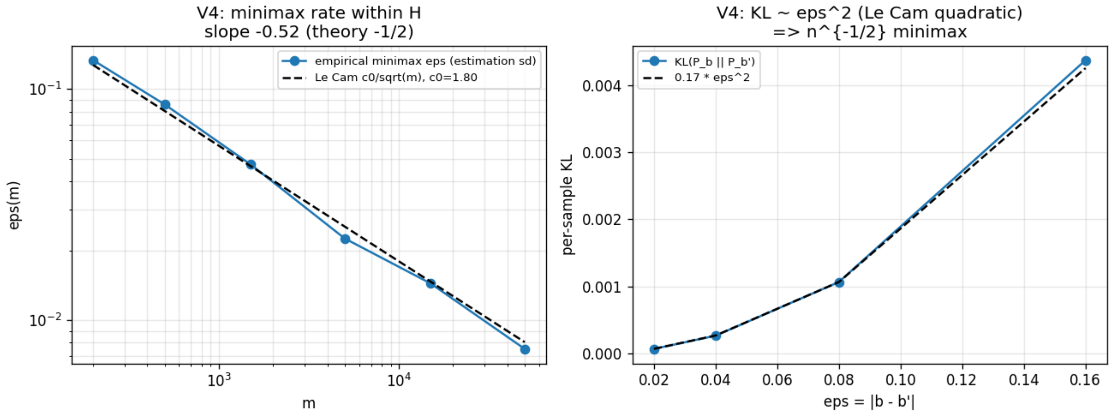

Figure C.11: The evidence-channel rate (Thm C.14). Empirical minimax error tracks the Le Cam curve $c_0/\sqrt{m}$ (slope -0.525); per-sample KL grows quadratically in the separation (KL$/\varepsilon^2$ constant); the labeled-vs-evidence efficiency ratio is constant in m but grows as $c_{\min} \to 0$.

Honest scope. Falsify-not-verify is the classical epistemic position of exchangeability testing (**Vovk et al.**) and assumption-alarms appear in the algebraic-evaluation program (**Corrada-Emmanuel**, NeurIPS 2024) and in dependent-classifier diagnostics (**Jaffe et al.** 2016); **Rosenfeld & Garg** (2023) bound error under an explicitly unverifiable assumption. The new content here is the *statistical* level-α rejection test for a *continuous budget* on a quantity that is unobservable yet point-identified up to a provably irreducible bit; the exact blind set $\mathcal{B}_\beta$; and the constructive incompleteness of τ. Within this framework the earlier budget devices (Propositions on η-transfer, β-drift, B-drift) stop being unexaminable: each budget premise acquires a sound falsifier, and what no falsifier can ever reach is characterized exactly ($\mathcal{B}_\beta$ within H; rank-one-preserving violations outside H, Theorem C.13(b)).

C.6.4 The evidence-channel rate

Theorem C.14 (Matching $m^{-1/2}$ bounds for the label-free channel)**.** *Within audited H with margins $(c_{\min}, b_{\min})$ and given the bit:* (a) *the product-ratio estimator satisfies, with probability $\geq 1-\delta$, $|\hat{b}_j - b_j| \leq C(c_{\min}, b_{\min})\sqrt{\log(K/\delta)/m}$ with $C = \Theta(1/(b_{\min}c_{\min}^2))$, so* $\operatorname{sign}(b_j - b_0)$ *is certified at sample complexity $m \asymp C^2(b_j - b_0)^{-2}\log(K/\delta)$;* (b) *a Le Cam two-point argument inside H (two models differing by ε in one coordinate; the per-sample KL between prediction-pattern laws is $\leq C'\varepsilon^2$) shows no estimator can beat $m^{-1/2}$;* (c) *consequently labels buy* constants and the bit, not the rate*: the labeled-vs-evidence efficiency ratio is constant in m but scales like $1/c_{\min}$ and diverges as pairwise agreement approaches chance.*

Validated (`V2_identification_rate`, `V4_rate`; Figs. C.9, C.11): recovery slope -0.510 (theory -0.5); empirical minimax error tracks $c_0/\sqrt{m}$; KL$/\varepsilon^2 \in [0.164, 0.171]$ across ε (quadratic separation, as required); efficiency ratio 2.15 at $c_{\min}$=0.21, 4.49 at 0.067, divergent at ≤ 0.022.

Honest scope. The tools are classical (Hoeffding/Bernstein/Le Cam; cf. **Jaffe et al.** 2015 for asymptotics of the same estimator); the contribution is the two-sided minimax statement *for the evidence channel within a falsifiable class*, answering the open question left by the knowability-rate propositions (Props. C.22ff.): those concerned mean estimation from *labeled* paired benefits, whereas Theorem C.14 is the genuinely label-free rate. The constant's $1/c_{\min}^2$ dependence is real and reported, not optimized; tightening it is open.

C.6.5 Consolidation and what remains open

Corollary C.4 (The budget propositions, unified)**.** *Under H, the calibration-transfer premise (Prop. 9), the β-drift premise (Prop. 11), the bounded-drift premise (Prop. 13), and the majority anchor (Prop. C.14) each decompose as* (testable structure) + (one bit)*: the structure is falsifiable by the audit of Theorem C.12 together with the residual* τ*, and the bit is irreducible by Theorem C.10. In this precise sense the paper's trichotomy is complete:* ADAPT/FREEZE *when the audited structure plus a declared bit certify the sign with margin;* ABSTAIN *when the audit rejects, or when the margin falls inside the blind set* $\mathcal{B}_\beta$*.*

C.6.6 Closing Conjecture 5.1: the testability dichotomy

Conjecture 5.1 asks for a label-free bracketing of the ordinal comparison on D "without target labels." The γ-meter section (§C.5) resolves it *relative to checkable agreement structure* and leaves the unconditional question open. We now close the unconditional question — negatively, with an exact characterization. Validated: `theory_v2/val_conj1_closure.py` → `conj1_closure_results.json`; Fig. C.12.

Definition C.7 (Evidence-definable class)**.** Let $L(P)$ denote the joint law of *all* label-free observables under target P (labeled source, fixed predictor maps, the unlabeled target stream — hence every label-free statistic of every order). A class $\mathcal{C}$ of targets is *evidence-definable* (label-free testable in the population limit) if membership depends on P only through $L(P)$.

Lemma C.4 (The swap)**.** *In each output space there is an involution* $P \mapsto P^\star$ *with* $\mathrm{TV}(L(P), L(P^\star)) = 0$ *and* $\Delta(P^\star) = -\Delta(P)$*:* (a) *multiclass (binary included): on D swap, pointwise in x, the conditional masses of the two disagreeing predicted values,* $P^\star(Y{=}f_a(x) \mid x) = P(Y{=}f_0(x) \mid x)$ *and vice versa;* (b) *regression: reflect Y about the prediction midpoint on D,* $Y^\star = f_0(X) + f_a(X) - Y$*;* (c) *for binary the swap is the D-local label complement and preserves H, mapping* $b \mapsto -b$*.*

Proof. Each $P^\star$ is a valid law preserving the X-marginal; predictions are fixed maps of X and the source is untouched, so $L(P^\star) = L(P)$ exactly, at every moment order. (a) exchanges $p_a(x) \leftrightarrow p_0(x)$ pointwise on D, so $\Delta^\star = P(D)(p_0 - p_a) = -\Delta$ (off D, $\delta = 0$). (b) negates $f_0{+}f_a{-}2Y$ pointwise on D, so the benefit density negates. (c) is (a) with $K{=}2$: $q_j \mapsto 1 - q_j$ preserves CEI and per-class symmetry. □

Theorem C.15 (Resolution of Conjecture 5.1; testability dichotomy)**.** (i) **No testable bracketing.** *If $\mathcal{C}$ is evidence-definable and contains any target with $\Delta \neq 0$, then* $\operatorname{sign}\Delta$ *is not identifiable on $\mathcal{C}$: by Lemma C.4, $P^\star \in \mathcal{C}$ shares P's evidence law with the opposite sign. The unconditional label-free bracketing sought by Conjecture 5.1 therefore* does not exist *— in any output space, at any evidence order, with unlimited labeled source data.* (ii) **The minimal supplement is one bit, and it suffices.** *The swap pairs each $\Delta \neq 0$ target with an opposite-sign twin on the same evidence fiber; identifying* $\operatorname{sign}\Delta$ *on a class is exactly selecting one member per pair — one bit, irreducible by (i). Conversely, falsifiable structure plus one declared bit identifies* $\operatorname{sign}\Delta$ *at the minimax $m^{-1/2}$ rate: binary under H by Theorems C.8 and C.14; multiclass one-coin/tensor-regular families by Theorem C.6. Hence every conditional resolution in this paper (calibration transfer, γ-meter, H+anchor) is of the* minimal possible form*: falsifiable structure + one bit.*

Proof. (i) Any decision rule with asymptotic guarantees on $\mathcal{C}$ is a functional of L; on the fiber $\{L(P)\}$ it outputs one (possibly randomized) action while $P, P^\star \in \mathcal{C}$ demand opposite committal actions; the committal-regret argument of Theorem 4.1 applies with the pair constructed *inside* $\mathcal{C}$. (ii) The swap is an involution reversing the sign, so $\{\Delta \neq 0\}$ partitions into orbits of size two with opposite signs; meeting each orbit at most once is a one-bit selection; sufficiency is the cited theorems. □

Remark (the algebraic heart). For binary Y write $v = 2Y-1$ (latent) and $u_j = 2g_j-1$ (observable); correctness signs are $s_j = u_j v$. Label-free observables generate the *even* subalgebra ($s_i s_j = u_i u_j$, $\ldots$); the decision target $\operatorname{sign}\Delta \propto \operatorname{sign}\mathbb{E}[s_a - s_0]$ is *odd*. The swap is $v \mapsto -v$ on D: it fixes every observable and negates every odd element. Odd-order correctness moments (e.g. $\mathbb{E}[s_1 s_2 s_3]$, §C.5) are label-carrying — estimating them presupposes an anchored frame — so the reach hierarchy $r_2 > r_3 > \cdots$ of §C.5 measures residual *magnitude* ambiguity within a declared frame, while the raw observable channel retains the sign bit at every order. This answers the question §C.5 leaves open: however fast r_k shrinks, its limit cannot break the swap orbit. *What is observable is even; what is decided is odd; the quotient is one bit.*

Honest scope and positioning. The swap constructions are elementary; the value is the characterization, which upgrades Corollary C.1 from "some supplied bound is needed" to "no testable condition suffices, and the untestable content is exactly one bit." **Rosenfeld & Garg** (NeurIPS 2023, Remark 3.9) assert the impossibility informally; (i) is its formalization, with the mechanism and the minimal residual. **Ben-David et al.** (AISTATS 2010) build observably indistinguishable tasks for learnability, not benefit-sign identification over testable classes; the testable-learning-with-shift line (**Klivans et al.**) certifies agnostic *learning*, a different estimand; **Manski**-style partial identification (sign of a treatment effect requires untestable monotonicity) is the conceptual precedent from causal inference — this is its exact analogue for label-free adaptation, with the supplement quantified. A targeted live search (June 2026) found no prior statement of the dichotomy as a theorem over testable classes.

Validated (`conj1_closure_results.json`): multiclass K=5 swap — evidence statistics (pairwise

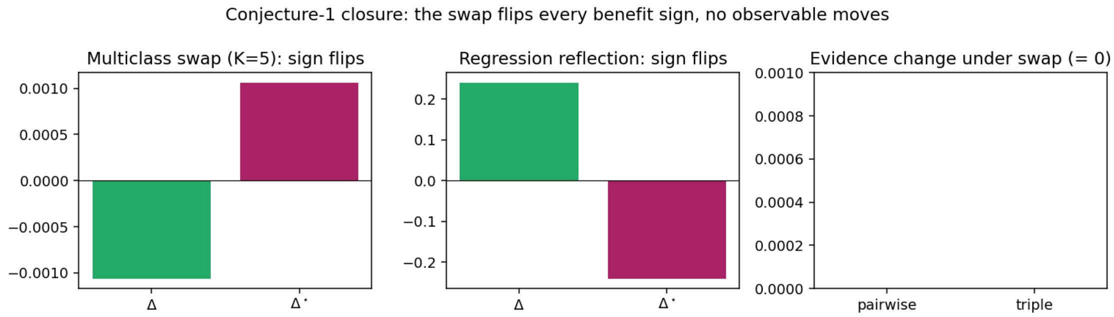

Figure C.12: Closing Conjecture 5.1 (Thm C.15): the swap flips every benefit sign (multiclass K=5, left; regression, middle) while no label-free observable of any order moves (right, exact zero) — so no testable assumption can bracket the sign, and the residual is exactly one bit.

+ triple agreements, marginals) change by exactly 0, accuracies exchange exactly ($p_a^\star = p_0$, $p_0^\star = p_a$), $\Delta^\star = -\Delta$ to 0; regression reflection — $\Delta^\star = -\Delta$ exactly; binary H-swap — agreement matrix invariant (0 change), $b^\star = -b$ to 1.1×10^{-16}, τ unchanged (≈ 0).

Open problems (the honest frontier). (1) A tight (non-conservative) finite-sample radius for the audit—the proven C_2 constant is loose; the bootstrap version is powerful but its validity is empirical. (2) With Conjecture 5.1 resolved as a dichotomy (Theorem C.15), the surviving structural question: the weakest *falsifiable* class beyond H on which one bit still suffices — any such class must exclude the rank-one-preserving stealth laws of Theorem C.13(b), and the r_k reach hierarchy of §C.5 measures its magnitude side. (3) Per-class asymmetric panels: quantitative recovery when $\delta \neq 0$ but $\|\delta\|$ is small, via the exact deficit of Theorem C.7.

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